MODULE 00 · EXAM STRATEGY - ALL DOMAINS

AWS Certified Solutions Architect – Professional

SAP-C02. What the exam actually tests, and how this course is built.

EXAM **SAP-C02** QUESTIONS **75 in 180 min** PASS **750 of 1000**

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Who this exam is for

- **AWS recommends two or more years of hands-on experience** designing and managing solutions on AWS. There are no formal prerequisites.
- **It tests judgement, not recall.** Most questions have two defensible answers and one that fits the stated constraint.
- **The stems are long.** A scenario runs to a paragraph or two, and the deciding detail is often one clause inside it.
- If you hold the Associate, expect a different exercise. This one is about trade-offs across an organisation, not single services.

> ON THE EXAM
>
> The Associate asks which service. The Professional asks which trade-off, given a constraint somebody else has already fixed.

The format, precisely

- **75 questions in 180 minutes.** 65 are scored; 10 are unscored trials and are not identified.
- **Scaled score of 100 to 1,000, with 750 to pass.** Scoring is compensatory: you pass overall, not per domain.
- **Unanswered scores as incorrect, and guessing carries no penalty.** Never leave a question blank.
- A list of AWS service short names is available through the **Help button** during the exam.

> ON THE EXAM
>
> 750 out of 1,000 is not 75%. Scaled scoring means you cannot convert the cut score to a number of questions.

The four domains

DOMAIN	WEIGHT	WHAT IT IS REALLY ABOUT
1 - Organisational Complexity	26%	Many accounts, many teams, one estate
2 - New Solutions	29%	Choosing between defensible designs
3 - Continuous Improvement	25%	Fixing what already runs
4 - Migration and Modernisation	20%	Moving it, and knowing when not to

ON THE EXAM

Domain 4 is 20% of the exam and under 4% of most question banks. It is the most under-prepared domain by a wide margin.

Warning one: this exam retires soon

- **The last day to take SAP-C02 is 16 November 2026.** Registration for SAP-C03 opens on 27 October.
- **SAP-C03 moves to five domains**, adding generative and agentic AI, post-quantum cryptography and resilience engineering.
- **If you are prepared, book now.** The certification is valid for three years whichever version you sit.
- Retakes need a 14-day wait, so book early enough to fail once and still sit it again before the deadline.

> ON THE EXAM
>
> Recorded on 4 September 2026. Check the AWS certification page before booking, because dates do move.

Warning two: some answers no longer exist

- **Migration Hub, Application Discovery Service, the Snow Family, App Runner and Simple AD are all closed to new customers.**
- **Every one of them is still on the exam's in-scope list**, so every one is still a correct answer.
- Answer the in-scope list on the day. Knowing more than the exam can genuinely cost you marks.
- Modules 42 and 43 cover this in full: what was renamed, what closed, and what is simply gone.

> ON THE EXAM
>
> If you study alongside current AWS documentation you will hit this contradiction fast. Better to hear it now than at module 33.

How this course was built

- **The exam guide was fetched live**, so the domain weights and the in-scope service list are the current ones, not remembered.
- **Every fact was checked against current AWS documentation** while writing. Where something had changed, the module says so.
- **A 492-question bank was used as a topic map only**, to weight coverage. No question from it is reproduced anywhere.
- **Every scenario in this course is original**, and most follow one fictional company so the architecture accumulates.

> ON THE EXAM
>
> Where AWS's own pages disagreed with each other, the course says so rather than picking one and sounding certain.

How to read the slides

- **The coloured marker on each bullet is a signal.** Amber for a key fact, red for a trap, teal for a nuance.
- **The boxed line at the foot of a slide is the exam-facing point**, usually the one property that separates two options.
- **The bar down the left edge shows which domain you are in**, and moves as the course progresses.
- Narration carries the reasoning. The slide carries the structure, so listening matters more than reading here.

> ON THE EXAM
>
> Nothing important is on a slide and absent from the narration. You can listen to this course without watching it.

The worked scenarios, and the pause

- **Every teaching module ends with six worked scenarios**, written at the difficulty of the real exam.
- **When the narration says pause, pause.** Commit to an answer before the walkthrough plays.
- **Every option gets an explanation**, including the wrong ones, because knowing why a good answer fails is the actual skill.
- Watching without answering first feels productive and teaches almost nothing.

> ON THE EXAM
>
> 343 worked scenarios and questions across the course. They are the part that moves your score, not the teaching slides.

The course, measured

SECTION	MODULES	RUNTIME
Domain 1 - Organisational Complexity	02 to 11	2h 53m
Domain 2 - New Solutions	12 to 22	2h 50m
Domain 3 - Continuous Improvement	23 to 32	2h 25m
Domain 4 - Migration and Modernisation	33 to 40	1h 53m
Cross-cutting: pairs, currency, technique	41 to 44	0h 53m
Five practice exams	45 to 49	1h 49m

ON THE EXAM

Teaching runs 10 hours 54 minutes; with the practice exams, about 12 hours 49 minutes. These are measured from the rendered video, not estimated.

Finding your way around

IF YOU WANT	GO TO
The policy foundations everything rests on	00:00 - Guardrails
Connectivity, DNS and private access	1:11 - Multi-VPC through 2:20
Choosing between compute, storage, databases	3:24 - Choosing Compute onward
Anything you already run and want to improve	5:43 - Observability Foundations
The whole migration domain	8:08 - Portfolio Assessment onward
Revision, currency and technique	10:01 - Confusion Pairs onward

ON THE EXAM

Chapter marks are on the video. The four cross-cutting modules from 10:01 are the ones to rewatch in the last week before the exam.

A study route

- **Watch modules 02 to 44 in order.** Later modules reference earlier ones deliberately, and the scenarios build.
- **Then sit practice exam 1.** Score it, and rewatch only the modules behind your misses.
- **Space the remaining exams out.** Taking all five in a weekend measures fatigue rather than knowledge.
- **Count anything you guessed correctly as a miss.** Otherwise the score flatters you and the gap stays.

> ON THE EXAM
>
> Each practice exam ends with a table mapping every question to the module that teaches it, so a miss is immediately actionable.

Two notes on language

- **The prose is British.** Organisation, optimise, analyse, licence, centre.
- **AWS product names keep their US spelling.** The pillar is Cost Optimization, and it is AWS Organizations.
- Mixing those would look like an error to an examiner, so the inconsistency is deliberate.
- Depth is set for passing the exam, not for implementation. Where more depth exists, the module says where to find it.

> ON THE EXAM
>
> Every module ends with a recap slide naming the four things worth carrying out of it. Those are the revision notes.

Module 02 opens Domain 1 at 00:00 with guardrails: SCPs, RCPs and how policies actually evaluate.

MODULE 02 · DOMAIN 1 - DESIGN SOLUTIONS FOR ORGANIZATIONAL COMPLEXITY

Guardrails: SCPs, RCPs and Policy Evaluation

Five policy types decide every request. This is the order they run in.

DOMAIN 1 of 4 EXAM WEIGHT 26% MODULE 2 of 44

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A service control policy never grants anything. It only **defines the ceiling** that grants must fit under.

The five policy types that can affect one request

- **Service control policy.** Attached in Organizations. Caps what IAM principals in an account may do.
- **Resource control policy.** Also in Organizations. Caps access **to** that account's resources, whoever asks.
- **Identity-based policy.** On a user, group or role. One of the two places a grant can come from.
- **Resource-based policy.** On the resource itself. The other place a grant can come from.
- **Permissions boundary** and **session policy.** Ceilings on one principal, never grants.
- Only two of the five can say yes. The other three can only say no.

> ON THE EXAM
>
> Two granters, three limiters. A limiter offered as the way to give access is always wrong.

How AWS actually decides

01

Explicit deny?

Any deny in any policy. Stop. Denied.

›

02

SCP allows?

Action must be inside the SCP ceiling.

›

03

RCP allows?

Resource must be reachable under the RCP.

›

04

Boundary allows?

Action must be inside the boundary.

›

05

Grant exists?

Identity or resource policy must allow it.

ON THE EXAM

One explicit deny anywhere ends the evaluation. Everything else must all agree before a request succeeds.

Think of it as four gates and one key. The gates can only be shut. Only the key opens anything.

SCP or RCP?

SERVICE CONTROL POLICY

Caps what your principals do

- Follows the **identity** making the request
- Applies to IAM users and roles here
- Skips the management account
- Use: deny unapproved Regions

RESOURCE CONTROL POLICY

Caps access to your resources

- Follows the **resource** being touched
- Applies to **any** caller, even outsiders
- Also skips the management account
- Use: stop a bucket being shared out

THE TIEBREAKER

Ask who the scenario is worried about. Our people doing the wrong thing means SCP. Outsiders reaching our data means RCP.

The management account exception

- **SCPs and RCPs do not apply to the management account.** Attach one at the root and it is still ignored.
- This is why the guidance is to run **no workloads** in the management account.
- It is also why sensitive functions get **delegated administrator** accounts instead.
- Service-linked roles are exempt too, so AWS services keep working under a restrictive policy.

> ON THE EXAM
>
> Nothing you attach in Organizations constrains the management account. Treat it as outside the guardrails, always.

Permissions boundary or session policy?

PERMISSIONS BOUNDARY

A durable ceiling on one principal

- Attached to a user or role and stays there
- Survives across every session
- Classic use: let developers create roles safely
- The delegation pattern the exam tests most

SESSION POLICY

A temporary ceiling on one session

- Passed at **AssumeRole** time, in the API call
- Applies only to that set of credentials
- Classic use: one broad role, narrowed per customer
- Common in SaaS multi-tenant designs

THE TIEBREAKER

Permanent limit on a principal means boundary. Limit that varies per session means session policy.

Cross-account access needs two yeses

REQUEST	WHAT MUST ALLOW IT	THE COMMON MISTAKE
Same account, identity policy	Identity policy alone	Resource policy is optional here
Same account, resource policy	Resource policy alone	Either one suffices in-account
Cross-account: S3, KMS, SQS	**Both** sides	Assuming the resource policy is enough
Cross-account via AssumeRole	Trust policy **and** caller	Forgetting sts:AssumeRole on the caller
Cross-account with SCPs	Both accounts' SCPs	Checking only their own account

ON THE EXAM

In-account, either policy suffices. Across accounts, both sides must say yes.

Where SCPs are actually placed

- **Policies inherit downward and intersect.** An action must be allowed at root, OU and account.
- Inheritance is not a union. A narrow policy above cannot be widened below.
- AWS attaches **FullAWSAccess** by default. Detach it and you have denied everything.
- Common design: **allow-list at the root, deny-list at the OU**, so exceptions sit near the workloads.

> ON THE EXAM
>
> An action must survive every level. The effective permission is the intersection, never the union.

An SCP that **allows** a service does nothing on its own. Someone still has to be granted the permission.

Reading the scenario for the answer

THE SCENARIO SAYS	IT IS POINTING AT
Prevent any account from doing X, ever	SCP with an explicit deny
Stop data being shared outside the organisation	RCP, or aws:PrincipalOrgID
Let developers create roles but not escalate	Permissions boundary
One role, different limits per tenant	Session policy at AssumeRole
Third party needs access to our account	Cross-account role with external ID
The control did not apply to one account	That account is the management account

ON THE EXAM

Most policy questions are decided by one phrase. Spot the phrase rather than reasoning it out under time pressure.

Condition keys the exam expects you to know

- **aws:PrincipalOrgID** - the caller belongs to your organisation. The cleanest way to scope a resource policy without listing accounts.
- **aws:PrincipalOrgPaths** - narrower still. The caller sits under a specific OU.
- **aws:SourceArn** and **aws:SourceAccount** - prevent the confused deputy when a service calls on someone's behalf.
- **aws:RequestedRegion** - the Region restriction pattern, used in almost every landing zone.
- **aws:ResourceOrgID** - the resource belongs to your organisation. Blocks staff sending data to a personal account.

> ON THE EXAM
>
> PrincipalOrgID controls who reaches in. ResourceOrgID controls where your people can send things out.

> Meridian Freight has 140 accounts under AWS Organizations. A compliance rule states that no workload may run outside eu-west-1 and eu-west-2. The security team attaches an SCP at the organisation root denying all actions when aws:RequestedRegion is not one of those two Regions.
>
> A week later, an auditor finds EC2 instances running in us-east-1 in one account. The SCP is confirmed as attached at the root and correctly written.

What is the MOST likely explanation?

A SCPs take up to 24 hours to propagate to all accounts

B The account is the organisation's management account

C The instances were launched before the SCP was attached, and SCPs are not retroactive

D A permissions boundary in that account overrides the SCP

◆ PAUSE AND DECIDE

What is the MOST likely explanation?

A Propagation delay

SCP changes take effect in seconds, and a week has passed regardless. This option invents a limitation that does not exist.

B It is the management account

SCPs never apply to the management account, however they are attached. This is the only explanation consistent with a correct policy at the root.

C Not retroactive

True but irrelevant. Existing instances would keep running, but the scenario says instances are running, not that they were launched recently. And a genuinely blocked account could not have launched them at all.

D Boundary overrides SCP

Backwards. A boundary is another ceiling, not an override. Nothing below the SCP can widen it.

THE SIGNAL

Correct policy, correct attachment, one account unaffected. That combination has exactly one cause.

A platform team wants application teams to create their own IAM roles for their workloads, so the platform team is no longer a bottleneck. Security requires that a team can never create a role with more privilege than the team itself holds.

The company already uses SCPs at the OU level to restrict which services each business unit may use.

What should the platform team implement?

A Tighten the OU's SCP to deny iam:CreateRole for all application teams

B Grant iam:CreateRole but require that created roles carry a permissions boundary, enforced with a condition

C Grant iam:CreateRole and pass a session policy when application teams sign in

D Grant iam:CreateRole and review created roles daily with IAM Access Analyzer

◆ PAUSE AND DECIDE

What should the platform team implement?

A Deny role creation

Solves the security requirement by abandoning the business requirement. The whole point was to remove the bottleneck.

B Require a boundary on created roles

The canonical safe-delegation pattern. The condition means a role cannot be created at all unless it carries the boundary, so escalation is structurally impossible.

C Session policy at sign-in

Limits the team's own session. It places no limit on roles they create, which are separate principals with their own permissions.

D Detective review

Finds the escalation after it has happened. The requirement says never, which is a preventive word.

THE SIGNAL

Delegate creation, constrain the creation. Boundary plus a condition requiring it is the pattern AWS documents for exactly this.

> A data science account must read objects from an S3 bucket in a separate production account. An administrator adds a bucket policy in the production account allowing the data science role's ARN to perform s3:GetObject.
>
> Calls still fail with an access denied error. The role has the AmazonS3ReadOnlyAccess managed policy attached, and no SCP in either account restricts S3.

What is the MOST likely cause?

A. The bucket policy alone is insufficient; the identity policy must also allow the specific bucket

B. The bucket's KMS key policy does not grant the data science role decrypt permission

C. Cross-account S3 access requires AWS RAM to share the bucket first

D. The role needs sts:AssumeRole on the production account

◆ PAUSE AND DECIDE

What is the MOST likely cause?

A. Identity policy also required

True as a rule, but the scenario already satisfies it. AmazonS3ReadOnlyAccess allows GetObject on every bucket, so the caller's side is covered.

B. KMS key policy missing the role

With both S3 policies satisfied, the remaining gate is the key. A bucket encrypted with a customer managed key needs the key policy to permit Decrypt for the cross-account role.

C. Share via AWS RAM

RAM shares things like subnets, Transit Gateways and Route 53 rules. S3 buckets are not RAM shareable, and never were.

D. sts:AssumeRole needed

Only if the design used role assumption. This design grants the role directly in the bucket policy, so there is no role to assume.

THE SIGNAL

When both S3 policies are correct and access still fails, the encryption key is the next gate. KMS is a separate authorisation decision.

> A regulated company must guarantee that data in its S3 buckets can never be accessed by any AWS principal outside its own organisation, across 200 accounts. Buckets are created continuously by application teams.
>
> A previous attempt used a bucket policy template with an aws:PrincipalOrgID condition, but audits keep finding buckets created without it.

What should the solutions architect recommend?

A. An SCP denying s3:PutBucketPolicy unless the policy contains aws:PrincipalOrgID

B. A resource control policy at the organisation root denying S3 access where aws:PrincipalOrgID does not match

C. S3 Block Public Access enforced organisation-wide

D. An AWS Config rule that flags non-compliant bucket policies for remediation

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A SCP on PutBucketPolicy

Constrains how policies are written, not what access is permitted. Buckets with no policy at all, or access granted by other means, slip straight past.

B RCP at the root

Applies to every bucket in every account automatically, including ones created tomorrow, and it caps access regardless of what the bucket policy says.

C Block Public Access

Blocks anonymous and public access only. A named principal in an external AWS account is neither, so the stated risk remains.

D Config rule

Detective. The requirement says never, and buckets appear continuously, so there is always a window between creation and remediation.

THE SIGNAL

This is exactly what RCPs were built for. Per-bucket policy templates depend on discipline; an RCP does not.

A SaaS provider runs one shared application role that can read from any customer's DynamoDB table. Each customer's data is partitioned by a leading key value. The provider must guarantee a request handled for customer A can never read customer B's items.

There are 4,000 customers and the number grows weekly. Creating a role per customer has been ruled out as unmanageable.

Which approach meets the requirement?

A Attach a permissions boundary to the shared role limiting it to one customer's keys

B Pass a session policy at AssumeRole scoping dynamodb:LeadingKeys to that customer

C Add an SCP to the account restricting DynamoDB access by partition key

D Apply a resource-based policy on the table for each customer

◆ PAUSE AND DECIDE

Which approach meets the requirement?

A **Boundary on the shared role**

A boundary is fixed to the principal. One shared role can carry only one boundary, so it would pin every request to the same single customer.

B **Session policy per assumption**

The scope changes per session, which is exactly what varies here. Each set of credentials is limited to one customer's leading keys.

C **SCP by partition key**

SCPs are account-wide ceilings. They cannot vary per request, and would apply identically to all four thousand customers.

D **Resource policy per customer**

DynamoDB has no resource-based policy of the kind described, and 4,000 entries would exceed policy size limits even if it did.

THE SIGNAL

The limit changes per request while the principal stays the same. Only a session policy varies at that granularity.

A third-party monitoring vendor needs read access to a company's CloudWatch metrics. The vendor asks the company to create an IAM role trusting the vendor's AWS account.

The company's security team is concerned that the vendor serves many customers, and that another of the vendor's customers could be tricked into causing access to this company's account.

What should the security team require?

A An external ID in the role's trust policy, supplied by the company

B An IAM user with long-lived access keys instead of a role

C An SCP denying all actions from principals outside the organisation

D A permissions boundary on the vendor's role limiting it to CloudWatch

◆ PAUSE AND DECIDE

What should the security team require?

A External ID in the trust policy

The documented answer to the confused deputy problem. The vendor must present a secret only this company knows, so another customer cannot cause access here.

B Long-lived access keys

Replaces temporary credentials with permanent ones handed to a third party. Strictly worse, and does not address confusion between the vendor's customers at all.

C SCP denying external principals

SCPs constrain principals in your accounts, not callers from outside. It would also block the access the company wants to grant.

D Boundary limiting to CloudWatch

Correctly limits scope, and is good practice, but scope was never the concern. The concern is who can trigger the assumption.

THE SIGNAL

Third party, many customers, risk of the wrong one triggering access. That is the confused deputy, and external ID is its named fix.

What to carry out of this module

- **Four gates and one key.** Deny, SCP, RCP and boundary can only shut. Only a grant opens.
- **SCPs follow the identity, RCPs follow the resource.** Only an RCP stops an outsider.
- Nothing in Organizations constrains the management account. Run nothing there.
- Boundary for a permanent limit. Session policy when the limit changes per session.
- **Never** and **prevent** are preventive words. They eliminate detective options.

> ON THE EXAM
>
> This chain reappears in every domain. When an access question looks ambiguous, walk the gates in order rather than guessing.

MODULE 03 · DOMAIN 1 - DESIGN SOLUTIONS FOR ORGANIZATIONAL COMPLEXITY

Control Tower and Landing Zones

Opinionated Organizations. Know what it automates and what it refuses to.

DOMAIN 1 of 4 EXAM WEIGHT 26% MODULE 3 of 44

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Control Tower does nothing Organizations cannot do. It does it **the same way every time**.

What Control Tower actually orchestrates

- **AWS Organizations** for the account tree, SCPs and RCPs.
- **AWS Service Catalog** underneath Account Factory. Vended accounts are provisioned products.
- **IAM Identity Center** for single sign-on and pre-built permission sets.
- **AWS Config** and **CloudTrail**, configured organisation-wide with logs centralised.
- **CloudFormation**, both to build the landing zone and to run proactive control hooks.
- It is an orchestration layer. Nothing here is a new service you have not already met.

> ON THE EXAM
>
> Naming the underlying services makes the limits obvious. Control Tower inherits every constraint they have.

What you get on day one

- **A Security OU**, created automatically. This one is required.
- **A Log Archive account** holding immutable CloudTrail and Config history for every account.
- **An Audit account** for security teams, with cross-account read access and notifications.
- A **Sandbox OU**, offered during setup. Recommended but optional.
- The **Infrastructure OU** and **Workloads OU** are recommended but **not created for you**.

> ON THE EXAM
>
> Security OU with Log Archive and Audit is automatic. Everything else in the reference diagram is your job.

The three control types

01

Proactive

CloudFormation hooks. Blocks non-compliant resources before provisioning.

02

Preventive

SCPs, RCPs and declarative policies. Denies the API call outright.

03

Detective

Config rules. Reports non-compliance after the fact.

ON THE EXAM

Three answers to one question: at what moment do you want to be stopped?

The three types in detail

TYPE	BUILT ON	STOPS IT?	BLIND SPOT
Preventive	SCPs, RCPs, declarative policies	Yes, at the API call	Management account is exempt
Proactive	CloudFormation hooks	Yes, before provisioning	Only sees CloudFormation
Detective	AWS Config rules	No, reports only	Damage already done

ON THE EXAM

Only preventive controls stop every path. Proactive controls stop only the CloudFormation path.

A proactive control only inspects CloudFormation. The console, the CLI, the SDKs and Terraform **walk straight past it**.

The three guidance levels

- **Mandatory.** Always applied, cannot be disabled. These protect the landing zone itself.
- **Strongly recommended.** Enabled by choice, based on AWS best practice. Disable if you must.
- **Elective.** Optional controls for common but not universal requirements.
- Guidance level is about **whether you can turn it off**. It says nothing about how the control works.
- Controls apply to an **OU**, and reach every account inside it, including nested OUs.

> ON THE EXAM
>
> Do not confuse the three guidance levels with the three control types. They are unrelated axes.

Drift, and why the exam likes it

- **Drift is divergence from what Control Tower expects.** Someone changed something outside Control Tower.
- Editing a Control Tower SCP directly in Organizations causes drift. So does moving an account between OUs by hand.
- Removing an account from the organisation, or deleting the Log Archive configuration, both drift.
- Control Tower detects drift and reports it. **It does not silently repair it.**
- The fix is to repair the change, then **re-register the OU** or update the landing zone.

> ON THE EXAM
>
> Adopting Control Tower means changes go through Control Tower. Working around it in Organizations is the definition of drift.

Control Tower or plain Organizations?

CONTROL TOWER FITS WHEN

You want a repeatable environment

- Many accounts, provisioned continuously
- Standard logging and audit is the ask
- Compliance wants consistent baselines
- A small team governs many accounts

PLAIN ORGANIZATIONS FITS WHEN

You need something it will not do

- A bespoke structure that must not change
- Requirements outside supported controls
- Mature automation already does this
- You need governance at the Root

THE TIEBREAKER

Consistency at scale means Control Tower. One unusual requirement it cannot express means Organizations directly.

Three ways to vend an account

APPROACH	WHAT IT IS	REACH FOR IT WHEN
Account Factory	The built-in console template	Standard accounts, low volume, humans clicking
Account Factory for Terraform	Terraform pipeline on top of Account Factory	You want accounts defined as code, with customisation per account
Customizations for Control Tower	CloudFormation and SCP pipeline, triggered on account events	You need extra resources or policies deployed into every new account

ON THE EXAM

Account Factory creates the account. AFT and CfCT are how you make every new account arrive already configured.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Must be blocked however the resource is created	Preventive control
Block it during CloudFormation deployment	Proactive control
Report which accounts are non-compliant	Detective control
Someone changed an SCP directly in Organizations	Drift
Every new account must arrive with a VPC and roles	Customizations, or AFT
One control must apply at the Root	Control Tower cannot; use Organizations

ON THE EXAM

Most Control Tower questions turn on how the resource is created, or on stop versus report.

Meridian Freight uses Control Tower across 90 accounts. A compliance rule states that no EBS volume may be created unencrypted, anywhere, ever. The platform team enables the proactive control that requires EBS encryption on the Workloads OU.

Two months later an audit finds unencrypted volumes in three accounts. All three teams provision infrastructure with Terraform.

Why did the control fail to prevent this?

A Proactive controls only inspect CloudFormation deployments

B The control was applied to the OU rather than to each account

C Proactive controls report non-compliance but do not block it

D Terraform accounts must be enrolled in Control Tower separately

◆ PAUSE AND DECIDE

Why did the control fail to prevent this?

A Only inspects CloudFormation

A proactive control is a CloudFormation hook. Terraform calls the EC2 API directly, so the hook is never invoked.

B Applied to the OU

Backwards. Controls are designed to attach to an OU and reach every account inside it. That is the correct placement.

C Proactive controls only report

Describes a detective control. Proactive controls genuinely do block, but only on the path they can see.

D Separate enrolment

Enrolment is per account, not per tool. The accounts are enrolled; the deployment method is what differs.

THE SIGNAL

Three teams, all using Terraform, is the planted detail. It tells you the creation path avoids CloudFormation entirely.

> A company is moving 300 developer teams onto AWS over 18 months. Each team needs its own account. Security requires identical CloudTrail, Config and log centralisation in every account, and wants to demonstrate that to an auditor.
>
> The cloud platform team has four engineers.

What should the solutions architect recommend?

A Build the structure in AWS Organizations with custom SCPs and a CloudFormation pipeline

B Deploy a Control Tower landing zone and vend accounts through Account Factory

C Create a single account with 300 IAM roles, one per team

D Use AWS Config conformance packs applied to each account individually

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Hand-built Organizations**

Would work, and four engineers building and maintaining it for 300 accounts is exactly the overhead Control Tower removes.

B **Control Tower landing zone**

Standard baseline, identical in every vended account, with central logging and a dashboard the auditor can be shown.

C **One account, 300 roles**

Abandons account isolation, shared quotas, and any per-team billing separation. It solves none of the stated requirements.

D **Conformance packs per account**

Detective only, and applying them individually 300 times is the manual effort the scenario is trying to avoid.

THE SIGNAL

Identical baseline, many accounts, small team, and an auditor to satisfy. That combination is the landing zone use case exactly.

A security engineer needs a hard block on any account being able to disable CloudTrail. Control Tower is deployed with all mandatory controls active.

During testing, the engineer signs in to the management account and successfully stops a trail. The same attempt from a workload account is denied as expected.

What is the correct interpretation?

A The mandatory control is misconfigured and should be re-enabled

B Preventive controls do not apply to the management account; this is expected behaviour

C The management account has drifted and needs re-enrolment

D Mandatory controls apply only to accounts vended by Account Factory

◆ PAUSE AND DECIDE

What is the correct interpretation?

A Misconfigured control

The workload account was correctly denied, which proves the control is working. Only the management account behaved differently.

B Expected; management account is exempt

Preventive controls are SCPs and RCPs, and neither applies to the management account. Control Tower inherits that limitation exactly.

C Drift

Drift is a change made outside Control Tower. Nothing was changed here; a documented exemption was observed.

D Only vended accounts

Invented. Enrolled accounts are governed whether Account Factory created them or they were enrolled later.

THE SIGNAL

This is module two's exemption reappearing. Control Tower is built on Organizations, so it inherits every one of its limits.

> Every new account at a company must arrive with a standard VPC, three IAM roles for the operations team, and a budget alarm. Accounts are created roughly twice a week through Account Factory.
>
> Today an engineer applies these manually after each account is vended, and has forgotten twice this quarter.

What should the solutions architect recommend?

A Add the resources to a Control Tower proactive control

B Deploy Customizations for AWS Control Tower to apply the baseline on account creation

C Enable additional strongly recommended controls on the OU

D Write an AWS Config rule that flags accounts missing the resources

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A. Proactive control

Controls restrict what may be created. None of the three control types deploys resources into an account.

B. Customizations for Control Tower

Purpose-built for this. It triggers on account creation or enrolment and deploys CloudFormation templates into the new account.

C. More controls on the OU

Same category error as A. Controls govern behaviour; they do not provision a VPC.

D. Config rule

Detects the omission after it has happened, which is what the engineer already discovers manually. It creates nothing.

THE SIGNAL

Controls restrict. Customizations deploy. If the requirement is that something must exist, no control can deliver it.

> An organisation runs Control Tower. A platform engineer needed an urgent exception, so they edited one of the Control Tower service control policies directly in the AWS Organizations console.
>
> The Control Tower dashboard now reports the OU as out of compliance, although the SCP is functioning as the engineer intended.

What should the team do?

A. Ignore it; the SCP works, and the dashboard status is cosmetic

B. Revert the manual change, then re-register the OU in Control Tower

C. Delete and rebuild the landing zone

D. Remove the affected accounts from the organisation and re-enrol them

◆ PAUSE AND DECIDE

What should the team do?

A Ignore the drift

Tempting because nothing is visibly broken. But Control Tower will overwrite the change on the next landing zone update, silently removing the exception.

B Revert, then re-register the OU

The documented drift remediation. It restores the expected state and returns the OU to a governed condition.

C Rebuild the landing zone

Enormously disruptive for a single policy edit. Drift has a targeted remedy and this is not it.

D Re-enrol the accounts

The drift is at the OU and its policy, not in the accounts. Re-enrolling them does not touch the edited SCP.

THE SIGNAL

Drift is not cosmetic. The next landing zone update reconciles to Control Tower's expected state, and the manual change disappears.

> A regulator requires evidence, updated continuously, of which accounts have public S3 buckets. The company must also be able to prove the check has run without interruption for the past year.
>
> The security team has asked whether a preventive control would satisfy the regulator.

What should the solutions architect advise?

A. A preventive control, because preventing the problem is stronger than reporting it

B. A detective control, because the requirement is continuous evidence rather than prevention

C. A proactive control, because it stops the bucket before it exists

D. No control is needed; S3 Block Public Access covers the requirement

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A. Preventive control

Stops the bucket but produces no ongoing compliance record. A control that blocks everything has nothing to report.

B. Detective control

Config rules evaluate continuously and retain a compliance history, which is precisely the evidence the regulator asked for.

C. Proactive control

Also preventive in effect, also produces no evidence trail, and only sees CloudFormation.

D. Block Public Access alone

A sound configuration, and worth having, but it generates no per-account compliance record over time.

THE SIGNAL

Evidence is a detective requirement. Prevention is a strong instinct and the wrong answer when the ask is to demonstrate, not to stop.

What to carry out of this module

- **Proactive controls only see CloudFormation.** Console, CLI, SDK and Terraform bypass them.
- **Preventive stops at the API. Detective reports. Proactive stops at deployment.**
- Preventive controls are SCPs and RCPs, so the management account exemption applies here too.
- Controls restrict. **Customizations and AFT deploy.** No control creates a resource.
- Control Tower creates the Security OU. Infrastructure and Workloads OUs are yours.

> ON THE EXAM
>
> Two questions answer most Control Tower scenarios: how is the resource being created, and does the requirement say stop or say report?

MODULE 04 · DOMAIN 1 - DESIGN SOLUTIONS FOR ORGANIZATIONAL COMPLEXITY

Identity at Scale

Identity Center, federation, and the four directory answers.

DOMAIN 1 of 4 EXAM WEIGHT 26% MODULE 4 of 44

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Identity Center does not replace IAM. It replaces **IAM users**. What people receive is still a role.

What actually happens at sign-in

01
Authenticate
The identity source proves who the user is.

›

02
Look up
Identity Center finds their group assignments.

›

03
Match
A permission set is assigned to that group for that account.

›

04
Assume
The user assumes the role the permission set created.

›

05
Expire
Temporary credentials, which lapse on their own.

ON THE EXAM

A permission set is not a new kind of policy. It is a template that provisions an IAM role into every assigned account.

Facts about Identity Center the exam checks

- **An organization instance lives in the management account.** One per organisation, with administration delegable.
- **Account instances** are single-account, for AWS managed applications only. Not for multi-account access.
- **An instance runs in one Region.** Changing it means rebuilding, so decide early.
- **SCIM** provisions users and groups automatically. Assign permission sets to **groups**, never to individuals.

> ON THE EXAM
>
> Delegated administration moves the day-to-day work out of the management account, which is what module two told you to do.

Three identity sources

SOURCE	USE IT WHEN	THE CATCH
Identity Center directory	No existing corporate directory, or a small organisation	You now run a second user store
Active Directory	Corporate AD already holds the workforce	Needs Managed Microsoft AD or AD Connector underneath
External identity provider	Okta, Entra ID, Ping or similar is already the source of truth	Use SCIM, or provisioning becomes manual

ON THE EXAM

The right answer is almost always whichever directory the company already has. Creating a second source of truth is the anti-pattern.

Managed Microsoft AD or AD Connector?

AWS MANAGED MICROSOFT AD

A real directory running in AWS

- Real Windows Server AD, not an emulation
- Group Policy, schema extensions, **trusts**
- **Shareable across accounts**, 25 or 500
- Needed for FSx and RDS for SQL Server

AD CONNECTOR

A proxy to your existing directory

- **Stores nothing in AWS**
- Your password and lockout policies apply
- **Cannot be shared across accounts**
- Not multi-VPC aware; fails with the link

THE TIEBREAKER

No directory data may live in AWS means AD Connector. Sharing across accounts, or trusts, means Managed Microsoft AD.

Simple AD, and an honest warning

- **Samba 4 based**, not real Active Directory. Small up to 500 users, Large up to 5,000.
- **No trust relationships. No schema extensions. No multi-factor authentication.** Those three absences decide most questions.
- **Closed to new customers since 30 July 2026.** AWS points at Managed Microsoft AD or AD Connector instead.
- Still on the in-scope service list, so answer with it if a question describes a small, standalone, low-cost directory.
- Any mention of a **trust to an existing forest** eliminates it instantly, whatever the budget says.

> ON THE EXAM
>
> Teach yourself the examinable answer and know the reality. You cannot buy this today, and it is still a correct exam answer.

Workforce or customers?

IAM IDENTITY CENTER

Your employees

- Hundreds or thousands of people you employ
- They need the console, the CLI and AWS applications
- Access is by job role, managed centrally
- Free of charge

AMAZON COGNITO

Your application's users

- Thousands to millions of people you do not employ
- They need your app, not the AWS console
- Sign-up, sign-in, social and enterprise federation
- Priced per monthly active user

THE TIEBREAKER

Ask who the people are. Employees mean Identity Center. Customers of an application mean Cognito. The scale in the question usually gives it away.

Cognito: two components, two jobs

- **User pool.** A directory for your application. Sign-up, sign-in, MFA, social and SAML federation. It returns tokens.
- **Identity pool.** Exchanges a token for **temporary AWS credentials** from an IAM role.
- User pool answers who are you. Identity pool answers what may you touch in my AWS account.
- An identity pool can issue credentials to **unauthenticated** guests as well, with a separate, narrower role.
- If users never touch an AWS service directly, you may need only a user pool.

> ON THE EXAM
>
> App users uploading straight to S3 or writing to DynamoDB means an identity pool. Signing in to your app alone means a user pool.

Thirty teams needing thirty different scopes does not mean thirty permission sets. It means **one**, plus attributes.

Trusted identity propagation

- **The user's identity travels into the data service**, rather than stopping at the application's role.
- Without it, CloudTrail records the role that queried the data. Every analyst looks identical in the audit trail.
- Targets include **Redshift, Athena, EMR, Lake Formation, S3 Access Grants, OpenSearch** and Transfer Family.
- Permissions are then granted to existing groups, so entitlements live in one place.
- Requires Identity Center, and an organization instance is recommended.

> ON THE EXAM
>
> The audit trail names the role rather than the person is the signature phrase. That is what this feature exists to fix.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Employees need console access across many accounts	Identity Center with permission sets
No directory data may be stored in AWS	AD Connector
Domain join across several accounts, or a forest trust	AWS Managed Microsoft AD
Millions of application users signing in	Cognito user pool
App users must upload directly to S3	Cognito identity pool
Audit shows the role, not the person	Trusted identity propagation

ON THE EXAM

Identify who the humans are first. Employee, application user, or a service. That single question eliminates most options.

> Meridian Freight has 210 AWS accounts. Engineers currently hold an IAM user in each account they work in, with access keys stored locally. Offboarding takes a week and the security team cannot confirm it is ever complete.
>
> The company already runs Entra ID for all staff. Auditors have asked for a single, provable list of who can reach production.

What should the solutions architect recommend?

A. A central identity account holding IAM users, with cross-account roles into each account

B. IAM Identity Center with Entra ID as the identity source, SCIM provisioning, and permission sets assigned to groups

C. An IAM SAML identity provider configured in each of the 210 accounts, trusting Entra ID

D. Rotate all IAM user access keys every 30 days using Secrets Manager

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Central identity account

The pre-Identity Center pattern. Better than today, but it still means IAM users, long-lived keys, and trust policies maintained in 210 places.

B Identity Center with Entra ID

Entra ID stays the source of truth, SCIM removes the leaver problem, and permission sets give auditors one place to read.

C SAML provider per account

Federation without central management. You would configure and maintain the same trust 210 times, which is the overhead being complained about.

D Rotate the keys

Treats a symptom. A rotated key belonging to someone who left last month is still a key belonging to someone who left.

THE SIGNAL

A directory already exists, offboarding is unreliable, and auditors want one list. All three point the same way.

> A company has 40 delivery teams sharing several AWS accounts. Each team must administer only the resources it owns, which are tagged with the team name. New teams form roughly monthly.
>
> The platform team has created 40 permission sets and finds maintaining them unsustainable.

What should the solutions architect recommend?

A. Consolidate to one permission set using ABAC, matching a team attribute passed as a session tag against resource tags

B. Give each team its own AWS account and one shared administrator permission set

C. Replace the permission sets with an SCP per team

D. Automate permission set creation with CloudFormation StackSets

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A One permission set with ABAC

The team attribute arrives as a session tag and is compared against the resource tag. A new team needs no AWS change at all.

B An account per team

Defensible in general, and a large restructuring that the scenario did not ask for. It also does not scale better when a team needs two environments.

C An SCP per team

SCPs are account-wide ceilings. They cannot vary by user within a shared account, which is exactly the requirement here.

D Automate the 40

Makes the sprawl faster to create rather than smaller. Forty permission sets still have to be reviewed, audited and eventually deleted.

THE SIGNAL

Many groups needing the same shape of access with a different scope is the ABAC signature. It replaces N policies with one.

A bank runs Active Directory on-premises. Regulation forbids replicating or storing employee directory data outside its own data centres. It wants to deploy Amazon WorkSpaces in a single AWS account so staff can sign in with existing credentials.

A high-bandwidth Direct Connect link to the data centre already exists.

Which directory option should the architect choose?

A. AWS Managed Microsoft AD with a two-way forest trust to on-premises

B. AD Connector pointing at the on-premises directory

C. Simple AD with users imported from the on-premises directory

D. An Amazon Cognito user pool federated to Active Directory

◆ PAUSE AND DECIDE

Which directory option should the architect choose?

A. Managed Microsoft AD with a trust

A trust avoids copying accounts, but the service still holds directory objects in AWS. It is also more service than a single account needs.

B. AD Connector

A proxy that caches nothing. Authentication is forwarded on-premises, so no directory data resides in AWS, and WorkSpaces is supported.

C. Simple AD with an import

Importing is replicating, which the regulation forbids. Simple AD also cannot trust an existing forest, and is closed to new customers.

D. Cognito user pool

Cognito is for application users. WorkSpaces needs a directory, and bank staff are workforce rather than customers.

THE SIGNAL

No directory data may leave the data centre is a phrase with exactly one answer. AD Connector is the only option that stores nothing.

> The same bank now needs domain-joined EC2 instances, Amazon FSx for Windows File Server, and RDS for SQL Server across eleven AWS accounts. Staff must still authenticate with their existing on-premises credentials.
>
> The architect proposes reusing the AD Connector from the WorkSpaces project.

What should the architect be told?

A. AD Connector cannot be shared across accounts; use AWS Managed Microsoft AD with a trust to on-premises

B. Deploy an AD Connector in each of the eleven accounts

C. AD Connector supports this once VPC peering is configured between the accounts

D. Replicate the on-premises directory into Simple AD and share that

◆ PAUSE AND DECIDE

What should the architect be told?

A Managed Microsoft AD with a trust

It can be shared across accounts, and a trust means credentials are still validated on-premises rather than copied.

B One AD Connector per account

Technically possible and operationally poor. Eleven connectors, eleven sets of credentials to maintain, and no shared domain across them.

C Peering fixes it

Peering is a network path. The limitation is a service one: AD Connector is not shareable and is not multi-VPC aware.

D Replicate into Simple AD

Replication is still forbidden, Simple AD cannot trust a forest, and it is closed to new customers.

THE SIGNAL

One constraint changed from scenario three: eleven accounts instead of one. That alone flips the answer.

> A media company is launching a mobile app expecting three million users. People will sign up with email or Google, and upload video directly from the device to Amazon S3, into a prefix belonging only to them.
>
> The team's first design routes every upload through an EC2 fleet holding one IAM role.

What should the solutions architect recommend instead?

A. IAM Identity Center with a permission set for application users

B. A Cognito user pool for sign-in, and an identity pool issuing scoped credentials for direct upload

C. A Cognito user pool alone, with the app calling S3 using an embedded access key

D. Keep the EC2 proxy but add an Application Load Balancer for scale

◆ PAUSE AND DECIDE

What should the solutions architect recommend instead?

A **Identity Center**

Built for a workforce. Nobody assigns permission sets to three million consumers, and they have no business in an AWS account.

B **User pool plus identity pool**

The user pool handles email and Google sign-in. The identity pool exchanges the token for credentials scoped to that user's own prefix.

C **User pool with an embedded key**

An access key shipped inside a mobile app is extractable by anyone who downloads it. This is the worst option in the list.

D **Scale the proxy**

Pays compute and data transfer to relay video that could go straight to S3, and one shared role cannot isolate three million prefixes.

THE SIGNAL

Application users needing to touch an AWS service directly is the identity pool signature. The user pool alone only proves who they are.

A data platform lets 400 analysts query a lake through Amazon Athena and Amazon Redshift. The query tool assumes a single shared IAM role on their behalf.

Compliance has rejected the design: CloudTrail records the shared role for every query, so it is impossible to prove which analyst read which dataset.

What should the solutions architect implement?

A Create 400 IAM roles, one per analyst, and have the tool assume the right one

B Enable trusted identity propagation through IAM Identity Center

C Turn on CloudTrail data events for the S3 buckets behind the lake

D Pass the analyst's username as a session tag on the shared role

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A 400 individual roles

Solves attribution and creates 400 roles to maintain, plus per-dataset policies in each. The scale problem ABAC and propagation exist to remove.

B Trusted identity propagation

The user identity reaches Athena and Redshift, so authorisation runs against the person and CloudTrail records them.

C S3 data events

Records more calls made by the same shared role. More volume, identical attribution problem, and a noticeable bill.

D Session tag the username

The closest miss. It does surface the name in CloudTrail, but it is asserted by the tool rather than proven, so it is not evidence.

THE SIGNAL

The audit trail names the role rather than the person, on an analytics service, is this feature's exact signature.

What to carry out of this module

- **A permission set provisions a real IAM role.** Module two's evaluation chain still applies to it.
- **AD Connector stores nothing and cannot be shared.** Managed Microsoft AD can be shared and holds trusts.
- Employees mean Identity Center. App users mean Cognito: user pool proves identity, identity pool grants AWS access.
- Same access shape, different scope, many groups means **ABAC**. Never one policy per team.

> ON THE EXAM
>
> Identify the humans first. Employee, application user, or service. Then the directory question usually answers itself.

MODULE 05 · DOMAIN 1 - DESIGN SOLUTIONS FOR ORGANIZATIONAL COMPLEXITY

Cross-Account Access and Resource Sharing

Three ways across an account boundary, and when each is the wrong one.

DOMAIN 1 of 4 EXAM WEIGHT 26% MODULE 5 of 44

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Two of the three mechanisms grant **access to** a resource. RAM makes the resource **appear in** another account.

The three mechanisms

MECHANISM	WHAT CROSSES	TYPICAL USE
Role assumption	The caller, briefly	A human or workload acting in another account
Resource-based policy	Permission, attached to the thing	S3, KMS, SQS, SNS, Lambda, Secrets Manager
AWS RAM	The resource itself	Subnets, Transit Gateways, Resolver rules

ON THE EXAM

If the question is about infrastructure being duplicated in every account, it is a RAM question, not an IAM one.

Role assumption details the exam tests

- **Role chaining caps the session at one hour**, and it cannot be extended. This breaks long jobs.
- The trust policy names who may assume. The caller still needs **sts:AssumeRole** in their own policy.
- **External ID** for any third party, to defeat the confused deputy.
- Session tags let one role behave differently per caller, which is how ABAC reaches across accounts.

> ON THE EXAM
>
> A batch job dying at exactly sixty minutes is role chaining, every time. It is the most recognisable symptom in this domain.

What RAM can and cannot share

- **Networking:** subnets, Transit Gateways, Route 53 Resolver rules, prefix lists, Network Firewall rule groups.
- **Capacity:** Capacity Reservations, Dedicated Hosts, Aurora DB clusters, License Manager configurations.
- **Not S3 buckets.** Use a bucket policy. Not KMS keys either; use the key policy.
- Not security groups, and never the VPC itself. You share subnets out of a VPC, not the VPC.

> ON THE EXAM
>
> Anything with its own resource-based policy is shared with that policy, not with RAM. RAM covers what has no policy of its own.

RAM or a resource-based policy?

AWS RAM

The resource appears elsewhere

- For things with no policy of their own
- Share to an OU or a whole organisation
- Owner keeps control and pays
- Participants use it as if it were local

RESOURCE-BASED POLICY

Permission to reach across

- For S3, KMS, SQS, SNS, Lambda
- Scope with aws:PrincipalOrgID
- Both sides must allow the call
- The caller stays in their own account

THE TIEBREAKER

Ask whether the resource has a policy of its own. If it does, that is the tool. If it does not, RAM is the only option.

VPC sharing: the ownership model

- **You share subnets, never the VPC.** The VPC stays wholly owned by the owner account.
- **Owner keeps** route tables, network ACLs, gateways, endpoints and flow logs. Participants cannot change any of them.
- **Participants launch resources** into shared subnets, and own what they launch.
- Participants cannot use a security group owned by the owner or another participant. The default group belongs to the owner.
- Same organisation only, and default VPCs cannot be shared.

ON THE EXAM

Owner controls the network. Participants control their workloads. Nearly every shared-VPC question turns on which side of that line the problem falls.

Sixty accounts with their own VPCs means sixty NAT gateways, sixty endpoint sets and **sixty Transit Gateway attachments**.

The honest limits of a shared VPC

- **One IP pool.** A participant that fills a subnet blocks every other participant in it.
- Troubleshooting splits awkwardly. Security group problems are the participant's; route table and NACL problems are the owner's.
- Flow logs belong to the owner, so participants must ask for network evidence.
- Dedicated per-team subnets inside the shared VPC contain the address exhaustion risk.

> ON THE EXAM
>
> The blast radius moves rather than disappearing. You trade many small isolated failures for fewer, wider ones.

IAM Access Analyzer has three analyzers

- **External access.** Which resources are reachable from outside your zone of trust. Free.
- **Internal access.** Which principals **inside** your organisation can reach a chosen resource.
- **Unused access.** Unused roles, access keys, console passwords, and unused actions on a role.
- They are separate products. Enabling one in 2020 gave you nothing for the other two.

> ON THE EXAM
>
> Nobody outside was granted anything, yet the whole organisation can read it, means an internal access analyzer. External finds nothing there.

The rest of Access Analyzer

- **Policy generation** writes a least-privilege policy from what CloudTrail shows a principal actually did.
- **Custom policy checks** compare a proposed policy against your standards, so a pipeline can refuse a merge.
- **The zone of trust** is what counts as internal. An organisation-level analyzer treats the whole organisation as trusted.
- Set it up in a delegated administrator account, following module two's advice about the management account.

> ON THE EXAM
>
> Policy generation is the answer to we do not know what permissions this needs. It reads the evidence instead of guessing.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
The same infrastructure is rebuilt in every account	AWS RAM
Too many NAT gateways, endpoints, TGW attachments	Shared VPC subnets
A job fails at exactly one hour	Role chaining
Who inside the org can reach this resource?	Internal access analyzer
We do not know what permissions this app needs	Access Analyzer policy generation
A third party must access our account	Cross-account role with an external ID

ON THE EXAM

Ask whether the problem is duplication or permission. Duplication is a sharing question; permission is an IAM one.

> Meridian Freight gives each of its 60 delivery teams a dedicated AWS account with its own VPC. Every VPC has two NAT gateways, a set of interface endpoints, and a Transit Gateway attachment.
>
> Finance has flagged that networking now costs more than the compute it serves. The network team also cannot keep routing consistent across 60 VPCs.

What should the solutions architect recommend?

A. Replace the Transit Gateway with VPC peering between the 60 VPCs

B. Share subnets from a central VPC with the team accounts using AWS RAM

C. Consolidate the 60 accounts into one account with 60 VPCs

D. Replace the NAT gateways with NAT instances to reduce cost

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Peering instead of TGW**

Sixty VPCs fully meshed needs 1,770 peering connections, none of them transitive. It makes the routing problem far worse.

B **Share subnets via RAM**

One VPC means one set of NAT gateways, endpoints and one attachment. Accounts stay separate, so isolation and billing survive.

C **One account, 60 VPCs**

Destroys account isolation, merges quotas and blast radius, and keeps all sixty sets of networking charges.

D **NAT instances**

Trades a managed service for 120 instances to patch, size and make highly available. It also leaves the endpoint and attachment costs untouched.

THE SIGNAL

Duplicated networking with separate accounts still required is the shared-VPC signature. The sprawl is the problem, not the account count.

After adopting a shared VPC, a participant team reports that they cannot add a route to a new on-premises range, and cannot see VPC Flow Logs while troubleshooting.

They have full administrator permissions in their own account.

What is the correct explanation?

A Their administrator permissions need an explicit allow for ec2:CreateRoute

B Route tables and flow logs belong to the VPC owner, and participants cannot modify or read them

C An SCP on their OU is denying the actions

D The resource share must be re-created with write permissions

◆ PAUSE AND DECIDE

What is the correct explanation?

A. Add an explicit allow

Administrator already allows the action. The block is not in their account's permissions at all.

B. Owner keeps the network

The sharing model gives participants the subnet and nothing else. Route tables, NACLs, gateways and flow logs stay with the owner.

C. An SCP is denying it

Plausible in general, and it would deny the action everywhere rather than only on someone else's resource. Nothing here suggests one exists.

D. Re-share with write

Invented. RAM shares do not have a write mode that would hand over the owner's route tables.

THE SIGNAL

Full admin in your own account and still blocked means the resource is not in your account. That is the shared VPC model working correctly.

> A nightly data pipeline runs in a tooling account. It assumes a role there, then assumes a second role in each of 30 data accounts to copy files. The job takes about ninety minutes.
>
> It has begun failing partway through, always with expired credentials, and always at roughly the same elapsed time.

What is the cause, and what should be changed?

A. Role chaining caps the session at one hour; assume the target roles directly from the source identity

B. Raise the MaxSessionDuration on the target roles to twelve hours

C. The trust policies are missing a condition and should be widened

D. Request a quota increase for STS session length

◆ PAUSE AND DECIDE

What is the cause, and what should be changed?

A Role chaining, so assume directly

Chained sessions are capped at one hour regardless of settings. Removing the intermediate hop removes the cap.

B Raise MaxSessionDuration

The near miss. The setting exists and is ignored when a role assumes a role, so the session still ends at sixty minutes.

C Widen the trust policies

The credentials expired rather than being refused. Widening trust weakens security and changes nothing about duration.

D Quota increase

The one-hour chaining limit is a hard behaviour, not an adjustable quota. Support cannot raise it.

THE SIGNAL

Failure at a consistent elapsed time, with expired rather than denied credentials, points at a session limit rather than a policy.

A security team runs IAM Access Analyzer with an organisation-level external access analyzer, and reports zero findings on a sensitive S3 bucket holding customer records.

An internal review then finds that a broad identity policy lets every principal in the organisation read that bucket.

Why did Access Analyzer report nothing, and what should be enabled?

A. The analyzer was in the wrong Region; create one per Region

B. The organisation is the zone of trust, so internal grants are not external findings; enable an internal access analyzer

C. External access analyzers do not evaluate S3 buckets

D. The finding was archived automatically and should be unarchived

◆ PAUSE AND DECIDE

Why did Access Analyzer report nothing, and what should be enabled?

A Wrong Region

External analyzers are per Region, so this is a genuine operational trap. It does not explain a bucket that was in scope and returned nothing.

B Zone of trust; add internal analyzer

Everything inside the organisation is trusted by definition, so an organisation-wide grant is not external. Internal access analysis is the one that sees it.

C S3 is not evaluated

S3 buckets are among the most commonly analysed resources. This is simply false.

D Unarchive the finding

Archive rules suppress findings that were generated. No finding was ever produced here.

THE SIGNAL

An external analyzer is silent by design when the over-permission is internal. Enabling it once and assuming full coverage is the real error.

A central networking account has built Route 53 Resolver rules that forward queries for the company's private on-premises domains. Forty application accounts need the same resolution.

An engineer proposes copying the rules into each account and keeping them in step with a scheduled script.

What should the solutions architect recommend instead?

A Share the Resolver rules with the organisation using AWS RAM

B Keep the script but add a Config rule to detect drift

C Create a private hosted zone in each account and associate it with each VPC

D Grant each account a cross-account role into the networking account

◆ PAUSE AND DECIDE

What should the solutions architect recommend instead?

A Share the rules with RAM

Resolver rules are RAM shareable. One rule set, shared to the organisation, and a new account inherits it with no copying.

B Script plus drift detection

Keeps forty copies and adds monitoring of the copies. It detects the problem faster instead of removing it.

C Private hosted zone per account

A hosted zone answers for a domain AWS hosts. These queries must be forwarded to on-premises resolvers, which is a different job.

D Cross-account role

Lets someone administer the networking account. DNS resolution inside a VPC is not performed by assuming a role.

THE SIGNAL

One artefact needed identically in many accounts is the RAM signature. Copying and reconciling is the anti-pattern it removes.

> A data engineering team wants every account in the organisation to read from a central S3 data lake bucket. Account numbers change as accounts are added and closed.
>
> The team has opened a request to share the bucket using AWS RAM.

What should the solutions architect tell them?

A RAM cannot share S3 buckets; use a bucket policy with an aws:PrincipalOrgID condition

B Share the bucket with RAM, selecting the organisation as the principal

C Create an access point per account and share those with RAM

D List every account number in the bucket policy and update it monthly

◆ PAUSE AND DECIDE

What should the solutions architect tell them?

A Bucket policy with PrincipalOrgID

S3 has its own resource-based policy, so that is the tool. The condition key matches current membership, so the list maintains itself.

B Share via RAM

S3 buckets are not a RAM shareable resource type. The request cannot be fulfilled as asked.

C Access points via RAM

Access points are also not RAM shareable, and one per account reintroduces the maintenance the team is trying to avoid.

D List the accounts

Works today and is wrong by next month. A policy that must be edited whenever an account is created will eventually be forgotten.

THE SIGNAL

If a resource has its own policy, RAM is not involved. Pair this with scenario five: same shape of request, opposite answer.

What to carry out of this module

- **RAM shares the resource. Policies grant access to it.** Its own policy means RAM is not involved.
- **Share subnets, never the VPC.** Owner keeps routing, NACLs, gateways and flow logs.
- **Role chaining caps sessions at one hour.** MaxSessionDuration is ignored. Failure at sixty minutes is the tell.
- Access Analyzer is three analyzers. External finds nothing when the over-permission is internal.

> ON THE EXAM
>
> Ask first whether the problem is duplication or permission. That single question routes you to the right half of this domain.

MODULE 06 · DOMAIN 1 - DESIGN SOLUTIONS FOR ORGANIZATIONAL COMPLEXITY

Multi-VPC Connectivity and Transit Gateway

A wire, a router, or a doorway. Three answers, and they are not interchangeable.

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Peering is a **wire**. Transit Gateway is a **router**. PrivateLink is a **doorway** to one service.

Choosing between the three

APPROACH	REACH FOR IT WHEN	DEAL BREAKER
VPC peering	Two VPCs, high volume, no transit needed	Not transitive; CIDRs must not overlap
Transit Gateway	Many VPCs, hybrid, segmentation needed	Per-GB processing charge on everything
PrivateLink	Expose one service, keep networks separate	One service only; needs an NLB or GWLB

ON THE EXAM

Overlapping CIDRs eliminate peering and Transit Gateway together. Only PrivateLink is indifferent to the address space.

VPC peering: four limits that decide questions

- **Not transitive.** If A peers with B and B with C, A still cannot reach C.
- **No edge-to-edge routing.** You cannot use a peer's internet gateway, NAT device, VPN, Direct Connect or endpoints.
- **CIDRs must not overlap**, and only one peering connection can exist between the same two VPCs.
- **50 active connections per VPC** by default, raisable to 125. Cross-region and cross-account both work.
- **No per-GB processing charge.** This is peering's real advantage and the exam does test it.

> ON THE EXAM
>
> Edge-to-edge is the forgotten one. Routing your traffic out through a peer's NAT gateway simply does not work.

Ten VPCs fully meshed is 45 connections. Fifty VPCs is 1,225, each with route table entries on both sides.

Transit Gateway fundamentals

- **Regional.** Attach VPCs, VPNs, Direct Connect gateways, and other Transit Gateways.
- **Route tables do the work.** One TGW can hold many, and that is how segmentation happens.
- **5,000 attachments** per gateway. One VPC attachment per VPC per gateway; up to 5 gateways per VPC.
- **50 Gbps burst** per VPC attachment. MTU 8500 inside, but only 1500 over VPN.
- Share it across accounts with **RAM**, exactly as in module five. Do not build one per account.

> ON THE EXAM
>
> The bandwidth ceiling is per attachment, not per gateway. A single very hungry VPC is the case where it bites.

Follow the packet through a Transit Gateway

01

Subnet route

The subnet route table sends the packet to the TGW attachment.

02

Association

The attachment is associated with exactly one TGW route table. That one is consulted.

03

Lookup

That table decides which attachment the packet leaves by.

04

Delivery

The destination VPC's own subnet route table takes over.

ON THE EXAM

Association picks which table an attachment reads. Propagation decides which tables learn about it. They are different questions.

Association or propagation?

ASSOCIATION

Which table I read

- Exactly one table per attachment
- Decides where my traffic may go
- Controls what this VPC can reach
- Change it to move a VPC's segment

PROPAGATION

Which tables learn about me

- Can propagate into many tables
- Decides who can reach me
- Controls this VPC's visibility
- Or add static routes by hand

THE TIEBREAKER

Segmentation is built by controlling propagation. Spokes associate with a table holding only shared services routes.

Appliance mode, and why flows break without it

- **By default a TGW keeps traffic in the Availability Zone it arrived in**, choosing per direction.
- With a firewall in the path, the return packet can land on a different appliance, which has no record of the flow, and drops it.
- **Appliance mode** pins both directions of a flow to the same AZ, so the same appliance sees both.
- Enable it on the **inspection VPC attachment**. Symptom: intermittent failures that look like packet loss.

> ON THE EXAM
>
> Stateful appliance plus multiple AZs plus mysterious asymmetric drops equals appliance mode, every time.

Newer capabilities worth knowing

- **Security group referencing across a TGW**, since late 2024. Reference a group in another attached VPC.
- It does **not** work across TGW peering connections. Both VPCs must attach to the same gateway.
- **Network function attachments** connect a TGW straight to AWS Network Firewall, with no inspection VPC.
- **Encryption support** enforces encryption in transit for all TGW traffic.

> ON THE EXAM
>
> Security group referencing removed the main reason teams stayed on peering. Older study material still says it is impossible.

The cost model, because it decides answers

- **TGW charges twice:** an hourly fee per attachment, and a charge per GB processed.
- **Peering charges no processing fee at all.** You pay only normal data transfer.
- So for two VPCs moving very large volumes, peering is meaningfully cheaper.
- For many VPCs, the TGW fee buys away the mesh, and usually wins on both cost and operations.

> ON THE EXAM
>
> Peering is cheaper per byte. Transit Gateway is cheaper per relationship. The VPC count tells you which one the question is about.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Dev must not reach Prod, both reach shared services	TGW route table segmentation
Intermittent drops through a firewall across AZs	Appliance mode
Overlapping CIDRs, one service to expose	PrivateLink
Two VPCs, enormous traffic, nothing else needed	VPC peering
VPC count keeps growing, plus on-premises	Transit Gateway
Route out through the other VPC's NAT gateway	Impossible over peering

ON THE EXAM

Count the VPCs first, then check the CIDRs. Those two facts decide most of these before you read the options.

> Meridian Freight has 14 VPCs connected by a partial mesh of peering connections. Two new VPCs are planned each quarter. Every VPC also needs to reach the on-premises data centre, currently through a VPN terminating in one VPC.
>
> Teams report that some VPC pairs cannot reach each other, and that on-premises traffic works from only one VPC.

What should the solutions architect recommend?

A. Complete the mesh so every VPC peers with every other VPC, and add a VPN in each VPC

B. Deploy a Transit Gateway, attach every VPC, and move the VPN onto it

C. Configure transitive routing on the existing peering connections

D. Route all inter-VPC traffic through the VPC that terminates the VPN

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Complete the mesh**

Ninety-one connections today, growing every quarter, plus sixteen VPN connections to pay for and maintain. It scales in the wrong direction.

B **Transit Gateway**

One attachment per VPC, one VPN, and a new VPC is one attachment rather than fourteen peerings.

C **Enable transitive routing**

There is no such setting. Peering is non-transitive by design, and no configuration changes that.

D **Hub through the VPN VPC**

This is exactly what non-transitive means. Traffic cannot pass through a peered VPC to reach a third.

THE SIGNAL

Both reported symptoms are the same limitation. Non-transitivity explains the unreachable pairs and the single-VPC on-premises access.

> A company runs Production, Development and Shared Services VPCs attached to one Transit Gateway. Security requires that Development can never reach Production, while both must reach Shared Services, and Shared Services must reach both.
>
> The current design uses one Transit Gateway route table with all attachments propagating into it.

How should the architect meet the requirement?

A. Add network ACLs in the Production VPC denying the Development CIDR

B. Use separate TGW route tables: spokes associate with a table holding only Shared Services routes

C. Detach Development and connect it by VPC peering instead

D. Apply an SCP denying network access between the accounts

◆ PAUSE AND DECIDE

How should the architect meet the requirement?

A Network ACLs

Works, and depends on a deny list staying correct as CIDRs change. The route still exists, so a mistake restores reachability silently.

B Separate TGW route tables

The spoke table contains no route to Production, so the path does not exist. Nothing to misconfigure later.

C Peering for Development

Solves isolation by leaving the design. Development would then lose Shared Services and on-premises reachability.

D An SCP

SCPs govern API calls, not packets. No policy in Organizations can stop one instance reaching another.

THE SIGNAL

Removing the route beats denying the traffic. If the path does not exist, there is no rule for anyone to get wrong.

An inspection VPC runs a fleet of stateful third-party firewalls behind a Gateway Load Balancer, spread across three Availability Zones, with all inter-VPC traffic routed through it by the Transit Gateway.

Long-lived connections fail intermittently. Short requests usually succeed. Firewall logs show packets arriving for sessions the appliance has no record of.

What is the cause?

A The Gateway Load Balancer is unhealthy and needs deregistration delay tuning

B Appliance mode is not enabled on the inspection VPC attachment

C The firewall instances need larger instance types

D MTU mismatch is fragmenting packets between VPCs

◆ PAUSE AND DECIDE

What is the cause?

A Load balancer health

An unhealthy appliance fails every flow sent to it, not a subset. It also would not produce sessions with no record.

B Appliance mode not enabled

Without it the TGW picks an AZ per direction, so the return packet reaches a different appliance that never saw the session.

C Bigger instances

A capacity answer to a symptom that is not about capacity. Load is never mentioned.

D MTU mismatch

Real and worth knowing, since TGW is 8500 and VPN is 1500. It would affect large packets, not whole sessions the appliance does not recognise.

THE SIGNAL

Packets arriving for unknown sessions is the giveaway. That is asymmetry, and asymmetry across AZs is what appliance mode fixes.

> Following an acquisition, the parent company must let the acquired company's applications call an internal payments API. Both estates use 10.0.0.0/16 with heavily overlapping subnets, and renumbering either is out of the question for at least two years.
>
> Only that one API needs to be reachable.

What should the solutions architect recommend?

A VPC peering between the two environments

B Attach both estates to a Transit Gateway

C Expose the API through PrivateLink, with an endpoint service behind a Network Load Balancer

D A Site-to-Site VPN between the two VPCs

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **VPC peering**

Refused outright. Peering requires non-overlapping CIDRs, so the connection cannot even be created.

B **Transit Gateway**

Also routing, so it needs unambiguous addresses. Overlapping ranges cannot be routed between.

C **PrivateLink endpoint service**

Not routing at all. The consumer reaches a local endpoint IP, so the two address spaces never meet.

D **Site-to-Site VPN**

Routing again, with the same overlap problem, plus tunnel cost and complexity for a single API.

THE SIGNAL

Overlapping CIDRs eliminate every routing answer at once. PrivateLink is the only option that never routes between the two networks.

A team is migrating from a peering mesh to a Transit Gateway. Their security groups rely heavily on referencing security groups in peered VPCs rather than CIDR blocks.

After cutting the first two VPCs over, connectivity fails. An engineer proposes rewriting every rule to use CIDR blocks.

What should the solutions architect advise?

A Enable security group referencing on the Transit Gateway and the VPC attachments

B Rewrite the rules to CIDR blocks, as referencing is unsupported over a Transit Gateway

C Keep peering alongside the Transit Gateway purely for the security group references

D Replace security groups with network ACLs referencing the peer VPC

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Enable SG referencing on the TGW

Supported since late 2024 for VPCs attached to the same gateway. It is a setting, not a rewrite.

B Rewrite to CIDRs

Was correct before late 2024 and is now outdated. It also discards the maintainability the references were giving them.

C Keep peering too

Retains the mesh they are migrating away from, and doubles the paths to reason about during troubleshooting.

D Network ACLs

NACLs are stateless and only understand CIDR ranges. They cannot reference a security group at all.

THE SIGNAL

This answer changed. Material written before late 2024 says option B, and it was right at the time.

> A media company runs a rendering VPC and a storage VPC in the same Region. They exchange roughly 900 TB per month and need nothing else on the network. The team has attached both to the company Transit Gateway.
>
> Finance has queried a large and growing Transit Gateway line on the bill.

What should the solutions architect recommend?

A. Leave it; Transit Gateway is the AWS-recommended pattern for VPC connectivity

B. Add a VPC peering connection between the two VPCs and route that traffic over it

C. Move both workloads into a single VPC

D. Enable a Transit Gateway multicast domain to reduce data processing

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Leave it as is

Pays a per-GB processing charge on 900 TB every month to solve a problem that has only two endpoints.

B Peer the two VPCs

Peering carries no processing charge. Two VPCs needing nothing else is exactly the case where the mesh objection does not apply.

C Merge into one VPC

Would remove the charge and also the isolation, quotas and blast-radius separation the two VPCs provide.

D Multicast domain

Multicast is for one-to-many distribution. It has no bearing on point-to-point volume or on processing charges.

THE SIGNAL

Pair this with scenario one. Fourteen growing VPCs made Transit Gateway right. Two static VPCs and huge volume reverses it.

What to carry out of this module

- **Wire, router, doorway.** Peering, Transit Gateway, PrivateLink. Name the shape the scenario needs.
- **Overlapping CIDRs eliminate every routing answer.** Only PrivateLink is indifferent to addresses.
- **Association is which table I read. Propagation is which tables learn about me.** Segmentation is built from propagation.
- Stateful appliances across AZs need **appliance mode**. Unknown-session drops are the signature.
- Peering is cheaper per byte. Transit Gateway is cheaper per relationship.

> ON THE EXAM
>
> Count the VPCs, then check the CIDRs. Those two facts decide the majority of questions in this area before you read the options.

MODULE 07 · DOMAIN 1 - DESIGN SOLUTIONS FOR ORGANIZATIONAL COMPLEXITY

Hybrid Connectivity: Direct Connect and VPN

One in five questions on this exam has an on-premises data centre in it.

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Direct Connect is private but **not encrypted**. VPN is encrypted but rides the **public internet**. Most real designs use both.

Direct Connect or VPN?

DIRECT CONNECT

A private circuit

- 1, 10, 100 or 400 Gbps ports
- Consistent latency, no internet
- Weeks to months to provision
- Not encrypted by default

SITE-TO-SITE VPN

Encrypted over the internet

- 1.25 Gbps per tunnel, 5 with LBT
- Latency varies with the internet
- Up today, in minutes
- IPsec encrypted by default

THE TIEBREAKER

Urgency favours VPN. Sustained throughput and predictability favour Direct Connect. Most answers pair them.

The three virtual interface types

VIF TYPE	CONNECTS TO	USE IT FOR
Private	A VGW, or a DX gateway	Reaching VPCs over private addresses
Public	AWS public endpoints	S3, DynamoDB and public services over the circuit
Transit	A DX gateway only	Reaching a Transit Gateway, and everything behind it

ON THE EXAM

Transit VIF is the only way to reach a Transit Gateway. If a scenario has a TGW and a Direct Connect, it needs one.

Direct Connect gateway

- **A global object.** One circuit reaches VPCs in any Region, except China.
- **20 virtual private gateways** or **6 Transit Gateways** per DX gateway. Not both at once.
- It also removes the account problem: associate VGWs from many accounts with one gateway.
- **VGWs attached to a DX gateway cannot route to each other.** It is a hub for reaching AWS, not for VPC-to-VPC.

> ON THE EXAM
>
> A DX gateway does not give you transitive VPC connectivity. If VPCs must reach each other, you still need a Transit Gateway.

Follow the packet from the data centre

01

Customer router

Your router at the Direct Connect location, speaking BGP.

›

02

VIF

A VLAN on the circuit. Its type decides what is reachable.

›

03

DX gateway

Global. Routes on to a VGW or a Transit Gateway.

›

04

TGW route table

Module six's association decides where it goes next.

›

05

Subnet route

The destination VPC's own route table delivers it.

ON THE EXAM

Five hops, five places a route can be missing. Troubleshooting questions are almost always asking which one.

The four resiliency models

MODEL	WHAT IT IS	FITS
Maximum	Separate connections at two or more locations, separate devices	Critical production
High	One connection at each of two locations	Most production
Development and Test	Two connections, separate devices, one location	Non-critical only
Classic	Added without the toolkit, no resiliency	Existing single links

ON THE EXAM

The word to hunt for is location. Two connections in one building survive a device failure and not a building failure.

Circuit-level details worth knowing

- **A hosted connection supports exactly one VIF.** A dedicated connection supports 51. This decides scenarios.
- **LAG** bundles connections into one logical link. Four below 100G, two at 100G.
- **MACsec** encrypts at layer 2 on supported dedicated ports, between your router and the AWS router only.
- **SiteLink** sends traffic between two Direct Connect locations over the AWS backbone, with no VPC in the path.

> ON THE EXAM
>
> MACsec protects the circuit, not the journey. If a partner sits in the path, it does not cover that segment.

VPN numbers that decide answers

- **Every connection has two tunnels.** Standard is 1.25 Gbps each; Large Bandwidth Tunnels reach 5 Gbps.
- **Large Bandwidth Tunnels need a Transit Gateway.** They do not work on a virtual private gateway.
- **A VGW caps at about 1.25 Gbps in aggregate**, however many VPN connections you attach to it.
- **ECMP** aggregates tunnels for more bandwidth. It needs BGP, and it also needs a Transit Gateway.

> ON THE EXAM
>
> Every route past the VPN bandwidth ceiling runs through a Transit Gateway. A VGW is the ceiling itself.

Encrypting traffic over Direct Connect

MACSEC

Layer 2 on the circuit

- Supported dedicated ports only
- Line rate, no throughput penalty
- Your router to the AWS router
- Does not cover a partner segment

PRIVATE IP VPN

IPsec inside the circuit

- Runs over a transit VIF
- Subject to VPN tunnel bandwidth
- End to end, whoever is in the path
- Needs a Transit Gateway

THE TIEBREAKER

Line-rate encryption on your own circuit means MACsec. Encryption end to end, or over a hosted connection, means a VPN.

Steering traffic with BGP

- **Longest prefix match wins first**, before anything else. Advertise a more specific route to pull traffic.
- **AS_PATH prepending** makes a path look longer, so AWS prefers the other one. The usual way to mark a backup.
- **Local preference BGP communities** let you set preference on AWS's side of the connection.
- **Direct Connect is preferred over VPN** to the same destination by default, so failover is automatic.

> ON THE EXAM
>
> Traffic taking the wrong circuit is almost always a prefix that is more specific somewhere it should not be.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Connectivity needed this week	Site-to-Site VPN, possibly then DX
Direct Connect must reach a Transit Gateway	Transit VIF plus a DX gateway
Survive a Direct Connect location outage	Maximum or High resiliency
Regulator requires encryption over the circuit	MACsec, or private IP VPN
More VPN throughput than 1.25 Gbps	Transit Gateway, then LBT or ECMP
Traffic prefers the wrong circuit	A more specific prefix somewhere

ON THE EXAM

Check what terminates the connection first. A virtual private gateway is the constraint behind several of these answers.

> Meridian Freight has one 10 Gbps Direct Connect at a single location, terminating on a virtual private gateway. A recent outage at that Direct Connect location took the data centre offline from AWS for six hours.
>
> The business now requires that no single facility failure can sever connectivity, and wants an encrypted fallback path.

What should the solutions architect recommend?

A. Add a second connection on a different device at the same location

B. Add a connection at a second Direct Connect location, and a VPN as backup

C. Increase the existing connection to 100 Gbps

D. Add a LAG bundling two connections at the existing location

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Second device, same location

This is the Development and Test model. It survives a device failure and not the facility failure that actually happened.

B Second location plus VPN

Two locations meet the facility requirement, and the VPN adds the encrypted fallback. DX is preferred over VPN by default, so failover is automatic.

C Upgrade to 100 Gbps

Answers a bandwidth question nobody asked. One faster circuit in one building is still one building.

D LAG at one location

A LAG is one logical link for capacity and device redundancy. Every member terminates at the same location.

THE SIGNAL

No single facility failure eliminates every same-location answer at once. Three of the four options are in one building.

> A company has a Direct Connect with a private VIF to a virtual private gateway in one VPC. It has since built a Transit Gateway with 40 VPCs attached, and wants the data centre to reach all of them.
>
> An engineer proposes creating 40 private VIFs, one per VPC.

What should the solutions architect recommend?

A. Create the 40 private VIFs as proposed

B. Create a transit VIF to a Direct Connect gateway, associated with the Transit Gateway

C. Associate the existing virtual private gateway with the Transit Gateway

D. Peer the 40 VPCs with the VPC that holds the virtual private gateway

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A. 40 private VIFs

Within the 50 VIF limit on a dedicated connection, and forty BGP sessions to operate. It also ignores the Transit Gateway that already exists.

B. Transit VIF and DX gateway

One VIF, one BGP session, and every VPC behind the Transit Gateway becomes reachable, including ones added later.

C. Associate the VGW

A DX gateway holds virtual private gateways or Transit Gateways, never both. A VGW cannot be associated with a Transit Gateway.

D. Peer 40 VPCs

Peering is not transitive, from module six. On-premises traffic cannot pass through the peered VPC to reach the others.

THE SIGNAL

A transit VIF is the only route from Direct Connect to a Transit Gateway. If both appear in a question, that is the connection.

> A bank buys a 1 Gbps hosted connection from a Direct Connect partner. Regulation requires all traffic between the data centre and AWS to be encrypted in transit, end to end.
>
> The network team has asked whether MACsec on the connection satisfies the requirement.

What should the solutions architect advise?

A Yes, enable MACsec on the hosted connection

B No; run a private IP VPN over a transit VIF, so IPsec covers the whole path

C No; encryption over Direct Connect is not possible, so use a Site-to-Site VPN over the internet

D Yes, provided the application also uses TLS

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Enable MACsec

MACsec runs on supported dedicated ports, not hosted connections, and covers only router to router. The partner segment stays unprotected.

B Private IP VPN over transit VIF

IPsec inside the circuit protects the traffic regardless of who is in the path, which is what end to end requires.

C Encryption over DX impossible

Simply false. Both MACsec and private IP VPN exist for this, and abandoning the circuit gives up its predictability for nothing.

D MACsec plus TLS

TLS protects application traffic, and the requirement covers all traffic. It also does not fix MACsec's unavailability here.

THE SIGNAL

Hosted connection plus end to end is the pair that decides it. A partner in the path is exactly what MACsec does not cover.

A company runs four Site-to-Site VPN connections from its data centre to a virtual private gateway, expecting roughly 5 Gbps in total. Measured throughput from AWS to on-premises sits stubbornly near 1.2 Gbps regardless of how many connections are added.

All connections use BGP and are healthy.

What is the cause, and what should change?

A. A virtual private gateway has an aggregate throughput limit; terminate on a Transit Gateway instead

B. Enable ECMP on the virtual private gateway

C. Switch the connections to static routing to reduce BGP overhead

D. Request a quota increase for VPN tunnel bandwidth

◆ PAUSE AND DECIDE

What is the cause, and what should change?

A VGW aggregate limit; move to TGW

A VGW caps at roughly 1.25 Gbps in aggregate however many connections attach. Only a Transit Gateway lifts that ceiling.

B ECMP on the VGW

The closest miss. ECMP is real and is the right idea, and it is supported only on a Transit Gateway, never on a VGW.

C Static routing

Backwards. Static routing would remove ECMP as an option entirely, since it requires BGP.

D Quota increase

The aggregate limit is service behaviour rather than an adjustable quota. Support cannot raise it.

THE SIGNAL

Adding connections changing nothing is the giveaway. A per-connection problem would improve with more connections; a ceiling does not.

A company has Direct Connect at two locations, intending one as primary and one as backup. Both advertise the same 10.10.0.0/16 prefix, except that the backup site also advertises 10.10.5.0/24 for a legacy application.

Traffic destined for 10.10.5.0/24 always takes the backup circuit, even when the primary is healthy.

Why, and what should be changed?

A AS_PATH prepending on the backup is missing; add it to fix the /24

B Longest prefix match wins before any BGP attribute; advertise the /24 from both sites

C The primary connection has a lower bandwidth and is being deprioritised

D Direct Connect always load balances across all available circuits

◆ PAUSE AND DECIDE

Why, and what should be changed?

A **Add AS_PATH prepending**

Prepending influences choice between routes of equal specificity. A /24 beats a /16 before AS_PATH is ever consulted.

B **Longest prefix wins; advertise from both**

Specificity is decided first. Once both sites advertise the /24, the usual attributes take over and the primary is preferred.

C **Bandwidth deprioritisation**

BGP does not select paths by circuit bandwidth. Nothing in the scenario mentions differing speeds anyway.

D **Always load balances**

Not true, and it would not explain traffic consistently choosing one circuit.

THE SIGNAL

Traffic ignoring your carefully tuned preferences is nearly always specificity. Nothing outranks longest prefix match.

> A retailer must connect a new distribution centre to AWS. The site opens in three weeks. Steady-state traffic is expected to be around 400 Mbps, rising sharply during seasonal peaks.
>
> A Direct Connect order has been placed and the partner quotes ten to twelve weeks.

What should the solutions architect recommend?

A. Delay the site opening until the Direct Connect is delivered

B. Establish a Site-to-Site VPN to the Transit Gateway now, and keep it as backup once DX is live

C. Cancel the Direct Connect order and rely on VPN permanently

D. Route the site's traffic over the existing data centre's Direct Connect

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A. Delay the opening

Subordinates a business event to a network order when a same-week option exists. Architecture rarely gets to set the opening date.

B. VPN now, backup later

Up in minutes, comfortably above 400 Mbps, and it becomes the encrypted backup path once the circuit arrives.

C. Cancel the DX order

Tempting given the numbers, and the scenario warns of sharp seasonal peaks that a single tunnel may not absorb.

D. Backhaul via the data centre

Adds a hop and a dependency, and makes the existing data centre a single point of failure for a site that does not need one.

THE SIGNAL

Pair this with scenario one. There the VPN was the backup; here it is the bridge. Same service, different job.

What to carry out of this module

- **A transit VIF is the only path from Direct Connect to a Transit Gateway.** Both in a question means that VIF.
- **A hosted connection supports one VIF.** A dedicated connection supports 51.
- **A VGW caps near 1.25 Gbps in aggregate.** ECMP and 5 Gbps tunnels both need a Transit Gateway.
- **Longest prefix match beats every BGP attribute.** Wrong-circuit traffic is nearly always specificity.
- MACsec protects the circuit. A private IP VPN protects the journey, partners included.

> ON THE EXAM
>
> Check what terminates the connection before anything else. A virtual private gateway is the hidden ceiling behind several answers here.

MODULE 08 · DOMAIN 1 - DESIGN SOLUTIONS FOR ORGANIZATIONAL COMPLEXITY

Hybrid DNS and Route 53 at Scale

Two resolvers that know nothing about each other, and how to introduce them.

DOMAIN 1 of 4 EXAM WEIGHT 26% MODULE 8 of 44

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Your VPC has a resolver. Your data centre has a resolver. Neither knows the other exists until you **introduce them, in each direction separately**.

How a VPC resolves a name

01

Instance asks

The query goes to the VPC resolver at base plus two.

02

Private zone

Any private hosted zone associated with this VPC is checked.

03

Resolver rule

A forwarding rule may send the query to on-premises.

04

Public DNS

Otherwise it resolves publicly, as normal.

ON THE EXAM

A private hosted zone only answers for VPCs it is associated with. Association is the step people forget.

Inbound or outbound endpoint?

INBOUND ENDPOINT

On-premises asks AWS

- Gives on-prem an IP to query
- Resolves private hosted zone names
- Point your data centre DNS at it
- Traffic comes in over DX or VPN

OUTBOUND ENDPOINT

AWS asks on-premises

- Lets the VPC resolver forward out
- Paired with a forwarding rule
- The rule names the domain to forward
- Traffic goes out over DX or VPN

THE TIEBREAKER

Name the direction of the question, not the traffic. Who needs an answer decides which endpoint you need.

Resolver rules and sharing them

- **A forwarding rule says: for this domain, ask these servers.** It needs an outbound endpoint.
- Rules are **shareable with AWS RAM**, so one networking account can define them once for the organisation.
- A rule must still be **associated with each VPC** that should use it. Sharing alone changes nothing.
- The most specific matching domain wins, in the same spirit as longest prefix match.

> ON THE EXAM
>
> Sharing and associating are two steps. A shared rule that nobody associated is the commonest hybrid DNS failure.

Private hosted zones

- **Answers only for associated VPCs.** Association can cross accounts, and is authorised through the API.
- **Split-horizon** works: a public and a private zone for the same name, answering differently by where you ask from.
- The VPC needs **DNS hostnames and DNS resolution enabled**, or the zone is silently ignored.
- In a shared VPC, the **owner** associates the zone. Participants cannot.

> ON THE EXAM
>
> Split-horizon is how one hostname reaches a private IP internally and a public endpoint externally, with no client changes.

Route 53 Profiles bundle private zones, Resolver rules and firewall groups into **one shareable object**.

The routing policies

POLICY	ROUTES BY	CLASSIC USE
Simple	One answer, no logic	A single endpoint
Weighted	Proportions you set	Canary releases, A/B splits
Latency	Lowest measured latency	Fastest experience globally
Failover	Health of the primary	Active-passive disaster recovery
Geolocation	Where the user is	Data residency, language, licensing
Geoproximity	Distance, with a bias dial	Shifting load between Regions

ON THE EXAM

Multivalue answer also exists: several healthy IPs returned at random. It is not a load balancer, but it is health-aware.

Latency or geolocation?

LATENCY

Fastest, wherever that is

- Optimises the user experience
- Which Region is fastest can change
- No guarantee about jurisdiction
- Use when speed is the goal

GEOLOCATION

Deterministic by location

- Guarantees which endpoint answers
- Continent, country or US state
- Set a default for unmatched queries
- Use when the rule is legal

THE TIEBREAKER

Compliance and data residency mean geolocation. Latency optimises speed and promises nothing about where data lands.

Health checks, and the private resource trap

- **Endpoint checks** probe an IP or domain. **Calculated checks** combine other checks. **CloudWatch alarm checks** follow a metric.
- **Route 53 health checkers live on the public internet.** They cannot reach a private IP address.
- So for a private resource, publish a metric and use a **CloudWatch alarm health check**.
- Failover only works if the health check is genuinely measuring the thing users depend on.

> ON THE EXAM
>
> Failover configured against a private endpoint that never goes unhealthy is the classic broken DR design.

Resolver DNS Firewall

- **Filters outbound DNS queries from your VPCs** against domain lists, allowing, blocking or alerting.
- AWS-managed lists cover malware and botnet command and control. Custom lists cover your own policy.
- The point is **DNS exfiltration**: data smuggled out inside query names, where no other control looks.
- Rule groups can be bundled into a Route 53 Profile and applied across the organisation.

> ON THE EXAM
>
> Security groups and NACLs never inspect the query name. DNS-layer exfiltration is invisible to both.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
On-premises must resolve AWS private names	Inbound endpoint
AWS workloads must resolve on-premises names	Outbound endpoint plus a forwarding rule
Same DNS config repeated across many VPCs	Route 53 Profiles
EU users must be served from the EU	Geolocation, not latency
Failover for a private, internal endpoint	CloudWatch alarm health check
Data leaving over DNS queries	Resolver DNS Firewall

ON THE EXAM

For every hybrid DNS question, say out loud who is asking and who holds the answer. The endpoint follows from that.

Meridian Freight runs applications in AWS that must resolve hostnames in corp.meridian.internal, held by DNS servers in the data centre. Separately, an on-premises batch system must resolve names in a private hosted zone for aws.meridian.internal.

Direct Connect is already in place and routing works in both directions.

What should the solutions architect deploy?

A. An outbound endpoint with a forwarding rule for corp.meridian.internal

B. An inbound endpoint, and on-premises DNS pointed at it

C. Both an inbound and an outbound endpoint, with a forwarding rule for corp.meridian.internal

D. A private hosted zone for corp.meridian.internal associated with every VPC

◆ PAUSE AND DECIDE

What should the solutions architect deploy?

A Outbound only

Solves the first requirement and leaves the batch system unable to resolve the AWS names. Half the scenario.

B Inbound only

The other half. On-premises can now resolve AWS names, and AWS still cannot resolve the corporate domain.

C Both endpoints

Two directions, two endpoints. The forwarding rule sends corporate queries out; the inbound endpoint gives on-premises somewhere to ask.

D Copy the zone into AWS

Duplicates a zone somebody else maintains. It goes stale the moment a record changes on-premises, and silently.

THE SIGNAL

Two requirements pointing opposite ways need two endpoints. Options A and B are each exactly half the answer.

> A company has 70 VPCs across 40 accounts. Every VPC needs the same three Resolver forwarding rules, two private hosted zone associations, and one DNS Firewall rule group. New VPCs appear weekly.
>
> A platform engineer currently runs a script after each VPC is created, and it has been missed twice.

What should the solutions architect recommend?

A. Share the rules with AWS RAM and keep the script for associations

B. Build a Route 53 Profile containing all of it, share it with RAM, and associate it with each VPC

C. Add an AWS Config rule that flags VPCs missing the associations

D. Consolidate the 70 VPCs into one shared VPC

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A. RAM plus the script

RAM shares the rules and does not associate them. The script, which is the thing being missed, survives.

B. A Route 53 Profile

One object holding zones, rules and firewall groups. Associate it once, and later additions to the Profile reach every VPC automatically.

C. Config rule to flag gaps

Detective. It reports the omission after the fact, which is roughly what happened both times it was missed.

D. One shared VPC

An enormous network restructuring in answer to a DNS configuration problem, discarding isolation the company chose deliberately.

THE SIGNAL

Option A is the pre-Profiles answer and still half right. Sharing is not associating, which is exactly the gap the script was filling.

A company operates identical application stacks in eu-west-1 and us-east-1. New data protection rules require that personal data of EU residents is processed only in the EU, with no exceptions.

The current design uses latency-based routing, and traffic analysis shows some EU users being served from us-east-1 during European network events.

What should the solutions architect change?

A Keep latency-based routing and add health checks on the EU stack

B Switch to geolocation routing, with a default record for unmatched queries

C Switch to geoproximity routing with a strong positive bias toward eu-west-1

D Use weighted routing, sending 100% of traffic to eu-west-1

◆ PAUSE AND DECIDE

What should the solutions architect change?

A Latency plus health checks

Health checks change which endpoints are eligible. They do nothing about latency choosing a non-EU Region that is perfectly healthy.

B Geolocation with a default

Deterministic by the user's location, which is what a legal boundary requires. The default record catches queries matching no location.

C Geoproximity with bias

The closest miss. Bias shifts proportions and still permits routing outside the EU, so it cannot satisfy no exceptions.

D Weighted 100%

Sends every user in the world to the EU, abandoning the US stack. It over-delivers on compliance by breaking the architecture.

THE SIGNAL

No exceptions is a phrase demanding a guarantee. Only geolocation gives one; the others express preference.

> An internal application runs behind an internal Application Load Balancer in two Regions, reachable only over private addresses. The team has configured Route 53 failover records with endpoint health checks against the load balancer DNS names.
>
> The health checks report Unhealthy permanently, and failover has never worked in testing.

What is the cause, and what should be done?

A. The health check interval is too aggressive; increase it to 30 seconds

B. Route 53 health checkers are public and cannot reach private endpoints; use CloudWatch alarm health checks

C. Add the Route 53 health checker IP ranges to the load balancer security group

D. Switch the records from failover to multivalue answer routing

◆ PAUSE AND DECIDE

What is the cause, and what should be done?

A Increase the interval

Tuning frequency on a check that can never succeed. Permanently unhealthy is not a flapping problem.

B CloudWatch alarm health checks

The checkers are on the public internet and an internal ALB has no public address. An alarm on ALB metrics can be evaluated instead.

C Allow the checker IPs

The sharpest distractor. Those ranges are published and this is real advice for public endpoints, and no security group rule makes a private IP routable from the internet.

D Multivalue answer

Also health-check driven, so it inherits the identical problem and additionally abandons the active-passive design.

THE SIGNAL

Permanently unhealthy rather than intermittently means the checker never reached it at all. Ask whether it could have, before tuning anything.

> An internal portal is published at portal.meridian.com. Staff in the office and in AWS should reach it over private addresses. Field engineers on the public internet must reach the same hostname through a public load balancer.
>
> Changing the hostname for either group is not acceptable.

What should the solutions architect implement?

A. A public hosted zone and a private hosted zone, both for meridian.com, with different records

B. One public hosted zone, with the portal record pointing at the private IP

C. Two hostnames, internal and external, documented for each audience

D. A Resolver forwarding rule sending portal.meridian.com to on-premises DNS

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Public and private zones

Split-horizon DNS. Queries from associated VPCs hit the private zone; everyone else gets the public one. Same name, two answers.

B Public zone, private IP

Publishes internal addressing to the world and gives field engineers an address they cannot route to.

C Two hostnames

Explicitly ruled out by the scenario. It also pushes the problem onto every user and every bookmark.

D Forwarding rule

Forwarding changes which server answers, not what the answer is. Both audiences would still receive the same record.

THE SIGNAL

One hostname, two audiences, two answers is the split-horizon signature. The refusal to change the hostname is the planted detail.

> A security review finds that a compromised instance in a private subnet has been exfiltrating data encoded inside DNS query names to an attacker-controlled domain. The subnet has no internet gateway and no NAT gateway.
>
> Security groups, network ACLs and VPC Flow Logs were all in place and showed nothing unusual.

Which control would have detected or prevented this?

A. A stricter outbound security group on the instance

B. Route 53 Resolver DNS Firewall with a rule group blocking unapproved domains

C. AWS Network Firewall inspecting the subnet's outbound traffic

D. Enabling VPC Flow Logs at the subnet level as well

◆ PAUSE AND DECIDE

Which control would have detected or prevented this?

A. Stricter security group

Security groups filter addresses and ports, never query names. The traffic went to the VPC resolver, which every instance legitimately needs.

B. Resolver DNS Firewall

The only control that inspects the queried domain. It blocks or alerts on lookups to domains outside your policy.

C. Network Firewall

Would inspect traffic leaving the VPC. The queries never left this way; they went to the VPC resolver, which sits outside that path.

D. Subnet Flow Logs

Records the flow to the resolver as ordinary DNS. Flow logs carry addresses and ports, and no query names at all.

THE SIGNAL

No internet gateway and no NAT is the planted detail. The data left through the one path everybody permits: DNS resolution.

What to carry out of this module

- **Inbound means on-premises asks AWS. Outbound means AWS asks on-premises.** Two directions need two endpoints.
- **Sharing is not associating.** Rules and zones must be associated with each VPC, which is what Profiles automate.
- **Legal requirement means geolocation. Performance requirement means latency.** Bias is not a guarantee.
- **Health checkers are public.** Private endpoints need a CloudWatch alarm health check.
- Only DNS Firewall inspects the query name. Security groups and flow logs never do.

> ON THE EXAM
>
> Say who is asking and who holds the answer. That one sentence resolves most hybrid DNS questions before the options matter.

MODULE 09 · DOMAIN 1 - DESIGN SOLUTIONS FOR ORGANIZATIONAL COMPLEXITY

Private Service Access and Traffic Inspection

Endpoints, PrivateLink, and putting a firewall in the middle of everything.

DOMAIN 1 of 4 EXAM WEIGHT 26% MODULE 9 of 44

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A gateway endpoint is a **route**. An interface endpoint is an **address**. Almost every difference follows from that.

Gateway or interface endpoint?

GATEWAY ENDPOINT

A route table entry

- S3 and DynamoDB only
- No hourly or per-GB charge
- Only works inside the VPC
- Not reachable from on-premises

INTERFACE ENDPOINT

An ENI with a private IP

- Almost every AWS service
- Per AZ per hour, plus per GB
- Reachable over peering and TGW
- Works from on-premises

THE TIEBREAKER

On-premises, a peered VPC or a Transit Gateway in the question eliminates the gateway endpoint immediately.

Gateway endpoint details

- **It adds a prefix list to a route table.** Choose which route tables, and the subnets using them are covered.
- **Free.** It also removes NAT gateway processing charges for that traffic, which is often the real saving.
- Not reachable from on-premises, a peered VPC, or over a Transit Gateway. It is a local route.
- An **endpoint policy** limits what can be done through it, independently of IAM.

> ON THE EXAM
>
> The saving is rarely the endpoint fee, which is zero anyway. It is the NAT gateway processing charge you stop paying.

Interface endpoint details

- **An ENI per subnet you select**, each with a private IP. You pay per ENI per hour and per GB.
- **Private DNS** makes the normal service hostname resolve to the endpoint, so applications need no change.
- It has a **security group**, which a gateway endpoint does not.
- **Cross-Region PrivateLink** now reaches AWS services in other Regions, which was impossible until late 2025.

> ON THE EXAM
>
> Private DNS is why applications never notice. Turn it off and clients keep resolving the public endpoint instead.

Locking a service to your network

- **An endpoint policy** controls what can pass through the endpoint. It grants nothing on its own.
- **aws:SourceVpce** in a bucket policy means: only requests arriving through this endpoint.
- **aws:SourceVpc** is the broader form, restricting to a whole VPC rather than one endpoint.
- An endpoint alone does not stop anyone reaching the bucket publicly. The resource policy does that.

> ON THE EXAM
>
> Creating an endpoint changes the path traffic takes. Only a resource policy condition makes it the only permitted path.

Publishing your own service with PrivateLink

- **You create an endpoint service** in front of a Network Load Balancer or Gateway Load Balancer.
- Consumers create an interface endpoint. **You accept or reject each connection**, and name permitted principals.
- **Address space never has to agree**, which is why it survives overlapping CIDRs.
- It is unidirectional. The consumer reaches you; you do not reach the consumer.

> ON THE EXAM
>
> One service, one direction, no shared routing. That combination is what makes PrivateLink the answer for SaaS and acquisitions.

Two newer endpoint types, and an honest scope note

- **Resource endpoints** reach an RDS instance, an IP, or a domain in another VPC or on-premises, with **no load balancer**.
- **Service network endpoints** reach many shared services and resources through one endpoint.
- Both are **Amazon VPC Lattice** features, and **VPC Lattice is not on the exam's in-scope list**.
- Know they exist. Answer with an NLB-fronted endpoint service, which is what the exam expects.

> ON THE EXAM
>
> This is a real gap between current AWS and the exam. In production these often beat an endpoint service; on the exam they are not the answer.

Gateway Load Balancer

- **For third-party appliances**: firewalls, intrusion prevention, deep packet inspection.
- **Encapsulates traffic in GENEVE on port 6081**, so the appliance sees the original packet unaltered.
- **Transparent.** Traffic is routed to a GWLB endpoint, inspected, and continues. No address changes.
- Scales the appliance fleet and health checks it, so one dead firewall does not black-hole traffic.

> ON THE EXAM
>
> GWLB is the answer whenever a scenario names a commercial appliance. Network Firewall is the answer when it does not.

Centralised inspection and egress

01

Spoke VPC

Default route points at the Transit Gateway.

02

Transit Gateway

A spoke route table sends everything to the inspection VPC.

03

Inspect

GWLB endpoint or Network Firewall examines the traffic.

04

Egress

One NAT gateway set serves every VPC.

ON THE EXAM

Enable appliance mode on the inspection VPC attachment, or asymmetric routing breaks stateful inspection. Module six.

Network Firewall or Gateway Load Balancer?

AWS NETWORK FIREWALL

Managed, AWS-operated

- No instances to run or patch
- Suricata-compatible rules
- Domain filtering and IPS
- Scales without your involvement

GATEWAY LOAD BALANCER

Your chosen appliance

- Run a specific vendor's product
- Reuse existing rules and expertise
- You own patching and capacity
- Needed when the vendor is named

THE TIEBREAKER

A named vendor, or existing rules that must be reused, means GWLB. Anything else favours the managed service.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Large NAT charges for S3 or DynamoDB traffic	Gateway endpoint
On-premises needs private access to an AWS service	Interface endpoint
The bucket must only be reachable from our VPC	aws:SourceVpce in the bucket policy
Expose one service to customers, overlapping CIDRs	Endpoint service behind an NLB
Inspect with a named vendor appliance	Gateway Load Balancer
Endpoint costs repeated in dozens of VPCs	Centralise, plus a private hosted zone

ON THE EXAM

Ask where the caller sits. Inside the VPC allows a gateway endpoint; anywhere else forces an interface endpoint.

> Meridian Freight runs a nightly analytics job on EC2 in private subnets that reads about 40 TB per month from Amazon S3 in the same Region. Traffic reaches S3 through a NAT gateway.
>
> Finance has queried the NAT gateway line, which is dominated by data processing charges rather than hours.

What should the solutions architect recommend?

A Create a gateway endpoint for S3 and add it to the private subnets' route tables

B Create an interface endpoint for S3 in each Availability Zone

C Move the instances to public subnets with Elastic IPs

D Replace the NAT gateway with NAT instances

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Gateway endpoint

Free, and it takes 40 TB a month off the NAT gateway entirely. Same Region, same VPC, so every condition is met.

B Interface endpoint

Works, and charges per ENI per hour plus per GB. You would pay for 40 TB when a gateway endpoint carries it for nothing.

C Public subnets

Removes the NAT charge by putting analytics instances on the internet. A security regression to solve a billing problem.

D NAT instances

Still processes the same 40 TB, now on instances you patch and scale. Module six's false economy, repeated.

THE SIGNAL

Same Region, same VPC, S3. All three conditions for the free option hold, so paying per GB would be a choice.

> The same company now needs its on-premises backup software to write directly to Amazon S3 across the existing Direct Connect, without traversing the internet.
>
> An engineer suggests reusing the gateway endpoint created for the analytics workload.

What should the solutions architect advise?

A Reuse the gateway endpoint and add a route on-premises

B Create an interface endpoint for S3; gateway endpoints cannot be reached from on-premises

C Use a public VIF and rely on the bucket policy for security

D Deploy a proxy fleet in the VPC that forwards requests to the gateway endpoint

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Reuse the gateway endpoint

A gateway endpoint is a route inside one VPC. On-premises routers have no way to use somebody else's route table.

B Interface endpoint

It has a private IP that on-premises can reach over Direct Connect, which is exactly what a gateway endpoint lacks.

C Public VIF

Genuinely reaches S3 over the circuit rather than the internet, and it uses public addressing, which the requirement rules out.

D Proxy fleet

Builds and operates infrastructure to emulate a service AWS already sells, and inserts a bottleneck into a backup path.

THE SIGNAL

One constraint changed from scenario one: the caller moved outside the VPC. That alone flips a free answer to a paid one.

> An auditor requires that a bucket holding customer records can be reached only from the company's own VPCs, and never from the internet, even with valid credentials.
>
> The team has created interface endpoints in every VPC and believes the requirement is met.

What should the solutions architect tell them?

A The requirement is met; endpoint traffic does not traverse the internet

B Add a bucket policy denying requests unless aws:SourceVpce matches the approved endpoints

C Enable S3 Block Public Access on the bucket

D Add an endpoint policy on each interface endpoint allowing only that bucket

◆ PAUSE AND DECIDE

What should the solutions architect tell them?

A Already met

Endpoints change the path their traffic takes. The bucket's public endpoint is untouched and still answers anyone with credentials.

B Deny unless aws:SourceVpce

Makes the endpoint the only permitted path. Requests arriving any other way are denied regardless of credentials.

C Block Public Access

Blocks anonymous and public access. A stolen credential is neither, and the scenario says even with valid credentials.

D Endpoint policies

Limits what may pass through the endpoint. It says nothing about requests that never use the endpoint.

THE SIGNAL

Creating a path is not restricting a path. Only a condition on the resource policy closes the others.

A software company sells a fraud-scoring API. Customers run it from their own AWS accounts, many of which use 10.0.0.0/16, as does the provider. Customers must not be able to reach anything else in the provider's VPC.

The current design asks each customer to accept a VPC peering connection.

What should the solutions architect recommend?

A Continue with peering and use security groups to restrict access

B Create a PrivateLink endpoint service behind a Network Load Balancer

C Publish the API on the internet behind an Application Load Balancer with IP allow-lists

D Attach every customer to a shared Transit Gateway

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A. Peering plus security groups

Peering refuses overlapping CIDRs outright, so most customers cannot connect at all. Security groups never get a chance to help.

B. PrivateLink endpoint service

Exposes one service, tolerates identical address space, and gives the provider no route into customer networks.

C. Public ALB with allow-lists

Puts a fraud API on the internet and asks customers to maintain address lists that change whenever they scale.

D. Shared Transit Gateway

Routing again, so overlapping CIDRs still fail, and it would join customers' networks to each other.

THE SIGNAL

Overlapping CIDRs plus one service plus mutual distrust is the PrivateLink signature. All three point the same way.

A bank must inspect all traffic between its VPCs using the same commercial firewall product it runs on-premises, because its rule base and its regulators' approvals are tied to that vendor.

The proposed design routes inter-VPC traffic through an inspection VPC with the appliances spread across three Availability Zones.

What should the design use, and what must be enabled?

A. AWS Network Firewall in the inspection VPC, with appliance mode on its attachment

B. Gateway Load Balancer fronting the appliances, with appliance mode on the inspection VPC attachment

C. A Network Load Balancer fronting the appliances, with cross-zone load balancing enabled

D. An Application Load Balancer fronting the appliances, with sticky sessions

◆ PAUSE AND DECIDE

What should the design use, and what must be enabled?

A. Network Firewall

Excellent and managed, and it is not the named vendor. The regulatory approval and the rule base both attach to a specific product.

B. GWLB plus appliance mode

GWLB runs the chosen vendor transparently over GENEVE, and appliance mode keeps both directions of a flow in one AZ.

C. NLB, cross-zone

An NLB is a load balancer, not a transparent bump in the wire. Traffic would need to be addressed to it rather than routed through.

D. ALB, sticky sessions

Layer 7 only, so it cannot inspect arbitrary protocols, and it terminates connections rather than passing them through.

THE SIGNAL

A named vendor forces GWLB. Stateful appliances across AZs forces appliance mode. The question tests both at once.

> A company has 55 VPCs, each with interface endpoints for Secrets Manager, Systems Manager, ECR and CloudWatch Logs, in three Availability Zones. The endpoint charges have become a noticeable monthly line.
>
> All VPCs already attach to a Transit Gateway.

What should the solutions architect recommend?

A. Reduce each VPC's endpoints to a single Availability Zone

B. Centralise the endpoints in a shared services VPC, and use a private hosted zone shared across accounts

C. Remove the endpoints and route the traffic through NAT gateways instead

D. Replace the interface endpoints with gateway endpoints

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A One AZ per VPC

Cuts the bill by two thirds and creates a single point of failure in every VPC for secrets and logging.

B Centralise plus private hosted zone

660 endpoints become 12. The TGW already provides the path, and a shared private hosted zone points clients at the central ENIs.

C Back to NAT gateways

Trades endpoint charges for NAT processing charges and puts service traffic across the internet, losing the privacy the endpoints bought.

D Gateway endpoints

Only S3 and DynamoDB have them. None of these four services does, so the option is not available.

THE SIGNAL

Interface endpoints are reachable over a Transit Gateway, which is what makes centralisation possible. Gateway endpoints never could be.

What to carry out of this module

- **Gateway endpoint is a route; interface endpoint is an address.** Every difference follows.
- **On-premises, peering or a TGW rules out gateway endpoints.** They cover only S3 and DynamoDB.
- **Creating an endpoint is not restricting access.** Only aws:SourceVpce closes the other paths.
- A named vendor appliance means **GWLB**. Stateful inspection across AZs means **appliance mode**.

> ON THE EXAM
>
> Ask where the caller sits, then whether the requirement is a new path or a closed one. Those two questions cover most of this area.

MODULE 10 · DOMAIN 1 - DESIGN SOLUTIONS FOR ORGANIZATIONAL COMPLEXITY

Org-Wide Logging, Auditing and Security Aggregation

Aggregate, make immutable, detect, query. Most designs do the first and forget the second.

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An audit log an administrator can delete is not an audit log. **Immutability is the requirement**, not collection.

The organisation trail

- **One trail, created once, covering every account** including accounts created afterwards.
- **Member accounts cannot alter or delete it.** They can see it and nothing more.
- Delegate administration to a security account. **Up to three delegated administrators.**
- The **management account always owns it**, whoever created it, so removing a delegated admin leaves it intact.

> ON THE EXAM
>
> A per-account trail is only as trustworthy as that account's administrator. That is the whole argument for an organisation trail.

Making the logs tamper-evident

- **A separate Log Archive account** that workload administrators have no access to at all.
- **S3 Object Lock in compliance mode.** Nobody can delete or overwrite before the retention date. Not even the root user.
- **Log file integrity validation** produces digest files, so alteration is detectable.
- An SCP denying deletion is a good layer, and the management account is exempt from it. Module two.

> ON THE EXAM
>
> Governance mode can be overridden by a privileged user. Compliance mode cannot. Regulators mean compliance mode.

AWS Config across the organisation

- **The recorder** captures configuration state and every change, per account and Region.
- **An aggregator** collects it all into one account for a single compliance view.
- **Organisation rules and conformance packs** deploy from a delegated administrator to every account at once.
- **Advanced queries** answer questions across the whole estate, such as which buckets are public.

> ON THE EXAM
>
> Config knows what a resource looked like at any moment. CloudTrail knows who changed it. You usually need both.

CloudTrail, Config or CloudWatch?

QUESTION	SERVICE	BECAUSE
Who deleted the bucket?	CloudTrail	It records API calls and callers
What did this look like on Tuesday?	AWS Config	It records state over time
Are any buckets public right now?	AWS Config	Current state, across the estate
Alert me when errors spike	CloudWatch	Alarms evaluate metrics

ON THE EXAM

Who did it means CloudTrail. What it looked like means Config. How it is performing means CloudWatch.

The delegated administrator pattern

- **Nearly every security service supports it**: CloudTrail, Config, GuardDuty, Security Hub, Macie, Detective, IAM Access Analyzer.
- It moves daily security work **out of the management account**, which module two said to keep empty.
- The delegated admin sees findings from every member account, and can enable the service in new accounts automatically.
- Only the **management account** can appoint or remove one.

> ON THE EXAM
>
> Security team needs a full view but must not hold management account credentials is the delegated administrator signature.

The detection stack: who finds what

- **GuardDuty** finds threats from CloudTrail, DNS and flow logs. Behaviour, not configuration.
- **Inspector** finds software vulnerabilities and unintended network exposure.
- **Macie** finds sensitive data in S3. **Detective** investigates a finding after the fact.
- **Security Hub** aggregates all of them and runs posture checks against standards.

> ON THE EXAM
>
> GuardDuty detects. Detective explains. Security Hub aggregates. Naming which job a scenario needs picks the service.

A naming warning on Security Hub

- **The old service is now called AWS Security Hub CSPM.** Standards, controls, and aggregated findings.
- **A new, larger AWS Security Hub** now exists, with attack-path analysis and paid plans. CSPM sits inside it.
- The exam's in-scope list says **AWS Security Hub**, and means the posture-management behaviour described here.
- Answer with aggregation, standards and controls. That is true of both, and it is what questions test.

> ON THE EXAM
>
> Another rename that study material has not caught up with, like QuickSight becoming Amazon Quick. The exam behaviour is unchanged.

Querying and responding

- **Athena over the log archive bucket** is the standard way to query years of CloudTrail cheaply.
- **CloudWatch Logs Insights** for recent operational logs. **OpenSearch** when you need dashboards and full-text search.
- **EventBridge** is how findings become action: a GuardDuty finding triggers a Lambda or a runbook.
- CloudTrail Lake exists and entered maintenance in 2026. Do not reach for it as a new design.

> ON THE EXAM
>
> Querying archived logs occasionally is Athena. Searching live logs constantly is OpenSearch. The frequency decides it.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
An account admin disabled logging	Organisation trail, plus an SCP
Logs must be provably unaltered for years	Object Lock compliance mode, plus validation
Who made this change?	CloudTrail
What did it look like before the change?	AWS Config
Security team needs a view without management access	Delegated administrator
Respond automatically to a finding	EventBridge to Lambda or a runbook

ON THE EXAM

Ask whether the gap is collection, protection, detection or query. Naming the missing job picks the service.

> During an incident review, Meridian Freight discovers that an administrator in a workload account disabled CloudTrail for two hours and deleted the log files from that account's S3 bucket. Each account currently configures its own trail.
>
> The security team must guarantee this cannot happen again in any of the 140 accounts.

What should the solutions architect implement?

A. A Config rule that detects when CloudTrail is disabled and re-enables it

B. An organisation trail delivering to a separate Log Archive account, with an SCP denying trail changes

C. Enable S3 versioning on each account's log bucket

D. Require MFA before CloudTrail can be disabled

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A **Config rule with remediation**

Detective. It notices after the gap exists, and those two hours are already unlogged and gone.

B **Organisation trail plus SCP**

Member accounts cannot alter an organisation trail, the logs land where they cannot reach, and the SCP closes the rest.

C **Versioning**

Protects against overwrite by keeping old versions, and an administrator with delete permission can remove versions too.

D **Require MFA**

Raises the bar for an attacker and none at all for the legitimate administrator who did this, who has MFA.

THE SIGNAL

Guarantee and cannot happen again are preventive words. They eliminate every detective option, however fast it remediates.

> A financial regulator requires seven years of audit logs, and requires the company to demonstrate that no log could have been altered or deleted during that period, by anyone, including the cloud team.
>
> Logs are already delivered to a dedicated Log Archive account.

What should the solutions architect add?

A. S3 Object Lock in governance mode with a seven-year retention period

B. S3 Object Lock in compliance mode, plus CloudTrail log file integrity validation

C. A lifecycle policy transitioning logs to Glacier Deep Archive after 90 days

D. Cross-Region replication of the log bucket to a second Region

◆ PAUSE AND DECIDE

What should the solutions architect add?

A. Governance mode

The sharpest distractor. Governance mode can be overridden by a user with the right permission, which is precisely the cloud team.

B. Compliance mode plus validation

Compliance mode cannot be overridden by anyone, root included. Digest files then let the company demonstrate integrity rather than assert it.

C. Lifecycle to Deep Archive

Sensible for cost over seven years and orthogonal to the requirement. Cheap storage is still deletable storage.

D. Cross-Region replication

Protects against losing a Region. A deletion request replicates as a deletion, so it does not protect against deliberate destruction.

THE SIGNAL

Including the cloud team is the phrase that separates the two Object Lock modes. Governance exists to be overridden; compliance does not.

> A production outage is traced to a security group that stopped allowing database traffic. The team needs to know what the rules were before the change, who made it, and whether any other security group in the estate has the same problem now.
>
> CloudTrail, Config and CloudWatch are all enabled organisation-wide.

Which services answer those three questions, in order?

A Config, CloudTrail, Config

B CloudTrail, Config, CloudWatch

C CloudWatch, CloudTrail, Config

D Config, Config, CloudTrail

◆ PAUSE AND DECIDE

Which services answer those three questions, in order?

A Config, CloudTrail, Config

Prior state is Config's configuration history. Who made the call is CloudTrail. Estate-wide current state is a Config advanced query.

B CloudTrail, Config, CloudWatch

Inverts the first two, and CloudWatch holds metrics and logs rather than a searchable inventory of resource configuration.

C CloudWatch, CloudTrail, Config

CloudWatch cannot tell you what a security group's rules were. It never recorded them.

D Config, Config, CloudTrail

Right about state twice, and CloudTrail is the identity question rather than the inventory one.

THE SIGNAL

Prior state and current state are both Config. Only the who question belongs to CloudTrail.

> A security team of six must monitor findings across 200 accounts. Policy forbids them from holding credentials in the management account. They also want GuardDuty and Security Hub enabled automatically in every new account.
>
> Today they are emailed a spreadsheet of findings each week.

What should the solutions architect recommend?

A. Grant the security team a cross-account role into all 200 accounts

B. Register a security account as delegated administrator for GuardDuty and Security Hub, with auto-enable for new accounts

C. Give the security team read-only access to the management account

D. Export findings from each account to a central S3 bucket and query with Athena

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Cross-account roles**

Six people switching between 200 accounts to read findings. It works and it does not aggregate anything.

B **Delegated administrator**

One console showing every account's findings, no management account credentials, and auto-enable covers accounts created next month.

C **Read-only in the management account**

Directly contradicts the stated policy, and read-only there still would not aggregate member account findings.

D **Export and query**

Rebuilds aggregation the services already do, and swaps a live console for a query the team must remember to run.

THE SIGNAL

Three requirements: one view, no management credentials, automatic for new accounts. Delegated administration is the only option meeting all three.

> An auditor asks for a list of every S3 bucket across all accounts and Regions that currently allows public access, and asks how the company would produce the same list weekly without manual work.
>
> An organisation trail and a Config aggregator are both in place.

What should the solutions architect propose?

A Query CloudTrail in Athena for PutBucketAcl and PutBucketPolicy calls

B Run a Config advanced query against the aggregator, and deploy an organisation Config rule for ongoing evaluation

C Enable Macie in every account and review its findings

D Write a Lambda function that lists every bucket in every account weekly

◆ PAUSE AND DECIDE

What should the solutions architect propose?

A **CloudTrail in Athena**

Finds calls that changed permissions, not buckets that are public now. A bucket made public before the trail existed never appears.

B **Config query plus org rule**

The aggregator already holds current state for every account and Region, and the organisation rule keeps evaluating it continuously.

C **Macie**

Finds sensitive data inside buckets. It answers what is in them, not who can reach them.

D **Custom Lambda**

Builds and maintains a weekly job to reproduce a query the aggregator can already answer.

THE SIGNAL

Currently allows is a state question, so it is Config. CloudTrail would only find buckets whose permissions changed while it was watching.

> GuardDuty in the delegated administrator account raises findings for compromised EC2 credentials several times a year. The current process is that an analyst reads the finding the next working day and manually isolates the instance.
>
> The security team wants isolation to happen within minutes, at any hour, across all accounts.

What should the solutions architect implement?

A. A CloudWatch alarm on the GuardDuty findings metric, notifying an on-call rota

B. An EventBridge rule matching the finding type, invoking a Lambda function that applies an isolation security group

C. Enable Detective so analysts can investigate faster

D. Increase GuardDuty finding frequency to fifteen minutes

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Alarm and on-call rota

Wakes a human faster. Within minutes at any hour is a machine requirement, and paging six people several times a year is expensive.

B EventBridge to Lambda

Findings raise events immediately. A rule can match the type and act automatically, in every account, without anyone awake.

C Enable Detective

Speeds up investigation after isolation. Valuable, and it takes no action, so nothing is contained any sooner.

D Raise finding frequency

Changes how often findings are published, not what happens next. The delay is the human, not the detection.

THE SIGNAL

Within minutes at any hour is an automation requirement. Any option ending in a human reading something cannot meet it.

What to carry out of this module

- **An organisation trail cannot be altered by member accounts.** A per-account trail can.
- **Governance mode can be overridden; compliance mode cannot.** Regulators mean compliance.
- **Config for state, CloudTrail for identity, CloudWatch for performance.** Past and present state are both Config.
- GuardDuty detects, Detective explains, Security Hub aggregates, EventBridge acts.

> ON THE EXAM
>
> Ask which of the four jobs is missing: aggregate, make immutable, detect, or query. Collection is usually already there.

MODULE 11 · DOMAIN 1 - DESIGN SOLUTIONS FOR ORGANIZATIONAL COMPLEXITY

Enterprise Cost Visibility and Purchasing

Closing Domain 1. Twenty-two per cent of the bank mentions cost.

DOMAIN 1 of 4 EXAM WEIGHT 26% MODULE 11 of 44

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You cannot allocate what you never tagged, and tagging is **not retroactive**.

The four visibility tools

TOOL	ANSWERS	REACH FOR IT WHEN
Cost Explorer	What have we spent, roughly?	Analysis, trends, forecasting, in a console
Cost and Usage Report	What did every resource cost, hourly?	Chargeback, audit, anything needing detail
AWS Budgets	Are we about to overspend?	Thresholds, alerts, and automated actions
Cost Anomaly Detection	Has something changed unexpectedly?	Spotting spikes nobody planned

ON THE EXAM

Cost Explorer looks backwards and summarises. Budgets looks forwards. The Cost and Usage Report is the only one with per-resource detail.

Cost Explorer or the Cost and Usage Report?

COST EXPLORER

Fast answers, summarised

- Console and API, no setup
- Up to 13 months back, 12 forward
- Grouping and filtering built in
- Not per-resource, per-hour detail

COST AND USAGE REPORT

Every line item

- Delivered to S3, queried with Athena
- Hourly, with resource IDs
- CUR 2.0 via Data Exports is current
- You build the reporting yourself

THE TIEBREAKER

Chargeback, audit, or showback per team means the Cost and Usage Report. Nothing else has the granularity.

Tagging for cost allocation

- **Tags must be activated** in the billing console before they appear in cost data. Applying a tag is not enough.
- **Activation is not retroactive.** Spend before activation is never attributed to that tag.
- **AWS-generated** tags such as createdBy, and **user-defined** tags that you apply and activate.
- Enforce with a **tag policy** for consistency, and an **SCP** to deny creation without the tag.

> ON THE EXAM
>
> Tag policies standardise spelling and case. Only an SCP actually prevents an untagged resource being created.

Consolidated billing and discount sharing

- **Consolidated billing aggregates usage**, so volume tiers are reached faster across the whole organisation.
- **A Savings Plan applies to the purchasing account first.** Only unused commitment spills to other accounts.
- Spillover goes to the accounts with the **largest calculated saving**, not the ones that need it most.
- **The management account can deactivate sharing per account**, and both ends must have it on.

> ON THE EXAM
>
> If the account owning a Savings Plan leaves the organisation, the discount stops applying to the consolidated bill entirely.

The purchase options

OPTION	COMMIT TO	FITS
On-Demand	Nothing	Unpredictable or short-lived work
Savings Plans	Dollars per hour, 1 or 3 years	Steady compute spend, changing shape
Reserved Instances	Capacity or usage, 1 or 3 years	Stable workloads, and services SPs miss
Spot	Nothing, tolerate interruption	Fault-tolerant, flexible timing work
Capacity Reservations	Capacity in one AZ	Guaranteed capacity, not a discount

ON THE EXAM

A Capacity Reservation reserves capacity and gives no discount. A Savings Plan gives a discount and reserves no capacity.

Compute or EC2 Instance Savings Plan?

COMPUTE SAVINGS PLAN

Flexible, up to 66%

- Any family, size, Region, OS
- Also covers Lambda and Fargate
- Survives re-architecture
- Choose when the shape may change

EC2 INSTANCE SAVINGS PLAN

Deeper, up to 72%

- One family, in one Region
- EC2 only, no Lambda or Fargate
- Flexible on size, OS, tenancy
- Choose when the shape is settled

THE TIEBREAKER

You are buying flexibility with the six-point difference. A workload moving to Graviton or containers needs the Compute plan.

Traps in the commitment model

- **A Compute Savings Plan covers EC2, Lambda and Fargate only.** It touches no database spend.
- **Database Savings Plans** arrived Dec 2025: Aurora, RDS, DynamoDB, ElastiCache for Valkey, DocumentDB, Neptune, Keyspaces, DMS.
- **Not Redshift.** That still needs RIs. Database plans are 1-year, no-upfront, latest-generation only.
- Application order: **RIs first, then EC2 Instance plans, then Compute plans.** Spot gets no further discount.

> ON THE EXAM
>
> A database-heavy bill needs a Database Savings Plan or RIs. A Compute plan touches none of it, however large.

Finding the waste

- **Compute Optimizer** recommends rightsizing for EC2, EBS, Lambda, ECS on Fargate, and RDS, from real utilisation.
- **Trusted Advisor** flags idle and underused resources, and unassociated Elastic IPs.
- **S3 Storage Lens** gives storage analytics across every account, showing where the objects and the money are.
- Cost Optimization Hub consolidates all of this and **is not on the exam's in-scope list**. Answer with the two above.

> ON THE EXAM
>
> Compute Optimizer measures utilisation and recommends a size. Trusted Advisor checks against a fixed set of rules.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Charge costs back per team, per resource	Cost and Usage Report plus Athena
Alert before we exceed the budget	AWS Budgets
Stop spend automatically at a threshold	Budget actions
Compute shape will change over three years	Compute Savings Plan
The bill is mostly RDS and Redshift	Database Savings Plan, plus Redshift RIs
Instances are oversized	Compute Optimizer

ON THE EXAM

Decide first whether the question is about seeing cost, controlling cost, or committing to spend. They use different services.

> Meridian Freight must charge AWS costs back to 40 business units, showing each unit its exact spend by resource and by hour. Resources have been tagged with a CostCentre tag for two years, but the tag was never activated in the billing console.
>
> Finance wants the last two years attributed, and a repeatable monthly process.

What should the solutions architect tell them?

A. Activate the tag; Cost Explorer will backfill the last two years

B. Activate the tag now and use the Cost and Usage Report going forward; historical spend cannot be attributed by that tag

C. Use Cost Explorer grouped by tag for both the history and the monthly process

D. Re-apply the tags to all resources, which will trigger a billing backfill

◆ PAUSE AND DECIDE

What should the solutions architect tell them?

A Activation backfills

It does not. Activation begins inclusion from that point forward, and no AWS mechanism attributes prior spend to a newly activated tag.

B Activate now, CUR forward

Honest and correct. It also picks the right tool, since per-resource hourly chargeback is beyond what Cost Explorer reports.

C Cost Explorer for both

Wrong on history for the same reason as A, and wrong on granularity, since Cost Explorer does not give per-resource hourly detail.

D Re-apply to backfill

Invented. Re-applying a tag updates the resource, and creates no historical billing records.

THE SIGNAL

The right answer here is partly bad news. An option that refuses part of the request can still be correct.

> A company spends about $90,000 monthly on compute. Over the next three years it plans to migrate much of it from x86 to Graviton, move several services from EC2 to Fargate, and open a second Region.
>
> Finance wants a commitment discount on the steady portion of that spend.

What should the solutions architect recommend?

A. Three-year Standard Reserved Instances for the current instance families

B. A three-year Compute Savings Plan sized to the steady baseline

C. A three-year EC2 Instance Savings Plan for the largest family in the current Region

D. On-Demand Capacity Reservations across both Regions

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Standard RIs**

Locked to instance family. Every one of the three planned changes would strand the commitment for three years.

B **Compute Savings Plan**

Follows the spend across families, Regions and into Fargate. All three planned changes stay covered.

C **EC2 Instance Savings Plan**

Deeper discount, and locked to one family in one Region. It covers neither Fargate nor the second Region.

D **Capacity Reservations**

Reserve capacity and provide no discount at all. The requirement was explicitly a commitment discount.

THE SIGNAL

Three planned changes, each of which a narrower commitment would strand. That is what you are paying the six points for.

> A company's monthly bill is 20% EC2, 55% Amazon RDS and Amazon Redshift, and 25% storage and networking. A new FinOps lead proposes a large three-year Compute Savings Plan to reduce the total.
>
> Usage across all services has been stable for two years.

What should the solutions architect advise?

A. Proceed; a Compute Savings Plan discounts all compute-related spend

B. A Compute Savings Plan for EC2, a Database Savings Plan for RDS, and Reserved Instances for Redshift

C. Buy a larger Compute Savings Plan to cover the databases through spillover

D. Move the databases to EC2 so the Savings Plan applies

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Proceed as proposed

Savings Plans cover EC2, Lambda and Fargate. RDS and Redshift are untouched, so 55% of the bill gets nothing.

B The right instrument for each

Each service gets the commitment that actually applies to it. Redshift is excluded from Database Savings Plans, so it still needs RIs.

C Oversize for spillover

Spillover shares unused commitment between accounts, not between services. Unused commitment on a database bill is simply wasted.

D Move databases to EC2

Abandons managed services to qualify for a discount, adding patching, backups and failover work worth more than the saving.

THE SIGNAL

Over half this bill is invisible to a Compute Savings Plan. Check what a commitment actually covers before sizing it.

> A shared services account purchased a large Compute Savings Plan on behalf of the organisation. Six months later, that account was moved out of the organisation into a new organisation during a corporate restructure.
>
> Finance reports that every other account's compute bill rose sharply the following month.

What is the explanation?

A Savings Plans stop applying to the consolidated bill when the owning account leaves the organisation

B The Savings Plan expired coincidentally

C Discount sharing was deactivated for the member accounts

D Savings Plans only ever apply to the purchasing account

◆ PAUSE AND DECIDE

What is the explanation?

A Owner left the organisation

Documented behaviour. If the account owning the plan leaves, the discount stops applying to the consolidated bill, and the commitment leaves with it.

B Coincidental expiry

A plan bought six months ago has a one or three year term. It cannot have expired.

C Sharing deactivated

Would produce the same symptom, and the scenario names a different event. Nothing suggests a preference was changed.

D Only the purchasing account

Half true and misleading. The plan applies to the purchaser first, and unused commitment does spill to sharing accounts.

THE SIGNAL

Account moves during restructures break commitments in ways nobody plans for. The commitment belongs to the account, not the organisation.

> A sandbox OU repeatedly overruns its monthly allowance. Alerts are sent at 80% and 100% of budget, and by the time anyone acts the spend is already committed.
>
> The business wants new resource creation to stop automatically once the budget is exhausted.

What should the solutions architect implement?

A Additional AWS Budgets alerts at 50% and 90%

B A budget action that applies a restrictive IAM policy when the threshold is reached

C Cost Anomaly Detection on the sandbox accounts

D A weekly Cost Explorer report circulated to team leads

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A. More alert thresholds

More of what already failed. The problem is that alerts require a human to act, not that there were too few of them.

B. Budget actions

Budgets can attach an IAM policy, or stop supported resources, automatically at a threshold. No human in the loop.

C. Cost Anomaly Detection

Finds unexpected changes. A sandbox reliably overspending every month is entirely expected, so there is no anomaly to find.

D. Weekly report

Slower than the alerts that already exist, and still ends with a person deciding to act.

THE SIGNAL

Automatically is the word. Every option that ends with a person reading something has already been tried and failed here.

> A platform team is asked to reduce EC2 spend without affecting performance. They have 900 instances across 30 accounts, sized by developers two years ago based on guesswork.
>
> The team wants evidence-based recommendations rather than opinions, and needs to review them centrally.

What should the solutions architect recommend?

A. AWS Compute Optimizer, with the organisation-level view enabled

B. AWS Trusted Advisor's cost optimisation checks alone

C. A Cost Explorer report grouped by instance type

D. Ask each team to review and resize their own instances

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Compute Optimizer, org-wide

Analyses actual utilisation and recommends a specific size, with a projected performance risk. Exactly evidence rather than opinion.

B Trusted Advisor alone

Flags idle and low-utilisation resources against fixed rules. Useful, and it does not recommend which size to move to.

C Cost Explorer by instance type

Shows what each type costs, and nothing about whether any instance is the wrong size.

D Ask the teams

Returns to the guesswork that produced the problem, now spread across thirty accounts.

THE SIGNAL

Trusted Advisor says this looks idle. Compute Optimizer says move this to that size. The word evidence separates them.

What to carry out of this module

- **Cost allocation tags are not retroactive.** Activation starts the clock; earlier spend is never attributed.
- **Chargeback needs the Cost and Usage Report.** Cost Explorer summarises only.
- **A Compute Savings Plan covers EC2, Lambda and Fargate only.** Databases need a Database Savings Plan, or RIs.
- A Savings Plan gives a discount and no capacity. A Capacity Reservation gives capacity and no discount.

> ON THE EXAM
>
> Decide whether the scenario wants to see cost, control cost, or commit to spend. Those are three different toolsets.

MODULE 12 · DOMAIN 2 - DESIGN FOR NEW SOLUTIONS

Infrastructure as Code

Domain 2 begins. CloudFormation appears in 8% of the question bank.

DOMAIN 2 of 4 EXAM WEIGHT 29% MODULE 12 of 44

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The template is not the valuable part. The **stack** is, because it remembers what it created.

The deployment safety chain

- **A change set** shows what an update will do before it does it, including which resources will be **replaced**.
- **Drift detection** finds resources changed outside CloudFormation, which the next deployment would overwrite.
- **Stack policies** deny updates to named resources, protecting a database from a careless template change.
- Failed updates roll back automatically, and **rollback triggers** can watch CloudWatch alarms during deployment.

> ON THE EXAM
>
> Replacement is the word to hunt for in a change set. It means a new resource and a deleted one, which for a database means data loss.

Controlling what happens on delete

- **DeletionPolicy: Retain** keeps a resource when the stack is deleted. Snapshot takes a final snapshot first.
- **UpdateReplacePolicy** does the same when an **update** would replace the resource. They are separate settings.
- Setting only DeletionPolicy leaves you exposed on updates, which is where most accidental data loss happens.
- **Termination protection** on the stack itself stops the whole stack being deleted.

> ON THE EXAM
>
> Deletion and replacement are different events with different settings. Databases need both policies, plus termination protection.

Nested stacks or cross-stack references?

NESTED STACKS

One lifecycle, many templates

- Child stacks owned by a parent
- Deployed and deleted together
- Good for reusable components
- Update the parent to change a child

CROSS-STACK REFERENCES

Separate lifecycles

- Export an Output, import elsewhere
- Stacks deploy independently
- Good for shared network or IAM
- An export cannot change while imported

THE TIEBREAKER

Ask whether the pieces live and die together. Shared infrastructure with its own lifecycle needs exports, not nesting.

StackSets across the organisation

- **One operation deploys to many accounts and Regions.** Target whole OUs, not lists of accounts.
- **Service-managed permissions** use Organizations and need no roles per account. Self-managed means you create them.
- **Automatic deployment** means an account joining a target OU receives the stack without anyone acting.
- Delegate administration to a member account, so this does not run from the management account.

> ON THE EXAM
>
> Service-managed plus automatic deployment is the combination. Without the second, every new account is a task somebody must remember.

Extending CloudFormation

- **Custom resources** call a Lambda function or SNS topic during deployment, for anything CloudFormation cannot do natively.
- They must signal success or failure, or the stack hangs until it times out. This is the classic custom resource failure.
- **Hooks** inspect a deployment before it proceeds and can block it. Control Tower's proactive controls are built on them.
- **Modules** package a reusable block of resources for use inside templates.

> ON THE EXAM
>
> A custom resource does work. A hook makes a judgement. If a scenario wants a deployment stopped, that is a hook.

Bringing existing resources under control

- **Resource import** brings an existing resource into a stack without recreating it.
- **The IaC generator** scans an account and writes a template from resources that already exist.
- **Stack refactoring** moves resources between stacks and renames logical IDs, with no replacement.
- The old retain-and-reimport dance is no longer the answer for splitting a stack. Older material still teaches it.

> ON THE EXAM
>
> Splitting a monolithic stack used to mean retain, delete, re-import. Stack refactoring does it in one operation now.

Newer safety and workflow features

- **Drift-aware change sets** show a three-way diff, so a deployment cannot silently overwrite a manual fix.
- **Git sync** deploys a stack from a repository on every commit, and posts the change set into a pull request.
- **StackSets deployment ordering** handles dependencies between accounts, such as networking before workloads.
- Sequencing and concurrency controls limit blast radius during a large rollout.

> ON THE EXAM
>
> Git sync makes the repository the source of truth. A change made in the console is drift by definition afterwards.

An honest scope note on the alternatives

- **AWS CDK** writes CloudFormation from a real programming language, and is what many teams now use.
- **AWS SAM** is a serverless-focused transform over CloudFormation. **Terraform** is the multi-cloud alternative.
- **None of the three is on the exam's in-scope service list.** CloudFormation is.
- Expect them named in scenarios as context. Expect the answer to be expressed in CloudFormation.

> ON THE EXAM
>
> This is a gap between the modern default and the examinable one. CDK compiles to CloudFormation, so the concepts transfer.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
A baseline in every account, including new ones	StackSets, service-managed, auto-deploy
See what an update will change first	A change set
Someone changed a resource in the console	Drift detection
Split one large stack without downtime	Stack refactoring
Do something CloudFormation cannot do natively	A custom resource
Block a non-compliant deployment	A hook

ON THE EXAM

Separate three questions: what is being deployed, how widely, and what must not go wrong. Each has its own feature.

Meridian Freight must deploy a standard set of IAM roles, a Config recorder and a logging configuration into every account. New accounts are vended weekly into one of four OUs.

The current process is a CloudFormation stack deployed by hand after each account is created, and it has been missed twice.

What should the solutions architect recommend?

A. A StackSet with self-managed permissions, targeting the account list

B. A StackSet with service-managed permissions targeting the four OUs, with automatic deployment enabled

C. A CodePipeline that deploys the stack to each account on a schedule

D. A Config rule that detects accounts missing the resources

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Self-managed, account list

Works, and requires roles created in every target account and the list updated for each new one. The list is the thing being forgotten.

B Service-managed, OU targets, auto-deploy

Targets OUs rather than accounts, and a new account in a target OU receives the stack automatically.

C Scheduled pipeline

Builds a scheduler to reproduce something StackSets does on account creation, and leaves a window between vending and baseline.

D Config rule

Detective again. It reports the account that was missed, which is how they found out the last two times.

THE SIGNAL

Automatic deployment is the feature being tested. Without it, a new account is still a task somebody must remember.

> A routine stack update intended to add a tag to an RDS instance instead destroyed and recreated it, losing four hours of transactions. The template change modified the DB subnet group.
>
> The team wants to guarantee this can never happen again on any database.

What should the solutions architect implement?

A. Review change sets before every update, checking the Replacement column

B. Set UpdateReplacePolicy and DeletionPolicy to Snapshot, and add a stack policy denying updates to the database

C. Enable termination protection on the stack

D. Increase the RDS backup retention period to 35 days

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A. Review change sets

Good practice and essential, and it depends on a human noticing a column under time pressure. The requirement says guarantee.

B. UpdateReplacePolicy plus stack policy

The stack policy denies the update outright, and the snapshot policies protect the data if a replacement is ever deliberately allowed.

C. Termination protection

Protects against the stack being deleted. The stack was never deleted here; a resource inside it was replaced.

D. Longer backup retention

More restore points and the same four hours lost. It shortens recovery, and does not prevent the event.

THE SIGNAL

Deletion, replacement and stack deletion are three different events. Only UpdateReplacePolicy and a stack policy address the middle one.

> During an incident, an engineer widened a security group rule in the console to restore service. Two weeks later a routine stack update silently reverted it and the incident recurred.
>
> The team wants deployments to surface manual changes rather than quietly discarding them.

What should the solutions architect recommend?

A Deny console access to security groups with an SCP

B Run drift detection before deployments, and use drift-aware change sets to surface the difference

C Add the security group to a stack policy denying updates

D Disable automatic rollback so the update does not revert

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Deny console access

Would have prevented the emergency fix that restored service. It solves the symptom by removing the incident response.

B Drift detection and drift-aware change sets

Exactly what was asked: surface the manual change so a human decides whether to keep it or fold it into the template.

C Stack policy on the group

Freezes the resource so CloudFormation never manages it again, which quietly makes the template wrong forever.

D Disable rollback

Rollback was not involved. The update applied the template successfully; the template was simply out of date.

THE SIGNAL

Read what the team actually asked for. They wanted visibility, not prevention, and only one option provides it.

A single CloudFormation stack has grown to hold a VPC, three databases and eleven application services. Deployments now take 90 minutes and one team's change blocks everyone.

The company wants networking, data and application resources in separate stacks, with no downtime and no resource replacement.

What should the solutions architect use?

A Set DeletionPolicy to Retain, delete the stack, then import resources into three new stacks

B CloudFormation stack refactoring to move resources into the new stacks

C Convert the stack into nested stacks with a parent template

D Recreate the resources in three new stacks and migrate data across

◆ PAUSE AND DECIDE

What should the solutions architect use?

A. Retain, delete, re-import

The pre-2025 technique, and it works. It is many manual steps with a window where nothing manages the resources.

B. Stack refactoring

Purpose-built to move resources between stacks and rename logical IDs without replacement, in one operation.

C. Nested stacks

Keeps one lifecycle, so one team's change still blocks everyone. That is the actual complaint.

D. Recreate and migrate

Replaces every resource and requires data migration, contradicting both stated constraints.

THE SIGNAL

Option A was the correct answer until February 2025. Study material written before then still teaches it.

> A deployment must register the new environment's hostname with a third-party monitoring vendor's API, and store the returned identifier for later stages of the same template.
>
> There is no CloudFormation resource type for that vendor.

What should the solutions architect implement?

A. A custom resource backed by a Lambda function that calls the vendor API and returns the identifier

B. A CloudFormation hook that calls the vendor API before deployment

C. A CodePipeline stage that runs after the stack completes

D. A Systems Manager Automation runbook triggered by an EventBridge rule on stack completion

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Custom resource

Runs during the deployment and returns values other resources in the same template can reference. Both requirements met.

B A hook

Hooks make a judgement about whether a deployment may proceed. They do not create things or return values into the template.

C Pipeline stage after

Runs after the stack finishes, so the identifier cannot be used by later resources in the same template.

D Runbook on completion

Same timing problem as C, with the stack already complete before the registration happens.

THE SIGNAL

The phrase for later stages of the same template is what forces a custom resource. Anything running afterwards is too late.

> A networking stack exports subnet IDs, which eleven application stacks import. The network team needs to change the subnet layout, and CloudFormation refuses to update the export.
>
> Application teams deploy on independent schedules and cannot all change at once.

What is happening, and what should the architect recommend?

A. An exported value cannot change while it is imported; publish the values to Systems Manager Parameter Store instead

B. Convert the eleven stacks into nested stacks under the networking stack

C. Delete and recreate the networking stack during a maintenance window

D. Increase the export limit with a service quota request

◆ PAUSE AND DECIDE

What is happening, and what should the architect recommend?

A Parameter Store instead of exports

Correct diagnosis, and it decouples the lifecycles. Consumers read the current value at deploy time with no export lock.

B Nested stacks

Forces eleven independent teams onto one lifecycle, which contradicts the stated constraint about independent schedules.

C Delete and recreate

Deleting a networking stack takes the VPC with it, along with everything the eleven applications are running in.

D Quota increase

There is no quota involved. The refusal is a dependency protection, not a limit.

THE SIGNAL

The export lock is a safety feature doing its job. The fix is a looser coupling, not a way around the protection.

What to carry out of this module

- **Service-managed StackSets plus automatic deployment** is the baseline pattern. OUs are the target, not account lists.
- **Deletion, replacement and stack deletion are three events.** Databases need DeletionPolicy, UpdateReplacePolicy and termination protection.
- **Stack refactoring replaced retain-and-reimport** for splitting stacks. Older material still teaches the old way.
- Custom resources do work and return values. Hooks judge whether a deployment may proceed.

> ON THE EXAM
>
> An exported value cannot change while imported. That is a protection, and the fix is looser coupling rather than a workaround.

MODULE 13 · DOMAIN 2 - DESIGN FOR NEW SOLUTIONS

Deployment Strategies and Rollback

Every strategy trades cost against blast radius. None of them is simply best.

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You are always trading three things: **cost during deployment, time to deploy, and blast radius on failure**.

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The strategies

STRATEGY	HOW IT WORKS	FAILURE COSTS YOU
All at once	Replace everything, immediately	Full outage until rollback
Rolling	Batch by batch, in place	Reduced capacity, mixed versions
Rolling with batch	Adds temporary instances first	Mixed versions, full capacity kept
Immutable	New instances, then swap	Terminate the new group. Nothing else
Blue/green	Whole parallel environment	Shift traffic back. Near instant
Canary	A slice of traffic first	Only that slice, briefly

ON THE EXAM

The last column is what the exam actually asks. Read the scenario for what a failure must not cost, not for what sounds modern.

Rolling or immutable?

ROLLING

Update in place, in batches

- No extra instances to pay for
- Two versions run at once
- A half-failed deploy leaves a mess
- Rollback means deploying again

IMMUTABLE

Build new, then swap

- Doubles instances during deploy
- Never mixes versions
- Failure leaves the old set untouched
- Rollback is terminating a group

THE TIEBREAKER

Mixed versions must never be visible, or rollback must be fast, means immutable. Cost sensitivity means rolling.

How traffic actually shifts

- **Weighted target groups** on an ALB shift a percentage between old and new. The usual mechanism.
- **Lambda alias weights** shift invocations between two function versions.
- **Route 53 weighted records** also work, and DNS caching means clients may ignore you for the TTL.
- **Bake time** is the pause after shifting, during which alarms can still trigger a rollback.

ON THE EXAM

DNS-based shifting cannot guarantee when traffic moves. Load balancer and alias weights take effect immediately.

CodeDeploy

- **In-place or blue/green.** EC2 and on-premises can do either; Lambda and ECS are always blue/green.
- **Canary** shifts a percentage, waits, then the rest. **Linear** shifts equal increments on a timer.
- **Lifecycle hooks** run Lambda functions at defined points, and a failing hook triggers a rollback.
- **Automatic rollback on a CloudWatch alarm** is the feature that makes the rest of it safe.

> ON THE EXAM
>
> Named configurations like Canary10Percent5Minutes encode the whole strategy. If a scenario names percentages and minutes, this is it.

Containers: a changed recommendation

- **ECS rolling** is the default controller: replace tasks in place, governed by minimum and maximum healthy percent.
- **The CODE_DEPLOY controller** was the established way to get blue/green on ECS, with two target groups.
- **ECS now has built-in blue/green**, with lifecycle hooks, bake time and managed rollback. Canary and linear followed.
- AWS now recommends the **ECS-native** path for new deployments. Most study material still says CodeDeploy.

> ON THE EXAM
>
> Both are in scope and both are defensible. If a scenario names CodeDeploy, use it; if it asks for the simplest current approach, ECS-native.

Serverless deployment

- **Publish a version, point an alias at it**, and shift alias weights. Versions are immutable, which makes rollback trivial.
- **PreTraffic and PostTraffic hooks** run tests before and after the shift, and can fail the deployment.
- **Provisioned concurrency attaches to a version or alias**, so a new version starts cold unless you configure it.
- Alarms on errors, throttles and duration are what make automatic rollback meaningful.

> ON THE EXAM
>
> Rollback for Lambda is repointing an alias, which is close to instantaneous. Nothing is rebuilt or redeployed.

Elastic Beanstalk and Auto Scaling

- **Beanstalk has five policies:** all at once, rolling, rolling with additional batch, immutable, traffic splitting.
- **Rolling needs a manual redeploy to roll back.** Immutable and traffic splitting roll back on their own.
- Replacing all instances **loses accumulated EC2 burst balances**, which surprises people on burstable instance types.
- **Auto Scaling instance refresh** does the same job for a plain ASG, with checkpoints and a minimum healthy percentage.

> ON THE EXAM
>
> Beanstalk's traffic splitting is canary under another name: a second ASG, a percentage of traffic, and an evaluation period.

The pipeline around it

- **CodePipeline** orchestrates stages. **CodeBuild** builds and tests. **CodeDeploy** performs the release.
- **Manual approval actions** gate a stage, which is how a change reaches production only with a human decision.
- Cross-account deployment uses a role in the target account and a **KMS key shared with it** for the artefact bucket.
- Test in the pipeline, not after it. A deployment strategy is a safety net, not a substitute for tests.

> ON THE EXAM
>
> The classic cross-account pipeline failure is the artefact bucket's KMS key not being shared with the target account.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Two versions must never run at once	Immutable or blue/green
Rollback in seconds, cost is acceptable	Blue/green
Catch regressions on real traffic first	Canary or traffic splitting
No extra capacity may be provisioned	Rolling
A human must approve production	Manual approval action
Roll back automatically on errors	CloudWatch alarm plus bake time

ON THE EXAM

The constraint sentence is usually about money or about mixed versions. Find it before you compare the options.

> Meridian Freight's tracking API must be updated weekly with no downtime. A bad release must be reversible within one minute. The platform runs on EC2 behind an Application Load Balancer, and the business accepts doubled compute cost for the duration of a deployment.
>
> The current rolling deployment takes 40 minutes and rollback takes another 40.

What should the solutions architect recommend?

A. Reduce the rolling deployment batch size so failures affect fewer instances

B. A blue/green deployment with weighted target groups and CodeDeploy

C. An immutable deployment

D. All at once, during a scheduled maintenance window

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Smaller batches

Makes each batch safer and the whole deployment longer. Rollback is still another full deployment, so one minute is impossible.

B Blue/green with weighted target groups

The old environment stays running, so rollback is shifting weights back. Seconds, not minutes.

C Immutable

The closest miss. Rollback is fast once the swap has happened, and after the old group is terminated you are redeploying again.

D All at once

Directly contradicts no downtime, and a maintenance window does not make an outage into not-an-outage.

THE SIGNAL

One minute to reverse is the constraint. Only a strategy that keeps the old environment running can meet it.

A rolling deployment failed at batch three of six. Half the fleet runs the new version and half the old, and a schema change means the two are not compatible. The team spent two hours restoring service.

They want failures to leave the environment in a known state, and cannot justify permanently doubling capacity.

What should the solutions architect recommend?

A Rolling with an additional batch, to maintain full capacity

B An immutable deployment

C All at once, so both versions never coexist

D Increase the health check grace period so batches are validated for longer

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Rolling with additional batch**

Fixes capacity during deployment and still mixes versions batch by batch, which is the actual failure described.

B **Immutable**

Builds a complete new set and swaps only when healthy. Versions never mix, and a failure terminates the new group.

C **All at once**

True that versions never coexist, achieved through an outage on every release. It also has no rollback beyond redeploying.

D **Longer grace period**

Validates each batch more thoroughly, and does nothing about the incompatibility between batches that have deployed and those that have not.

THE SIGNAL

Doubling capacity permanently is not the same as doubling it during a deployment. Read which one the scenario refuses.

> A Lambda function behind an API handles 40 million invocations a day. A recent release introduced an error affecting roughly 2% of requests, which passed all pre-production tests and was only noticed after four hours.
>
> The team wants real traffic to validate a release, with automatic reversal.

What should the solutions architect implement?

A. CodeDeploy canary traffic shifting on the function alias, with a CloudWatch alarm on the error metric

B. Increase pre-production test coverage and add a staging environment

C. Deploy to a second function and switch the API Gateway integration when confident

D. Enable provisioned concurrency on the new version

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Canary shifting plus alarm

A small slice of real traffic, an alarm watching errors, and automatic reversal by repointing the alias. Every requirement met.

B More testing

The bug already passed all pre-production tests. More of the thing that missed it is not a plan.

C Second function, manual switch

Builds blue/green by hand with a human deciding when. The requirement said automatic.

D Provisioned concurrency

Addresses cold starts, which is a latency concern. It has no bearing on functional errors.

THE SIGNAL

Passed all pre-production tests is the planted detail. It exists to eliminate every answer that adds more testing.

> A team is building a new ECS service on Fargate behind an Application Load Balancer, and wants blue/green deployments with validation hooks and automatic rollback. They have no existing CodeDeploy configuration and no preference.
>
> They ask which approach involves the fewest moving parts.

What should the solutions architect recommend?

A. The CODE_DEPLOY deployment controller, with CodeDeploy managing two target groups

B. The built-in ECS blue/green deployment strategy with ECS lifecycle hooks

C. The default ECS rolling controller with a low minimum healthy percent

D. The EXTERNAL deployment controller with a custom deployment system

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A CODE_DEPLOY controller

The established answer and still fully supported. It adds a second service to configure, which the fewest-moving-parts requirement counts against.

B ECS built-in blue/green

Hooks, bake time and managed rollback inside ECS itself, with no CodeDeploy application or deployment group to maintain.

C Rolling controller

Replaces tasks in place, so it is not blue/green and offers none of the validation the team asked for.

D EXTERNAL controller

Hands the entire deployment process to software you write. The opposite of fewest moving parts.

THE SIGNAL

This answer changed in 2025. Material written before then teaches option A, and option A is still correct where CodeDeploy already exists.

A CodePipeline in a tooling account deploys to a production account. The source and build stages succeed. The deploy stage fails with an access denied error when the target account tries to read the artefact from S3.

The cross-account role exists and has S3 read permission on the artefact bucket.

What is the most likely cause?

A The KMS key encrypting the artefact bucket is not shared with the production account

B The pipeline needs to run in the production account instead

C The artefact bucket policy must allow s3:ListBucket as well

D CodePipeline does not support cross-account deployment

◆ PAUSE AND DECIDE

What is the most likely cause?

A KMS key not shared

The artefact bucket is encrypted with a customer managed key, and reading an object needs Decrypt on that key as well as S3 access.

B Run it locally

Abandons the separation of tooling and production that the design deliberately has. Cross-account pipelines are the recommended pattern.

C Add ListBucket

Needed for listing, not for reading a known object key. It would not produce this failure.

D Not supported

It is explicitly supported and widely used. Two stages already succeeded.

THE SIGNAL

This is module two's third door reappearing. S3 access and KMS access are two separate authorisation decisions.

> A blue/green deployment completed successfully. All health checks passed, traffic shifted to green, and the blue environment was terminated. Twenty minutes later, error rates climbed as a memory leak in the new version began to bite.
>
> The team wants this class of failure caught automatically in future.

What should the solutions architect change?

A Add deeper health checks on the target group

B Configure a bake time before terminating blue, with a CloudWatch alarm on error rate as a rollback trigger

C Switch to a rolling deployment so fewer instances run the new version

D Run load tests in pre-production before every release

◆ PAUSE AND DECIDE

What should the solutions architect change?

A Deeper health checks

Health checks pass or fail at a moment. A leak that takes twenty minutes to matter looks healthy at every check until it does not.

B Bake time plus alarm rollback

Keeps blue alive through a monitoring window, so an alarm firing at twenty minutes can still shift traffic back instantly.

C Rolling deployment

Reduces how many instances run the new version, and every instance eventually runs it. The leak arrives regardless, more slowly.

D Pre-production load tests

Worth doing and not automatic detection in production. The requirement was to catch this class of failure automatically.

THE SIGNAL

Terminating blue immediately is the mistake. A rollback plan that has already deleted its target is not a rollback plan.

What to carry out of this module

- **Find the constraint sentence.** It is usually about extra cost or about mixed versions, and it eliminates half the options.
- **Immutable doubles capacity during a deployment, not permanently.** Scenarios test whether you read that correctly.
- **Bake time plus a CloudWatch alarm** is what turns a deployment strategy into automatic rollback.
- ECS blue/green is now native as well as CodeDeploy. Cross-account pipelines fail on the KMS key.

> ON THE EXAM
>
> A rollback plan that terminates the old environment immediately is not a rollback plan. Keep it alive for the monitoring window.

MODULE 14 · DOMAIN 2 - DESIGN FOR NEW SOLUTIONS

Choosing Compute

EC2 appears in half the question bank. Rarely as the topic.

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At professional level you rarely choose an instance type. You choose a **shape**, and let AWS pick the types.

Instance families, by what they are for

FAMILY	OPTIMISED FOR	TYPICAL USE
T	Burstable, cheap baseline	Small or idle-heavy workloads
M	Balanced	The default when nothing dominates
C	Compute	Batch, encoding, HPC, gaming servers
R and X	Memory	In-memory caches, large databases
I and D	Storage throughput and IOPS	NoSQL, data warehouses, local scratch
P and G	Accelerators	Training, inference, graphics

ON THE EXAM

T instances accrue credits and can throttle when exhausted. Sustained load on a T instance is a classic misdiagnosed performance problem.

Graviton

- **AWS-designed arm64 processors**, consistently better price-performance than equivalent x86 families.
- Generations run from Graviton2 to **Graviton5**, which reached general availability in mid-2026.
- **arm64 and Linux only.** No Windows, and no GPU-attached instance types.
- Managed services run on it too, including RDS, Aurora, ElastiCache, OpenSearch and Lambda, with no porting work.

> ON THE EXAM
>
> Interpreted and JIT languages port with a rebuild. Compiled binaries and native dependencies are where migrations actually stall.

Describing a fleet rather than listing it

- **Attribute-based instance selection** expresses requirements as vCPUs, memory and storage, and AWS picks matching types.
- New instance families qualify automatically as AWS releases them. A hard-coded list never does.
- **Mixed instances policies** combine On-Demand and Spot in one group, with a base count and a percentage above it.
- Both together let one Auto Scaling group span x86 and arm64, which massively widens the capacity you can draw on.

> ON THE EXAM
>
> Diversity is availability. The more instance types and Availability Zones a group can use, the less a shortage in any one pool matters.

The scaling policy types

POLICY	REACTS TO	CHOOSE IT WHEN
Target tracking	A metric against a target	The default. Keep CPU near 60%
Step scaling	Alarm breach size	Different responses to different severities
Simple scaling	One alarm, with a cooldown	Legacy. Prefer the two above
Scheduled	The clock	Known events: market open, a sale
Predictive	Forecast from history	Recurring daily or weekly patterns

ON THE EXAM

Reactive policies scale after demand arrives. Only scheduled and predictive scaling can have capacity ready beforehand.

Reactive or ahead of demand?

TARGET TRACKING

Responds to what happened

- Simple to configure and reason about
- Always lags the demand curve
- Lag equals detection plus boot time
- Fine for gradual change

PREDICTIVE SCALING

Acts on what usually happens

- Forecasts from at least 24h of history
- Provisions before the rise
- Needs a repeating pattern to work
- Runs alongside target tracking

THE TIEBREAKER

Use both. Predictive handles the pattern it has seen; target tracking catches the day the pattern is wrong.

Making instances ready sooner

- **Warm pools** keep pre-initialised instances stopped or hibernated, so scale-out skips boot and application startup.
- Stopped instances cost only storage. This is the answer when boot time, not scaling logic, is the problem.
- **Lifecycle hooks** pause an instance entering or leaving service, so you can bootstrap or drain.
- A **golden AMI** built with EC2 Image Builder removes most startup work in the first place.

> ON THE EXAM
>
> If the complaint is that scaling reacts too slowly, ask what is slow. A long boot needs a warm pool; a late trigger needs a different policy.

Placement groups

- **Cluster** packs instances close together in one AZ for lowest latency and highest throughput between them.
- Cluster concentrates risk. Everything is in one AZ, and a capacity shortfall can fail the whole launch.
- **Spread** places each instance on distinct hardware. Limited to seven per AZ, for small critical sets.
- **Partition** groups instances into partitions on separate racks. For HDFS, Cassandra, Kafka.

> ON THE EXAM
>
> Cluster optimises for speed between nodes. Spread and partition optimise for not losing them together. Read which the scenario wants.

Spot without pain

- **Use price-capacity-optimized.** AWS now advises against lowest-price, because replacements land in pools that get interrupted again.
- **Diversify hard.** Many instance types across many AZs is what actually reduces interruptions.
- **Capacity Rebalancing** acts on a rebalance recommendation, often arriving before the two-minute notice.
- Handle the interruption notice: drain, checkpoint, and hand work back to a queue.

> ON THE EXAM
>
> Spot Fleet is now legacy. New designs use an Auto Scaling group or EC2 Fleet, which get new capabilities first.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Scaling is too slow, boot takes minutes	Warm pool, or a golden AMI
Traffic spikes at the same time daily	Predictive or scheduled scaling
Drain connections before termination	A lifecycle hook
Nodes need low latency between them	Cluster placement group
Frequent Spot interruptions	price-capacity-optimized, plus diversity
Cut compute cost, Linux, no GPU	Graviton

ON THE EXAM

Separate the trigger from the delay. Many scaling complaints are about how long an instance takes to be useful, not about the policy.

> Meridian Freight's tracking service scales on CPU with target tracking at 60%. During sudden load, requests time out for roughly seven minutes before new capacity helps. Instances take about six minutes to boot, install dependencies and warm a local cache.
>
> The scaling alarm itself fires within a minute.

What should the solutions architect recommend?

A. Lower the target tracking threshold from 60% to 40%

B. Add a warm pool of pre-initialised stopped instances, and bake dependencies into a golden AMI

C. Switch from target tracking to step scaling

D. Increase the Auto Scaling group's maximum size

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Lower the target**

Buys headroom by running more instances permanently, and every scale-out still takes six minutes. It pays continuously to hide a delay.

B **Warm pool plus golden AMI**

Attacks the six minutes directly. A stopped, pre-initialised instance joins in seconds instead.

C **Step scaling**

Changes how much capacity is added, not how fast it becomes useful. The six-minute boot is unchanged.

D **Raise the maximum**

The group was never at its ceiling. Nothing in the scenario suggests it was capped.

THE SIGNAL

One minute to trigger, six minutes to boot. The scenario hands you the arithmetic so you attack the right number.

> A retail platform sees traffic rise sharply at 09:00 every weekday and fall after 18:00, with a much larger peak on Mondays. Target tracking scales correctly but always about ten minutes late, and the first ten minutes of each morning are degraded.
>
> The pattern has been stable for two years.

What should the solutions architect recommend?

A. Enable predictive scaling alongside the existing target tracking policy

B. Replace target tracking with a scheduled action at 08:45 every weekday

C. Replace target tracking with step scaling on a lower threshold

D. Run the peak instance count all day

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Predictive plus target tracking

Predictive provisions ahead of a pattern it has learned, including the larger Monday peak, and target tracking still covers unexpected days.

B Scheduled action only

The strong near miss. It handles 09:00, and one fixed number cannot express a bigger Monday, and replacing target tracking removes the safety net.

C Step scaling

Still reactive. It changes the size of the response, not whether capacity exists before demand arrives.

D Peak all day

Solves it with money, paying for the evening and weekend peak that never comes.

THE SIGNAL

The Monday detail is planted. A scheduled action is one number; a forecast varies by day, which is exactly what the pattern needs.

> A batch rendering fleet runs on Spot across two instance types in one Availability Zone, using the lowest-price allocation strategy. Interruptions are frequent, and replacements are often interrupted within minutes of launching.
>
> Jobs take about 40 minutes and currently restart from the beginning.

What should the solutions architect change?

A Move the fleet to On-Demand instances

B Use price-capacity-optimized, diversify across many types and AZs, enable Capacity Rebalancing, and checkpoint work

C Set a higher maximum Spot price so instances are not outbid

D Use a cluster placement group to improve capacity availability

◆ PAUSE AND DECIDE

What should the solutions architect change?

A Move to On-Demand

Removes interruptions and most of the saving. Rendering is the archetypal Spot workload once it checkpoints.

B Strategy, diversity, rebalancing, checkpoints

Fixes the cause and the symptom. Deeper pools mean fewer interruptions, and checkpoints mean an interruption costs minutes rather than the job.

C Higher maximum price

Spot pricing has not worked by bidding for years. Interruptions come from capacity demand, not from being outbid.

D Cluster placement group

Optimises network latency between instances and concentrates them in one AZ, making a capacity shortfall more likely, not less.

THE SIGNAL

Replacements interrupted within minutes is the giveaway for lowest-price. It refills from the cheap pool that just emptied.

> A company runs a large Java microservices estate on Linux on M6i instances and wants to reduce compute cost. The services are containerised on ECS and have no native dependencies. A separate reporting service runs on Windows, and a nightly training job uses GPU instances.
>
> Finance wants the largest saving with the least risk.

What should the solutions architect recommend?

A. Migrate everything to Graviton instances

B. Migrate the Java services to Graviton; leave the Windows and GPU workloads on x86

C. Keep everything on x86 and buy a Compute Savings Plan instead

D. Move the Java services to T-family instances to cut cost

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Everything to Graviton

Impossible. Graviton has no Windows support and no GPU-attached types, so two of the three workloads cannot move.

B Java to Graviton, rest on x86

Java on Linux with no native dependencies is the lowest-risk Graviton migration there is. The other two stay where they must.

C Savings Plan instead

Worth doing as well, and it is a pricing change rather than an efficiency one. The two combine; they are not alternatives.

D T-family

Burstable instances under sustained microservice load exhaust credits and throttle. Cheaper per hour and slower when it matters.

THE SIGNAL

The Windows and GPU workloads are planted to test whether you know Graviton's two hard limits.

A computational fluid dynamics workload runs across 40 instances that exchange data continuously using MPI. Performance is far below expectations and profiling shows the bottleneck is inter-node network latency.

The instances are currently spread across three Availability Zones for resilience.

What should the solutions architect recommend?

A. A spread placement group across the three Availability Zones

B. A cluster placement group in a single Availability Zone, with Elastic Fabric Adapter

C. A partition placement group

D. Larger instances in the same three Availability Zones

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Spread placement group**

Deliberately separates instances onto distinct hardware, which is the opposite of what a latency-bound workload needs.

B **Cluster group plus EFA**

Packs the nodes close together in one AZ and gives MPI a low-latency bypass of the normal network stack.

C **Partition group**

Separates instances into racks for distributed data stores. It reduces correlated failure rather than latency.

D **Larger instances**

More CPU and memory per node, and the bottleneck is the network between nodes. It may worsen the ratio.

THE SIGNAL

Spreading across AZs is the cause here, not the solution. Tightly coupled HPC trades availability for latency, deliberately.

> When the Auto Scaling group scales in, in-flight requests are dropped and the terminating instance's local log files are lost before they are shipped.
>
> The team needs up to three minutes to drain connections and flush logs before an instance goes away.

What should the solutions architect implement?

A. An EC2 termination lifecycle hook that holds the instance in Terminating:Wait while a script drains and flushes

B. Increase the load balancer's deregistration delay to 300 seconds

C. Enable instance scale-in protection on all instances

D. Suspend the Terminate process on the Auto Scaling group

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Termination lifecycle hook

Holds the instance in a wait state until the script completes or the timeout expires, covering both draining and log flushing.

B Deregistration delay

The strong near miss. It genuinely fixes the dropped requests, and does nothing for the logs, because that is a load balancer setting.

C Scale-in protection

Stops instances being chosen for termination at all, so the group can never scale in. It prevents the event rather than handling it.

D Suspend Terminate

Breaks scale-in entirely, and also breaks replacement of unhealthy instances, which is worse.

THE SIGNAL

Two problems are described and only one option addresses both. Deregistration delay is half an answer.

What to carry out of this module

- **Describe the fleet, do not list it.** Attribute-based selection plus mixed instances is the professional-level answer.
- **Separate the trigger from the delay.** A slow boot needs a warm pool; a late trigger needs a different policy.
- **Predictive runs alongside target tracking**, not instead of it. Scheduled actions cannot vary by day.
- **price-capacity-optimized, and diversify.** Graviton is Linux and arm64 only: no Windows, no GPU.

> ON THE EXAM
>
> Cluster placement groups trade availability for latency deliberately. That is the one place where spreading across AZs is the wrong instinct.

MODULE 15 · DOMAIN 2 - DESIGN FOR NEW SOLUTIONS

Containers and Serverless

ECS, EKS, Fargate and Lambda. The question is never simply which one.

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Ask what the **unit of scale** is. A request, a container, or a node. That answers most of the question.

The four options

OPTION	YOU OPERATE	FITS
Lambda	Nothing. Just code	Event-driven, spiky, under 15 minutes
ECS on Fargate	Containers only	Long-running services, no cluster to run
ECS on EC2	Containers and instances	GPU, special instance types, cost at scale
EKS	Kubernetes and its ecosystem	Portability, Helm, operators, existing skills

ON THE EXAM

ECS has no control plane charge. EKS charges hourly per cluster, and costs far more in people.

ECS or EKS?

AMAZON ECS

AWS-native, simpler

- No control plane charge
- Deep IAM and VPC integration
- Small operational surface
- Nothing transfers off AWS

AMAZON EKS

Kubernetes, with its costs

- Helm, operators, CRDs, the ecosystem
- Skills and manifests are portable
- Hourly cluster charge
- You own upgrades and add-ons

THE TIEBREAKER

Existing Kubernetes skills or artefacts means EKS. Anything else, and the simpler operational surface usually wins.

Fargate, and capacity providers

- **Fargate removes the instances**, and charges a premium per vCPU and GB for that.
- **Fargate Spot** gives large savings on interruption-tolerant tasks, with two minutes' notice.
- **Capacity providers** mix Fargate, Fargate Spot and EC2 in one service, with a base and a weight.
- A new Fargate task takes tens of seconds to start. That is not a Lambda cold start, and it is not instant either.

ON THE EXAM

Fargate is not always cheaper. At steady high utilisation, EC2 with reservations often wins; Fargate wins on operational cost.

EKS specifics worth knowing

- **Managed node groups** handle provisioning and updates. **Karpenter** provisions right-sized nodes on demand instead.
- **Fargate profiles** run pods with no nodes at all, selected by namespace and labels.
- **IRSA**, or EKS Pod Identity, maps a Kubernetes service account to an IAM role. This is the answer for pod-level permissions.
- **EKS Anywhere** and **EKS Distro** run Kubernetes outside AWS. Both are on the exam's in-scope list.

> ON THE EXAM
>
> Pods needing distinct AWS permissions is an IRSA question. Giving the node role everything is the anti-pattern it exists to fix.

Reserved or provisioned concurrency?

RESERVED CONCURRENCY

A ceiling, and a floor

- Caps this function's concurrency
- Also guarantees it that much
- Protects the rest of the account
- No extra charge

PROVISIONED CONCURRENCY

Environments kept warm

- Pre-initialised, ready to serve
- Removes cold starts
- Charged for whether used or not
- Scheduled scaling is common

THE TIEBREAKER

One function starving the others is reserved concurrency. Latency spikes on first invocation is provisioned concurrency.

Lambda's edges

- **Fifteen minutes maximum.** Ten gigabytes of memory, ten of ephemeral storage, six megabyte synchronous payload.
- **SnapStart** restores a pre-initialised snapshot. Java first, now Python and .NET too.
- **Since August 2025 the INIT phase is billed** for every configuration. Cold starts now cost money as well as latency.
- Functions in a VPC use Hyperplane ENIs, so the old cold start penalty for VPC access is long gone.

> ON THE EXAM
>
> Fifteen minutes is the hardest boundary in serverless. A scenario mentioning a longer job has already excluded Lambda.

An honest note on App Runner

- **App Runner entered maintenance in March 2026** and closed to new customers on 30 April 2026.
- **It remains on the exam's in-scope service list**, so answer with it where a scenario describes its shape.
- Its shape: a container or repository in, an HTTPS service out, with no load balancer or scaling to configure.
- **Amazon ECS Express Mode** is the successor, giving the same simplicity on ordinary ECS and Fargate.

> ON THE EXAM
>
> Another examinable service you cannot buy. Teach the answer, know the reality, exactly as with Simple AD in module four.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Existing Helm charts, or multi-cloud	EKS
No Kubernetes experience, AWS only	ECS on Fargate
A job runs for several hours	Not Lambda. Fargate or AWS Batch
One function is exhausting account concurrency	Reserved concurrency
p99 latency spikes on first request	SnapStart or provisioned concurrency
Each pod needs different AWS permissions	IRSA or EKS Pod Identity

ON THE EXAM

Check the duration and the trigger first. Those two facts eliminate whole services before you weigh anything else.

> Meridian Freight's platform team of four is containerising eleven internal services. They have no Kubernetes experience, run entirely on AWS with no plans to change, and want to spend as little time as possible on platform operations.
>
> A consultant has recommended EKS because it is the industry standard.

What should the solutions architect advise?

A. EKS with managed node groups, as recommended

B. ECS on Fargate, with capacity providers for cost control

C. EKS with Fargate profiles, to avoid managing nodes

D. Run the eleven services on EC2 with Docker and a load balancer

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A EKS as recommended

Buys a Kubernetes learning curve, upgrade cycle and add-on ecosystem for four people with no Kubernetes experience and no portability requirement.

B ECS on Fargate

No control plane to run, no instances, no Kubernetes to learn, and capacity providers give cost control later.

C EKS with Fargate profiles

The closest miss. It removes the nodes and leaves the entire Kubernetes control surface, which is the actual overhead here.

D EC2 with Docker

Hand-rolls orchestration, health checking and deployment that both managed services provide.

THE SIGNAL

Industry standard is not a requirement. The stated requirements are no Kubernetes skills, AWS only, and minimal operations.

A company acquired a business running forty microservices on Kubernetes on-premises, with a large library of Helm charts, several custom operators, and a team of Kubernetes engineers. They must run in AWS within nine months, and keep the option of returning workloads on-premises later.

The parent company standardises on ECS.

What should the solutions architect recommend?

A Rewrite the workloads as ECS task definitions to match the parent standard

B Run them on EKS, accepting a documented exception to the ECS standard

C Run them on ECS on EC2, since both use containers

D Run them as Lambda functions to remove orchestration entirely

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Rewrite for ECS**

Discards Helm charts, operators and team skills, and spends the nine months rewriting rather than migrating. The portability requirement dies too.

B **EKS with an exception**

Charts and operators run unchanged, the team's skills apply, and Kubernetes manifests remain portable back on-premises.

C **ECS on EC2**

Both run containers, and ECS does not run Helm charts or Kubernetes operators. The artefacts do not transfer.

D **Lambda**

Forty microservices with operators are not functions. This is a rewrite disguised as modernisation.

THE SIGNAL

Pair this with scenario one. Same choice, opposite answer, because existing artefacts and a portability requirement changed everything.

> An account runs 40 Lambda functions. During a marketing campaign, one function scaled to thousands of concurrent executions and consumed almost all the account's concurrency. Unrelated functions, including a payment webhook handler, began throttling.
>
> The team wants the payment handler always to have capacity.

What should the solutions architect implement?

A. Request a concurrency quota increase for the account

B. Set reserved concurrency on the campaign function to cap it, and on the payment handler to guarantee it

C. Enable provisioned concurrency on the payment handler

D. Move the payment handler to a separate AWS account

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Quota increase

Raises the ceiling everyone shares. A bigger pool that one function can still consume entirely is the same failure, later.

B Reserved concurrency both sides

Reserved concurrency caps the noisy function and simultaneously guarantees the payment handler its own share.

C Provisioned concurrency

The classic confusion. It keeps environments warm to remove cold starts, and does nothing to protect a share of account concurrency.

D Separate account

Would work, by adding an account boundary, cross-account plumbing and operational overhead to solve a per-function setting.

THE SIGNAL

Reserved concurrency is a ceiling and a floor at the same time. That dual nature is exactly what this scenario needs.

> A customer-facing API runs on Lambda in Java. Median latency is 90 milliseconds, but p99 exceeds 4 seconds because of JVM cold starts. Traffic is steady during the day and near zero overnight.
>
> The team wants predictable latency without paying for idle capacity all night.

What should the solutions architect recommend?

A. Provisioned concurrency at the daytime peak, kept on permanently

B. Enable SnapStart, and add scheduled provisioned concurrency for business hours if needed

C. Increase the function's memory allocation

D. Rewrite the function in Python

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Permanent provisioned concurrency

Fixes latency and pays for the overnight hours the scenario explicitly rules out. It is right about the mechanism, wrong about the schedule.

B SnapStart, plus scheduled PC

SnapStart restores a pre-initialised snapshot, which is what JVM cold starts need, and costs nothing while idle.

C More memory

More memory means more CPU, which shortens init a little. It does not remove a four-second JVM start.

D Rewrite in Python

Rewriting a working API to change its cold-start profile, when a configuration setting addresses it.

THE SIGNAL

Without paying for idle capacity all night is the constraint. SnapStart costs nothing when the function is not running.

> A nightly job reconciles ledgers across systems. It currently runs on Lambda and has begun failing as data volumes grew, timing out at fifteen minutes. It is CPU-bound, single-threaded, and takes about three hours locally.
>
> The team has asked whether the timeout can be raised.

What should the solutions architect advise?

A Request a Lambda timeout increase through AWS Support

B Split the job into many Lambda invocations coordinated by Step Functions

C Move the job to AWS Batch or an ECS scheduled task on Fargate

D Increase Lambda memory to 10 GB so it completes faster

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Raise the timeout

Fifteen minutes is a hard service limit, not an adjustable quota. Support cannot raise it.

B Split with Step Functions

Excellent when work is divisible. This job is single-threaded and sequential, so there is nothing to split it along.

C Batch or scheduled Fargate task

No timeout, more CPU available, and a nightly batch job is exactly what these services are for.

D 10 GB memory

More memory does mean more CPU, and turning three hours into under fifteen minutes needs a twelve-fold speedup. Not plausible.

THE SIGNAL

Single-threaded and sequential is the planted detail. It exists to eliminate the Step Functions answer, which would otherwise be right.

> A team runs three internal web tools on AWS App Runner, deployed from container images. They are planning next year's roadmap and ask whether to keep investing in that platform.
>
> They want the same simplicity: no load balancer or scaling policy to configure.

What should the solutions architect advise?

A Keep App Runner; it is on the AWS service list and works well

B App Runner is in maintenance and closed to new customers; plan a move to ECS Express Mode on Fargate

C Migrate to EKS for long-term flexibility

D Migrate to Lambda behind API Gateway

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A. Keep investing

It does still work, and a service in maintenance receives no new features and eventually no service. Roadmap questions need that answered honestly.

B. Plan a move to ECS Express Mode

Preserves the simplicity they value, on ECS and Fargate, which are not going anywhere.

C. Migrate to EKS

Answers a simplicity requirement with the most operationally demanding option available.

D. Lambda behind API Gateway

A rewrite from containers to functions, changing the programming model to solve a platform-lifecycle problem.

THE SIGNAL

Note the question is about next year's roadmap. That is a lifecycle question, and lifecycle questions need lifecycle answers.

What to carry out of this module

- **Ask what the unit of scale is.** A request means Lambda, a process means a container, a node means Kubernetes.
- **Reserved concurrency is capacity. Provisioned concurrency is latency.** The names mislead; the jobs do not overlap.
- **Fifteen minutes is hard.** Not a quota, not raisable. Split it, or move it to Batch or Fargate.
- Existing Helm charts, operators or portability means EKS. Otherwise ECS costs less to operate.

> ON THE EXAM
>
> App Runner is examinable and closed to new customers. Teach the answer, know the reality, as with Simple AD in module four.

MODULE 16 · DOMAIN 2 - DESIGN FOR NEW SOLUTIONS

Choosing Storage

Object, block, file or archive. The access pattern decides before any product does.

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Decide the **category** before the product. Object, block, file, or archive is a question about how the data is reached.

The four categories

CATEGORY	REACHED BY	SERVICE
Object	HTTP GET by key, whole objects	Amazon S3
Block	A disk, attached to one instance	Amazon EBS, or instance store
File	A mounted share, many clients	Amazon EFS, Amazon FSx
Archive	Restored before use, or slow reads	S3 Glacier classes

ON THE EXAM

Many clients writing to the same files at once eliminates EBS immediately, whatever the performance requirement says.

The S3 storage classes

CLASS	CHOOSE IT WHEN	WATCH FOR
Standard	Frequent access, unknown pattern	Costs most per GB
Intelligent-Tiering	Access pattern varies or is unknown	Small monitoring fee per object
Standard-IA	Infrequent but immediate	30-day minimum, retrieval fee
One Zone-IA	Infrequent and reproducible	One AZ. Losing it must be acceptable
Glacier Instant Retrieval	Archive, but read in milliseconds	90-day minimum
Glacier Deep Archive	Cold, hours to restore acceptable	180-day minimum

ON THE EXAM

Glacier Flexible Retrieval sits between the last two, restoring in minutes to hours. Instant Retrieval is the one people forget exists.

The S3 charges that surprise people

- **Minimum storage durations:** 30 days for the IA classes, 90 for Glacier, 180 for Deep Archive. Delete early and you still pay.
- **128 KB minimum billable object size** in the IA classes. A million tiny objects costs as if each were 128 KB.
- **Retrieval fees** apply to IA and Glacier. Frequently read data in a cheap class costs more than Standard.
- **Intelligent-Tiering has no retrieval fee**, which is why it beats a badly guessed lifecycle policy.

> ON THE EXAM
>
> A lifecycle rule moving small, frequently read objects to IA can raise the bill. Three separate charges make that happen.

EBS volume types

- **gp3 is the default.** 3,000 IOPS and 125 MiB/s baseline, provisioned independently of size.
- **gp3 now reaches 64 TiB, 80,000 IOPS and 2,000 MiB/s** since late 2025. It was 16 TiB and 16,000 IOPS.
- **io2 Block Express** for sub-millisecond latency, 99.999% durability, Multi-Attach, or beyond 80,000 IOPS.
- **st1 and sc1** are HDD, for large sequential throughput. Neither can be a boot volume.

> ON THE EXAM
>
> The old rule that anything above 16,000 IOPS needs io2 is now wrong by a factor of five. Study material still teaches it.

gp3 or io2 Block Express?

GP3

The default, now far bigger

- Up to 80,000 IOPS, 2,000 MiB/s
- Up to 64 TiB per volume
- Cheaper per GB and per IOPS
- Single-digit ms latency

IO2 BLOCK EXPRESS

For what gp3 cannot meet

- Up to 256,000 IOPS
- Sub-millisecond latency
- 99.999% durability
- Multi-Attach across instances

THE TIEBREAKER

Four reasons remain for io2: latency below a millisecond, five-nines durability, Multi-Attach, or above 80,000 IOPS. Nothing else.

Amazon EFS

- **NFS for Linux**, mounted by thousands of clients across Availability Zones at once. It grows and shrinks automatically.
- **Standard** spans multiple AZs. **One Zone** costs far less and dies with its zone.
- **Elastic throughput** scales with demand and is the sensible default. Provisioned suits steady high throughput.
- Lifecycle management moves untouched files to infrequent access and archive tiers automatically.

> ON THE EXAM
>
> Many Linux instances writing to shared files is EFS. Windows or SMB in the same sentence means FSx instead.

The FSx family

FSX FOR	GIVES YOU	REACH FOR IT WHEN
Windows File Server	SMB, with Active Directory	Windows apps, home directories, AD auth
Lustre	Massive parallel throughput	HPC, ML training, S3-linked scratch
NetApp ONTAP	NFS, SMB and iSCSI together	Migrating an existing ONTAP estate
OpenZFS	NFS with ZFS snapshots	Linux workloads wanting ZFS features

ON THE EXAM

SMB plus Active Directory is FSx for Windows. High-throughput compute against S3 data is Lustre. Those two carry most questions.

Storage Gateway

- **S3 File Gateway** presents NFS or SMB on-premises, storing files as objects in S3.
- **Volume Gateway, cached** keeps the full dataset in S3 and a hot subset locally. **Stored** keeps it all locally and backs up to S3.
- **Tape Gateway** presents a virtual tape library, so existing backup software writes to S3 and Glacier unchanged.
- **FSx File Gateway** gives on-premises low-latency access to an FSx for Windows file system.

> ON THE EXAM
>
> Cached keeps the primary copy in the cloud. Stored keeps the primary copy on-premises. That is the whole distinction.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
SMB share with Active Directory authentication	FSx for Windows File Server
Thousands of Linux instances sharing files	Amazon EFS
Archive, but must be readable in milliseconds	S3 Glacier Instant Retrieval
Access pattern is unknown or changing	S3 Intelligent-Tiering
Local capacity exhausted, keep hot data fast	Volume Gateway, cached mode
Above 80,000 IOPS, or sub-millisecond	io2 Block Express

ON THE EXAM

Protocol and access pattern decide most storage questions. Performance numbers usually confirm a choice rather than making it.

> Meridian Freight is sizing storage for a PostgreSQL database on EC2. It needs about 45,000 IOPS sustained, 8 TiB of capacity, and single-digit millisecond latency. There is no clustering requirement.
>
> An architect has specified io2 Block Express, citing that gp3 caps at 16,000 IOPS.

What should the solutions architect advise?

A. Proceed with io2 Block Express as specified

B. Use gp3, which now supports up to 80,000 IOPS and 64 TiB

C. Stripe four gp3 volumes in RAID 0 to reach the required IOPS

D. Use st1 volumes, since databases are throughput-bound

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A io2 as specified

Would work, and costs more per GB and per IOPS for requirements gp3 now meets. The stated reason for choosing it is out of date.

B gp3

45,000 IOPS and 8 TiB both sit inside gp3's current limits, and single-digit millisecond latency is what gp3 delivers.

C RAID 0 across four volumes

The workaround from before the limits rose. It adds complexity and a wider failure surface for no benefit now.

D st1

HDD, capped near 500 IOPS. It is for sequential throughput and cannot serve a transactional database.

THE SIGNAL

The architect's reasoning was correct until September 2025. Check the numbers rather than the rule you remember.

> A company must retain seven years of medical imaging for compliance. Files are almost never read, but when a clinician requests one it must be available within milliseconds, not hours.
>
> Storage cost is the primary concern after that requirement.

Which storage class should the architect choose?

A. S3 Glacier Deep Archive, with expedited retrieval

B. S3 Glacier Instant Retrieval

C. S3 Standard-IA

D. S3 Intelligent-Tiering with the Deep Archive Access tier enabled

◆ PAUSE AND DECIDE

Which storage class should the architect choose?

A **Deep Archive**

Cheapest per GB and restores in hours. Expedited retrieval is not offered for Deep Archive, and the requirement says milliseconds.

B **Glacier Instant Retrieval**

Built for exactly this: archive pricing with millisecond access. Rarely read, immediately needed, cost-sensitive.

C **Standard-IA**

Also millisecond access, and costs considerably more per GB than Glacier Instant Retrieval for data read this rarely.

D **Intelligent-Tiering with Deep Archive**

Objects reaching the Deep Archive tier take hours to restore, which is the outcome the requirement forbids.

THE SIGNAL

Rarely read but instantly needed is the exact gap Glacier Instant Retrieval fills. Most people forget the class exists.

> A finance team's Windows application needs a shared file system for 300 users, authenticating against the company's existing Active Directory, using SMB, with Windows ACLs preserved and shadow copies for user self-service restore.
>
> The application cannot be modified.

What should the solutions architect recommend?

A Amazon EFS with an SMB gateway in front

B Amazon FSx for Windows File Server, joined to the existing Active Directory

C Amazon S3 with S3 File Gateway

D An EC2 Windows file server on an io2 volume

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A. EFS plus a gateway

EFS is NFS. Bolting on an SMB translation layer adds a component to maintain and still loses Windows ACL fidelity.

B. FSx for Windows File Server

Native SMB, AD-joined, Windows ACLs preserved, and shadow copies built in. Every requirement is a native feature.

C. S3 File Gateway

Presents a share backed by object storage. It is for extending on-premises capacity, not for hosting an AD-integrated share in AWS.

D. Self-managed Windows file server

Works, and you now own patching, availability, backups and capacity for something AWS offers managed.

THE SIGNAL

SMB plus Active Directory is the pair. Four requirements here are FSx for Windows features and nothing else's.

> A genomics team runs simulations across 500 EC2 instances. Each run reads a 40 TB reference dataset held in S3 and needs hundreds of gigabytes per second of aggregate read throughput. Results are written back to S3 and the scratch data is disposable.
>
> Runs happen twice a week and last about six hours.

What should the solutions architect recommend?

A. Amazon EFS with provisioned throughput

B. Amazon FSx for Lustre, linked to the S3 bucket, created for each run

C. Copy the dataset to an io2 Block Express volume on each instance

D. Read directly from S3 with a gateway endpoint

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A EFS provisioned

Genuinely shared and not built for hundreds of GB/s across 500 clients. Provisioning that throughput continuously would also be expensive.

B FSx for Lustre linked to S3

Purpose-built for this. It presents the S3 data as a file system, delivers the throughput, and is deleted after each run.

C io2 per instance

500 copies of a 40 TB dataset, and EBS volumes are not shared. The copy alone would take longer than the run.

D Direct from S3

Avoids a file system and gives object semantics and higher latency, which simulation code expecting POSIX files cannot use.

THE SIGNAL

Twice a week for six hours is the planted detail. It makes a transient file system created per run the economical answer.

> An analytics platform writes about 40 million small JSON files a day, averaging 12 KB each. A lifecycle rule moves them to S3 Standard-IA after seven days, and a separate job deletes them after twenty-one days. The bill has risen since the rule was introduced.
>
> Files are occasionally re-read during the first two weeks.

Why has the cost increased, and what should change?

A. Standard-IA has a 30-day minimum and a 128 KB minimum billable size; keep them in Standard, or use Intelligent-Tiering

B. Move the objects to One Zone-IA instead to halve the cost

C. Move them to Glacier Instant Retrieval after seven days

D. Enable S3 Versioning to reduce duplicate storage

◆ PAUSE AND DECIDE

Why has the cost increased, and what should change?

A Minimum duration and minimum size

Deleted at 21 days but billed for 30, and 12 KB objects billed as 128 KB. Retrieval fees on re-reads make it worse still.

B One Zone-IA

Has the same 30-day minimum and the same 128 KB minimum. It reduces the per-GB rate on a bill inflated by both penalties.

C Glacier Instant Retrieval

Worse on both counts: a 90-day minimum against a 21-day lifetime, and higher retrieval fees on data still being read.

D Versioning

Keeps additional copies of objects, which increases storage. It has nothing to do with storage class economics.

THE SIGNAL

Three penalties compound here: early deletion, minimum object size, and retrieval fees. A cheaper class is not a cheaper bill.

> A manufacturing site has 90 TB of design files on ageing on-premises NAS with only 20 TB of usable capacity left. Engineers need low-latency access to the roughly 8 TB they are actively working on, and the rest can tolerate slower access.
>
> Replacing the NAS is not budgeted, and the applications use iSCSI.

What should the solutions architect recommend?

A. AWS Storage Gateway, Volume Gateway in cached mode

B. AWS Storage Gateway, Volume Gateway in stored mode

C. AWS DataSync to move everything to S3, and mount it with S3 File Gateway

D. Amazon FSx for OpenZFS accessed over Direct Connect

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Volume Gateway, cached

The full 90 TB lives in S3 while the active subset stays local. iSCSI is preserved and local capacity stops being the ceiling.

B Volume Gateway, stored

The exact inverse. Stored mode keeps every byte on-premises, which is the capacity problem they already have.

C DataSync plus File Gateway

File Gateway presents NFS or SMB. The applications use iSCSI, so they would need changing.

D FSx for OpenZFS over Direct Connect

Puts every read across the link, so the active 8 TB loses the low latency the scenario requires.

THE SIGNAL

Cached keeps the primary copy in the cloud with a hot local subset. Stored keeps the primary copy local. Capacity pressure means cached.

What to carry out of this module

- **Pick the category first.** Object, block, file or archive is a question about how the data is reached.
- **gp3 now reaches 80,000 IOPS and 64 TiB.** Only four things still require io2 Block Express.
- **Minimum durations and the 128 KB minimum** mean a cheaper class can produce a larger bill.
- SMB plus Active Directory is FSx for Windows. S3-linked high throughput is Lustre. Capacity pressure is cached Volume Gateway.

> ON THE EXAM
>
> Glacier Instant Retrieval is the class people forget: archive economics with millisecond access. It answers more questions than it should.

MODULE 17 · DOMAIN 2 - DESIGN FOR NEW SOLUTIONS

Choosing a Database

Relational, key-value, or purpose-built. Read the access pattern, not the data.

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The question is not what the data looks like. It is **how it is queried**, and whether the pattern is known in advance.

Relational or DynamoDB?

AURORA OR RDS

Unknown queries, joins, SQL

- Ad hoc queries and reporting
- Joins and multi-table transactions
- Existing SQL applications
- Scales up, and read replicas out

DYNAMODB

Known access patterns, huge scale

- Single-digit ms at any volume
- Access patterns fixed at design time
- No servers, no version upgrades
- Scales horizontally without limit

THE TIEBREAKER

Unbounded scale with predictable queries means DynamoDB. Anything needing joins or unforeseen queries means relational.

RDS or Aurora?

- **Aurora separates compute from storage**, with a shared volume across three AZs and six copies. Replicas share it rather than copying.
- **Replica lag is milliseconds**, against seconds for RDS, and failover is typically under a minute.
- **RDS still wins** when you need Oracle, SQL Server, MariaDB or Db2. Aurora is MySQL and PostgreSQL compatible only.
- **RDS Multi-AZ cluster deployments** give two readable standbys, closing part of the gap.

> ON THE EXAM
>
> Aurora storage grows automatically to 128 TiB. Engine choice, not performance, is what usually forces RDS.

Aurora at multi-Region scale

- **Aurora Global Database** replicates to secondary Regions with typical lag under a second, and cross-Region failover in about a minute.
- **Write forwarding** lets a secondary Region accept writes and forward them to the primary, so applications need no routing logic.
- It is still **one writer**. Global Database is active-passive, whatever the write forwarding makes it look like.
- **Aurora Serverless**, formerly Serverless v2, scales by ACU and now down to zero with auto-pause.

> ON THE EXAM
>
> Write forwarding hides the round trip; it does not remove it. A write from Sydney still travels to the primary Region.

DynamoDB, and a changed default

- **On-demand throughput was cut by 50% in Nov 2024.** AWS now calls on-demand the default and recommended mode for most workloads.
- **Global tables** are multi-active: every Region accepts writes. Replicated write pricing fell by up to 67% at the same time.
- **DAX** is a DynamoDB-specific cache giving microsecond reads. **Streams** emit item changes for event-driven work.
- Global secondary indexes give alternative access patterns. Model the table around queries, not entities.

> ON THE EXAM
>
> The old advice was provisioned for steady workloads, on-demand for spiky ones. The 2024 price cut reversed the default.

Global tables or Aurora Global Database?

DYNAMODB GLOBAL TABLES

Every Region writes

- Multi-active, no failover needed
- Eventually consistent across Regions
- Strong consistency now optional
- Last writer wins on conflict

AURORA GLOBAL DATABASE

One Region writes

- Active-passive, promote to fail over
- Sub-second replication lag
- Write forwarding hides the routing
- Full SQL and transactions

THE TIEBREAKER

Writes accepted in every Region with no failover means DynamoDB. If the answer must be relational, accept active-passive.

The purpose-built line-up

SERVICE	DATA SHAPE	SIGNAL IN THE SCENARIO
ElastiCache	In-memory cache or store	Reduce database load, session state
Amazon Neptune	Graph, with relationships	Fraud rings, recommendations, social
Amazon DocumentDB	JSON documents, MongoDB API	Migrating an existing MongoDB estate
Amazon Keyspaces	Wide-column, Cassandra API	Existing Cassandra, CQL
Amazon Timestream	Time series	IoT sensors, metrics over time
Amazon Redshift	Columnar warehouse	Analytics over billions of rows

ON THE EXAM

Purpose-built questions name an API or a data shape. Cassandra, MongoDB or a graph in the wording is the whole answer.

ElastiCache, and a renamed engine

- **Three engines: Valkey, Redis OSS and Memcached.** Valkey is the open-source Redis fork and is API compatible.
- **Valkey is roughly 20% cheaper node-based and 33% cheaper serverless**, with a zero-downtime in-place upgrade path.
- **Redis OSS v4 and v5 left standard support on 31 January 2026** and now carry Extended Support charges.
- Memcached is multi-threaded and simple. Redis and Valkey add replication, persistence and data structures.

> ON THE EXAM
>
> Replication, persistence or sorted sets means Redis or Valkey. Pure horizontal cache with no features means Memcached.

Two notes on cost and scope

- **Database Savings Plans** now cover Aurora, RDS, DynamoDB, ElastiCache for Valkey, DocumentDB, Neptune, Keyspaces and DMS.
- **Not Redshift.** One year, no upfront, latest-generation instances only. Redis OSS and Memcached are excluded.
- **Aurora DSQL** gives active-active multi-Region SQL writes with strong consistency. It reached GA in mid-2025.
- **DSQL is not on the exam's in-scope list.** Know it exists; answer questions with Aurora Global Database.

> ON THE EXAM
>
> DSQL genuinely solves what Global Database cannot. The exam does not list it, so it is not the answer there.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Millions of requests, fixed access patterns	DynamoDB
Ad hoc reporting queries and joins	Aurora or RDS
Oracle, SQL Server, MariaDB or Db2	RDS, not Aurora
Writes accepted in several Regions at once	DynamoDB global tables
Relationships between entities are the query	Amazon Neptune
Database CPU spent on repeated identical reads	ElastiCache, or DAX

ON THE EXAM

Name the engine, the access pattern and the write topology. Most database questions are decided by one of those three.

> Meridian Freight is designing a parcel tracking service. Every lookup is by tracking number and returns one parcel's status history. Volume is expected to reach 400,000 requests per second at peak, growing yearly. Responses must be under 10 milliseconds.
>
> The team's existing skills are in PostgreSQL.

What should the solutions architect recommend?

A. Aurora PostgreSQL with read replicas, since the team knows PostgreSQL

B. DynamoDB, with the tracking number as the partition key

C. Aurora PostgreSQL with an ElastiCache layer in front

D. Amazon Keyspaces, for wide-column scale

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Aurora with read replicas**

Aurora allows fifteen read replicas. 400,000 requests per second and growing will exhaust that ceiling, and the ceiling is fixed.

B **DynamoDB**

One access pattern, always by key, at unbounded scale with a millisecond target. This is precisely what DynamoDB is for.

C **Aurora plus ElastiCache**

The strong near miss. It would work today, and it adds a cache to manage and invalidate, and Aurora remains the ceiling underneath.

D **Keyspaces**

Also scales, and is for teams with existing Cassandra applications and CQL. Nothing here suggests either.

THE SIGNAL

Team skills are real and they are not a requirement. The access pattern and the growth curve are.

A retailer runs a global storefront. Customers in Europe, North America and Asia all place orders, and an order written in any Region must be visible in the others within seconds. Order records are simple documents retrieved by order ID.

The business will not accept a failover step before writes can resume in a Region.

What should the solutions architect recommend?

A Aurora Global Database with write forwarding from the secondary Regions

B DynamoDB global tables across the three Regions

C Aurora in one Region with read replicas in the others

D Three independent DynamoDB tables, synchronised nightly

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A. Global Database with write forwarding

The sharpest distractor. Writes are accepted everywhere and forwarded to the single primary, so losing the primary still needs a promotion.

B. DynamoDB global tables

Genuinely multi-active. Every Region owns writes, so there is no primary to lose and no failover step at all.

C. Read replicas

Replicas are read-only. Two of the three Regions could not accept an order at all.

D. Nightly sync

Seconds was the requirement. Nightly is roughly fifty thousand times too slow, and invites conflicting order IDs.

THE SIGNAL

No failover step is the phrase. Write forwarding removes the routing logic, not the single primary underneath it.

A DynamoDB table serves an application whose traffic is steady at roughly 4,000 read units and 1,500 write units, all day, every day. It has used provisioned capacity with auto scaling since 2022.

A cost review asks whether the capacity mode is still the right choice.

What should the solutions architect advise?

A Keep provisioned capacity; it is cheaper for steady workloads

B Compare against on-demand, which AWS now recommends as the default after the 2024 price cut

C Switch to provisioned capacity with reserved capacity for a further discount

D Move the table to Aurora, which is cheaper at steady load

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A **Keep provisioned**

The correct answer until November 2024. AWS halved on-demand pricing and now states most provisioned workloads are cheaper on-demand.

B **Compare against on-demand**

The price cut changed the default. Modelling the actual numbers is the right advice, and on-demand also removes capacity management.

C **Reserved capacity**

Deepens a commitment to a mode that may no longer be the cheaper one. Measure before committing for a year.

D **Move to Aurora**

Replaces the database to change a billing mode, and Aurora would not serve this access pattern better.

THE SIGNAL

Since 2022 is the planted detail. The design was correct when it was made and the pricing moved underneath it.

> An insurer is migrating a claims system that runs on Oracle with PL/SQL packages, materialised views, and a third-party actuarial extension. It needs Multi-AZ resilience and automated backups, and the business will not fund an application rewrite.
>
> An architect has proposed Aurora PostgreSQL for its performance.

What should the solutions architect advise?

A. Aurora PostgreSQL, using Babelfish for compatibility

B. Amazon RDS for Oracle in a Multi-AZ deployment

C. Oracle on EC2 with a self-managed standby

D. Amazon DocumentDB, since claims are documents

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A **Aurora with Babelfish**

Babelfish translates SQL Server T-SQL, not Oracle PL/SQL. It is the wrong compatibility layer for this engine entirely.

B **RDS for Oracle, Multi-AZ**

Keeps the engine, so PL/SQL and the extension keep working, while AWS handles Multi-AZ and backups. No rewrite.

C **Oracle on EC2**

Preserves the engine and hands back patching, backup and standby management that RDS provides.

D **DocumentDB**

Claims being called documents in English does not make them MongoDB documents. This is a full rewrite.

THE SIGNAL

Babelfish is for SQL Server, not Oracle. Knowing which compatibility layer belongs to which engine is the whole question.

> A payments platform's Aurora writer is at 85% CPU. Analysis shows 70% of queries are identical reads of a small set of merchant configuration rows that change a few times a day.
>
> The team wants to reduce load without changing the schema or sharding.

What should the solutions architect recommend?

A. Add Aurora read replicas and route reads to them

B. Add an ElastiCache layer caching the configuration rows, invalidated on change

C. Scale the writer to a larger instance class

D. Add DynamoDB Accelerator in front of Aurora

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Read replicas**

The strong near miss. It moves the reads off the writer, and you still execute the same identical query millions of times, just elsewhere.

B **ElastiCache with invalidation**

Data read constantly and changed a few times a day is the textbook cache. The query stops executing rather than moving.

C **Bigger writer**

Buys headroom for the same wasteful pattern, and instance classes run out. It postpones rather than solves.

D **DAX in front of Aurora**

DAX is DynamoDB-specific. It cannot sit in front of Aurora at all.

THE SIGNAL

Read often, changed rarely, small dataset. Three properties that together describe a cache and nothing else.

> A bank is building fraud detection. The core query is finding chains of relationships between accounts, devices and addresses, up to six hops deep, in real time. The data is highly connected and the useful questions are about paths rather than rows.
>
> An architect proposes recursive SQL on Aurora PostgreSQL.

What should the solutions architect recommend?

A. Aurora PostgreSQL with recursive common table expressions

B. Amazon Neptune

C. DynamoDB with a global secondary index per relationship type

D. Amazon Redshift, since it handles large joins

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Recursive CTEs on Aurora

Expressible in SQL, and each hop is another self-join. At six hops in real time the query planner will not hold up.

B Amazon Neptune

A graph database traverses relationships as a first-class operation, so six hops costs roughly six steps rather than six joins.

C DynamoDB with GSIs

Each hop becomes a separate query the application stitches together. Traversal logic moves into your code and gets slower with depth.

D Redshift

Columnar and built for aggregation over many rows. It is an analytics warehouse, and this is a real-time traversal.

THE SIGNAL

Paths rather than rows is the phrase. When relationships are the query rather than an implementation detail, that is a graph.

What to carry out of this module

- **Ask how it is queried, not what it looks like.** Known access patterns at scale means DynamoDB.
- **Aurora Global Database is active-passive.** Write forwarding hides the routing, not the single primary.
- **DynamoDB on-demand is now the default** after the 2024 price cut. Older material still says provisioned.
- Babelfish is for SQL Server, not Oracle. DAX is for DynamoDB only. Both are precise knowledge checks.

> ON THE EXAM
>
> Engine support is what usually forces RDS over Aurora. Oracle, SQL Server, MariaDB and Db2 have no Aurora equivalent.

MODULE 18 · DOMAIN 2 - DESIGN FOR NEW SOLUTIONS

Decoupling and Integration

Queue, topic, stream or bus. Four shapes, and they are not interchangeable.

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A queue message is **consumed and gone**. A stream record **stays**, and everyone can read it.

The four shapes

SHAPE	SERVICE	USE IT WHEN
Queue	Amazon SQS	Work for exactly one consumer, buffered
Topic	Amazon SNS	Push the same message to many subscribers
Stream	Kinesis, MSK	An ordered log many consumers replay
Bus	Amazon EventBridge	Route events by content, with filtering

ON THE EXAM

SNS pushes and forgets. EventBridge routes on content and integrates with SaaS. Both fan out; only one filters richly.

SQS or Kinesis?

AMAZON SQS

Work, done once

- Message deleted after processing
- One consumer per message
- No ordering unless FIFO
- Scales with no shards to manage

KINESIS DATA STREAMS

A log, read many times

- Records persist for the retention period
- Many independent consumers
- Ordered within a partition key
- Replay from any point in time

THE TIEBREAKER

Several systems needing the same events, or replay after a bug, means a stream. A queue cannot give you either.

SQS details that decide questions

- **Visibility timeout** hides a message while it is processed. Too short means duplicate processing; too long means slow retries.
- **A dead-letter queue** catches messages that fail repeatedly, so one poison message cannot block a queue forever.
- **FIFO** gives ordering and exactly-once processing within a message group, at lower throughput than standard.
- 256 KB message limit, 14-day retention, and long polling to avoid paying for empty receives.

> ON THE EXAM
>
> The same message processed twice usually means visibility timeout shorter than processing time. It is the classic SQS bug.

SNS or EventBridge?

AMAZON SNS

Fan-out, fast and simple

- Very high throughput, low latency
- Filtering on message attributes
- SQS, Lambda, HTTP, email, SMS
- You control who publishes

AMAZON EVENTBRIDGE

Routing on content

- Rules match the event body
- AWS service events arrive free
- SaaS partners and custom buses
- Archive and replay events

THE TIEBREAKER

Reacting to an AWS service event, or routing on the event's content, means EventBridge. Raw fan-out speed means SNS.

Streams: Kinesis and the rest

- **Shards** set capacity: 1 MB/s in, 2 MB/s out each. On-demand mode removes the sizing decision.
- **Enhanced fan-out** gives each consumer its own 2 MB/s per shard, instead of sharing it.
- **Amazon Data Firehose** delivers to S3, Redshift or OpenSearch with no code, buffering for at least 60 seconds.
- Firehose is **delivery, not processing**. If records must arrive in seconds, use Data Streams with a consumer.

ON THE EXAM

Several consumers each needing full throughput is the enhanced fan-out signal. Without it they share 2 MB/s per shard.

When the answer is a lift and shift

- **Amazon MQ** runs ActiveMQ or RabbitMQ. Choose it when the application already speaks JMS, AMQP, MQTT or STOMP.
- **Amazon MSK** is managed Apache Kafka, for existing Kafka producers, consumers and Connect ecosystems.
- Neither is the answer for a new application. SQS and Kinesis are cheaper and need no brokers.
- The signal is always an existing protocol or ecosystem that cannot be changed.

> ON THE EXAM
>
> A named protocol like JMS or AMQP means Amazon MQ. Named Kafka tooling means MSK. Neither appears without that signal.

Step Functions: Standard or Express?

STANDARD WORKFLOWS

Long, durable, auditable

- Up to one year of execution
- Exactly-once execution
- Full history in the console
- Priced per state transition

EXPRESS WORKFLOWS

Short, cheap, very high volume

- Up to five minutes
- At-least-once execution
- History goes to CloudWatch Logs
- Priced per invocation and duration

THE TIEBREAKER

A human approval step, or anything measured in hours, means Standard. High-volume short-lived orchestration means Express.

Patterns worth naming

- **Fan-out** is SNS to several SQS queues, so each consumer gets its own buffer and its own retry behaviour.
- **Load levelling** puts a queue in front of a slow component, so a traffic spike becomes a backlog rather than an outage.
- **EventBridge Pipes** connects one source to one target with filtering and enrichment, replacing glue Lambda functions.
- **SQS fair queues**, from 2025, stop one noisy tenant starving others in a shared multi-tenant queue.

> ON THE EXAM
>
> Fan-out into queues rather than straight to consumers is the pattern the exam rewards. Each consumer then fails independently.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Several systems need the same events	SNS fan-out, or a stream
Replay events after fixing a bug	Kinesis, or EventBridge archive
Route on a field inside the event	EventBridge rules
Existing JMS or AMQP application	Amazon MQ
A human must approve, days may pass	Step Functions Standard
Spikes overwhelm a downstream system	SQS for load levelling

ON THE EXAM

Count the consumers first, then ask whether the message must survive being read. Those two facts decide most of these.

> Meridian Freight's booking service publishes a shipment-created event. Four systems must each receive it: billing, customer notification, warehouse allocation and analytics. Each processes at a different speed, and warehouse allocation is occasionally down for maintenance.
>
> No event may be lost while a consumer is unavailable.

What should the solutions architect design?

A SNS publishing directly to four Lambda functions

B SNS publishing to four SQS queues, one per consumer

C One SQS queue that all four systems poll

D Four separate publishes from the booking service, one per system

◆ PAUSE AND DECIDE

What should the solutions architect design?

A SNS to four Lambdas

Works while everything is healthy. SNS retries then gives up, so events are lost while warehouse allocation is down.

B SNS to four SQS queues

Each consumer gets its own buffer and retry behaviour, so a consumer being down means a backlog rather than lost events.

C One shared queue

A queue message goes to one consumer. Whichever system reads it first, the other three never see it.

D Four publishes

Couples the booking service to every consumer, so adding a fifth means changing the publisher.

THE SIGNAL

The fan-out pattern is SNS into queues, not SNS into consumers. The queue is what survives a consumer being offline.

A telematics platform ingests 200,000 vehicle position records per second. Three independent teams consume the data: a live map, a fraud model, and an hourly analytics job. After a bug in the fraud model, the team needs to reprocess the last three days of records.

Ordering per vehicle matters.

What should the solutions architect recommend?

A SQS FIFO queues, one per consuming team

B Kinesis Data Streams with the vehicle ID as partition key, retention set to seven days

C SNS fan-out to three SQS queues

D Amazon Data Firehose delivering to S3, with each team reading the bucket

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A SQS FIFO per team

FIFO throughput is far below 200,000 per second, and a consumed message cannot be replayed three days later.

B Kinesis with 7-day retention

Ordered per partition key, read independently by all three teams, and replayable from any point inside the retention window.

C SNS to three queues

Solves fan-out and not replay. Once each queue is drained, those messages are gone.

D Firehose to S3

The near miss. S3 gives replay, and Firehose buffers at least 60 seconds, which the live map cannot accept.

THE SIGNAL

Three requirements: many consumers, ordering, and replay. Only a stream provides all three at once.

> A security team wants to react automatically when any EC2 instance in the organisation is tagged with Environment=Production but launched without encryption. They also want to receive alerts from a third-party SaaS security product into the same workflow.
>
> The routing must depend on fields inside the event, not just on its type.

What should the solutions architect use?

A SNS with message attribute filtering

B EventBridge rules on a custom bus, with a SaaS partner event source

C CloudTrail delivering to an SQS queue that a Lambda function inspects

D Kinesis Data Streams with a Lambda consumer applying the logic

◆ PAUSE AND DECIDE

What should the solutions architect use?

A. SNS attribute filtering

Filters on attributes the publisher sets, not on the event body. AWS service events would also need routing into SNS first.

B. EventBridge with a partner source

AWS service events arrive natively, rules match fields inside the event body, and SaaS partners publish into the same bus.

C. CloudTrail to SQS and Lambda

Builds the routing engine yourself in code, and adds nothing for the third-party product.

D. Kinesis with logic in Lambda

Same objection, with a stream added. The filtering logic lives in code that must be maintained.

THE SIGNAL

Three signals: an AWS service event, routing on content, and a SaaS integration. All three are EventBridge and nothing else.

A manufacturer is migrating a plant control system to AWS. It uses ActiveMQ with JMS topics and queues, transactional sessions, and a message selector syntax the vendor supports but will not change. The migration must complete in four months with no application changes.

An architect has proposed rewriting it to use SQS and SNS.

What should the solutions architect advise?

A Rewrite to SQS and SNS, which are cheaper and fully managed

B Amazon MQ running ActiveMQ, so the application connects unchanged

C Amazon MSK, since Kafka can emulate JMS semantics

D Run ActiveMQ on EC2 instances the team manages

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Rewrite to SQS and SNS

Cheaper to run and impossible here. The vendor will not change the application, and four months is not a rewrite window.

B Amazon MQ with ActiveMQ

The same broker, managed. JMS, transactional sessions and message selectors all keep working with no code change.

C Amazon MSK

Kafka does not speak JMS. Adapters exist and they change the application, which is the one thing ruled out.

D Self-managed ActiveMQ

Preserves compatibility and hands back patching, failover and broker operations that Amazon MQ provides.

THE SIGNAL

A named protocol the application cannot change is the Amazon MQ signal. It exists almost entirely for this situation.

> An insurance claim workflow has eleven steps. Three call external services, one requires a human adjuster to approve, and claims frequently sit waiting for that approval for several days. The business needs to see exactly where any claim is at any moment.
>
> Volume is around 4,000 claims a day.

What should the solutions architect recommend?

A Step Functions Express workflows, for cost efficiency

B Step Functions Standard workflows, with a task token for the approval step

C A chain of Lambda functions triggered by SQS queues

D EventBridge rules chaining the eleven steps

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A. Express workflows

Express workflows stop at five minutes. A claim waiting days for a human cannot run inside one.

B. Standard with a task token

Runs up to a year, so it can wait for the adjuster, and the execution history shows exactly where every claim is.

C. Lambda chained by queues

Works, and the workflow now exists only implicitly across eleven functions. Nobody can see where a claim is without building that view.

D. EventBridge chaining

Same objection. EventBridge routes events; it does not track the state of a long-running process.

THE SIGNAL

Five minutes is the Express ceiling, and 4,000 a day is low volume, so the cost argument for Express does not apply anyway.

> A ticketing site's checkout writes to a legacy inventory system that handles 200 requests per second. During on-sale events, traffic reaches 5,000 per second for about ten minutes. The inventory system currently returns errors and customers see failures.
>
> Replacing the inventory system is a two-year programme already under way.

What should the solutions architect recommend?

A. Auto Scaling on the inventory system to handle the peak

B. An SQS queue between checkout and inventory, with consumers polling at a sustainable rate

C. An ElastiCache layer in front of the inventory system

D. API Gateway throttling, rejecting requests above 200 per second

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Auto Scaling the legacy system

Legacy systems are usually the ones that cannot scale horizontally, which is why it is being replaced. Nothing suggests it can.

B SQS load levelling

The spike becomes a backlog the queue absorbs, drained at 200 per second. Customers get an accepted order rather than an error.

C ElastiCache

Caching helps repeated reads. These are inventory writes, which cannot be served from a cache.

D Throttle at 200/s

Protects the inventory system by rejecting most customers during precisely the ten minutes that matter commercially.

THE SIGNAL

Load levelling changes what failure looks like. A ten-minute spike becomes a queue that drains, instead of errors customers see.

What to carry out of this module

- **A queue message is consumed and gone. A stream record stays.** Replay or multiple consumers forces a stream.
- **Fan-out means SNS into queues**, not SNS into consumers. The queue survives a consumer being offline.
- EventBridge routes on the event body and speaks to SaaS. SNS is faster fan-out with attribute filtering.
- **Express workflows stop at five minutes.** A human approval always means Standard.

> ON THE EXAM
>
> Amazon MQ and MSK appear only when a protocol or ecosystem cannot change. Without that signal, they are not the answer.

MODULE 19 · DOMAIN 2 - DESIGN FOR NEW SOLUTIONS

Business Continuity by Design

Two numbers decide the architecture. Everything else follows from them.

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RTO and RPO are **business decisions**. The architecture is the cheapest thing that meets them.

The continuum, drawn once

01

Backup and restore

Data copied off-site. Nothing else runs.

›

02

Pilot light

Data replicating. Compute defined but switched off.

›

03

Warm standby

A scaled-down copy, running and able to serve.

›

04

Multi-site

Full capacity in both places, both taking traffic.

ON THE EXAM

Cost rises left to right. RTO and RPO fall. Every question in this module is a position on this line.

The four strategies

STRATEGY	RTO / RPO	WHAT IS RUNNING
Backup and restore	Hours to a day / hours	Nothing but stored backups
Pilot light	Tens of minutes / minutes	Data replicating; compute off
Warm standby	Minutes / seconds	A scaled-down live copy
Multi-site active-active	Near zero / near zero	Full capacity, both serving

ON THE EXAM

The line between pilot light and warm standby is whether anything is serving requests. Not whether resources exist.

Pilot light or warm standby?

PILOT LIGHT

Ready to start, not running

- Database replicating continuously
- Compute defined but switched off
- You pay for data, not servers
- Recovery means provisioning first

WARM STANDBY

Running, just small

- A scaled-down copy already serving
- Failover means scaling up
- Costs more, every hour
- It works, because it is running

THE TIEBREAKER

Warm standby buys a second benefit: a running copy proves itself continuously. A pilot light is a hypothesis until tested.

AWS Elastic Disaster Recovery

- **Continuous block-level replication** from on-premises, other clouds or EC2 into a low-cost staging area.
- **RPO in seconds, RTO typically 5 to 20 minutes.** Pilot-light economics with warm-standby recovery.
- **Drill instances** test recovery without touching source servers or interrupting replication.
- **It recovers servers; it does not fail over traffic.** Route 53 or Global Accelerator moves the users.

ON THE EXAM

DRS is the answer for server-based workloads nobody may re-architect. It converts machines to run natively on AWS.

Replicating the data

- **Aurora Global Database:** typically under a second of lag, cross-Region failover in about a minute.
- **DynamoDB global tables:** multi-active, so there is no failover step at all.
- **RDS cross-Region read replicas** for engines with no global option, promoted manually.
- **S3 Cross-Region Replication**, with Replication Time Control giving a 15-minute SLA.

> ON THE EXAM
>
> Your RPO is whatever your slowest replicating datastore is. One nightly snapshot sets it for the whole system.

AWS Backup, and surviving an attacker

- **Backup plans** apply schedules and lifecycle by tag, across accounts and Regions, from a delegated administrator.
- **Vault Lock governance mode can be overridden** by a privileged user. Compliance mode cannot, by anyone.
- **Logically air-gapped vaults** are locked in compliance mode and shareable via RAM, so another account restores directly.
- **Restore testing** runs scheduled restores, turning recoverability from an assumption into a measurement.

> ON THE EXAM
>
> Sharing an air-gapped vault removes the copy step before restoring, which is exactly where recovery time is usually lost.

An untested disaster recovery plan is not a plan. It is a **hypothesis with a budget**.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Hours of downtime acceptable, cost matters most	Backup and restore
Tens of minutes, minimal data loss, cost-sensitive	Pilot light
Minutes, and it must be proven to work	Warm standby
On-premises servers, RPO seconds, no re-architecture	Elastic Disaster Recovery
Backups must survive a compromised admin	Air-gapped vault, compliance mode
The RTO has never been measured	Run a drill first

ON THE EXAM

Find the two numbers, then pick the cheapest strategy that meets both. Over-delivering on RTO is a design error.

> Meridian Freight's order system must recover within two hours of a Region failure, losing no more than five minutes of data. The business has been explicit that it will not fund a second full production environment.
>
> The system is Aurora PostgreSQL behind an Auto Scaling group and an Application Load Balancer.

What should the solutions architect design?

A Backup and restore, with hourly snapshots copied to the second Region

B Pilot light: an Aurora Global Database secondary, with launch templates and an ASG at zero desired capacity

C Warm standby with a scaled-down ASG running continuously

D Multi-site active-active across both Regions

◆ PAUSE AND DECIDE

What should the solutions architect design?

A Backup and restore

Hourly snapshots give an RPO of up to an hour against a five-minute requirement. It fails on data loss, not on time.

B Pilot light with Global Database

Sub-second replication meets the RPO, and provisioning compute fits comfortably inside two hours. Nothing idle is paid for.

C Warm standby

Meets both numbers, and costs more every hour to deliver an RTO of minutes when two hours was asked for.

D Multi-site active-active

The most expensive option available, and it is precisely the second full environment the business ruled out.

THE SIGNAL

Two numbers, and only one strategy meets both at the lowest cost. C and D fail on money rather than capability.

> A company must provide disaster recovery for 340 on-premises servers running a mix of Windows and Linux, including several vendor applications nobody may modify. The requirement is an RPO of seconds and an RTO under thirty minutes.
>
> There is no appetite to re-architect anything before the deadline.

What should the solutions architect recommend?

A. AWS Elastic Disaster Recovery replicating into a staging area in AWS

B. AWS Backup taking hourly backups of the on-premises servers

C. Rebuild the servers as AMIs, kept updated with EC2 Image Builder

D. AWS Application Migration Service to move them to AWS permanently

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Elastic Disaster Recovery

Continuous block replication gives an RPO of seconds, and recovery instances launch in five to twenty minutes. No application changes at all.

B Hourly AWS Backup

An RPO of up to an hour against a requirement of seconds. It is the wrong order of magnitude.

C AMIs plus Image Builder

An image captures a machine at build time, so the RPO is however old the image is. It also means rebuilding 340 servers.

D Application Migration Service

Its close cousin, and it migrates permanently. The requirement is disaster recovery, with production staying on-premises.

THE SIGNAL

MGN migrates; DRS protects. They share technology and answer different questions, which the exam tests directly.

A documented DR plan claims a four-hour RTO based on restoring snapshots into a second Region. It has never been executed. A tabletop review found that the CloudFormation templates reference a KMS key and an IAM role that exist only in the primary Region.

The board has asked whether the four-hour figure is credible.

What should the solutions architect recommend FIRST?

A Upgrade to warm standby to reduce the RTO

B Run a full recovery drill, measure the actual RTO, and fix what it exposes

C Create the missing KMS key and IAM role, then declare the plan complete

D Increase snapshot frequency to improve the RPO

◆ PAUSE AND DECIDE

What should the solutions architect recommend FIRST?

A **Upgrade to warm standby**

Spends money to improve a number nobody has measured. The current design might meet four hours once repaired.

B **Run a drill and measure**

The tabletop found two gaps without executing anything. A real drill finds the rest, and produces a measured figure rather than a claim.

C **Fix the two known gaps**

Fixes what one review happened to catch, and assumes it caught everything. A tabletop cannot find a slow restore.

D **More frequent snapshots**

Improves the RPO, which nobody questioned. The board asked about recovery time.

THE SIGNAL

The word FIRST matters. You cannot improve a number you have never measured, or trust one you have never tested.

> A company has implemented DRS and successfully tested launching recovery instances in the secondary Region. During a simulated Region failure the recovery instances came up in twelve minutes, but customers continued to receive errors for another forty minutes.
>
> Nothing was wrong with the recovery instances themselves.

What was missing?

A. DRS replication had fallen behind and needed to catch up

B. Automated traffic failover: DRS launches instances but does not redirect users

C. The recovery instances needed larger instance types

D. AWS Backup should have been used instead of DRS

◆ PAUSE AND DECIDE

What was missing?

A Replication lag

Would affect how current the recovered data is, not how long users see errors. The instances launched in twelve minutes.

B Traffic failover was never automated

DRS recovers servers and stops there. Something must move users to them, and that is Route 53 or Global Accelerator.

C Bigger instances

A performance answer to an availability symptom. The scenario says nothing was wrong with the instances.

D Use AWS Backup

Would make recovery slower rather than faster. DRS was the right service; the gap is elsewhere.

THE SIGNAL

Twelve minutes to recover and fifty-two before users are served is the signature of a plan that stops at the servers.

> After a ransomware incident at a competitor, a bank asks whether an attacker who compromised an administrator account could destroy both production data and its backups. Backups are currently written to an AWS Backup vault in the same account.
>
> The recovery team must be able to restore quickly even if that account is unusable.

What should the solutions architect implement?

A Enable versioning on the backup vault

B Copy backups to a logically air-gapped vault, shared with a recovery account through AWS RAM

C Apply Vault Lock in governance mode to the existing vault

D Increase backup frequency and retention

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Versioning

Not a backup vault feature, and an administrator who can delete the vault is not stopped by keeping more copies inside it.

B Air-gapped vault, shared via RAM

Locked in compliance mode so nobody can delete it, and shareable so a separate account restores directly with no copy step.

C Vault Lock, governance mode

The sharpest miss. Governance mode is designed to be overridden by a sufficiently privileged user, which is exactly the threat.

D More backups

More copies in the same account the attacker controls. Frequency was never the gap.

THE SIGNAL

Governance versus compliance mode appears again, exactly as with S3 Object Lock in module ten. The threat model decides it.

> A platform replicates Aurora with a Global Database, S3 with Cross-Region Replication and Replication Time Control, and DynamoDB with global tables. A configuration store on Amazon EFS is backed up nightly to the second Region.
>
> The team documents an RPO of one second and is confident it is achievable.

What should the solutions architect tell them?

A The documented RPO is correct; all three databases replicate in under a second

B The real RPO is up to 24 hours, set by the nightly EFS backup

C The real RPO is 15 minutes, set by S3 Replication Time Control

D RPO cannot be determined without measuring replication lag

◆ PAUSE AND DECIDE

What should the solutions architect tell them?

A One second is correct

True of the three databases and not of the system. The EFS configuration store is part of the workload.

B 24 hours, set by the EFS backup

A system's RPO is its slowest replicating component. A nightly copy means up to a day of configuration loss.

C 15 minutes, from RTC

Right reasoning, wrong component. RTC is the slowest of the three replicated stores; the nightly backup is far slower.

D Cannot be determined

It can. A nightly schedule bounds the worst case at 24 hours without measuring anything.

THE SIGNAL

RPO is a property of the whole system, not of its best component. The forgotten datastore usually sets it.

What to carry out of this module

- **Find the two numbers, then pick the cheapest strategy meeting both.** Over-delivering on RTO is a design error.
- **Pilot light versus warm standby is whether anything is serving.** Not whether resources exist.
- **DRS recovers servers, not traffic.** Route 53 or Global Accelerator moves the users.
- **Your RPO is your slowest replicating datastore.** One nightly backup sets it for everything.

> ON THE EXAM
>
> An untested plan is a hypothesis. When a claimed RTO has never been measured, running the drill is almost always step one.

MODULE 20 · DOMAIN 2 - DESIGN FOR NEW SOLUTIONS

Security Controls for New Solutions

Encryption is easy. Deciding who may decrypt is the architecture.

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Turning encryption on is a checkbox. The key policy is the architecture.

Three kinds of KMS key

- **AWS owned** keys are shared across customers, invisible to you, and free. The default where you configure nothing.
- **AWS managed** keys appear in your account with an aws slash prefix. You can see their use in CloudTrail and cannot edit their policy.
- **Customer managed** keys are yours: your key policy, your rotation setting, your grants. Charged monthly plus per request.
- Cross-account sharing, custom rotation and audit control all require a **customer managed** key.

ON THE EXAM

Any requirement to control, share or audit the key itself forces a customer managed key. That is the usual signal.

Key policy, IAM, and grants

- **The key policy is the root of authority.** If it does not delegate to IAM, no IAM policy in that account can grant access.
- The default key policy contains a statement enabling IAM. Remove it and only the key policy grants anything.
- **Grants** give temporary, narrow permissions, often created by AWS services on your behalf. They can be retired.
- Cross-account use needs the **key policy** to allow the other account, **and** IAM in that account to allow the caller.

> ON THE EXAM
>
> An administrator who cannot decrypt is nearly always a key policy that never delegated to IAM. Module two's chain, with an extra door.

Envelope encryption and multi-Region keys

- **Envelope encryption:** KMS encrypts a data key, the data key encrypts your data. KMS never sees the data.
- This is why a 5 GB object works. Only the small data key travels to KMS, not the payload.
- **Multi-Region keys** share key material across Regions, so ciphertext encrypted in one decrypts in another.
- Each replica needs **its own key policy**, and is charged separately in its Region. They are not one key.

> ON THE EXAM
>
> Ordinary KMS keys are Region-bound. Copying an encrypted snapshot cross-Region re-encrypts it with a key in the destination.

Rotation, and what changed

- **Automatic rotation is now configurable from 90 to 2,560 days.** It was a fixed 365 days before 2024.
- **On-demand rotation** exists, with a lifetime limit of ten per key.
- Rotation changes the key material, not the key ID. Old material is kept so existing ciphertext still decrypts.
- **Not supported** for asymmetric keys, HMAC keys, or keys in a CloudHSM custom key store.

ON THE EXAM

AWS managed keys rotate annually and you cannot change that. A specific rotation interval requires a customer managed key.

Secrets Manager or Parameter Store?

SECRETS MANAGER

Built for secrets that rotate

- Built-in rotation with Lambda
- Native RDS and Redshift integration
- Cross-Region replication of secrets
- Charged per secret per month

PARAMETER STORE

Configuration, and cheap secrets

- Standard tier is free
- SecureString encrypts with KMS
- No built-in rotation
- Advanced tier for larger values

THE TIEBREAKER

Automatic rotation is the separator. Without that requirement, Parameter Store SecureString does the job for nothing.

Certificates and hardware

- **ACM public certificates are free** and renew automatically, and can only be attached to AWS services such as ALB, CloudFront and API Gateway.
- **ACM cannot export a public certificate.** For a certificate on an EC2 instance you need Private CA or a third party.
- **AWS Private CA** issues internal certificates, at a monthly cost per CA plus per certificate.
- **CloudHSM** is single-tenant, FIPS 140-3 Level 3 hardware where you control the key material and AWS cannot see it.

> ON THE EXAM
>
> CloudHSM appears when a regulator demands sole control of key material, or a specific FIPS level KMS does not meet.

Shield Standard or Shield Advanced?

SHIELD STANDARD

On already, and free

- Automatic for every AWS customer
- Layer 3 and 4 volumetric attacks
- No console, no configuration
- Handles the large majority of attacks

SHIELD ADVANCED

$3,000 a month, per organisation

- DDoS Response Team, 24/7
- Cost protection for attack-driven scaling
- Standard WAF fees included
- Layer 7 protections and visibility

THE TIEBREAKER

Shield Advanced is one $3,000 charge for the whole organisation, and it absorbs standard WAF costs on protected resources.

AWS WAF

- **Managed rule groups** cover the OWASP top ten, known bad inputs and IP reputation, maintained by AWS.
- **Rate-based rules** block a source exceeding a request threshold. The usual answer for brute force and scraping.
- **Bot Control** and **Account Takeover Prevention** are paid additions, and are **not** covered by Shield Advanced.
- Attaches to CloudFront, ALB, API Gateway, AppSync and Cognito. **Not** to a Network Load Balancer.

> ON THE EXAM
>
> Use scope-down statements so expensive rule groups only inspect the pages that need them, such as login and checkout.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
An admin with full IAM cannot decrypt	The key policy never delegated to IAM
Encrypted snapshots usable in another Region	A multi-Region KMS key
Database credentials must rotate automatically	Secrets Manager
Config values, cost matters, no rotation needed	Parameter Store SecureString
Sole control of key material, FIPS 140-3 Level 3	AWS CloudHSM
Bill spiked during a DDoS attack	Shield Advanced cost protection

ON THE EXAM

Ask where the key lives and who the policy names. Most KMS questions are answered by those two facts alone.

A new engineer with the AdministratorAccess policy cannot read objects from an encrypted S3 bucket, receiving an access denied error on kms:Decrypt. Other engineers can. The bucket policy grants the whole account, and the engineer's IAM permissions are unrestricted.

The key is a customer managed key created two years ago by a security engineer who has since left.

What is the MOST likely cause?

A The key policy does not include the statement that delegates access control to IAM

B The engineer's IAM policy needs an explicit allow for kms:Decrypt

C The key has been scheduled for deletion

D The key needs rotating; the old material has expired

◆ PAUSE AND DECIDE

What is the MOST likely cause?

A Key policy does not delegate to IAM

Without that statement, only principals the key policy names can use the key. The other engineers are named; the new one is not.

B Add kms:Decrypt to IAM

AdministratorAccess already allows every KMS action. If IAM had any say, this would already work.

C Scheduled for deletion

A key pending deletion is unusable by everyone. Other engineers can decrypt, so the key is healthy.

D Old material expired

KMS retains old key material indefinitely so existing ciphertext still decrypts. Material does not expire.

THE SIGNAL

Some principals work and others do not, with identical IAM. That points at a policy naming principals individually.

> A company encrypts EBS snapshots with a customer managed key in eu-west-1 and copies them to eu-central-1 for disaster recovery. During a DR drill, the copied snapshots could not be used to launch instances.
>
> The team wants encrypted snapshots that are directly usable in either Region.

What should the solutions architect recommend?

A. Use a multi-Region KMS key, replicated to eu-central-1

B. Share the eu-west-1 key with eu-central-1 through its key policy

C. Disable encryption on the snapshots before copying them

D. Enable automatic key rotation on the existing key

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Multi-Region key

Replicas share key material, so ciphertext encrypted in one Region decrypts in another with no re-encryption step.

B Share the key by policy

A key policy grants access to principals, not to Regions. An ordinary KMS key cannot be used outside its own Region at all.

C Disable encryption

Solves a key problem by removing the protection. Unencrypted DR copies of production data are rarely acceptable.

D Enable rotation

Changes key material on a schedule. It has no bearing on which Region a key can be used in.

THE SIGNAL

KMS keys are Region-bound. That single fact is what makes cross-Region DR of encrypted data a design decision.

> An audit requires that database credentials for 40 RDS instances change every 30 days with no human involvement, and that the change is applied to both the database and the applications reading it.
>
> The team currently stores credentials in Parameter Store SecureString parameters and changes them by hand each quarter.

What should the solutions architect recommend?

A. Keep Parameter Store and write a Lambda function on a 30-day schedule

B. Move the credentials to Secrets Manager and enable managed rotation for RDS

C. Use Parameter Store Advanced tier, which adds rotation

D. Store the credentials in a KMS-encrypted S3 object and rotate that

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Custom Lambda rotation

Rebuilds Secrets Manager's rotation, including changing the password in RDS without breaking live connections. It is harder than it looks.

B Secrets Manager managed rotation

Rotation is built in, and the RDS integration changes the password in the database and the secret together.

C Parameter Store Advanced

Advanced tier adds larger values, more parameters and policies. It does not add credential rotation.

D Encrypted S3 object

Turns a solved problem into a bespoke one, with no rotation, no versioning semantics for secrets, and no database integration.

THE SIGNAL

Automatic rotation is the one thing Parameter Store does not do. That requirement alone decides between the two services.

> A regulated payments provider must generate and store the private keys used to sign transactions. The regulator requires that no third party, including the cloud provider, can access the key material, and mandates FIPS 140-3 Level 3 validated hardware.
>
> The team currently uses a KMS customer managed key.

What should the solutions architect recommend?

A. Keep the KMS customer managed key and enable automatic rotation

B. AWS CloudHSM, with the application using the HSM directly

C. A KMS key with imported key material

D. Store the private keys in Secrets Manager

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **KMS with rotation**

KMS is multi-tenant and AWS operates it. Rotation is irrelevant to a requirement about who can reach the key material.

B **CloudHSM**

Single-tenant, FIPS 140-3 Level 3 hardware where you hold the credentials and AWS has no access to the key material.

C **Imported key material**

The closest miss. You generate the material, and it still lives in and is operated by AWS KMS afterwards.

D **Secrets Manager**

A secret store, not a cryptographic module. It performs no signing and meets no FIPS hardware requirement.

THE SIGNAL

Two requirements: no provider access, and a specific FIPS level. Only CloudHSM satisfies both, which is why it exists.

> A retailer's login page is being attacked by credential stuffing from thousands of rotating IP addresses. Each individual address sends only a few requests per minute, so rate-based rules have not helped. Legitimate customers must not be blocked.
>
> The site already uses CloudFront with AWS WAF and managed rule groups.

What should the solutions architect recommend?

A. Lower the rate-based rule threshold until the attack stops

B. Add WAF Account Takeover Prevention, scoped down to the login endpoint

C. Subscribe to Shield Advanced, which covers this automatically

D. Block the attacking countries with a geo-match rule

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A. Lower the rate threshold

Each address is already below any sane threshold. Lowering it far enough to catch them starts blocking real customers.

B. Account Takeover Prevention, scoped down

Built for credential stuffing, inspecting login attempts rather than counting requests. Scoping it down keeps the cost to the login path.

C. Shield Advanced

Focused on DDoS, and Account Takeover Prevention is explicitly not covered by the subscription. It would not solve this.

D. Geo-blocking

Blunt and unreliable. Attackers move proxies, and legitimate customers travel.

THE SIGNAL

Rate-based rules count requests per source. An attack spread across thousands of sources is invisible to them by design.

> During a sustained DDoS attack last quarter, a media company's Auto Scaling groups and CloudFront usage scaled up sharply and the monthly bill rose by $180,000. The security team also spent eleven hours on the phone without expert assistance.
>
> The company runs 30 accounts under one organisation.

What should the solutions architect recommend?

A. Shield Advanced, for cost protection and DDoS Response Team access

B. Set AWS Budgets alerts so finance is warned during the next attack

C. Lower the Auto Scaling maximum capacity so costs cannot spike

D. Rely on Shield Standard, which already mitigates volumetric attacks

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Shield Advanced

Cost protection credits attack-driven scaling charges, and the response team addresses the eleven hours. $3,000/month is charged once for the whole organisation.

B Budgets alerts

Tells finance the bill is rising during an attack they already know about. It prevents no charge and mitigates nothing.

C Cap Auto Scaling

Converts a cost problem into an outage. The attack now succeeds in taking the site down, cheaply.

D Shield Standard only

Already in place, and it is what the company had during the attack described. Nothing would change.

THE SIGNAL

Two symptoms, one service. Cost protection and the response team are both Shield Advanced features and nothing else's.

What to carry out of this module

- **The key policy is the root of authority.** Without a delegation statement, no IAM policy in the account grants KMS access.
- **KMS keys are Region-bound.** Cross-Region use of ciphertext needs a multi-Region key.
- **Rotation is the Secrets Manager separator.** Without it, Parameter Store SecureString is free and sufficient.
- **Shield Advanced is $3,000 per organisation**, buying cost protection and the response team. Bot Control is extra.

> ON THE EXAM
>
> CloudHSM answers only two requirements: sole control of key material, and a FIPS level KMS does not meet. Otherwise it is over-engineering.

MODULE 21 · DOMAIN 2 - DESIGN FOR NEW SOLUTIONS

Designing for Performance

A hierarchy of caches. Each layer removes work from the one below it.

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The fastest request is the one that **never reaches your origin**. Work backwards from the user.

The hierarchy

01

Edge

CloudFront caches the response near the user.

02

Application

ElastiCache holds computed results and sessions.

03

Database

Read replicas absorb reads the cache missed.

04

Origin

Only genuinely new work arrives here.

ON THE EXAM

Each layer should absorb what the one below it cannot afford. Scaling the bottom layer to fix the top is the classic error.

CloudFront or Global Accelerator?

AMAZON CLOUDFRONT

A cache at the edge

- Caches content near the user
- HTTP and HTTPS, plus gRPC
- WAF attaches to it
- Cuts origin load, not just latency

GLOBAL ACCELERATOR

A shortcut to the backbone

- Caches nothing at all
- TCP and UDP, any protocol
- Two static anycast IPs
- Fast regional failover

THE TIEBREAKER

Cacheable HTTP means CloudFront. Non-HTTP, or a requirement for static IPs, means Global Accelerator.

CloudFront, and what changed

- **VPC origins** let CloudFront reach an ALB, NLB or instance in a **private subnet**. No public origin at all.
- That replaces the old pattern of an internet-facing ALB protected by a secret custom header.
- **Origin Access Control** is the current way to lock an S3 origin to CloudFront. It replaced Origin Access Identity.
- **Origin Shield** adds a central caching layer, so many edges do not all miss to your origin at once.

> ON THE EXAM
>
> Cache policies decide what forms the cache key. Forwarding every header and cookie destroys your hit rate silently.

CloudFront Functions or Lambda@Edge?

CLOUDFRONT FUNCTIONS

Sub-millisecond, at the edge

- Viewer request and response only
- No network access
- Restricted JavaScript, tiny runtime
- Roughly a sixth of the cost

LAMBDA@EDGE

A real runtime, further back

- Viewer and origin events
- Network calls and AWS SDK access
- Node.js or Python, more memory
- Higher latency and cost

THE TIEBREAKER

Needing to call another service forces Lambda@Edge. Header rewrites and redirects at scale belong in CloudFront Functions.

Caching patterns, and how they fail

- **Lazy loading** caches on a miss. Simple, and the first request after expiry is always slow.
- **Write-through** updates the cache on write. Data is always fresh, and you cache things nobody reads.
- **Cache stampede:** a popular key expires and thousands of requests hit the database at once.
- Fix it with **jittered TTLs**, so keys do not expire together, or by refreshing before expiry.

> ON THE EXAM
>
> A TTL is a correctness decision, not a performance one. It says how stale the answer may be, and everything else follows.

Relieving the database

- **Read replicas** take read traffic off the writer. The application must know to route reads to them.
- **RDS Proxy** pools connections, which is what Lambda at scale needs. Thousands of functions, a handful of connections.
- **DAX** is a DynamoDB-only cache giving microsecond reads. It cannot front anything else.
- **Buffering with a queue** converts a write spike into a backlog, as in module eighteen.

> ON THE EXAM
>
> Too many connections and too many queries are different problems. A proxy fixes the first; a cache fixes the second.

Quotas as a design input

- **Quotas are per account and per Region**, and default values are not sized for your workload.
- A new Region for disaster recovery starts at defaults, however large the primary has grown.
- **Service Quotas** shows current values, and CloudWatch can alarm as usage approaches a limit.
- Some quotas cannot be raised at all. Design around those rather than planning to request an increase.

> ON THE EXAM
>
> A quota reached under load looks exactly like a performance problem, and no amount of scaling fixes it.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Global users, cacheable web content	CloudFront
Global users, UDP or non-HTTP, static IPs	Global Accelerator
Simple rewrite or redirect, huge volume	CloudFront Functions
Edge logic that must call another service	Lambda@Edge
Lambda exhausting database connections	RDS Proxy
Origin overwhelmed when a cache entry expires	Jittered TTLs, or Origin Shield

ON THE EXAM

Ask which layer should have absorbed the request. The answer is usually one level above where the symptom appears.

> Meridian Freight runs a vehicle telemetry endpoint that receives UDP datagrams from 60,000 trucks worldwide. Latency varies badly for vehicles in Asia and South America, and the operations team needs two fixed IP addresses to configure in the vehicle firmware.
>
> The payloads are unique per vehicle and cannot be cached.

What should the solutions architect recommend?

A. CloudFront with a custom origin and caching disabled

B. AWS Global Accelerator in front of Network Load Balancers in several Regions

C. Route 53 latency-based routing to regional endpoints

D. An Application Load Balancer with cross-Region support

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A CloudFront, caching off

CloudFront handles HTTP and gRPC, not UDP. Disabling the cache also removes the only thing it was bringing.

B Global Accelerator with NLBs

Works with UDP, provides two static anycast IPs for the firmware, and moves traffic onto the AWS backbone early.

C Route 53 latency routing

Directs clients to a nearby Region and leaves the traffic on the public internet. It also gives DNS names, not fixed IPs.

D ALB cross-Region

An ALB is HTTP-only and Region-scoped. Neither the protocol nor the topology fits.

THE SIGNAL

UDP, uncacheable, and static IPs. Three separate signals, all pointing away from CloudFront.

A web application runs on an Application Load Balancer in private subnets, reached today through a second internet-facing ALB. Security dislikes the public ALB and protects it with a secret custom header that CloudFront adds.

The team asks whether there is now a cleaner option.

What should the solutions architect recommend?

A. Keep the secret header pattern; it is the standard approach

B. Use a CloudFront VPC origin pointing directly at the private ALB

C. Put the ALB behind a NAT gateway

D. Replace the ALB with an API Gateway private integration

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A. Keep the secret header

Was the standard answer until late 2024, and it depends on a shared secret and an origin that is still publicly reachable.

B. CloudFront VPC origin

CloudFront reaches the private ALB directly. No public origin, no shared secret, and nothing else can reach it.

C. ALB behind a NAT gateway

A NAT gateway is for outbound traffic from private subnets. It does not make an ALB reachable from outside.

D. API Gateway private integration

Replaces a working load balancer with a different service to solve a connectivity problem that has a direct answer.

THE SIGNAL

This answer changed in November 2024. Material written before then teaches option A, and taught it correctly at the time.

A media site must rewrite incoming URLs to a canonical form and redirect legacy paths, on every one of 8 billion monthly requests. The logic depends only on the request path and a small mapping table that changes weekly.

The team currently does this with Lambda@Edge and finds the cost significant.

What should the solutions architect recommend?

A Keep Lambda@Edge but increase its memory to reduce duration

B Move the logic to CloudFront Functions, with the mapping in CloudFront KeyValueStore

C Move the rewrites to the origin application

D Use an S3 redirect rules configuration

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A More memory

Trims duration on a per-invocation cost that is roughly six times the alternative. It optimises the wrong service.

B CloudFront Functions with KeyValueStore

Path-only logic is exactly the Functions use case, and KeyValueStore holds the weekly mapping with no network call and no redeploy.

C Rewrites at the origin

Every request now travels to the origin to be told it went to the wrong place. It removes the point of doing this at the edge.

D S3 redirect rules

Only applies to an S3 website endpoint, and cannot express a mapping table for a media site behind CloudFront.

THE SIGNAL

The logic needs no network call, which is what makes CloudFront Functions viable. KeyValueStore is what lets the table change weekly.

> During traffic peaks, an application backed by Aurora begins returning connection errors. Metrics show CPU on the writer at 35% and the connection count at its maximum. The application runs as several hundred concurrent Lambda functions.
>
> Query performance itself is fine.

What should the solutions architect recommend?

A Add Aurora read replicas and route reads to them

B Introduce RDS Proxy between Lambda and Aurora

C Scale the writer to a larger instance class

D Add an ElastiCache layer in front of Aurora

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Read replicas

Spreads queries, and each Lambda still opens its own connection. Replicas add capacity for a problem that is not query volume.

B RDS Proxy

Pools and reuses connections, so hundreds of concurrent functions share a small pool. Purpose-built for serverless against a relational database.

C Larger writer

Would raise the connection limit as a side effect, at considerable cost, while CPU sits at 35%.

D ElastiCache

Reduces query volume, and the scenario says query performance is fine. It is the wrong bottleneck.

THE SIGNAL

CPU at 35% with connections at maximum is the giveaway. The database is not busy; it has simply run out of sockets.

> A news site caches article pages in ElastiCache with a one-hour TTL, populated lazily. Every hour, on the hour, the database briefly saturates and response times spike for about forty seconds.
>
> The cache hit rate is otherwise above 98%.

What is happening, and what should change?

A. A cache stampede on synchronised expiry; add jitter to TTLs and refresh popular keys before expiry

B. The cache is undersized; move to a larger node type

C. Increase the TTL to 24 hours

D. Switch from lazy loading to Memcached

◆ PAUSE AND DECIDE

What is happening, and what should change?

A Stampede; jitter the TTLs

Keys written together expire together, so every miss hits the database at once. Jitter spreads expiry across the hour.

B Bigger cache node

A 98% hit rate says the cache is comfortably sized. Evictions are not the problem.

C 24-hour TTL

Moves the stampede to once a day and makes news articles up to a day stale. It trades correctness for a less frequent outage.

D Memcached

A different engine with the same expiry behaviour. The stampede is caused by TTL alignment, not by the engine.

THE SIGNAL

On the hour, every hour, for forty seconds is a schedule, not a load pattern. Something is expiring in unison.

> A company fails over to its disaster recovery Region during a drill. The CloudFormation deployment stops partway through, reporting that the maximum number of VPCs has been reached and that an EC2 instance type is not available at the requested count.
>
> The primary Region has grown steadily over four years.

What is the underlying issue?

A. Service quotas are per Region and the DR Region still sits at defaults

B. The CloudFormation templates are out of date

C. The DR Region lacks capacity and AWS must add hardware

D. Organizations policies are blocking resource creation

◆ PAUSE AND DECIDE

What is the underlying issue?

A Quotas are per Region

Four years of increases were requested in the primary Region only. The DR Region has never carried load and still holds default limits.

B Stale templates

The templates worked; the deployment was refused. A quota error is not a template error.

C Regional capacity

Possible in principle, and a quota message names a quota. AWS capacity shortfalls report differently.

D SCPs blocking creation

An SCP produces an access denied error naming the action, not a quota-exceeded message.

THE SIGNAL

Quotas grow invisibly in the Region you use. The DR Region only reveals its limits on the day you need it.

What to carry out of this module

- **Work backwards from the user.** Ask which layer should have absorbed the request before scaling the one that failed.
- **CloudFront caches; Global Accelerator does not.** Non-HTTP protocols or static IPs force Global Accelerator.
- **Network access is the edge compute separator.** CloudFront Functions cannot; Lambda@Edge can, at six times the cost.
- **Connections and queries are different problems.** RDS Proxy fixes one, a cache fixes the other.

> ON THE EXAM
>
> Quotas are per Region. A DR Region sits at defaults until the day you need it, which is the day you discover them.

MODULE 22 · DOMAIN 2 - DESIGN FOR NEW SOLUTIONS

Cost Optimisation by Design

Closing Domain 2. The charges that never appear on an architecture diagram.

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Every arrow on your diagram has a price.
Data transfer is the cost nobody draws.

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Who pays, and how much

PATH	ROUGHLY	NOTES
Into AWS	Free	Ingress is never charged
Same AZ, private IP	Free	Public IP in the same AZ is charged
Cross-AZ	$0.01/GB each way	So $0.02 round trip. The quiet killer
Inter-Region	$0.01 to $0.02/GB	Replication doubles a database bill
Out to the internet	~$0.09/GB	First 100 GB a month is free
Through a NAT gateway	+$0.045/GB	On top of egress, not instead of

ON THE EXAM

These are US list prices for orientation, not values to memorise. What matters is the shape: cross-AZ is charged both ways, and NAT stacks.

The NAT gateway trap

- **A NAT gateway charges per hour and per GB processed.** Three for redundancy costs roughly $100 a month before any traffic.
- **S3 traffic from a private subnet routed through NAT is billed per GB**, even though S3 in-Region transfer is free.
- **A gateway endpoint for S3 or DynamoDB is free** and removes that charge entirely. This is the single cheapest fix in AWS.
- Interface endpoints cost about a penny per GB plus an hourly fee, which still undercuts NAT for AWS service traffic.

> ON THE EXAM
>
> Backing up terabytes to S3 through a NAT gateway is the classic. The data transfer is free; the NAT processing is not.

Charges that arrive quietly

- **Every public IPv4 address costs about $0.005 an hour since 2024**, attached or idle. Roughly $3.60 a month each.
- **Cross-AZ chatter** between microservices is charged both ways, and does not appear on any diagram.
- **Multi-AZ RDS replication is free** between primary and standby. Cross-Region read replicas are not.
- **S3 or EC2 to CloudFront origin fetches are free**, which is why serving through CloudFront beats serving directly.

> ON THE EXAM
>
> A cluster spread across three AZs with fifty chatty services can generate several TB a month of cross-AZ traffic on its own.

Design levers that actually move the number

- **Serve through CloudFront.** The origin fetch is free and cached responses never touch your origin again.
- **Keep chatty components in one AZ**, and accept the availability trade deliberately rather than by accident.
- **Gateway endpoints for S3 and DynamoDB**, always. There is no reason not to.
- **Compress and batch.** Halving payload size halves every per-GB charge on its path at once.

> ON THE EXAM
>
> Compression is the lever people forget. It reduces egress, cross-AZ, NAT processing and inter-Region charges simultaneously.

Cost decisions made at design time

- **Managed versus self-managed** is a staff cost question. Compare the instance bill against the people running it.
- **Serverless is cheapest when idle and dearest when saturated.** At steady high volume, containers usually win.
- Right-size at design time using load tests, then let Compute Optimizer correct you once real traffic arrives.
- A commitment made before the architecture settles becomes a commitment to the wrong thing, as in module eleven.

> ON THE EXAM
>
> Serverless economics invert somewhere. Find the crossover for the workload rather than assuming it is always cheaper.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Large NAT charges, traffic goes to S3	A gateway endpoint
Network cost exceeds compute cost	Cross-AZ chatter, or NAT processing
Heavy egress serving files to users	CloudFront
Idle Elastic IPs across many accounts	The public IPv4 hourly charge
Cross-Region replication is expensive	Ask what RPO actually requires
Steady high-volume Lambda	Compare against Fargate or EC2

ON THE EXAM

Ask which boundary each byte crosses. Most surprise charges are a boundary somebody did not know was there.

> Meridian Freight's analytics platform writes 90 TB a month to Amazon S3 from EC2 instances in private subnets. The S3 bucket is in the same Region. Finance has flagged a NAT gateway line of roughly $4,000 a month, dominated by data processing rather than hours.
>
> The instances also need occasional internet access for OS patching.

What should the solutions architect recommend?

A. Add a gateway endpoint for S3 to the private subnets' route tables

B. Move the instances to public subnets to bypass the NAT gateway

C. Consolidate to a single NAT gateway to reduce hourly charges

D. Compress the data before writing it to S3

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A S3 gateway endpoint

Free, removes 90 TB from the NAT gateway entirely, and the gateway stays for patching traffic. Nothing else changes.

B Public subnets

Removes the NAT charge by putting analytics instances directly on the internet. A security regression to solve a billing problem.

C One NAT gateway

The bill is dominated by processing, not hours. Consolidating also creates a single point of failure for a saving of about $65.

D Compress the data

Genuinely helps and is worth doing anyway. It halves a charge that a gateway endpoint removes completely, and for free.

THE SIGNAL

Dominated by processing rather than hours is the sentence that decides it. Options C and D attack the wrong term.

> A company runs 60 microservices on EKS across three Availability Zones. Services call each other heavily. Cost analysis shows regional data transfer charges of about $9,000 a month, larger than the EC2 bill for the cluster itself.
>
> Availability requirements permit some services to run in fewer zones.

What should the solutions architect investigate FIRST?

A. Cross-AZ traffic between services, charged in both directions

B. Internet egress from the cluster to end users

C. The cost of the EKS control plane

D. EBS volume charges for the worker nodes

◆ PAUSE AND DECIDE

What should the solutions architect investigate FIRST?

A Cross-AZ service traffic

Sixty services scheduled randomly across three zones means most calls cross a zone boundary and are charged twice, once each way.

B Internet egress

Would appear as internet data transfer out, not as regional transfer. The scenario names the line item.

C EKS control plane

A fixed hourly charge per cluster. It is not data transfer and does not scale with chatter.

D EBS charges

Storage, not transfer. It would appear under EBS rather than under data transfer.

THE SIGNAL

Regional data transfer is the line item, and it names the boundary. Cross-AZ is the only one that grows with service chatter.

A training company serves 400 TB a month of video files directly from Amazon S3 to learners worldwide, using pre-signed URLs. The egress line is the single largest item on the bill. Files are large, popular, and rarely change.

The team believes CloudFront would only add cost.

What should the solutions architect advise?

A. They are correct; CloudFront adds a layer and a charge

B. Serve through CloudFront: the S3 origin fetch is free and cached objects are served once from origin

C. Move the files to S3 Glacier Instant Retrieval to reduce cost

D. Use S3 Transfer Acceleration for the downloads

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A They are correct

It feels right and is wrong. CloudFront replaces the S3 egress charge rather than adding to it, and its rate is lower at volume.

B Serve through CloudFront

S3 to CloudFront transfer is free, CloudFront egress undercuts S3 egress, and popular files are fetched from origin once.

C Glacier Instant Retrieval

Reduces storage cost for files that are popular, and adds retrieval fees. Storage was never the large line.

D Transfer Acceleration

Speeds long-distance uploads and downloads at an additional per-GB premium. It increases cost rather than reducing it.

THE SIGNAL

Large, popular, rarely changing files is the ideal CDN profile. The origin fetch being free is what makes it a saving rather than a layer.

> A cost review across 140 accounts finds 2,300 Elastic IP addresses. About 1,900 are attached to running instances and 400 are unattached. A junior engineer reports that only the 400 unattached addresses cost anything.
>
> Finance wants to know the real figure.

What should the solutions architect explain?

A. The engineer is correct; only unattached Elastic IPs are charged

B. All public IPv4 addresses have been charged hourly since 2024, so all 2,300 cost money

C. Elastic IPs are free; only data transfer through them is charged

D. The charge applies only to addresses in public subnets

◆ PAUSE AND DECIDE

What should the solutions architect explain?

A. Only unattached are charged

The rule until February 2024, and widely still believed. AWS extended the charge to every public IPv4 address.

B. All 2,300 are charged

About $0.005 an hour each, roughly $3.60 a month, so 2,300 addresses cost in the region of $8,000 a year.

C. Free, transfer only

Was never true even before 2024. Unattached addresses have carried an hourly charge for many years.

D. Public subnets only

Invented. The charge attaches to the address itself, not to where it is used.

THE SIGNAL

The engineer is quoting a rule that expired in 2024. Worth auditing whether every one of the 1,900 attached addresses is needed.

> A team replicates 10 TB of data daily from eu-west-1 to us-east-1 for disaster recovery, costing around $6,000 a month in inter-Region transfer. On review, the documented RPO for this dataset is 24 hours, and the data is derived and fully reproducible from a source system.
>
> The replication was configured three years ago and never revisited.

What should the solutions architect recommend?

A. Keep replicating; disaster recovery justifies the cost

B. Match the replication to the actual RPO, and consider regenerating derived data instead of replicating it

C. Switch to S3 Replication Time Control for a stronger SLA

D. Compress the data before replication

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Keep replicating

Treats DR as beyond questioning. The requirement is 24 hours and the design delivers far more, at $72,000 a year.

B Match the RPO, regenerate what is derived

Derived, reproducible data does not need replicating at all. Replicate the source and rebuild after failover.

C Replication Time Control

Adds a stronger SLA and additional per-GB cost, tightening an RPO that is already far ahead of the requirement.

D Compress it

Reduces the bill and keeps replicating data the business does not need replicated. It optimises the wrong decision.

THE SIGNAL

Two planted details: the RPO is 24 hours, and the data is reproducible. Either alone would make this over-engineered.

> An image-processing service runs on Lambda, handling a steady 900 million invocations a month at around 3 seconds and 3 GB each. Load is flat around the clock with no idle periods. The Lambda bill is roughly $95,000 a month.
>
> The team chose serverless because it is cheaper.

What should the solutions architect advise?

A. Keep Lambda; serverless is always cheaper than running servers

B. Model the same workload on Fargate or EC2, since serverless economics invert at sustained saturation

C. Reduce the memory allocation to cut cost

D. Buy a Compute Savings Plan to cover the Lambda spend

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A **Serverless is always cheaper**

True when load is spiky or idle. Flat, saturated load is the case where paying per millisecond stops being an advantage.

B **Model it on Fargate or EC2**

Continuous saturation is what always-on compute is priced for. Model it rather than assume, but the crossover is clearly worth checking.

C **Cut the memory**

Lambda CPU scales with memory, so less memory usually means longer duration. It often costs more, not less.

D **Compute Savings Plan**

Genuinely covers Lambda and is worth having. It discounts the wrong architecture rather than questioning it.

THE SIGNAL

Flat load, no idle, sustained saturation. Every property that makes serverless economical is absent from this scenario.

What to carry out of this module

- **Every arrow has a price.** Ask which boundary each byte crosses and who pays for it.
- **Cross-AZ is charged both ways**, and NAT processing stacks on top of egress rather than replacing it.
- **Gateway endpoints are free.** S3 traffic through a NAT gateway is the most common avoidable charge in AWS.
- **Serverless inverts at saturation.** Flat, always-busy load is where containers usually win.

> ON THE EXAM
>
> Every public IPv4 address has been charged hourly since 2024, attached or not. Most estates have never audited this.

MODULE 23 · DOMAIN 3 - CONTINUOUS IMPROVEMENT FOR EXISTING SOLUTIONS

Observability Foundations

Domain 3 begins. Improving what exists, starting with being able to see it.

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Monitoring tells you **that** something broke. Observability lets you ask **why** without shipping new code.

Metrics: resolution costs money

- **Default metrics are five-minute resolution and free.** EC2 detailed monitoring gives one minute, and is charged.
- **High-resolution custom metrics go to one second**, which is what makes sub-minute alarms possible.
- Publish custom metrics with PutMetricData, or with **embedded metric format** in a log line, which avoids a separate API call.
- **Memory and disk utilisation are not default EC2 metrics.** They need the CloudWatch agent installed.

> ON THE EXAM
>
> An alarm cannot be faster than the metric feeding it. A one-minute alarm on a five-minute metric is a five-minute alarm.

Logs, and where the money goes

- **Retention defaults to never expire.** This is the single most common CloudWatch cost leak in any estate.
- **The Infrequent Access log class** costs less for logs that are stored for compliance and rarely queried.
- **Metric filters** turn log text into a numeric metric, so you can alarm on something an application only prints.
- **Subscription filters** stream logs to Kinesis, Data Firehose or Lambda for real-time processing or archiving.

> ON THE EXAM
>
> Metric filters are how you alarm on log content. Without one, a log line saying payment failed cannot trigger anything.

Querying: three tools, one question

- **Logs Insights** for recent operational logs, queried occasionally. No setup, pay per query.
- **Athena over S3** for years of archived logs, queried rarely and cheaply.
- **OpenSearch** when people search constantly and need dashboards. You pay to index everything, always.
- The deciding question is **query frequency**, not data volume. Constant searching justifies an index; occasional searching does not.

> ON THE EXAM
>
> Same as module ten: occasional queries mean Athena or Logs Insights. Constant interactive search means OpenSearch.

The alarm types

TYPE	FIRES ON	USE IT WHEN
Metric alarm	A metric crossing a static threshold	You know what normal looks like
Anomaly detection	Deviation from a learned band	Normal varies by hour or day
Composite alarm	A boolean of other alarms	Reducing noise, or requiring agreement
Metric math alarm	An expression across metrics	Error rate, not error count

ON THE EXAM

Metric math is the underused one. Alarming on errors divided by requests survives traffic growth; alarming on error count does not.

Two settings that decide whether an alarm works

- **Missing data treatment.** The default is to treat missing data as missing, so an alarm may simply never evaluate.
- If a metric stops being published when the thing breaks, **treat missing data as breaching** or the alarm stays silent.
- **Datapoints to alarm** versus evaluation periods lets you require, say, three bad minutes out of five. That is how you stop flapping.
- **Composite alarms** also support suppression, so a known dependency failure does not page five teams.

> ON THE EXAM
>
> An alarm that has never fired is not evidence of health. A metric that disappears on failure is the classic silent alarm.

Seeing across accounts

- **Cross-account observability** nominates a monitoring account that sees metrics, logs and traces from source accounts.
- From there you can run **cross-account Logs Insights queries** and create **alarms on other accounts' metrics**.
- Anomaly detection also works on cross-account metrics, so one team can watch behaviour across a whole estate.
- Pair it with the delegated administrator pattern from module ten. The monitoring account is not the management account.

ON THE EXAM

Without this, an incident spanning four accounts means four consoles and no single timeline. That is the problem it removes.

From metrics to service level objectives

- **An SLI is a measurement.** An SLO is the target for it. An SLA is the contractual promise, usually looser than the SLO.
- **Application Signals** collects SLIs automatically from instrumented applications and tracks SLOs, with burn-rate alerts.
- It also builds a service map with drill-down to metrics, logs and traces, and works across accounts.
- Newer than most study material. Where a question is generic, **metric math on error rate and latency** is also a defensible answer.

> ON THE EXAM
>
> Set the internal SLO tighter than the external SLA. That gap is your warning time before a contractual breach.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Memory or disk metrics are missing	The CloudWatch agent
One failure triggers dozens of alarms	A composite alarm
The alarm never fired during the outage	Missing data treatment
Alarm on a message the app writes to a log	A metric filter
One view of an incident across many accounts	Cross-account observability
CloudWatch Logs cost is growing steadily	Retention, and the IA log class

ON THE EXAM

Ask what the team could not see, and why. Most Domain 3 questions start with a visibility gap rather than a broken component.

> Meridian Freight's fleet of EC2 instances runs out of memory roughly once a fortnight, causing a crash. The team wants to scale out before that happens, but CloudWatch shows no memory metric to alarm on. CPU stays below 30% throughout.
>
> The instances run a standard Amazon Linux AMI.

What should the solutions architect recommend?

A. Enable EC2 detailed monitoring to expose memory metrics at one-minute resolution

B. Install the CloudWatch agent to publish memory as a custom metric, and alarm on it

C. Alarm on CPU utilisation as a proxy for memory pressure

D. Enable Container Insights on the instances

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A. Detailed monitoring

The sharpest distractor. Detailed monitoring changes resolution from five minutes to one, and adds no new metrics at all.

B. Install the CloudWatch agent

Memory lives inside the guest operating system, which the hypervisor cannot see. The agent is the only way to publish it.

C. CPU as a proxy

The scenario says CPU stays below 30% while memory exhausts. It is explicitly not correlated here.

D. Container Insights

For ECS and EKS container metrics. Nothing suggests containers are involved.

THE SIGNAL

Detailed monitoring buys frequency, not coverage. Anything inside the guest OS needs an agent, and always has.

> When the primary database becomes slow, 47 alarms fire within two minutes: application latency, queue depth, error rates, load balancer 5xx counts and Lambda durations. The on-call engineer receives 47 pages and spends the first ten minutes working out which one matters.
>
> All 47 alarms are individually correct.

What should the solutions architect implement?

A Raise the thresholds on the 46 downstream alarms so they fire less often

B Composite alarms that group the downstream signals, with suppression when the database alarm is active

C Delete the downstream alarms and keep only the database alarm

D Route all alarms to a single SNS topic

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A **Raise the thresholds**

Makes 47 correct alarms less sensitive, so they also miss real problems. The alarms are not wrong; the paging is.

B **Composite alarms with suppression**

Groups the symptoms, and suppression means a known upstream cause pages once instead of 47 times.

C **Delete the downstream alarms**

Those alarms are the only signal when the database is fine and something downstream is not. It trades noise for blindness.

D **One SNS topic**

Delivers 47 messages to one place. The engineer still has 47 things to read and rank.

THE SIGNAL

The alarms are all correct, which is stated deliberately. That eliminates every option that makes them less sensitive.

> A critical batch job publishes a custom metric on each successful run. An alarm is configured to fire if that metric falls below one in an hour. Last week the job's host failed and no runs happened for eleven hours. The alarm never fired.
>
> The alarm configuration was reviewed and the threshold is correct.

What is the cause?

A The metric stopped being published, and missing data is treated as missing by default

B The alarm needed a longer evaluation period

C Custom metrics cannot be used in alarms

D The alarm needed detailed monitoring enabled

◆ PAUSE AND DECIDE

What is the cause?

A Missing data treated as missing

With no runs there is no data point, so the alarm has nothing to evaluate and stays in insufficient data. Set missing data to breaching.

B Longer evaluation period

Evaluating over longer still evaluates nothing. The absence of data points is the problem, not their number.

C Custom metrics cannot alarm

They can, and routinely do. The alarm existed and was correctly configured.

D Detailed monitoring

Applies to EC2 metric resolution. It has no bearing on a custom metric published by an application.

THE SIGNAL

The metric that proves a thing is working disappears when it stops working. That is the trap, and it needs missing-data set to breaching.

> An incident spans four accounts: a front-end account, an API account, a shared services account and a data account. Engineers currently switch roles between all four, comparing timestamps by hand, and typically take 40 minutes to build a coherent timeline.
>
> There are 200 accounts in the organisation in total.

What should the solutions architect implement?

A. CloudWatch cross-account observability, with a dedicated monitoring account

B. Copy all logs to a central S3 bucket and query them with Athena

C. Give every engineer a role in all 200 accounts

D. Run all workloads in a single account to simplify monitoring

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Cross-account observability

One monitoring account sees metrics, logs and traces from all sources, with cross-account Logs Insights queries and alarms.

B Central S3 plus Athena

Good for archived forensics and poor for a live incident. It also covers logs only, leaving metrics and traces behind.

C Roles everywhere

Faster switching between 200 consoles. The problem is the switching itself, not the permission to switch.

D One account

Discards the isolation, quota separation and blast-radius boundaries the organisation deliberately built.

THE SIGNAL

The complaint is a fragmented timeline, not missing data. Every account already has the data; nothing correlates it.

> CloudWatch Logs charges have grown to $38,000 a month. Investigation finds 4,200 log groups, of which 3,900 have no retention setting. Compliance requires seven years of application logs. Engineers query logs from the last fortnight almost daily, and older logs perhaps twice a year.
>
> No log group has ever been deleted.

What should the solutions architect recommend?

A. Set retention on all log groups to 14 days

B. Set short retention, and export or subscribe older logs to S3 with lifecycle policies for the seven-year requirement

C. Move every log group to the Infrequent Access class

D. Reduce application log verbosity

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A 14 days everywhere

Cuts the bill and destroys the seven-year compliance requirement. It solves cost by creating a much worse problem.

B Short retention plus S3 archive

Recent logs stay queryable in CloudWatch, older logs move to far cheaper S3 storage, and Athena serves the twice-a-year question.

C IA class everywhere

Helps, and IA is still CloudWatch Logs pricing. It is a discount, not the order-of-magnitude change S3 with lifecycle gives.

D Less verbose logging

Reduces future volume and does nothing about years of accumulated data with no retention set.

THE SIGNAL

Two requirements: cheap seven-year retention, and fast access to a fortnight. One store cannot serve both well.

> A company promises customers 99.9% availability contractually. Today it alarms when 5xx error count exceeds 100 in five minutes. As traffic has tripled, the alarm now fires during normal operation, and during a quiet-period outage it did not fire at all.
>
> The team needs alarming that reflects the promise as traffic changes.

What should the solutions architect recommend?

A. Raise the threshold from 100 to 300 to match the tripled traffic

B. Alarm on an error rate using metric math, and track it as an SLO against a tighter internal target

C. Use an anomaly detection alarm on 5xx count

D. Alarm on CPU utilisation of the web tier instead

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Raise to 300**

Correct today and wrong again next year. A count threshold has to be re-tuned every time traffic changes.

B **Error rate via metric math, tracked as an SLO**

A ratio is traffic-independent, so it works at any volume, and it maps directly onto the 99.9% promise.

C **Anomaly detection on count**

Learns normal and still measures the wrong thing. A small absolute error count during a quiet period can still breach 99.9%.

D **CPU utilisation**

A resource metric standing in for a customer-facing one. An outage returning errors quickly may use very little CPU.

THE SIGNAL

Both failures come from alarming on a count. The quiet-period miss and the busy-period noise are the same bug.

What to carry out of this module

- **Detailed monitoring buys frequency, not coverage.** Memory and disk need the CloudWatch agent.
- **Missing data defaults to missing**, so an alarm on a metric that disappears on failure stays silent.
- **Alarm on rates, not counts.** A ratio survives traffic growth; a count threshold needs re-tuning forever.
- Composite alarms with suppression turn many symptoms of one cause into one page.

> ON THE EXAM
>
> Query frequency, not data volume, decides between Logs Insights, Athena and OpenSearch. Constant searching justifies an index.

MODULE 24 · DOMAIN 3 - CONTINUOUS IMPROVEMENT FOR EXISTING SOLUTIONS

Advanced Monitoring

Tracing, canaries, and when the managed open-source stack is the answer.

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Metrics tell you **something** is slow. A trace tells you **which of your twelve services** is slow.

AWS X-Ray

- **A trace** is one request's journey. **Segments** are the work each service did, and **subsegments** break that down further.
- **The service map** is drawn from traces automatically, showing latency and error rates on every edge.
- **Sampling rules** control how much you record. The default records a first request each second, then a small percentage.
- **Annotations are indexed** and searchable. Metadata is stored and not searchable. That distinction gets tested.

> ON THE EXAM
>
> Sampling is why X-Ray stays affordable at scale. You trace enough requests to see the pattern, not every request.

The instrumentation shift

- **The X-Ray SDKs are in maintenance.** AWS now points new instrumentation at OpenTelemetry.
- **ADOT** is AWS's supported OpenTelemetry distribution. Instrument once, then choose the destination in collector config.
- One instrumentation can feed **CloudWatch, X-Ray, Managed Prometheus, OpenSearch** and third-party tools at the same time.
- Managed Lambda layers auto-instrument a function with **no code change**. EKS uses an add-on; ECS and EC2 run the collector.

> ON THE EXAM
>
> X-Ray the service is fine and fully supported. It is the SDKs that are in maintenance, and that distinction matters.

CloudWatch Synthetics

- **A canary is a scripted request run on a schedule**, from outside, whether or not any real user is trying.
- Blueprints cover heartbeat checks, API workflows, broken link checking, and visual comparison against a baseline.
- Canaries catch what internal health checks miss: expired certificates, DNS faults, a broken CDN, a bad third party.
- They also cover the quiet hours, when no real traffic would reveal a problem.

> ON THE EXAM
>
> A load balancer health check proves the target is up. A canary proves a user in another country can complete a purchase.

Synthetic or real user monitoring?

SYNTHETICS CANARIES

A robot, on a schedule

- Runs with no real traffic
- Covers nights and quiet periods
- Consistent, comparable results
- Only tests what you scripted

CLOUDWATCH RUM

Actual users, actual browsers

- Real devices, networks and geographies
- Page load and rendering performance
- Session replay for UX faults
- Nothing to see when nobody visits

THE TIEBREAKER

Detecting an outage before customers do means canaries. Explaining why some customers are slow means RUM.

Containers and the open-source stack

- **Container Insights** gives cluster, node, pod and container metrics for ECS and EKS, with a curated console view.
- **Amazon Managed Service for Prometheus** is a managed Prometheus back end. You keep PromQL, alert rules and recording rules.
- **Amazon Managed Grafana** visualises it, alongside CloudWatch, OpenSearch, Timestream and third-party sources.
- Neither is cheaper than CloudWatch. You choose them to keep existing dashboards, queries and skills.

> ON THE EXAM
>
> Existing PromQL alert rules or Grafana dashboards is the whole signal. Without those, CloudWatch is the simpler answer.

Where the money goes

- **Tracing everything is the expensive mistake.** Sampling rules exist so you can trace representatively.
- Canaries are charged per run, so a one-minute schedule on forty endpoints costs real money. Match frequency to the SLO.
- Container Insights ingests a great many metrics. Scope it to the clusters that matter rather than everything.
- Managed Prometheus charges per sample ingested and stored, so cardinality is the cost driver.

> ON THE EXAM
>
> High-cardinality labels are the Prometheus cost trap. A label containing a user ID multiplies series without limit.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Which of many services is causing the latency	X-Ray distributed tracing
Customers noticed the outage before we did	Synthetics canaries
Some users are slow, others are fine	CloudWatch RUM
Existing PromQL rules and Grafana dashboards	Managed Prometheus and Managed Grafana
Tracing cost has become significant	Sampling rules
Instrument once, send to several backends	ADOT

ON THE EXAM

Ask whether the gap is where, who, or how much. Tracing answers where, RUM answers who, sampling answers how much.

> Meridian Freight's booking API has a p99 latency of 4 seconds against a 500 millisecond target. The request passes through an API Gateway, six microservices on ECS, two databases and a third-party address validation service. Each team's own dashboards show acceptable latency.
>
> Incident calls have twice ended without identifying a cause.

What should the solutions architect implement?

A. Distributed tracing with X-Ray, instrumented through ADOT, and review the service map

B. Detailed monitoring on every ECS task

C. A CloudWatch dashboard combining all six services' latency metrics

D. Increase log verbosity across all six services

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A X-Ray tracing via ADOT

A trace follows one slow request through every hop and attributes the time, including the third party. The service map shows it directly.

B Detailed monitoring

Higher resolution on the same per-service averages. Every team already sees acceptable latency at five-minute granularity.

C A combined dashboard

The strong near miss. Six averages on one screen still cannot connect the slow requests to each other.

D More logging

Produces six streams with no shared request identifier, so correlating a single journey remains manual and slow.

THE SIGNAL

Every team's dashboard looking fine is the giveaway. Averages hide a tail; only a trace follows one actual request.

> A retailer's checkout was unavailable for 90 minutes overnight. Every internal health check passed throughout: instances healthy, target groups healthy, database responsive. The cause was an expired TLS certificate on a third-party payment widget the checkout page loads.
>
> Customers reported it in the morning.

What should the solutions architect implement?

A. A CloudWatch Synthetics canary that completes a checkout flow every five minutes

B. More aggressive ALB health check thresholds

C. CloudWatch RUM to capture real user experience

D. An X-Ray trace on the checkout service

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A A checkout-flow canary

Exercises the whole journey from outside, including the third-party widget, and runs overnight when no real traffic would reveal it.

B Tighter health checks

Health checks probe your targets, which were genuinely healthy. No threshold detects a broken third-party asset.

C CloudWatch RUM

The strong near miss. RUM would have shown it, and only once a real user visited. Overnight there were none.

D X-Ray tracing

Traces server-side request paths. A certificate failing in the browser on a third-party asset never reaches your services.

THE SIGNAL

Overnight is the planted word. It eliminates RUM, because real user monitoring needs real users.

> A company migrating 200 microservices from on-premises Kubernetes to EKS has four years of Prometheus recording rules, alerting rules and roughly 300 Grafana dashboards. The SRE team is fluent in PromQL. The migration must complete in six months.
>
> A consultant has proposed rewriting everything as CloudWatch dashboards and alarms.

What should the solutions architect recommend?

A. Rewrite as CloudWatch dashboards and alarms, as proposed

B. Amazon Managed Service for Prometheus and Amazon Managed Grafana, with ADOT collecting metrics

C. Self-manage Prometheus and Grafana on EKS

D. Container Insights only

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A. Rewrite for CloudWatch

Four years of rules and 300 dashboards rebuilt inside a six-month migration, discarding the team's PromQL fluency.

B. Managed Prometheus and Grafana

Rules and dashboards keep working, PromQL skills transfer, and AWS runs the back end instead of the team.

C. Self-manage both

Preserves compatibility and keeps the operational burden the migration was meant to reduce, including scaling Prometheus itself.

D. Container Insights only

Gives infrastructure metrics and none of the application recording rules or dashboards. It replaces a fraction of what exists.

THE SIGNAL

Pair this with module fifteen's ECS versus EKS pair. Existing artefacts and skills change the answer, and the reasoning is identical.

> X-Ray costs have reached $19,000 a month on a platform serving 40,000 requests per second. Tracing was enabled with a rule recording 100% of requests, because the team wanted complete visibility. They still mostly look at the service map and p99 latency.
>
> They do occasionally need to find a specific customer's failed request.

What should the solutions architect recommend?

A Disable X-Ray and rely on CloudWatch metrics

B Tune sampling rules: a low base rate, with a higher rate for errors and specific paths

C Reduce the X-Ray trace retention period

D Move tracing to a self-managed Jaeger deployment

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Disable X-Ray**

Removes the cost and the capability. Scenario one showed what happens to a distributed system with no tracing.

B **Tune sampling rules**

A service map and p99 need a representative sample, not every request. Higher sampling on errors preserves the failed-request use case.

C **Shorter retention**

X-Ray cost is dominated by traces recorded and scanned, not by how long they are kept.

D **Self-managed Jaeger**

Trades a service bill for cluster, storage and operations work, plus re-instrumentation. Rarely cheaper once staff time counts.

THE SIGNAL

The team's actual usage is stated: service map and p99. Both are statistical, and neither needs a complete record.

> Canaries pass consistently and server-side latency is well within target, yet support tickets report that the site is slow. The complaints cluster among users in Brazil and India, on mobile devices, and mention pages rendering incompletely.
>
> The team cannot reproduce it from the office.

What should the solutions architect implement?

A CloudWatch RUM, with session replay to see what those users experience

B More canaries, running from additional Regions

C X-Ray tracing on the backend services

D Increase the ALB idle timeout

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A RUM with session replay

Captures real devices, networks and geographies, and replay shows pages rendering incompletely. Exactly what nobody can reproduce.

B Canaries from more Regions

The strong near miss. It adds geography, and a canary still runs a scripted headless browser on a data-centre connection.

C X-Ray tracing

Server-side latency is already confirmed as within target. The problem lives in the browser.

D Longer idle timeout

A server-side connection setting for a client-side rendering symptom. Nothing suggests connections are being dropped.

THE SIGNAL

Pair this with scenario two. There, overnight ruled RUM out. Here, real devices on real networks rules canaries out.

> A company must send traces to X-Ray for AWS-native troubleshooting, metrics to Amazon Managed Service for Prometheus for the SRE team's alerting, and a copy of both to a third-party platform the security team has already licensed.
>
> The applications are being instrumented now, and the team does not want to instrument three times.

What should the solutions architect recommend?

A. Instrument with the X-Ray SDK and build exporters for the other two destinations

B. Instrument with ADOT, and configure the collector to export to all three destinations

C. Instrument three times, once per destination's native SDK

D. Send everything to CloudWatch and use subscription filters to forward it

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A. X-Ray SDK plus custom exporters

The X-Ray SDKs are in maintenance, and this means writing and maintaining exporters that ADOT already ships.

B. ADOT to all three

Instrument once with OpenTelemetry, then choose destinations in collector configuration. Adding a fourth is a config change.

C. Instrument three times

Triples the instrumentation work and the risk of the three views disagreeing about the same request.

D. CloudWatch plus subscriptions

Works for logs and does not carry traces or Prometheus metrics to those back ends in a usable form.

THE SIGNAL

Instrument once, choose destinations in configuration. That property is the whole argument for OpenTelemetry.

What to carry out of this module

- **Metrics say something is slow; a trace says which hop.** Healthy per-service dashboards hide a p99 problem.
- **Canaries run without users; RUM needs users.** Overnight detection means canaries. Explaining slow users means RUM.
- **Managed Prometheus and Grafana exist to preserve rules, dashboards and skills**, not to be cheaper.
- **Sampling is not a compromise.** Service maps and percentiles are statistical and need a representative sample.

> ON THE EXAM
>
> ADOT is now the instrumentation layer. The X-Ray SDKs are in maintenance, while X-Ray the service remains fully supported.

MODULE 25 · DOMAIN 3 - CONTINUOUS IMPROVEMENT FOR EXISTING SOLUTIONS

Automated Remediation

The goal is not fewer incidents. It is fewer incidents that need a human.

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Detection that ends in an email is not remediation. **Close the loop or accept the delay.**

The remediation loop

01

Detect

Config, GuardDuty, CloudWatch or AWS Health notices something.

›

02

Route

EventBridge matches the event and picks a target.

›

03

Act

An SSM Automation runbook or Lambda function does the work.

›

04

Verify

The result is itself an event you can alarm on.

ON THE EXAM

The fourth step is the one teams skip. A remediation that fails silently is worse than none, because everyone believes it worked.

AWS Config remediation

- **A Config rule detects; an SSM Automation document fixes.** The rule names the runbook and passes the resource ID.
- **Three modes: automatic, manual, or manual with approval.** Retry count and interval are configurable.
- **Organisation Config rules** deploy the whole arrangement to every account from a delegated administrator.
- It is still **detective**. The resource existed in a bad state for some period, however short.

> ON THE EXAM
>
> AWS ships many remediation runbooks already, such as disabling public access on a security group. You rarely write one from scratch.

EventBridge as the nervous system

- **Almost everything emits events.** Config compliance changes, GuardDuty findings, SSM state changes, EC2 state changes.
- **AWS Health events** are the underused source: scheduled EC2 retirement, degraded services, certificate expiry, quota warnings.
- **A rule can target an SSM Automation runbook directly**, with no Lambda in between.
- **EventBridge Scheduler** covers the time-based half: run this runbook every night at two.

> ON THE EXAM
>
> AWS Health telling you an instance is scheduled for retirement is a fix you can fully automate. Most organisations still do it by hand.

Systems Manager Automation

- **Runbooks are documents** with steps, branching and error handling. AWS publishes hundreds of them already.
- **The Approval action pauses execution** until named principals approve, which puts a human in the loop without losing the automation.
- Runbooks can run **across accounts and Regions** in one execution, driven from a central account.
- **OpsCenter** collects issues as OpsItems and suggests the runbooks that remediate them.

> ON THE EXAM
>
> The Approval action is the answer whenever automation is wanted but a change must not happen unattended. It is not all-or-nothing.

Guardrails on your automation

- **Run rules in detection mode first.** Enabling auto-remediation blind is how you discover the deliberate exceptions painfully.
- **Alarm on remediation failure.** Match SSM automation status changes of Failed or TimedOut and route them to a topic.
- **Exclude what is intentionally non-compliant**, rather than letting people disable the rule entirely.
- Consider blast radius. A remediation that runs everywhere at once can turn one bad rule into an estate-wide outage.

> ON THE EXAM
>
> Automation applies your judgement at machine speed, including when your judgement was wrong. Test in detection mode first.

Preventive still beats detective

- **An SCP stops the API call.** Nothing bad ever exists, so there is nothing to remediate.
- **Config plus remediation fixes it afterwards.** The window may be seconds, and it is not zero.
- Use both. Preventive controls for the absolute rules, detective plus remediation for everything else.
- **Honest note:** Systems Manager Change Manager is closed to new customers. Existing customers continue as normal.

ON THE EXAM

Never and prevent are still preventive words, from module two. Remediation is the answer when the requirement says detect and correct.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Detect and correct non-compliant resources	Config rule with SSM remediation
React to a scheduled instance retirement	AWS Health event to EventBridge
Automate, but a human must approve	SSM Automation Approval action
This must never happen at all	An SCP, not remediation
Remediation ran but nothing was fixed	Alarm on automation Failed or TimedOut
Apply the same rule in every account	Organisation Config rules

ON THE EXAM

Detect and correct means remediation. Prevent means a policy. The verb in the requirement chooses the mechanism.

> Meridian Freight finds that security groups allowing SSH from 0.0.0.0/0 keep appearing across 90 accounts, usually during troubleshooting and then forgotten. Security wants them closed within minutes, at any hour, with a record of what was changed.
>
> Some are created legitimately during incidents and closed afterwards.

What should the solutions architect implement?

A. An organisation Config rule for restricted-ssh with automatic SSM remediation, deployed from a delegated administrator

B. A weekly report of open security groups sent to account owners

C. An SCP denying all security group modifications

D. A Lambda function scanning all accounts nightly

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Org Config rule with remediation

Detects on change, remediates within minutes at any hour, records everything in Config and CloudTrail, and deploys to all 90 accounts at once.

B Weekly report

Within minutes was the requirement. A weekly report is roughly ten thousand times too slow, and ends with a human.

C SCP denying changes

The scenario says some are created legitimately during incidents. Denying all changes breaks incident response.

D Nightly Lambda scan

Builds what Config already does, on a schedule that misses the requirement by hours, across 90 accounts you must reach.

THE SIGNAL

Some are legitimate is the planted detail. It rules out prevention and makes detect-and-correct the right shape.

> AWS notifies the company that 40 EC2 instances are scheduled for retirement over the next fortnight because of degraded underlying hardware. An engineer currently reads each notification and schedules a stop and start manually, which takes about two days of effort each month.
>
> The instances are EBS-backed and can be stopped and started at any convenient time.

What should the solutions architect implement?

A. An EventBridge rule on AWS Health events, targeting an SSM Automation runbook that stops and starts the instance

B. A CloudWatch alarm on instance status checks

C. Enable EC2 Auto Scaling so instances are replaced automatically

D. Subscribe the engineer to AWS Health notifications by email

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Health event to SSM runbook

Health emits the retirement event, EventBridge matches it, and the runbook stops and starts the instance onto healthy hardware. No human.

B Status check alarm

Fires once the instance has already failed. The whole value of a retirement notice is that it arrives beforehand.

C Auto Scaling

Would help for stateless fleets, and it is a large architectural change proposed to handle a maintenance notification.

D Email the engineer

Exactly the current process, which costs two days a month. It automates nothing.

THE SIGNAL

AWS Health is the underused event source. A retirement notice arrives in advance, which is what makes it fully automatable.

> A team enabled automatic remediation on a Config rule requiring EBS volume encryption. The runbook detached and replaced volumes. It ran against a legacy database server whose unencrypted volume was a documented exception, causing a four-hour outage.
>
> Security still wants the rule enforced across the estate.

What should the solutions architect recommend?

A. Abandon automatic remediation and rely on reports

B. Run new rules in detection mode first, add explicit exclusions for documented exceptions, and use manual remediation with approval for destructive actions

C. Reduce the remediation retry count to one

D. Move the legacy server to a separate account with no Config rules

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Abandon automation**

Overcorrects. One badly scoped rule does not mean remediation is wrong; it means this one was enabled without testing.

B **Detection mode, exclusions, approval**

Keeps enforcement while fixing the three causes: no dry run, no exception handling, and a destructive action running unattended.

C **Fewer retries**

The remediation succeeded. It did exactly what it was told, once. Retries were never involved.

D **Exempt account**

Hides the exception by removing visibility of it, and creates an account where compliance is simply not checked.

THE SIGNAL

Read what actually failed. The automation worked perfectly; the scoping and the review process did not.

> An audit finds 340 resources that a Config remediation rule should have fixed months ago. Investigation shows the remediation runbook has been failing since a permissions change, because its IAM role lost a required action. No alert was ever raised.
>
> The compliance dashboard showed the rule as configured with automatic remediation throughout.

What should the solutions architect implement?

A. An EventBridge rule matching SSM Automation executions with status Failed or TimedOut, routed to SNS

B. Increase the remediation retry count

C. Review the Config dashboard weekly

D. Grant the remediation role AdministratorAccess to avoid permission gaps

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Alarm on failed automations

Closes the fourth step of the loop. A failing remediation becomes visible immediately rather than at the next audit.

B More retries

Retrying a call the role has no permission to make fails the same way, more often. The failure is deterministic.

C Weekly dashboard review

The dashboard showed remediation as configured throughout, which is precisely why nobody noticed. Reading it more often changes nothing.

D AdministratorAccess

Fixes this permission gap by removing least privilege from something that deletes and replaces production resources.

THE SIGNAL

This is the fourth step of the loop, skipped. Nobody verified that the automation was still working, and it failed for months.

> A regulator requires that customer data buckets can never be made public, and asks the company to demonstrate that the control cannot be bypassed. The team proposes a Config rule with automatic remediation that removes public access within 30 seconds of detection.
>
> The company runs Control Tower across all accounts.

What should the solutions architect advise?

A The proposal is sufficient; 30 seconds is fast enough

B Use a preventive control so the change is refused, and keep the Config rule as detection and evidence

C Reduce the remediation window to 10 seconds

D Enable S3 Block Public Access and remove the Config rule

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A 30 seconds is enough

Never and cannot be bypassed are absolute. A 30-second window is a 30-second window, however short it sounds.

B Preventive control, plus Config for evidence

The API call is refused so no public bucket ever exists, and the Config rule supplies the continuous evidence the regulator asked for.

C 10 seconds

Shrinks a window the requirement says must not exist. It is the same wrong answer with a smaller number.

D Block Public Access alone

The right preventive instinct, and removing the Config rule discards the evidence half of what the regulator asked for.

THE SIGNAL

Two requirements again: prevent it, and demonstrate it. Modules two and ten both said the same thing about that pair.

A platform team must apply a remediation that restarts a stuck agent on instances across 120 accounts and four Regions. Change policy requires that a named on-call engineer approves each execution before it runs, but the team does not want to abandon automation.

Executions must be recorded for audit.

What should the solutions architect recommend?

A. An SSM Automation runbook with an Approval action, run as a multi-account and multi-Region execution

B. A Lambda function that emails the on-call engineer with instructions

C. Fully automatic remediation, with the approval requirement waived

D. Run the fix manually in each of the 120 accounts

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Runbook with Approval action

The approval pauses the execution until the named principal approves, then it runs across all accounts and Regions, fully recorded.

B Email with instructions

Turns automation back into a runbook a human types. Nothing is recorded and nothing is consistent.

C Waive the approval

Solves the problem by ignoring the constraint. Change policy is an input, not an obstacle to route around.

D Manual in 120 accounts

Satisfies approval by making the work enormous and inconsistent, with no reliable audit trail.

THE SIGNAL

Automation and human approval are not opposites. The Approval action is the specific feature that makes both true at once.

What to carry out of this module

- **Detect, route, act, verify.** The fourth step is the one teams skip, and it fails silently for months.
- **AWS Health to EventBridge to a runbook** is the underused pattern. Retirement notices arrive before the failure.
- **The Approval action** keeps automation while requiring a human. Automation is not all-or-nothing.
- **Prevent means a policy; detect and correct means remediation.** A short window is still a window.

> ON THE EXAM
>
> Run new rules in detection mode first. That is how you find the deliberate exceptions before they find you.

MODULE 26 · DOMAIN 3 - CONTINUOUS IMPROVEMENT FOR EXISTING SOLUTIONS

Systems Manager Operations

Turning "log into the server" into an API call that leaves a record.

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If administration is an API call, it inherits **IAM, CloudTrail and automation**. If it is an SSH session, it inherits none of them.

Session Manager

- **Shell access with no inbound ports, no bastion host and no SSH keys.** The agent connects outbound to Systems Manager.
- **Every session is logged** to S3 or CloudWatch Logs, and can be recorded keystroke by keystroke.
- **Port forwarding** tunnels to a database or a web console without opening anything to the internet.
- Access is governed by IAM, so permissions can be scoped by instance tag rather than by network position.

> ON THE EXAM
>
> This is why bastion hosts are an anti-pattern now. A bastion is a server to patch, protect and audit that provides no capability Session Manager lacks.

Just-in-time node access

- **Zero standing privileges.** Operators request access to a node, rather than holding it permanently.
- **Approval policies decide: deny, auto-approve, or require a human.** Written in Cedar, with timeouts and session limits.
- Covers EC2, on-premises and multi-cloud nodes, and adds **RDP session recording** to S3.
- Enabling it changes nothing until you **remove ssm:StartSession** from your IAM policies. Both paths coexist until then.

> ON THE EXAM
>
> Session Manager removes the bastion. Just-in-time node access removes the standing permission that replaced it.

What a managed node needs

- **The SSM Agent installed and running**, plus outbound connectivity to the Systems Manager endpoints.
- **An instance profile** granting the managed instance permissions, or Default Host Management Configuration doing it for you.
- In a private subnet with no NAT, you need **interface endpoints** for ssm, ssmmessages and ec2messages.
- An instance missing from Fleet Manager is almost always one of those three. Agent, role, or network path.

> ON THE EXAM
>
> Three endpoints, not one. Missing ssmmessages is the classic cause of an instance that registers but cannot start a session.

Patch Manager

- **A patch baseline** defines which patches are approved, with auto-approval rules and an approval delay for testing.
- **Patch groups** are just a tag, letting you patch development a week before production with the same baseline.
- **Maintenance windows** control when. **Patch policies** configure scanning and patching across an organisation from one place.
- Scan-only mode reports compliance without installing anything, which is how you start.

> ON THE EXAM
>
> Patch compliance is reported into Systems Manager and Config, so it becomes evidence rather than an assertion.

Which Systems Manager tool?

YOU WANT TO	USE	BECAUSE
Run a command now, on many nodes	Run Command	One-off, targeted by tag or resource group
Keep configuration as you defined it	State Manager	Associations reapply on a schedule
Fix something with steps and branching	Automation	Runbooks handle multi-step workflows
Know what software is installed	Inventory	Collects applications, patches, network config
Patch a fleet on a schedule	Patch Manager	Baselines, groups and maintenance windows

ON THE EXAM

Run Command is now, once. State Manager is continuously, forever. That is the distinction the exam tests most.

Beyond AWS

- **Hybrid activations** register on-premises servers, virtual machines and other clouds as managed nodes.
- They then use the same Session Manager, Run Command, Patch Manager and Inventory as EC2 instances.
- This is often the strongest argument for Systems Manager: one operational model across a hybrid estate.
- Hybrid nodes beyond the free allowance are billed on the advanced tier, so it is not free at scale.

ON THE EXAM

A company halfway through a migration runs two estates. Systems Manager is one of the few things that treats them identically.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Remove bastion hosts and SSH keys	Session Manager
Nobody should hold standing access to nodes	Just-in-time node access
Patch thousands of instances on a schedule	Patch Manager with patch policies
Configuration keeps drifting back	State Manager associations
Run one command across a tagged fleet now	Run Command
Manage on-premises servers the same way	Hybrid activations

ON THE EXAM

Ask whether the need is once or continuous, and whether it is AWS-only or hybrid. Those two answers pick the tool.

> Meridian Freight runs four bastion hosts, one per environment, with SSH keys distributed to 60 engineers. Auditors have asked who connected to which server and what they typed, and nobody can answer. Key rotation has not happened in two years.
>
> The bastions themselves must be patched and monitored.

What should the solutions architect recommend?

A. Session Manager, with session logging to CloudWatch Logs, and decommission the bastions

B. Rotate the SSH keys quarterly and enable verbose sshd logging on the bastions

C. Move the bastions into a dedicated account with tighter security groups

D. Replace SSH with RDP for easier auditing

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Session Manager, drop the bastions

Removes the keys, the inbound ports and four servers to maintain, while giving IAM-controlled access and a full session record.

B Rotate keys, more logging

Improves a design that should not exist. Keys still need distributing, and bastion logs are on a host users can reach.

C Bastions in their own account

Relocates the problem. There are still four servers, sixty keys, and no record of what anyone typed.

D RDP instead of SSH

Changes the protocol, not the model. Standing access, distributed credentials and no central audit trail remain.

THE SIGNAL

Three requirements: who connected, what they typed, and no unrotated keys. One service satisfies all three and removes four servers.

> A bank has adopted Session Manager successfully. A new security policy now requires zero standing privileges: no engineer should be able to connect to a production node without an explicit, time-bound approval recorded at the moment of access.
>
> Operational efficiency must not suffer for non-production environments.

What should the solutions architect implement?

A. Just-in-time node access, with approval policies requiring human approval for production and auto-approval elsewhere

B. Remove ssm:StartSession from all IAM policies and grant it manually on request

C. A permissions boundary limiting engineers to non-production instances

D. Shorten the Session Manager idle timeout

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A JITNA with per-environment policies

Approval policies can deny, auto-approve or require a human, so production gets approval and non-production stays frictionless.

B Manual IAM grants on request

Achieves zero standing privileges by making a human edit IAM policies at 3am. It is JITNA, built badly, with no audit structure.

C Permissions boundary

Removes production access entirely rather than gating it. Engineers legitimately need production access sometimes.

D Shorter idle timeout

Ends idle sessions sooner. The permission to start a new one at any moment is untouched.

THE SIGNAL

Zero standing privileges plus different rules per environment is exactly what approval policies express. One mechanism, two behaviours.

> A company must patch 3,000 instances across 40 accounts monthly. Development should receive patches a week before production. Compliance requires evidence of patch status per instance, and a security team wants to confirm coverage without visiting each account.
>
> Patching must occur only within approved maintenance windows.

What should the solutions architect implement?

A. Patch Manager with a baseline, patch groups by tag, maintenance windows, and patch policies deployed organisation-wide

B. A Run Command execution issued monthly against all instances

C. A State Manager association running the patch document hourly

D. Rebuild instances monthly from a freshly patched AMI

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Patch Manager, full arrangement

The baseline's approval delay gives development its week ahead, groups are tags, windows control timing, and compliance data is reported centrally.

B Monthly Run Command

A human must remember to run it in 40 accounts, with no maintenance window control and no compliance reporting.

C Hourly State Manager

Continuous patching ignores the maintenance window requirement entirely, which is the one constraint stated most firmly.

D Rebuild from AMIs

Defensible for immutable stateless fleets, and a very large change proposed to solve a patching problem, with no per-instance evidence.

THE SIGNAL

Four requirements, four Patch Manager concepts. Baseline delay, tag-based groups, maintenance windows, and compliance reporting.

> The CloudWatch agent configuration on a fleet keeps reverting. Engineers troubleshooting incidents edit the config locally, then forget to revert. Weeks later, instances are found publishing the wrong metrics, and nobody knows which ones are correct.
>
> The desired configuration is well documented.

What should the solutions architect implement?

A. A State Manager association applying the desired configuration on a schedule

B. A Run Command execution applying the configuration once to all instances

C. Deny engineers shell access to production instances

D. Rebuild the instances from a corrected AMI

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A State Manager association

Reapplies the documented configuration on a schedule, so a local edit survives only until the next run. Drift corrects itself.

B Run Command once

Fixes every instance today. The next incident produces the same drift, because nothing reapplies the configuration.

C Deny shell access

Prevents the edits and the troubleshooting. Engineers change the config during incidents for a reason.

D Rebuild from an AMI

A heavy correction that also only holds until the next incident. It has the same weakness as B, more expensively.

THE SIGNAL

The word is keeps. A recurring drift needs something continuous, which is exactly what separates State Manager from Run Command.

> Newly launched instances in a private subnet do not appear in Fleet Manager. The AMI includes the SSM Agent and the agent process is running. The instances have an instance profile with the AmazonSSMManagedInstanceCore policy. There is no NAT gateway in the VPC.
>
> Instances in a public subnet with identical configuration register normally.

What is the MOST likely cause?

A The private subnet lacks VPC interface endpoints for ssm, ssmmessages and ec2messages

B The instance profile needs AdministratorAccess

C The SSM Agent version is out of date

D Default Host Management Configuration is not enabled

◆ PAUSE AND DECIDE

What is the MOST likely cause?

A Missing interface endpoints

The agent connects outbound to Systems Manager. With no NAT and no endpoints, a private subnet has no path at all.

B AdministratorAccess

The managed instance core policy is exactly what AWS provides for this. More permission does not create a network path.

C Old agent version

Would likely still register, and the identical AMI works in a public subnet, so the agent is not the variable.

D Default Host Management

An alternative way to grant permissions. The instances already have an instance profile, so permissions are covered.

THE SIGNAL

Same AMI, same role, different subnet. When one variable differs, that variable is the answer.

> A company is two years into a migration. It runs 1,400 EC2 instances and 900 on-premises servers, and expects to run both for another three years. Patch compliance, software inventory and remote access are handled by different tools in each estate, and auditors receive two incompatible reports.
>
> The security team wants one operational model.

What should the solutions architect recommend?

A. Register the on-premises servers as managed nodes using hybrid activations

B. Accelerate the migration so everything runs on EC2

C. Deploy the third-party on-premises tool into AWS as well

D. Keep both tools and write a script that merges the two reports

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Hybrid activations

On-premises servers become managed nodes, using the same patch baselines, inventory and Session Manager as EC2. One model, one report.

B Accelerate the migration

The scenario says both estates run for three more years. Changing the business plan is not an architectural answer.

C Third-party tool into AWS

Defensible, and it means running and licensing that tool across a much larger estate, plus losing the AWS-native integrations.

D Merge the reports

Produces one report from two operational models. The auditors are satisfied and the operations teams are not.

THE SIGNAL

Three more years of both estates is the planted constraint. It rules out any answer that depends on the hybrid period ending.

What to carry out of this module

- **Session Manager removes the bastion.** Just-in-time node access removes the standing permission that replaced it.
- **Run Command is now, once. State Manager is continuously, forever.** The word keeps in a scenario chooses State Manager.
- **Three endpoints**, not one: ssm, ssmmessages and ec2messages. Missing the second is the classic silent failure.
- **Hybrid activations** give one operational model across AWS and on-premises, which matters most mid-migration.

> ON THE EXAM
>
> Administration through an API inherits IAM and CloudTrail. Administration over SSH inherits neither, which is the whole argument.

MODULE 27 · DOMAIN 3 - CONTINUOUS IMPROVEMENT FOR EXISTING SOLUTIONS

Improving Deployment Processes

Auditing what exists, and retrofitting safety onto it.

DOMAIN 3 of 4 EXAM WEIGHT 25% MODULE 27 of 44

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Start by naming what the current process **cannot do**. Adding tools before that is how you end up with two problems.

Five questions to audit a pipeline

- **Can it roll back?** If rollback means running the pipeline again, the recovery time is the deployment time.
- **What stops a bad change?** Tests, approvals, or nothing but the deployer's judgement.
- **What was deployed, and when?** If nobody can answer, incident timelines are guesswork.
- **Who can deploy to production?** And is that a permission or a convention?
- **Does anything watch after the deployment finishes?** Success at deploy time is not success.

> ON THE EXAM
>
> The last question catches the most designs. A pipeline that reports green and then stops watching has done half the job.

Retrofitting safety

- **CodeDeploy can be added to an existing Auto Scaling group** without rewriting the application, giving you rollback and hooks.
- **Rollback triggers on CloudWatch alarms** turn a monitored metric into an automatic reversal, as in module thirteen.
- **Manual approval actions** add a gate without slowing anything that does not need one.
- **CodeArtifact** pins dependencies, so a build is reproducible and a compromised upstream package cannot walk in.

> ON THE EXAM
>
> Retrofit in the order the gaps hurt. Rollback first, then a gate, then provenance. Each step is independently useful.

EC2 Image Builder

- **A recipe** is a source image plus components. **A pipeline** builds, tests and distributes on a schedule or a trigger.
- **Distribution is gated on tests.** An image that fails its tests is never distributed to anyone.
- **Amazon-managed STIG hardening components** exist, so a compliance baseline is a component rather than a script.
- It builds **container images to ECR** too, and copies AMIs across Regions and accounts.

> ON THE EXAM
>
> The test gate is what makes this a pipeline rather than a build script. Nobody can launch an image that failed its checks.

Making the golden image actually used

- **Building a golden AMI does not mean anyone launches from it.** Enforcement is a separate problem.
- **Config rules** approved-amis-by-tag and approved-amis-by-id detect instances running unapproved images.
- **An SCP denying ec2:RunInstances** unless the AMI carries the approved tag is the preventive version.
- **Image lifecycle policies** deprecate and delete old images automatically, which is the answer to AMI sprawl.

> ON THE EXAM
>
> Detect with Config, prevent with an SCP. The same pairing as modules three and twenty five, applied to images.

Closing the loop on images

- **An Inspector finding of High or Critical can trigger the image pipeline** through EventBridge, so a CVE rebuilds the image.
- The rebuilt image is tested, distributed, and then Auto Scaling instance refresh rolls it out.
- Recipes and components are **shareable through RAM** and versioned, so one platform team serves the organisation.
- Patching running instances and rebuilding images are different answers. Long-lived fleets need both.

> ON THE EXAM
>
> A vulnerability found by Inspector, rebuilt by Image Builder, rolled out by instance refresh, is the fully closed loop.

Separating deployment from release

- **Deploying code and enabling a feature can be different events.** Feature flags make the second one a configuration change.
- **AWS AppConfig**, part of Systems Manager, delivers configuration and flags with validation and gradual rollout.
- It can roll back automatically on a CloudWatch alarm, exactly like a deployment.
- The trade is that flags accumulate. Untidied flags become branches nobody dares remove.

> ON THE EXAM
>
> A feature that can be switched off without a deployment has a rollback measured in seconds rather than in pipeline runs.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Rollback means running the pipeline again	CodeDeploy with alarm rollback
AMIs are built by hand and drift	An Image Builder pipeline
Teams launch unapproved images	Config rule to detect, SCP to prevent
A new CVE must rebuild every image	Inspector finding to EventBridge
Turn a feature off without deploying	Feature flags with AppConfig
Hundreds of old AMIs and snapshots	Image lifecycle policies

ON THE EXAM

Ask what the process cannot do, then fit a mechanism to that gap. Do not propose a rewrite unless the scenario asks for one.

> Meridian Freight deploys a fleet application by running a shell script from an engineer's laptop, which copies artefacts to instances and restarts services. Deployments take 25 minutes. Rolling back means checking out the previous commit and running the script again, which takes another 25.
>
> Nobody can say which version is running where.

What should the solutions architect recommend FIRST?

A Rewrite the application as containers on ECS with blue/green deployments

B Move the deployment into CodePipeline with CodeDeploy against the existing Auto Scaling group, with alarm-based rollback

C Document the shell script and add it to version control

D Schedule deployments during a nightly maintenance window

◆ PAUSE AND DECIDE

What should the solutions architect recommend FIRST?

A **Rewrite as containers**

Might be right eventually, and it is a rewrite proposed as a first step. It fixes deployment by rebuilding the application.

B **CodePipeline and CodeDeploy retrofit**

Attaches to the Auto Scaling group that already exists. Rollback, deployment history and alarm-based reversal, with no application change.

C **Version control the script**

A real improvement and it addresses none of the three complaints. Rollback is still 25 minutes.

D **Nightly window**

Makes a slow, unreversible deployment happen when nobody is watching. It hides the problem rather than fixing it.

THE SIGNAL

The word FIRST matters again. A retrofit delivers all three fixes this month; a rewrite delivers them next year.

> A security team maintains a hardened base AMI by launching an instance from last quarter's image, applying updates and a hardening script by hand, and creating a new AMI. It takes two days, happens quarterly, and the resulting images differ in ways nobody can explain.
>
> Compliance requires evidence that each image meets a STIG baseline.

What should the solutions architect recommend?

A. An EC2 Image Builder pipeline with managed STIG hardening components and tests gating distribution

B. Document the manual steps as a runbook so they are followed consistently

C. Use AWS Marketplace images and skip the hardening step

D. Patch running instances with Patch Manager instead of rebuilding images

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Image Builder with STIG components

Repeatable recipe, managed hardening components, and distribution gated on tests, which is the evidence compliance asked for.

B Document the manual steps

Makes the ritual more consistent while keeping two days of human work and no test gate. Humans still deviate.

C Marketplace, no hardening

Abandons the compliance requirement. A pre-hardened Marketplace image can be the source, and not a replacement for the baseline.

D Patch running instances

Answers a different question. New instances would still launch from an unhardened, out-of-date image.

THE SIGNAL

Images that differ in unexplainable ways is the signature of a manual process. A recipe produces the same image every time.

> The platform team now produces a golden AMI monthly. An audit finds that only 40% of running instances were launched from it. Teams are using whichever AMI they had last time, including images two years old.
>
> The company runs AWS Organizations with SCPs already in place.

What should the solutions architect recommend?

A Email the AMI ID to all teams monthly

B A Config rule detecting unapproved AMIs, plus an SCP denying RunInstances unless the AMI carries the approved tag

C Delete all non-golden AMIs from every account

D Rely on Patch Manager to bring the old instances up to date

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Email the AMI ID

Depends on every team reading and acting on an email every month. The audit shows that 60% already do not.

B Config rule plus SCP

The SCP prevents launching from an unapproved image, and the Config rule reports anything already running. Detect and prevent together.

C Delete other AMIs

Breaks running Auto Scaling groups whose launch templates reference them, turning a compliance gap into an outage.

D Patch Manager

Patches the operating system and does nothing about hardening, agents or configuration that the golden image carries.

THE SIGNAL

Producing a standard and enforcing it are different projects. The scenario mentions existing SCPs so you reach for the preventive half.

> A critical vulnerability is announced in a library present in the company's base image. Today, somebody notices the advisory, opens a ticket, and the platform team rebuilds the image within about a week. Instances are then replaced over the following fortnight.
>
> Security wants this to happen without a human noticing the advisory.

What should the solutions architect implement?

A. An EventBridge rule on Amazon Inspector High or Critical findings, triggering the Image Builder pipeline, with Auto Scaling instance refresh to roll out

B. Schedule the Image Builder pipeline to run nightly

C. Subscribe the platform team to vendor security mailing lists

D. Run Patch Manager hourly across the fleet

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Inspector to EventBridge to pipeline

Inspector finds the CVE, EventBridge triggers a rebuild, tests gate distribution, and instance refresh rolls it out. No human reads anything.

B Nightly rebuild

The near miss. It removes the human, and rebuilds every night whether or not anything changed, and still waits up to 24 hours.

C Mailing lists

Exactly the current process. Somebody must read, recognise and act, which is what security asked to remove.

D Hourly Patch Manager

Patches running instances and leaves the base image vulnerable, so every new instance launches with the CVE.

THE SIGNAL

Without a human noticing is the requirement. Only an event-driven trigger removes the human; a schedule merely replaces them with a clock.

> A retailer deploys weekly. A new recommendation engine is ready but the business wants it enabled for 5% of customers first, and switched off within seconds if conversion drops. The current process would require a full deployment to enable it and another to disable it, each taking 30 minutes.
>
> The code is already written and tested.

What should the solutions architect recommend?

A. Deploy the code with the feature behind an AppConfig feature flag, rolled out gradually with automatic rollback on a CloudWatch alarm

B. Use a canary deployment to release the new version to 5% of instances

C. Deploy to a separate environment and split traffic with Route 53 weighted records

D. Speed up the pipeline so deployments take 10 minutes

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Feature flag with AppConfig

Deploy and release become separate events. Enabling and disabling are configuration changes measured in seconds, not deployments.

B Canary deployment

The strong near miss. It reaches 5% of instances rather than 5% of customers, and reverting is still a deployment.

C Second environment plus DNS

Duplicates the environment, and DNS caching means you cannot guarantee when a change takes effect. Module eight.

D Faster pipeline

Ten minutes is still not seconds, and it accepts that release must equal deployment rather than questioning it.

THE SIGNAL

Within seconds is the constraint. Any answer where turning the feature off is a deployment fails it, however fast the deployment is.

> Two years of monthly image builds across four Regions and 30 accounts have produced roughly 2,900 AMIs and 11,000 EBS snapshots. Storage costs are climbing and engineers cannot tell which images are current. Deleting them by hand has been attempted twice and abandoned.
>
> Some older images must be retained for a documented rollback window.

What should the solutions architect recommend?

A. EC2 Image Builder lifecycle policies that deprecate and delete images beyond the retention window

B. A Lambda function scanning all accounts and deleting AMIs older than 90 days

C. Stop building images monthly and build quarterly instead

D. Move the snapshots to S3 Glacier Deep Archive

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Image lifecycle policies

Built for exactly this. Deprecate first, delete after the retention window, across Regions and accounts, with no code to maintain.

B Custom Lambda cleanup

Rebuilds the feature, and a blunt age rule risks deleting an image still referenced by a launch template or needed for rollback.

C Build quarterly

Slows accumulation and leaves 2,900 existing images in place, while making the base image three months stale.

D Glacier for snapshots

EBS snapshots have their own archive tier and are not moved to S3 storage classes. It also does not answer which image is current.

THE SIGNAL

Deprecate then delete is the useful distinction. Deprecation hides an image from selection without removing it during the rollback window.

What to carry out of this module

- **Name what the process cannot do first.** Retrofit the gap; do not propose a rewrite unless asked.
- **Image Builder gates distribution on tests.** A failed image never reaches anybody, which is what makes it evidence.
- **Producing a golden image is not enforcing it.** Config rule to detect, SCP to prevent.
- **Separate deployment from release.** A feature flag turns rollback from a pipeline run into a configuration change.

> ON THE EXAM
>
> Inspector finding to EventBridge to Image Builder to instance refresh is the fully closed loop, built entirely from earlier modules.

MODULE 28 · DOMAIN 3 - CONTINUOUS IMPROVEMENT FOR EXISTING SOLUTIONS

Improving Security Posture

Mostly subtraction. Removing permissions and credentials nobody needs.

DOMAIN 3 of 4 EXAM WEIGHT 25% MODULE 28 of 44

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Nobody remembers why a permission was granted. **Measure what is used, then remove the rest.**

Finding what is not used

- **Access Advisor** shows the services a principal was granted and when each was last accessed. Never accessed is a strong signal.
- **IAM Access Analyzer unused access** finds unused roles, keys, passwords, and unused actions within a role.
- **Access Analyzer policy generation** writes a least-privilege policy from CloudTrail activity, so you start from evidence.
- **The credential report** lists every user, key age, last use and MFA status, as a single CSV.

> ON THE EXAM
>
> Policy generation is the practical answer to "nobody knows what this needs". It reads the log rather than asking anyone.

The root user problem, and what changed

- **Every account historically had root credentials**, each needing a password, an MFA device and somewhere safe to store both.
- **Centralized root access management** lets you delete root passwords, access keys and certificates across member accounts.
- **New accounts in Organizations now have no root credentials by default**, and members cannot recover them unless you allow it.
- **Privileged root actions** cover the narrow set that genuinely needed root, notably deleting a deny-all S3 bucket or SQS queue policy.

> ON THE EXAM
>
> This is newer than most material. Two hundred accounts once meant two hundred root credentials and two hundred MFA devices.

Credential hygiene

- **Long-lived access keys on IAM users are the recurring finding.** Roles and Identity Center replace them, as in module four.
- **Secrets Manager rotation** can be retrofitted onto existing credentials without the application knowing, using its RDS integration.
- **Access Analyzer unused access** is the tool that finds keys nobody has used, which is where you start.
- Rotating a key that should not exist is the wrong fix. Ask whether the identity needs a key at all.

> ON THE EXAM
>
> An access key that has not been used in 400 days does not need rotating. It needs deleting, and that is a different conversation.

Data protection posture

- **Macie** discovers sensitive data you did not know you had, which is where a classification programme starts.
- **S3 default encryption and Block Public Access** can be applied account-wide, retrospectively, in one action.
- **Data residency** is an SCP on aws:RequestedRegion, with a Config rule as the evidence, exactly as in module two.
- Retention is a lifecycle policy plus Object Lock. Deleting data on time is part of posture too.

> ON THE EXAM
>
> You cannot classify data you cannot find. Macie answers what and where before anybody argues about controls.

Measuring whether it improved

- **Security Hub security standards** give a score per standard, so posture becomes a number that moves.
- **Config conformance packs** bundle rules into a named baseline, deployable across the organisation.
- Track the trend, not the absolute. A score of 71 rising is healthier than a score of 88 falling.
- A control with many exceptions is not a control. Count the exceptions as part of the measurement.

> ON THE EXAM
>
> Domain 3 is about improvement, and improvement needs a baseline. Without a number, nobody can tell whether anything got better.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Roles have permissions nobody can justify	Access Advisor and unused access findings
Nobody knows what permissions this app needs	Access Analyzer policy generation
Hundreds of root credentials and MFA devices	Centralized root access management
A bucket policy locked everyone out	Privileged root action to delete it
Data must not leave a Region	SCP on aws:RequestedRegion
We do not know where sensitive data is	Amazon Macie

ON THE EXAM

Most of these start with a question nobody can answer. Find the tool that answers it before choosing a control.

> Meridian Freight has 140 accounts. Each has root credentials with a password stored in a shared vault and a physical MFA device in a safe. An audit found 12 accounts where the MFA device could not be located and 4 where nobody knew the password.
>
> The company runs AWS Organizations with a dedicated security account.

What should the solutions architect recommend?

A. Enable centralized root access management, delete member account root credentials, and register the security account as delegated administrator

B. Reset all 140 root passwords and buy replacement MFA devices

C. Apply an SCP denying all actions by the root user

D. Document the root credentials in a central register reviewed quarterly

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Centralize and delete root credentials

Removes 140 sets of credentials rather than managing them. New accounts then have none by default, and the security account handles privileged tasks.

B Reset and re-buy

Restores the problem to a working state. There are still 140 passwords and 140 devices, and the same audit finding next year.

C SCP denying root

SCPs do not apply to the management account, from module two, and the credentials still exist to be stolen or misused.

D Central register

Improves record-keeping of credentials that should not exist. Documentation is not a control.

THE SIGNAL

The best security improvement is usually deletion. 140 credentials managed well are still 140 credentials.

> A security review finds 60 IAM roles carrying AdministratorAccess. They were granted between 2019 and 2022. Nobody can explain most of them, and the owning teams refuse removal because they cannot prove nothing would break.
>
> The company needs to reduce this without causing outages.

What should the solutions architect recommend?

A. Use Access Advisor and Access Analyzer unused access findings to identify what is actually used, then generate least-privilege policies from CloudTrail

B. Remove AdministratorAccess from all 60 roles and restore it where things break

C. Apply a permissions boundary to all 60 roles

D. Leave them and monitor with GuardDuty

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Measure use, then generate policies

Replaces an argument about intent with evidence about use. Policy generation produces a starting policy from what each role actually called.

B Remove and see what breaks

Discovers dependencies through outages. It is also exactly why the teams are refusing, so it will not be agreed.

C Permissions boundary on all 60

A useful ceiling, and it does not reduce what the roles are granted. The audit finding stands.

D Monitor with GuardDuty

Detects misuse after it happens. Sixty administrator roles remain, which was the finding.

THE SIGNAL

The teams' objection is reasonable and evidence answers it. That is the whole argument for measuring before removing.

> An engineer applies a bucket policy intended to restrict access, but the Deny statement is written incorrectly and now denies every principal, including administrators and the account itself. The bucket holds production data and nobody can modify the policy.
>
> The company has centralized root access management enabled.

What should the solutions architect do?

A. Use a privileged root action from the management account or delegated administrator to delete the bucket policy

B. Recover the member account's root password and sign in as root

C. Open a support case to have AWS remove the policy

D. Recreate the bucket and restore the data from backup

◆ PAUSE AND DECIDE

What should the solutions architect do?

A Privileged root action

Deleting a deny-all S3 bucket policy is one of the specific privileged actions this feature supports, done centrally with temporary credentials.

B Recover root and sign in

The old answer, and it works. It also reinstates a credential you deliberately removed, for a task the feature now handles centrally.

C Support case

Slower, and unnecessary. The account holds the capability to fix this itself.

D Recreate and restore

Destroys and rebuilds a bucket because of a policy error. Data loss risk and downtime for a problem with a direct fix.

THE SIGNAL

This exact lockout is why privileged root actions exist. S3 and SQS deny-all policies are the two named examples.

> A European bank must guarantee that customer data is never processed or stored outside eu-west-1 and eu-central-1. Engineers currently have access to all Regions. Compliance also needs continuous evidence for the regulator.
>
> The company runs Control Tower across 60 accounts.

What should the solutions architect implement?

A. An SCP denying actions where aws:RequestedRegion is outside the two Regions, plus a Config rule providing continuous evidence

B. An IAM policy on each user restricting Regions

C. A Config rule that detects and deletes resources in other Regions

D. Disable unused Regions in the account settings

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A SCP plus Config rule

The SCP refuses the API call so nothing is ever created elsewhere, and the Config rule supplies the continuous record the regulator wants.

B IAM policy per user

Applies to the identities you remember to attach it to, and any new role or user starts unrestricted. It is a ceiling in the wrong place.

C Detect and delete

The resource existed and processed data before deletion. Never is a preventive word, from module two.

D Disable Regions

A real and useful control, and it applies per account and does not produce compliance evidence for the regulator.

THE SIGNAL

Guarantee plus continuous evidence is the same pair as module twenty five. Prevent with a policy, demonstrate with Config.

> The credential report shows 210 IAM users with access keys. 90 keys have not been used in over a year. 40 belong to applications running on EC2. 30 belong to people who have left. The remainder are used by CI systems outside AWS.
>
> Security has asked for a 90-day key rotation policy.

What should the solutions architect recommend?

A Implement 90-day rotation across all 210 keys

B Delete the unused and leaver keys, replace application keys with instance roles, and use IAM Roles Anywhere or OIDC federation for the external CI systems

C Move all 210 keys into Secrets Manager with automatic rotation

D Apply an SCP denying use of access keys

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Rotate all 210

Rotates 160 keys that should not exist, forever. It satisfies the letter of the request and improves very little.

B Delete, replace, federate

Removes 120 keys outright, replaces 40 with roles that need no rotation, and federates the rest. Only 50 keys ever needed a policy.

C All into Secrets Manager

Manages keys well that should have been deleted. It is option A with better storage.

D SCP denying key use

Would break the CI systems that legitimately need to authenticate, with no replacement offered.

THE SIGNAL

The report is broken into four groups deliberately. Three of them need deletion or replacement, not rotation.

> A regulator asks the company to state which of its 4,000 S3 buckets contain personal data. Nobody knows. Buckets have accumulated over nine years across 60 accounts, and naming conventions were never enforced.
>
> The answer must be kept current, not produced once.

What should the solutions architect recommend?

A. Amazon Macie, run from a delegated administrator across the organisation, with scheduled sensitive data discovery jobs

B. Ask each account owner to complete a data classification questionnaire

C. A Config rule checking bucket tags for a data classification value

D. Enable S3 Block Public Access on all buckets

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Macie across the organisation

Inspects the objects themselves rather than asking anybody, works organisation-wide from a delegated administrator, and can run on a schedule.

B Questionnaires

Asks people who do not know, about buckets created by people who have left. It produces a document, not a finding.

C Config rule on tags

Checks whether a tag exists, not whether it is true. An untrue tag passes the rule perfectly.

D Block Public Access

A good control, and it answers a different question. The regulator asked what the data is, not who can reach it.

THE SIGNAL

Nobody knows is the whole problem. Only inspecting the objects produces an answer; every other option asks somebody.

What to carry out of this module

- **Measure use, then remove.** Access Advisor, unused access findings and policy generation turn intent into evidence.
- **Centralized root access management deletes root credentials** across member accounts. New accounts have none by default.
- **Rotating a credential that should not exist is the wrong fix.** Ask whether it needs to exist at all.
- **Macie answers what and where.** You cannot classify data you cannot find, and questionnaires do not find it.

> ON THE EXAM
>
> Prevent with a policy, demonstrate with Config. That pairing has now appeared in modules two, ten, twenty five and here.

MODULE 29 · DOMAIN 3 - CONTINUOUS IMPROVEMENT FOR EXISTING SOLUTIONS

Threat Detection and Response

Detection is the easy half. Correlation and containment are the exam.

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Attackers do not produce one finding. They produce **eight, over twenty hours**, and the sequence is the incident.

GuardDuty protection plans

- **Foundational sources are always on:** CloudTrail management events, VPC flow logs and DNS logs.
- **Protection plans add sources:** S3, EKS, RDS, Lambda, Malware Protection for EC2 and for S3, and Runtime Monitoring.
- **On first enabling GuardDuty, every plan is switched on except Runtime Monitoring.**
- **A plan launched after you enabled GuardDuty is not enabled for you.** Older accounts quietly lag behind.

> ON THE EXAM
>
> That second rule is the one that bites. An account that turned GuardDuty on in 2023 is missing everything AWS has added since.

Extended Threat Detection

- **Correlates many signals into one critical-severity attack sequence finding**, rather than eight medium ones.
- Each finding carries an incident summary, an events timeline, **MITRE ATT&CK mapping** and remediation advice.
- **Automatically enabled at no extra cost.** Coverage grew to EKS in 2025, and to EC2 and ECS later that year.
- It correlates **within one account**, over roughly a 24-hour window. Cross-account attacks may not join up.

> ON THE EXAM
>
> Breadth of protection plans determines how much it can see. Without S3 Protection it cannot build a sequence involving S3 at all.

What a suppression rule actually does

- **A suppressed finding is still generated**, marked archived, kept 90 days and visible in the console.
- **It is not sent to Security Hub, S3, Detective or EventBridge**, so no automation ever sees it.
- **It does not trigger a malware scan, and it is not used as a signal for attack sequences.**
- So suppressing a noisy finding type quietly reduces what Extended Threat Detection can correlate.

> ON THE EXAM
>
> Suppression looks like tidying and behaves like removing a sensor. Suppress deliberately, and know what you have blinded.

Who does what

SERVICE	ANSWERS	SIGNAL IN THE SCENARIO
GuardDuty	Is something malicious happening?	Behaviour, credentials, network, runtime
Inspector	What is vulnerable?	CVEs in EC2, containers and Lambda
Macie	Where is sensitive data?	Classification and discovery in S3
Detective	What actually happened?	Investigation after a finding exists
Security Hub	How are we doing overall?	Aggregation and standards scoring
Firewall Manager	Are our controls applied everywhere?	WAF, Shield and Network Firewall at scale

ON THE EXAM

Detective never detects. It explains a finding that something else produced, which is the distinction most often tested.

From finding to contained

01

Detect

GuardDuty raises a finding, or an attack sequence.

02

Route

EventBridge matches the type and severity.

03

Contain

Isolate the resource without destroying it.

04

Investigate

Detective and the preserved evidence explain the rest.

ON THE EXAM

Containment and investigation pull against each other. Terminating the instance stops the attack and destroys the evidence.

Containing without destroying

- **Replace the security group with an isolation group** that permits nothing. The instance stays running and reachable to forensics.
- **Snapshot the volumes and capture memory before anything else.** Memory is gone the moment you stop the instance.
- **Revoke the instance role's sessions**, since stolen temporary credentials work from anywhere until they expire.
- Copy evidence to a separate forensics account that the compromised account cannot reach.

> ON THE EXAM
>
> Revoking sessions is the step people forget. Isolating the instance does nothing about credentials already exfiltrated from it.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Many related findings across a long period	Extended Threat Detection attack sequences
GuardDuty has been on for years, gaps found	Protection plans not auto-enabled since
We need to know what else the attacker touched	Amazon Detective
Vulnerabilities before images reach production	Inspector scanning in ECR
Contain the instance but keep the evidence	Isolation SG, snapshot, revoke sessions
WAF rules missing in some accounts	AWS Firewall Manager

ON THE EXAM

Ask whether the gap is detection, correlation, investigation or containment. Each has a different service.

> Meridian Freight enabled GuardDuty organisation-wide in 2022 and has left it running since. A security review finds no malware findings for S3 uploads and no runtime findings for containers, despite both features being described in current AWS documentation.
>
> GuardDuty shows as enabled and healthy in every account.

What is the MOST likely explanation?

A Protection plans released after GuardDuty was enabled are not turned on automatically for existing customers

B GuardDuty must be disabled and re-enabled to pick up new features

C Those features require Security Hub to be enabled first

D The findings exist but are below the severity threshold shown by default

◆ PAUSE AND DECIDE

What is the MOST likely explanation?

A New plans are not auto-enabled

Documented behaviour. A plan launched after you enabled GuardDuty stays off until somebody enables it, and the console still reports healthy.

B Disable and re-enable

Would lose finding history and is unnecessary. Plans can be enabled individually at any time.

C Requires Security Hub

Security Hub aggregates GuardDuty findings. It is not a prerequisite for GuardDuty to generate them.

D Severity filtering

Would hide low-severity findings, not remove entire categories. Malware findings are not low severity.

THE SIGNAL

Enabled and healthy does not mean complete. A four-year-old GuardDuty deployment is missing four years of protection plans.

> Over 20 hours, GuardDuty raised eight medium-severity findings in one account: anomalous credential use, unusual API discovery calls, creation of new access keys, a deleted CloudTrail trail, and several S3 object operations. Each was triaged separately and closed as probably benign.
>
> It was later confirmed as a genuine data exfiltration incident.

What would have surfaced this as a single critical incident?

A GuardDuty Extended Threat Detection attack sequence findings

B Raising the severity threshold on all GuardDuty findings

C Amazon Detective, reviewed daily

D A CloudWatch alarm on GuardDuty finding count

◆ PAUSE AND DECIDE

What would have surfaced this as a single critical incident?

A Attack sequence findings

Correlates signals over time into one critical-severity finding with a timeline and MITRE mapping. Exactly the eight-signals-over-a-day case.

B Raise severity thresholds

The findings were genuinely medium individually. Inflating severity produces noise and does not express the relationship.

C Detective daily

Excellent once you know to investigate. Every finding was closed as benign, so nobody would have opened it.

D Alarm on finding count

Eight findings in 20 hours across a busy account is unremarkable. Volume is not the signal; the pattern is.

THE SIGNAL

The scenario lists the signals in MITRE order: credential access, discovery, persistence, evasion, exfiltration. That shape is the finding.

A team was overwhelmed by GuardDuty findings about a vulnerability scanner they run themselves, so they created a suppression rule for that finding type. Six months later, an attack sequence finding that should have included those signals was never generated.

The suppressed findings are still visible in the GuardDuty console, marked archived.

What is the explanation, and what should change?

A. Suppressed findings are excluded from Security Hub, EventBridge, Detective and attack sequence correlation; narrow the rule instead

B. The suppression rule deleted the findings permanently

C. Attack sequences require Security Hub, which was not enabled

D. The suppression rule expired and needs recreating

◆ PAUSE AND DECIDE

What is the explanation, and what should change?

A Suppression excludes downstream, so narrow it

Suppressed findings are archived but not forwarded anywhere, and are not used as attack sequence signals. Scope the rule to the scanner's own resources.

B Findings deleted

The scenario states they are still visible, marked archived. Suppression archives; it does not delete.

C Security Hub required

Extended Threat Detection runs inside GuardDuty and is automatically enabled. Security Hub is a destination, not a prerequisite.

D Rule expired

Suppression rules do not expire. The rule worked exactly as designed, which is the problem.

THE SIGNAL

Suppression is not filtering a view. It removes the finding from every downstream consumer, including the correlation engine.

> GuardDuty reports that an EC2 instance is communicating with a known command-and-control server. The security team's runbook says to terminate the instance immediately. Legal has asked that evidence be preserved for a possible prosecution, and the incident response team needs to know what the attacker did.
>
> The instance has an instance profile with access to several S3 buckets.

What should the response be?

A Terminate the instance immediately to stop the communication

B Apply an isolation security group, snapshot the volumes, revoke the instance role's active sessions, and copy evidence to a forensics account

C Stop the instance and leave it for the forensics team

D Remove the instance profile and leave the instance running normally

◆ PAUSE AND DECIDE

What should the response be?

A Terminate immediately

Stops the attack and destroys volumes, memory and every trace of what happened. Legal's requirement is not met.

B Isolate, snapshot, revoke, preserve

Stops the communication while keeping the instance intact, and revoking sessions handles credentials already stolen from it.

C Stop the instance

The near miss. Volumes survive, and memory is lost the moment it stops, which is where much of the useful evidence lives.

D Remove the profile

Leaves the instance talking to the attacker, and existing temporary credentials keep working until they expire.

THE SIGNAL

Revoking the role's sessions is the step people miss. Isolating the network does nothing about credentials already taken off the box.

> Following a confirmed compromise of an IAM role, the security team must determine which resources the role touched over the previous 30 days, which other principals interacted with those resources, and whether the behaviour was unusual for that role.
>
> CloudTrail, GuardDuty and Security Hub are all enabled.

Which service should the team use?

A Amazon Detective

B Amazon GuardDuty, filtered by the role

C AWS Security Hub, filtered by resource

D Amazon Inspector

◆ PAUSE AND DECIDE

Which service should the team use?

A Amazon Detective

Built for exactly this: a behaviour graph over CloudTrail, VPC flow logs and GuardDuty, showing what a principal touched and what is unusual for it.

B GuardDuty by role

Shows findings involving that role. It does not build the surrounding picture of everything the role did or who else was involved.

C Security Hub by resource

Aggregates findings from other services. It is a queue of findings, not an investigation tool.

D Amazon Inspector

Finds software vulnerabilities. It has nothing to say about what a principal did.

THE SIGNAL

Detective is the only one of the four that answers what happened. The others report what was found.

> A company deploys container images to ECS from ECR. A critical vulnerability reached production last month inside a base image. The security team wants vulnerabilities identified before images are deployed, and continuously afterwards as new CVEs are published.
>
> There are 400 images across 30 repositories.

What should the solutions architect recommend?

A. Amazon Inspector with enhanced scanning on ECR, and a pipeline gate on scan results

B. GuardDuty Runtime Monitoring on the ECS tasks

C. Amazon Macie scanning the image layers in S3

D. A weekly manual review of base image versions

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Inspector enhanced scanning plus a gate

Scans on push and rescans continuously as new CVEs appear, so an image clean last week is re-flagged today. The gate stops the deploy.

B Runtime Monitoring

Detects malicious behaviour at runtime, which is after deployment and after exploitation. The requirement is before.

C Macie on image layers

Macie classifies sensitive data. It does not identify software vulnerabilities, and image layers are not its target.

D Weekly manual review

400 images across 30 repositories, reviewed by hand, weekly. It misses the before-deployment requirement entirely.

THE SIGNAL

Continuously afterwards is the second requirement. A scan at build time alone goes stale the moment a new CVE is published.

What to carry out of this module

- **A protection plan launched after you enabled GuardDuty stays off.** Enabled and healthy is not the same as complete.
- **Extended Threat Detection turns eight medium findings into one critical incident**, with a timeline and MITRE mapping.
- **Suppression removes a finding from every downstream consumer**, including attack sequence correlation.
- **Contain without destroying:** isolate, snapshot, revoke sessions. Terminating ends the incident and the investigation.

> ON THE EXAM
>
> Detective does not detect. It explains a finding something else produced, and that is the most commonly confused name here.

MODULE 30 · DOMAIN 3 - CONTINUOUS IMPROVEMENT FOR EXISTING SOLUTIONS

Improving Reliability

Find what fails, then ask what else fails with it.

DOMAIN 3 of 4 EXAM WEIGHT 25% MODULE 30 of 44

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Recovery that needs AWS to **create** something during an outage may be waiting in the same queue as everyone else.

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Control plane and data plane

- **The control plane creates and changes things**: launching an instance, creating a bucket, modifying a route table.
- **The data plane uses what exists**: serving a request, reading an object, routing a packet already provisioned for.
- **Data planes are designed to be more available than control planes.** That asymmetry is deliberate and documented.
- So a recovery path that depends on control plane actions is less reliable than the thing it is recovering.

> ON THE EXAM
>
> AWS's own example: in Kubernetes, adding pods is a data plane action and adding nodes is control plane. Failover should add pods.

Static stability: the system keeps working during a failure **without needing anything new to be created**.

Self-healing, and what it does not cover

- **Simplified automatic recovery is on by default**, moving an instance off a failing host when a system status check fails.
- **It is not initiated for instances in an Auto Scaling group.** The group replaces the instance instead.
- **It needs capacity to be available**, so during a large outage recovery attempts can fail. AWS says so plainly.
- ASG health checks, ELB health checks and Route 53 health checks each remove a failed component at a different layer.

> ON THE EXAM
>
> Automatic recovery is a control plane action too. It works well for one failed host and less well when a whole zone is short of capacity.

Limiting blast radius

- **Cell-based architecture** runs many independent copies of the stack, each serving a subset of customers. A failure takes out one cell.
- **Shuffle sharding** assigns each customer a different combination of cells, so no two customers share the same full set.
- **Bulkheads** separate resources so one workload cannot exhaust what another needs, such as separate connection pools.
- The cost is operational. Twenty cells means twenty deployments, twenty sets of metrics, and a routing layer that must never fail.

> ON THE EXAM
>
> Shuffle sharding is the elegant one. With eight cells and two per customer, a bad tenant affects a small fraction rather than everyone.

Client-side reliability

- **Naive retries amplify an outage.** A struggling service receiving three times the traffic does not recover.
- **Exponential backoff with jitter** spreads retries out. Without jitter, every client retries at the same moment.
- **Circuit breakers** stop calling a failing dependency entirely for a period, so it gets space to recover.
- **Timeouts must be shorter than the caller's timeout**, or work continues on a request nobody is waiting for.

> ON THE EXAM
>
> Jitter is the part people omit. Backoff alone synchronises every client onto the same retry schedule, which is a thundering herd.

Testing that it works

- **A game day exercises a failure deliberately**, in a controlled window, with the team that would respond.
- **Start with a dependency map.** You cannot test failure modes you have not identified.
- Chaos engineering is the discipline of injecting failure continuously rather than annually.
- **Scope note:** Fault Injection Service, Resilience Hub and Application Recovery Controller are not on the exam's in-scope list.

> ON THE EXAM
>
> Module nineteen said an untested plan is a hypothesis. The same is true of a resilience design that has never lost anything on purpose.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Failover depends on launching instances	Static stability: pre-provision capacity
One customer's load affects everyone	Cell-based architecture, shuffle sharding
Retries made the outage worse	Exponential backoff with jitter
A single NAT gateway or single AZ component	A single point of failure
The instance did not auto-recover	It is in an Auto Scaling group
Failover has never actually been tested	Run a game day

ON THE EXAM

Ask what has to happen during the failure. If the answer includes creating something, that is the weakness.

> Meridian Freight runs an Auto Scaling group across three Availability Zones, sized so all three together exactly meet peak demand. The plan for an AZ failure is that Auto Scaling launches replacement instances in the surviving zones.
>
> During a recent AZ event, those launches were throttled and the service was degraded for 50 minutes.

What should the solutions architect recommend?

A. Pre-provision capacity so any two zones can serve full demand, making the design statically stable

B. Increase the Auto Scaling group's maximum size

C. Add a fourth Availability Zone to the group

D. Reduce the health check grace period so replacements start sooner

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Pre-provision for N-1

Nothing needs creating when a zone fails, so the control plane is not on the recovery path at all. That is static stability.

B Raise the maximum

The group was not at its ceiling. It was throttled, which is a different failure, and a higher maximum does not get you capacity.

C A fourth AZ

Reduces the share lost to one zone, and still requires launching instances during the event. The dependency remains.

D Shorter grace period

Speeds up instances that did launch. The problem was launches being throttled, not being slow to become healthy.

THE SIGNAL

Throttled during an AZ event is the giveaway. Every customer's recovery plan fired at once, and yours was in the queue.

> An operations team reports that a standalone EC2 instance recovered automatically when its host failed, but an identical instance inside an Auto Scaling group did not, and stayed impaired for 20 minutes until someone noticed.
>
> The Auto Scaling group uses EC2 status checks only.

What is the explanation, and what should change?

A. Automatic recovery is not initiated inside an Auto Scaling group; enable ELB health checks so the group replaces impaired instances

B. Automatic recovery must be enabled manually on instances in a group

C. The instance type does not support automatic recovery

D. The Auto Scaling group needs a lifecycle hook

◆ PAUSE AND DECIDE

What is the explanation, and what should change?

A ASG replaces instead; use ELB health checks

Documented behaviour: auto recovery is not initiated inside an ASG. Replacement is the group's job, and ELB health checks catch application failures too.

B Enable it manually

It is on by default on supported types. The group's presence is what suppresses it, not a missing setting.

C Unsupported type

The scenario says an identical instance outside the group recovered. Same type, different outcome.

D Lifecycle hook

Hooks pause instances entering or leaving service. They do not detect or replace an impaired instance.

THE SIGNAL

Identical instance, different result, one variable: membership of an Auto Scaling group. That is the controlled experiment again.

A SaaS platform serves 900 customers from one shared stack. Twice this year, a single customer's unusual query pattern exhausted the database connection pool and every customer experienced an outage.

The business wants one customer's behaviour to affect as few others as possible, without running 900 copies of the stack.

What should the solutions architect recommend?

A. A cell-based architecture with shuffle sharding, assigning each customer a different combination of cells

B. Rate limiting per customer at the API layer

C. A larger database instance with a bigger connection pool

D. A dedicated stack for each of the 900 customers

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Cells with shuffle sharding

A handful of cells, each customer on a different combination, so one bad tenant affects a small overlapping fraction rather than everyone.

B Per-customer rate limiting

Worth doing and it limits request volume, not resource consumption. A small number of expensive queries still exhausts the pool.

C Bigger database

Raises the ceiling that was hit twice. The next unusual query pattern finds the new ceiling.

D 900 stacks

Perfect isolation and explicitly ruled out by the scenario. It is also 900 deployments to operate.

THE SIGNAL

Without running 900 copies is the planted constraint. Shuffle sharding gets most of the isolation from a handful of cells.

> A downstream service became slow for 90 seconds. Clients retried immediately, three times each. The retry traffic tripled load on the already-struggling service, which then failed completely and stayed down for 40 minutes.
>
> Every client is configured to retry three times with no delay.

What should the solutions architect recommend?

A. Exponential backoff with jitter, plus a circuit breaker that stops calling the dependency after repeated failures

B. Increase retries from three to five so requests eventually succeed

C. Scale the downstream service to handle the retry load

D. Remove retries entirely

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Backoff with jitter, plus a breaker

Backoff spreads retries over time, jitter stops them synchronising, and the breaker gives the dependency space to recover.

B More retries

Amplifies the amplification. Five retries each is five times load on a service that failed under three.

C Scale for retry load

Sizes the system for its own failure mode, permanently, so you pay every hour for capacity that exists only for retry storms.

D No retries

Removes the amplification and also the resilience. A transient blip now fails every request that hits it.

THE SIGNAL

Ninety seconds of slowness became forty minutes of outage. That gap is the retry storm, not the original fault.

> A resilience review of a three-tier application finds: instances across three AZs, an RDS Multi-AZ database, an Application Load Balancer, and one NAT gateway in a single AZ serving all private subnets. All outbound traffic to a payment provider goes through it.
>
> Everything else is considered redundant.

What is the most significant single point of failure?

A The single NAT gateway, which fails with its Availability Zone

B The Application Load Balancer

C The RDS Multi-AZ database

D There is no single point of failure in this design

◆ PAUSE AND DECIDE

What is the most significant single point of failure?

A The single NAT gateway

A NAT gateway is zonal. Losing that AZ removes outbound connectivity for every private subnet, so payments fail from all three tiers.

B The load balancer

An ALB is a regional service with nodes in each enabled subnet. It is not a single point of failure by design.

C The database

Multi-AZ provides a standby with automatic failover. It is one of the redundant parts here.

D No SPOF

The review's own conclusion, and it missed the NAT gateway because it is easy to read as shared infrastructure rather than zonal.

THE SIGNAL

NAT gateways are zonal, not regional. One per AZ with per-AZ route tables is the fix, and the extra cost is the trade.

> A company has invested heavily in resilience: multi-AZ, cross-Region replication, health checks and documented runbooks. None of it has ever been exercised. Leadership asks for assurance before an upcoming regulatory review.
>
> The team is nervous about testing in production.

What should the solutions architect recommend?

A. Run a scheduled game day in a controlled window, starting with a dependency map and a limited failure, with the on-call team responding

B. Produce architecture documentation demonstrating the redundancy

C. Test only in the staging environment, permanently

D. Assume the design works, since each component is individually redundant

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A A controlled game day

Produces evidence rather than assertion, and a controlled window with a limited failure addresses the team's nervousness directly.

B Documentation

Describes the intended behaviour. A regulator asking for assurance is asking what happened when you tried, not what you drew.

C Staging only, forever

Useful and never conclusive. Staging differs in scale, data, traffic and configuration, which is where failures hide.

D Assume it works

Individually redundant components frequently share a dependency nobody mapped. That is what a game day finds.

THE SIGNAL

Module nineteen said an untested DR plan is a hypothesis. A resilience design that has never lost anything on purpose is the same thing.

What to carry out of this module

- **Static stability:** if recovery requires creating something, it depends on the control plane at the worst moment.
- **Automatic recovery is not initiated inside an Auto Scaling group**, and it needs available capacity to work at all.
- **Shuffle sharding** buys most of the isolation of per-customer stacks from a handful of cells.
- **Jitter, not just backoff.** Backoff alone synchronises every client onto the same retry moment.

> ON THE EXAM
>
> NAT gateways are zonal. Three AZs of compute routing out through one gateway is one AZ of availability.

MODULE 31 · DOMAIN 3 - CONTINUOUS IMPROVEMENT FOR EXISTING SOLUTIONS

Improving Performance

Measure first. The bottleneck is rarely where the team assumes it is.

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Optimising the wrong component costs money and changes nothing. **Find the constraint before you spend.**

Four resources, four signals

IF THE CONSTRAINT IS	LOOK AT	COMMON MISDIAGNOSIS
CPU	Utilisation, and run queue depth	Blamed for everything by default
Memory	Swap, and the agent's memory metric	Invisible without the CloudWatch agent
Disk	IOPS, throughput, queue length, burst balance	Read as a CPU problem, since waiting looks idle
Network	Packets per second, bandwidth, connection count	Ignored until a specific instance ceiling is hit

ON THE EXAM

A process waiting on disk shows low CPU. Low CPU with poor performance is a signal, not an all-clear.

Percentiles, not averages

- **An average hides the tail.** A service can average 90 ms while one request in a hundred takes four seconds.
- **An SLA is about the experience of individual users**, and every user in that unlucky one per cent had a bad one.
- **Track p50, p90, p99.** The gap between p50 and p99 tells you how consistent the system is.
- At scale, p99 is not rare. A million requests a day means ten thousand people in the tail.

> ON THE EXAM
>
> Averages are for capacity planning. Percentiles are for user experience, and an SLA is a statement about user experience.

Diagnosing a database

- **Database load, measured as average active sessions**, is the metric that matters. It shows sessions doing work at any moment.
- **Broken down by wait event**, it tells you whether sessions are waiting on CPU, I/O, locks or the network.
- It also breaks down by SQL statement, user and host, which usually identifies the offending query directly.
- **Currency note:** the Performance Insights console has been replaced by CloudWatch Database Insights. The concepts and the API are unchanged.

> ON THE EXAM
>
> Wait events are the whole point. Sessions waiting on locks and sessions waiting on I/O need completely different fixes.

Rightsizing, properly

- **Compute Optimizer analyses real utilisation** and recommends a type, with a projected performance risk attached.
- **Fourteen days of data minimum.** A recommendation from a quiet fortnight will undersize you for month-end.
- **Change one dimension at a time** and measure. Changing family and size together tells you nothing about which helped.
- Consider a different family rather than a smaller size. A memory-bound workload on a compute family is oversized by design.

> ON THE EXAM
>
> Rightsizing is not only shrinking. A workload constrained on the wrong dimension needs a different shape, not less of it.

Gotchas that look like application problems

- **T-family credit exhaustion.** Fast for twenty minutes, then throttled. The classic misdiagnosed performance fault.
- **gp3 provisions IOPS independently of size**, so growing the volume does nothing. That was gp2 thinking.
- **Instance network and EBS bandwidth ceilings are per instance type.** A small instance caps throughput regardless of the volume.
- Connection limits, not query speed, are often the database constraint. Module twenty one's RDS Proxy case.

> ON THE EXAM
>
> Each of these presents as "the application is slow" and none is fixed in the application. Check the platform before the code.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Average latency is fine, users complain	Look at p99, not the mean
Fast at first, then slow, on small instances	T-family credit exhaustion
Database CPU low, performance poor	Wait events: I/O or locks, not CPU
Volume enlarged, IOPS unchanged	gp3 provisions IOPS separately
Teams guessed instance sizes years ago	Compute Optimizer, 14 days of data
Read replicas added, no improvement	The bottleneck is writes, or connections

ON THE EXAM

Every row is a metric that rules something out. Find the number in the scenario that eliminates the obvious answer.

> Meridian Freight's booking API reports average response time of 110 ms against a 500 ms target, and the dashboard has been green for months. Customer complaints about slowness have risen steadily, and the support team can reproduce it occasionally.
>
> Traffic is roughly 2 million requests a day.

What should the solutions architect do FIRST?

A Examine p99 and p99.9 latency, since the average is hiding a tail

B Scale the fleet up, since demand has clearly grown

C Add an ElastiCache layer to reduce response times

D Increase the response time target to 1 second

◆ PAUSE AND DECIDE

What should the solutions architect do FIRST?

A Look at the percentiles

A green average with rising complaints is the signature of a tail. At 2 million requests a day, 1% is 20,000 people.

B Scale up

Spends money before identifying the constraint. If the tail is caused by a lock or one slow dependency, more capacity changes nothing.

C Add caching

A plausible fix applied before diagnosis. It may help, and nobody yet knows which requests are slow or why.

D Loosen the target

Makes the number compliant by lowering the bar. The customers experiencing four-second responses are unaffected.

THE SIGNAL

Green dashboard plus rising complaints is a contradiction that resolves one way: the metric is not measuring what users feel.

> A reporting application on t3.medium instances performs well each morning and becomes very slow by mid-afternoon, recovering overnight. CPU utilisation shows 100% during the slow period. Adding instances helps briefly, then the new instances slow down too.
>
> The workload is steady throughout the day.

What is the cause?

A. CPU credit exhaustion on the burstable instance family

B. A memory leak developing through the day

C. Database connection pool exhaustion

D. Network bandwidth limits on the instance type

◆ PAUSE AND DECIDE

What is the cause?

A CPU credit exhaustion

Burstable instances spend credits above baseline and throttle when exhausted. Overnight idleness rebuilds them, which is why mornings are fine.

B Memory leak

Would fit the daily pattern, and a fresh instance would start clean and stay fast. The scenario says new instances also slow down.

C Connection pool

Would show as waiting rather than 100% CPU, and would not follow a daily recovery cycle.

D Network limits

Would appear as throughput plateauing, not sustained full CPU. The workload is steady, so bandwidth is too.

THE SIGNAL

New instances slowing down too is the planted detail. It rules out anything that accumulates inside one instance.

> An Aurora PostgreSQL cluster is reported as slow. CPU utilisation on the writer sits at 22%. Database load is high, and the breakdown shows most sessions waiting on lock events rather than CPU or I/O. One particular UPDATE statement dominates.
>
> The team has proposed a larger instance class.

What should the solutions architect recommend?

A Scale to a larger instance class as proposed

B Investigate and fix the contended UPDATE, since sessions are waiting on locks rather than resources

C Add read replicas to spread the load

D Increase the provisioned IOPS on the storage

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Larger instance**

More CPU for a database using 22% of what it has. Sessions waiting on a lock wait exactly as long on a bigger machine.

B **Fix the contended UPDATE**

Lock waits are contention, not capacity. The breakdown has already named the statement, so the fix is in the query or the transaction.

C **Read replicas**

Move read traffic. The contention is on an UPDATE, which only ever runs on the writer.

D **More IOPS**

The wait events are locks, not I/O. Faster storage does not shorten a wait for another transaction to commit.

THE SIGNAL

Lock waits are the one bottleneck no amount of hardware fixes. That is why the wait-event breakdown matters more than CPU.

> A workload needs 12,000 IOPS. It runs on a 500 GiB gp3 volume delivering 3,000 IOPS. An engineer has proposed enlarging the volume to 2 TiB, reasoning that a larger volume gives more IOPS.
>
> The instance is an m6i.4xlarge.

What should the solutions architect advise?

A gp3 provisions IOPS independently of size; provision 12,000 IOPS on the existing volume

B Enlarge the volume to 2 TiB as proposed

C Switch to io2 Block Express, since gp3 cannot reach 12,000 IOPS

D Stripe four gp3 volumes in RAID 0

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Provision IOPS directly

gp3 decouples IOPS from capacity. You provision 12,000 on the existing 500 GiB volume and pay only for the extra IOPS.

B Enlarge the volume

gp2 thinking. Under gp2, IOPS scaled with size, so this instinct was correct until gp3. On gp3 it buys storage and no performance.

C io2 Block Express

gp3 reaches 80,000 IOPS since late 2025, from module sixteen. 12,000 is comfortably inside it, and io2 costs more.

D RAID 0 striping

The workaround from when a single volume could not deliver enough. Unnecessary complexity and a wider failure surface.

THE SIGNAL

This is gp2 muscle memory. Under gp2, 3 IOPS per GiB meant capacity was the lever; gp3 removed that coupling entirely.

> A team added three read replicas to an RDS database after users reported slowness. Six weeks later performance is unchanged. Investigation shows the application sends all traffic to the writer endpoint, and 80% of database time is spent on write operations.
>
> The replicas sit at 3% utilisation.

What is the most useful conclusion?

A. The replicas were the wrong fix; the constraint is write throughput, which replicas do not address

B. Add more read replicas

C. Increase the replica instance size

D. Enable Multi-AZ on the writer

◆ PAUSE AND DECIDE

What is the most useful conclusion?

A Wrong fix; writes are the constraint

Two facts confirm it: the application never uses the replicas, and 80% of database time is writes. Replicas address neither.

B More replicas

More of a resource sitting at 3% utilisation. Three unused replicas do not become useful by becoming six.

C Bigger replicas

Enlarges idle capacity. The replicas are not the constraint, and were never receiving traffic.

D Multi-AZ

An availability improvement, not a performance one. The standby serves no read or write traffic.

THE SIGNAL

Six weeks of paying for three idle replicas is the cost of not measuring first. The scenario is this module's thesis.

> A platform team runs 200 EC2 instances sized by developers three years ago. They want evidence-based rightsizing recommendations, and are concerned that a recommendation made now would be based on the quiet summer period rather than the December peak.
>
> The business will not accept degraded performance at peak.

What should the solutions architect recommend?

A. Use Compute Optimizer, but check the observation period and consider its longer look-back before acting

B. Apply Compute Optimizer's recommendations immediately across all 200 instances

C. Halve every instance size and monitor for problems

D. Wait until December and rightsize then

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Compute Optimizer, with the period checked

Uses real utilisation, and the team's concern is legitimate. A longer look-back covers the peak, and the projected performance risk guides how far to go.

B Apply immediately

Ignores a concern the team has correctly identified. Recommendations from a quiet fortnight will undersize the December peak.

C Halve everything

Guessing in the opposite direction from the original guess. It is the behaviour that produced the problem.

D Wait for December

Delays the saving by months, and rightsizing during peak trading is the worst possible window to change instance types.

THE SIGNAL

The team's worry is correct and the answer honours it. A recommendation is only as good as the period it observed.

What to carry out of this module

- **Measure before you spend.** Several options will make something faster; only one addresses the constraint.
- **Low CPU with poor performance is a signal.** A process waiting on disk or a lock consumes no CPU.
- **Percentiles, not averages.** An SLA is about individual users, and the tail is where they live.
- **gp3 decouples IOPS from size.** Enlarging the volume is gp2 muscle memory and buys nothing.

> ON THE EXAM
>
> Wait events are the most useful database signal. Lock contention is the one bottleneck that no amount of hardware fixes.

MODULE 32 · DOMAIN 3 - CONTINUOUS IMPROVEMENT FOR EXISTING SOLUTIONS

Cost Optimisation of Running Systems

Closing Domain 3. Three kinds of waste, and they need different fixes.

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Three different problems wear the same name. **Unused, oversized, and undiscounted** need different fixes.

The three kinds of waste

KIND	LOOKS LIKE	THE FIX
Unused	Idle instances, unattached volumes, old snapshots, idle load balancers	Delete it
Oversized	Instances at 8% CPU, over-provisioned Lambda memory	Rightsize it
Undiscounted	Steady On-Demand usage with no commitment	Commit to it
Architectural	NAT processing, cross-AZ chatter, egress	Redesign it, as in module 22

ON THE EXAM

Do them in that order. Committing to usage you are about to delete is the most expensive mistake on this list.

Finding the unused

- **Trusted Advisor** flags idle instances, low-utilisation resources, idle load balancers, unassociated Elastic IPs and over-provisioned Lambda memory.
- **Unattached EBS volumes and orphaned snapshots** accumulate invisibly, because nothing fails when they are left behind.
- **S3 Storage Lens** shows where objects and money have accumulated across every account.
- Check before deleting. A volume detached for a reason, or a snapshot referenced by a launch template, still matters.

> ON THE EXAM
>
> Nothing breaks when waste is left behind, which is exactly why it accumulates. Deletion needs a process, not a spring clean.

Utilisation or coverage?

UTILISATION

Am I using what I bought?

- Percentage of commitment consumed
- Low means you bought too much
- Target above roughly 90%
- A multi-day drop is an early warning

COVERAGE

Am I discounting what I run?

- Percentage of eligible usage covered
- Low means paying On-Demand needlessly
- There is no single right number
- Raise it only for stable baseline usage

THE TIEBREAKER

High utilisation with low coverage means buy more. Low utilisation means you already bought too much. They point opposite ways.

Rightsize or commit first?

- **Committing first** applies a discount immediately across everything eligible, with no engineering work at all.
- **But rightsizing afterwards shrinks the usage**, and the commitment stays where it was. Utilisation falls and you owe the difference.
- **Commit to the baseline that will still exist after the planned rightsizing**, rather than to today's inflated usage.
- Cost Explorer recommendations use 30 to 60 days of history and refresh roughly every 72 hours, so recent changes are not yet reflected.

> ON THE EXAM
>
> This is a genuine trade-off, not a rule. What decides it is whether a rightsizing programme is actually going to happen.

Noticing sooner

- **Cost Anomaly Detection** learns your spending pattern and alerts on deviation, at service, account, tag or cost category level.
- It is detective. It tells you after the spend has happened, so pair it with budgets and budget actions.
- **Budget actions** from module eleven are the preventive half: apply a policy or stop resources at a threshold.
- Monitor at the right granularity. An anomaly in one account is invisible in an organisation-wide total.

> ON THE EXAM
>
> A monthly bill review means the average surprise is discovered two weeks after it started. Anomaly detection closes that gap.

Attributing it

- **The Cost and Usage Report is the only source with per-resource, hourly detail.** Cost Explorer summarises.
- Query it with Athena to answer which team, which resource, and which hour, from module eleven.
- Attribution depends entirely on tags, and cost allocation tags are **not retroactive**. Module eleven again.
- Showback changes behaviour on its own. Teams that can see their own spend reduce it without being asked.

> ON THE EXAM
>
> Optimising centrally has a ceiling. Showing each team its own number moves more spend than any central programme.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Savings Plan utilisation is 70%	Over-committed. You bought too much
Utilisation 99%, coverage 45%	Under-committed. Buy more for the baseline
Rightsizing dropped utilisation	Committed before shrinking the usage
A spike was found on the monthly bill	Cost Anomaly Detection
Unattached volumes and old snapshots	Trusted Advisor, plus a deletion process
Which team caused this increase?	CUR with Athena, and tags

ON THE EXAM

Decide first which of the three wastes you are looking at. Unused, oversized and undiscounted have nothing in common but the symptom.

> Meridian Freight reports Savings Plans utilisation of 68% and coverage of 82%. The commitment was purchased 14 months ago for a three-year term. Finance asks why the savings are lower than projected.
>
> Compute usage has fallen since a containerisation project moved several workloads to Fargate.

What is the situation, and what should be done?

A Over-committed: 32% of the commitment is unused. Grow eligible usage onto the plan, and size the next purchase to the reduced baseline

B Under-committed: buy an additional Savings Plan to raise coverage

C Cancel the Savings Plan and return to On-Demand

D Nothing; 68% utilisation is within normal range

◆ PAUSE AND DECIDE

What is the situation, and what should be done?

A Over-committed

Utilisation of 68% means nearly a third of the committed spend buys nothing. The containerisation project shrank the usage under a fixed commitment.

B Buy more

Reads coverage and ignores utilisation. Buying more commitment when a third of the existing one is unused makes it worse.

C Cancel it

Savings Plans cannot be cancelled. The seven-day return window applies only to a recent purchase, not one 14 months old.

D 68% is fine

Practice targets above roughly 90%. At 68%, almost a third of every committed dollar is producing no discount at all.

THE SIGNAL

Fargate is eligible for a Compute Savings Plan, so the containerisation should not have reduced utilisation. Unless the plan was EC2 Instance type.

> A company reports Savings Plans utilisation of 99% and coverage of 41%. Compute usage has been stable for two years and is forecast to remain so. The finance team believes the Savings Plan strategy is working well and no action is needed.
>
> The uncovered usage is steady, round-the-clock production workload.

What should the solutions architect advise?

A Agree; 99% utilisation shows the commitment is correctly sized

B Coverage of 41% means most eligible usage is paying On-Demand rates; increase the commitment for the stable baseline

C Reduce the commitment, since utilisation cannot exceed 100%

D Switch the remaining usage to Spot Instances

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Agree, 99% is good

Utilisation of 99% is excellent and only says the small commitment is fully used. It says nothing about the 59% paying full price.

B Coverage is low; commit more

Stable, round-the-clock, forecast to continue. That is exactly the profile a commitment exists for, and 59% of it is undiscounted.

C Reduce the commitment

Backwards. High utilisation with low coverage means the commitment is too small, not too large.

D Move to Spot

Steady round-the-clock production is the workload least suited to interruption. Spot suits fault-tolerant, flexible work.

THE SIGNAL

Pair this with scenario one. Same two metrics, opposite conclusions, because they measure different things.

> A company bought a large three-year Compute Savings Plan in January, sized to its On-Demand spend at the time. In March it completed a rightsizing programme that reduced compute consumption by 30%. Savings Plans utilisation has since sat at 71%.
>
> Both projects were run by different teams.

What went wrong, and what should be done differently next time?

A. The commitment was sized to pre-rightsizing usage; size commitments to the baseline that will remain after planned changes

B. The rightsizing programme was a mistake and should be reversed

C. Convert the Compute Savings Plan to an EC2 Instance Savings Plan

D. Nothing went wrong; utilisation recovers naturally over a three-year term

◆ PAUSE AND DECIDE

What went wrong, and what should be done differently next time?

A Sized to pre-rightsizing usage

Two teams optimised independently and the second undercut the first. Size to the baseline that survives the planned change.

B Reverse the rightsizing

Runs oversized instances for three years to justify a commitment. It optimises the metric rather than the bill.

C Convert the plan type

Savings Plans cannot be converted between types. Convertible Reserved Instances can be exchanged; Savings Plans cannot.

D It recovers naturally

Only if usage grows back to the pre-rightsizing level, which is the opposite of the intent. Nothing recovers on its own.

THE SIGNAL

Two cost programmes run independently can subtract from each other. That coordination failure is the real lesson here.

> A misconfigured job began writing 40 TB a day to S3 in a development account on the 3rd of the month. It was discovered on the 14th during a routine bill review, by which time it had cost roughly $28,000. Budgets are set at the organisation level only.
>
> The development account normally spends about $900 a month.

What should the solutions architect implement?

A. Cost Anomaly Detection monitors at account level, with alerts, plus per-account budgets and budget actions

B. Review the organisation bill weekly instead of monthly

C. Lower the organisation-level budget threshold

D. Apply an SCP preventing development accounts from writing to S3

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Account-level anomaly monitors and budgets

A jump from $900 to thousands is glaring against that account's own baseline, and invisible in the organisation total. Budget actions can stop it.

B Weekly bill review

Reduces discovery from 11 days to about 7. Still a week of spend, and it still depends on somebody reading a report.

C Lower the org threshold

$28,000 may be noise in an organisation-wide budget. The granularity is the problem, not the threshold.

D Block S3 in dev

Prevents legitimate development work to stop a misconfiguration. It solves the symptom by removing the capability.

THE SIGNAL

The $900 baseline is the planted detail. It tells you the anomaly is enormous locally and invisible globally.

> An audit across 80 accounts finds 3,400 unattached EBS volumes, 19,000 snapshots older than two years, 240 unassociated Elastic IPs and 60 load balancers with no healthy targets. Estimated waste is $47,000 a month.
>
> Two previous manual clean-up attempts stalled because nobody could confirm what was safe to delete.

What should the solutions architect recommend?

A. Use Trusted Advisor and Storage Lens to identify candidates, then a tag-based process: tag, notify owners, wait, then delete automatically

B. Delete everything older than two years immediately

C. Ask each account owner to review and clean up their own account

D. Accept the waste, since $47,000 is small relative to total spend

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Identify, tag, notify, wait, delete

Addresses why the previous attempts stalled. The waiting period converts silence into consent and makes deletion automatic rather than a decision.

B Delete immediately

A snapshot referenced by a launch template, or a volume detached for a forensic hold, breaks something. This is why attempts one and two stalled.

C Ask the owners

Exactly what failed twice. Owners cannot confirm what is safe either, and the request competes with their own work.

D Accept it

Over half a million pounds a year, growing, for resources nobody uses. It also normalises accumulation.

THE SIGNAL

Two failed attempts is the planted detail. It tells you the obstacle is decision-making, not identification.

> The monthly bill rose by $60,000. Finance asks which team caused it. Cost Explorer shows the increase is in EC2 and data transfer across three accounts, but cannot attribute it to a team or workload.
>
> Resources carry a Team tag, activated as a cost allocation tag 18 months ago.

What should the solutions architect do?

A Query the Cost and Usage Report in Athena, grouped by the Team tag and by resource, at hourly granularity

B Filter Cost Explorer by the Team tag

C Enable Cost Anomaly Detection and wait for the next occurrence

D Ask each of the three account owners to investigate

◆ PAUSE AND DECIDE

What should the solutions architect do?

A CUR in Athena, by tag and resource

The only source with per-resource hourly detail. The tag was activated 18 months ago, so the data exists for this period.

B Filter Cost Explorer

The near miss. Grouping by tag works, and Cost Explorer will not reach the individual resource and hour that identify the change.

C Anomaly Detection and wait

Worth enabling for next time, and it answers nothing about the $60,000 already spent. Finance asked about this month.

D Ask the owners

Delegates an investigation the data can answer directly, to people who will look at the same Cost Explorer view.

THE SIGNAL

The tag was activated 18 months ago, which is the detail that makes this answerable. Activation is not retroactive.

What to carry out of this module

- **Unused, oversized, undiscounted.** Delete, resize, commit. Doing them out of order commits you to waste.
- **Low utilisation means you bought too much. Low coverage means you bought too little.** They point opposite ways.
- **Size a commitment to the baseline that survives your planned rightsizing**, not to today's usage.
- Monitor anomalies at a granularity where they are visible. A big spike in a small account vanishes in an org total.

> ON THE EXAM
>
> When a clean-up has stalled twice, the obstacle is decision-making. Tag, notify, wait, then delete automatically.

MODULE 33 · DOMAIN 4 - ACCELERATE WORKLOAD MIGRATION AND MODERNIZATION

Portfolio Assessment and Wave Planning

Domain 4 begins. You cannot migrate what nobody can describe.

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An honest warning, up front

- **AWS Migration Hub closed to new customers on 7 November 2025.** Existing customers continue; no new features.
- **AWS Application Discovery Service closed on the same date.** Same terms.
- **Both remain on the exam's in-scope service list**, so both remain correct exam answers.
- **AWS Transform** is the stated replacement, carrying discovery, dependency mapping, wave planning and strategy recommendations. It is **not** on the in-scope list.

> ON THE EXAM
>
> Simple AD and App Runner had this problem individually. Here it affects the two central services of an entire domain.

Nobody knows what they are running.
Discovery is not a formality; it is the reason migrations slip.

Which tool answers which question

QUESTION	TOOL	NOTE
What servers do we have?	Application Discovery Service	Agent or agentless
What talks to what?	Discovery Agent	Connections and processes
What will it cost on AWS?	Migration Evaluator	TCO and what-if scenarios
How should each app move?	Strategy Recommendations	A path per application
How is it going?	Migration Hub	Aggregates. Moves nothing

ON THE EXAM

Migration Hub is a free dashboard that performs no migration. It shows MGN, DMS and partner status in one place.

Agent or agentless discovery?

AGENTLESS COLLECTOR

A VMware appliance

- One appliance, no software on servers
- VM inventory, profile and utilisation
- Database metadata and network traffic
- Cannot see inside the guest OS

DISCOVERY AGENT

Software on each server

- Network connections, per process
- Running processes and performance
- This is what gives dependency mapping
- Must be installed and permitted

THE TIEBREAKER

Dependency mapping needs the agent. If a scenario asks what talks to what, agentless collection alone will not answer it.

Building the business case

- **Migration Evaluator** produces a directional TCO comparison against the current on-premises cost.
- It models **what-if purchasing scenarios**, so the board sees On-Demand against commitments rather than one number.
- It analyses **Windows and SQL Server licensing** specifically, which is often the largest single line in the case.
- It can consume existing third-party inventory data, or Application Discovery Service data, instead of its own collector.

> ON THE EXAM
>
> The business case is a different artefact from the technical plan. A board approves money; an architect approves a design.

Planning the waves

- **A wave is a group of applications migrated together**, chosen because they depend on each other more than on anything outside.
- **Wave one should be low risk and low dependency**, because its real purpose is to prove the process and train the team.
- **Splitting a dependency across waves** means running traffic between on-premises and AWS for however long the gap lasts.
- Sequence by dependency first, then by business risk. Technical difficulty is a poor primary sort.

> ON THE EXAM
>
> The first wave's deliverable is a working process, not a migrated application. Choose something you can afford to get wrong.

Migration Hub mechanics

- **It aggregates, it does not migrate.** Status arrives from MGN, DMS and supported partner tools.
- **The home region is set once and cannot be changed.** All Migration Hub data lives there, whatever Region you migrate into.
- Group servers into applications, then track progress per application rather than per server.
- It is free. The cost is in the tools it tracks, not in the tracking.

> ON THE EXAM
>
> The home region catches people. Choosing it casually on day one is a decision you live with for the whole programme.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
We need to know what talks to what	Discovery Agent, not agentless
No software may be installed on servers	Agentless Collector
The board needs a cost comparison	Migration Evaluator
Track progress across several migration tools	Migration Hub
Which applications should move together?	Dependency-driven wave planning
Recommend a migration path per application	Strategy Recommendations

ON THE EXAM

Assessment questions are usually about which question is being asked, not which tool is best. Match the tool to the question.

> Meridian Freight is planning to migrate 1,200 on-premises servers. The existing CMDB is four years old and known to be inaccurate. Before planning waves, the team must establish which servers communicate with which, including undocumented dependencies.
>
> Security has approved installing software on the servers.

What should the solutions architect recommend?

A. Deploy the Application Discovery Agent to the servers, to collect network connections and running processes

B. Deploy the Agentless Collector into vCenter

C. Export the CMDB and validate it in a workshop with application owners

D. Run Migration Evaluator to produce a dependency map

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Discovery Agent

Only the agent sees network connections per process on each server, which is what dependency mapping requires. Agents are permitted here.

B Agentless Collector

Excellent for inventory and utilisation, and it cannot see inside the guest OS. Process-level connections are exactly what it misses.

C Validate the CMDB

Asks people about undocumented dependencies. Undocumented means nobody wrote them down, usually because nobody knew.

D Migration Evaluator

Builds a business case and a TCO model. It answers what it will cost, not what connects to what.

THE SIGNAL

Two details decide it: undocumented dependencies, and agents are permitted. The second exists to remove the usual objection.

> A bank must assess 600 VMware virtual machines before migration. Its change policy forbids installing any new software on production servers, and the approval process for an exception would take nine months.
>
> The team needs inventory, utilisation data and database metadata.

What should the solutions architect recommend?

A. Deploy the Application Discovery Service Agentless Collector as a vCenter appliance

B. Begin the exception process so agents can be installed

C. Deploy the legacy Discovery Connector

D. Estimate the inventory from purchase records and vendor documentation

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Agentless Collector in vCenter

One appliance, nothing on the servers, and it collects exactly what is asked for: inventory, utilisation and database metadata.

B Pursue an exception

Nine months before discovery starts, to obtain detail the scenario does not ask for. It does not need dependency mapping.

C Discovery Connector

The predecessor, and it reached end of support in November 2025. Application Discovery Service no longer accepts its data.

D Estimate from records

Purchase records show what was bought, not what runs or how heavily. Utilisation cannot be inferred from an invoice.

THE SIGNAL

Pair this with scenario one. Same choice, opposite answer, because agents are forbidden and no dependency mapping is required.

> A board will not approve a migration programme without a comparison of current on-premises total cost against projected AWS cost, including Windows and SQL Server licensing options. The company already has two years of inventory and utilisation data from a third-party monitoring tool.
>
> The CFO wants to see more than one purchasing scenario.

What should the solutions architect recommend?

A. Migration Evaluator, importing the existing third-party data, producing TCO with what-if purchasing scenarios

B. The AWS Pricing Calculator, sized from the inventory

C. Application Discovery Service, then manual cost modelling in a spreadsheet

D. Migrate a pilot application and extrapolate its cost

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Migration Evaluator with imported data

Built for the business case: TCO against current cost, multiple purchasing scenarios, and Windows and SQL licensing analysis. It accepts third-party data.

B Pricing Calculator

Prices an AWS design. It does not know the on-premises cost, so it cannot produce the comparison the board asked for.

C Discovery plus a spreadsheet

Rebuilds Migration Evaluator by hand, slowly, and a hand-built model carries no external credibility with a board.

D Pilot and extrapolate

One application's cost profile rarely generalises across a portfolio, and it delays approval by the length of the pilot.

THE SIGNAL

The comparison is the point. Anything that prices only the AWS side answers half the question a board is actually asking.

> A migration is under way using Application Migration Service for servers, Database Migration Service for databases, and a partner tool for a large file estate. The programme manager cannot answer how many of the 340 applications are complete, and builds a spreadsheet by hand each Friday.
>
> Each tool has its own console.

What should the solutions architect recommend?

A. AWS Migration Hub, grouping servers into applications and aggregating status from all three tools

B. A CloudWatch dashboard combining metrics from each tool

C. Continue the weekly spreadsheet, but automate its generation

D. Standardise on Application Migration Service for everything

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Migration Hub

Exactly its purpose: aggregate status from MGN, DMS and supported partner tools, grouped into applications rather than servers. Free.

B CloudWatch dashboard

CloudWatch shows metrics, not migration state. It has no concept of an application being migrated.

C Automate the spreadsheet

Builds the aggregation Migration Hub provides for nothing, and still reports per server rather than per application.

D One tool for everything

MGN does not migrate databases or large file estates well. The tools are different because the problems are.

THE SIGNAL

Grouping into applications is the part that matters. Per-server progress does not answer whether an application has moved.

> Dependency analysis of 340 applications shows one cluster of 14 applications sharing a database and messaging bus, and 60 applications with no dependencies beyond Active Directory. Leadership wants the most complex, highest-value application migrated first to prove the programme's value.
>
> The team has never performed a migration.

What should the solutions architect advise?

A. Begin with low-dependency applications to prove the process, and keep the 14-application cluster in a single later wave

B. Begin with the most complex application, as leadership requested

C. Split the 14-application cluster across several waves to spread the risk

D. Migrate all 340 applications simultaneously to avoid hybrid running costs

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Low dependency first, cluster together

Wave one's deliverable is a working process and a trained team. The tightly coupled cluster moves together, in one later wave.

B Most complex first

Puts the highest-value application in the hands of a team that has never migrated anything. It proves value by risking the most.

C Split the cluster

Spreads risk by guaranteeing it. Fourteen tightly coupled applications split across waves run chatty traffic over the hybrid link for months.

D All at once

Avoids hybrid cost by concentrating every risk into one event, with a team that has no experience of the process.

THE SIGNAL

Splitting a tightly coupled cluster is the trap. It sounds like risk management and creates a long hybrid period instead.

> A team sets up Migration Hub and selects us-east-1 as the home region during initial configuration, without discussion. The migration will move workloads into eu-west-1 and eu-central-1, and the company has a policy that migration metadata must be stored in the EU.
>
> Discovery data has already been collected.

What should the solutions architect advise?

A The home region cannot be changed; the account must be reconfigured or a different account used, and discovery data re-collected

B Change the home region to eu-west-1 in the Migration Hub console

C Migration Hub data is automatically replicated to the migration target Regions

D Set a second home region for the EU

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A It cannot be changed

The home region is set once per account and is permanent. Meeting the policy means a different account, and re-collecting discovery data.

B Change it in the console

There is no such operation. This is the assumption the scenario is testing, and it is the reason the choice matters on day one.

C It replicates

Migration Hub data stays in the home Region regardless of where migrations run. That is precisely the problem here.

D A second home region

One home region per account, by definition. There is no second.

THE SIGNAL

A one-off, irreversible configuration choice made casually on day one. Worth knowing before somebody clicks it.

What to carry out of this module

- **Migration Hub and Application Discovery Service are closed to new customers** and still examinable. Answer with them.
- **Dependency mapping needs the agent.** Agentless gives inventory and utilisation, not what talks to what.
- **Migration Evaluator builds the business case**, comparing current cost to projected cost with licensing analysis.
- **Wave one proves the process.** Keep tightly coupled applications in one wave; never split them to spread risk.

> ON THE EXAM
>
> Migration Hub's home region is set once and never changed. It is the irreversible decision people make casually.

MODULE 34 · DOMAIN 4 - ACCELERATE WORKLOAD MIGRATION AND MODERNIZATION

The Seven Rs

Each R is chosen by a constraint, not by a preference.

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The question is never which R is best. It is **which constraint the scenario has already given you**.

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Three Rs that avoid the work entirely

- **Retire.** Switch it off. Discovery routinely finds 10 to 20 per cent of an estate with no traffic and no owner.
- **Retain.** Leave it where it is, deliberately, and revisit later. A conscious decision, not a failure.
- **Repurchase.** Replace it with a SaaS product. The workload leaves your migration scope completely.
- Do these first. Every application removed from scope is one you never plan, test, cut over or support.

> ON THE EXAM
>
> Retire is the cheapest migration there is. Application Discovery Service finding servers with no network traffic is how you locate them.

Four Rs that move it

STRATEGY	WHAT CHANGES	CHOOSE IT WHEN
Relocate	The hypervisor. Nothing above it	Large VMware estate, exit the data centre fast
Rehost	Where it runs. Not the application	Speed matters; a deadline drives the plan
Replatform	One or two components, managed	A clear win is available for modest effort
Refactor	The architecture itself	Strategically important, and the design is the constraint

ON THE EXAM

Effort and long-term value both rise down this list. Refactoring gives the most and costs the most, in time and money.

The constraint that decides each

- **A hard deadline** means rehost or relocate. Neither requires application change, so neither depends on developer availability.
- **A vendor application nobody may modify** means rehost, retain or repurchase. Refactoring is not available to you.
- **An obvious managed-service equivalent** means replatform. Self-managed SQL Server to RDS is the standard example.
- **The architecture itself is the problem** means refactor, and only when the application is strategically worth it.

> ON THE EXAM
>
> Read for who is allowed to change the code. If nobody is, four of the seven Rs disappear before you compare anything.

Honest notes

- **Relocate's usual vehicle is VMware Cloud on AWS**, which is now procured through Broadcom rather than the AWS resell channel.
- **Relocate preserves the operating model**, and therefore the inefficiencies, licensing costs and technical debt that came with it.
- A typical portfolio lands near **40 to 50 per cent rehost, 20 to 30 per cent replatform, 10 to 15 per cent refactor**.
- Most portfolios use several Rs. A plan proposing one strategy for everything has not been assessed.

> ON THE EXAM
>
> If a proposed plan is 90% refactor, the question is not whether it is technically better. It is whether it will ever finish.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Data centre lease expires in months	Rehost, or relocate
No network traffic, no known owner	Retire
Self-hosted CRM, email or HR system	Repurchase
Vendor application, no modification allowed	Rehost or retain
Self-managed database on EC2	Replatform to a managed service
The monolith is holding the business back	Refactor, if it is strategic

ON THE EXAM

Two questions settle most of these: is there a deadline, and is anyone allowed to change the code?

> Meridian Freight's data centre lease expires in eight months and will not be renewed. There are 840 servers running 210 applications. The development teams are already committed to a regulatory programme for the next year and cannot take on migration work.
>
> Leadership has asked for a modernisation-first approach.

What should the solutions architect recommend?

A Refactor the applications to cloud-native services during the migration

B Rehost the majority to meet the deadline, then modernise selected applications afterwards

C Extend the data centre lease to allow time for modernisation

D Retain everything on-premises until developers are available

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Refactor during migration

Refactoring needs developers, and the scenario says there are none for a year. Leadership's preference does not create capacity.

B Rehost now, modernise later

Rehosting needs no application change, so it is unblocked by the developer constraint and fits the deadline. Modernisation follows once in AWS.

C Extend the lease

The scenario says it will not be renewed. Changing a fixed constraint is not an architectural recommendation.

D Retain everything

There will be no data centre in eight months. Retain requires somewhere to retain it.

THE SIGNAL

Leadership wants modernisation and the developers are unavailable. When a preference meets a constraint, the constraint wins.

> Discovery across 1,100 servers finds 180 with no inbound or outbound network traffic in 90 days, no logged-in users, and no owner listed in the CMDB. Several have hostnames referring to projects that ended years ago.
>
> The migration team has budgeted to migrate all 1,100.

What should the solutions architect recommend?

A. Investigate the 180 for retirement, removing them from scope after a confirmation period

B. Rehost all 1,100, since migrating is safer than deleting

C. Retain the 180 on-premises indefinitely

D. Power off the 180 immediately and migrate the rest

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Investigate for retirement

Retire is the cheapest strategy. Sixteen per cent of the estate leaves scope, with a confirmation period covering quarterly or annual jobs.

B Migrate everything

Pays to plan, test, cut over and then run 180 servers nobody uses, indefinitely. Migrating waste preserves it.

C Retain indefinitely

Keeps them powered on-premises forever. Retain is for workloads you will revisit, not for ones nobody has identified.

D Power off immediately

Ninety days misses quarterly and annual processes. The right instinct, applied without the confirmation period.

THE SIGNAL

Retire is the only R that reduces cost permanently and immediately. Sixteen per cent off the scope is a large result.

> A company self-hosts an email platform, a CRM and an HR system on 40 servers. Each requires patching, licensing and a specialist administrator. Mature SaaS equivalents exist for all three, and the business has no requirement that differentiates its usage from any other company's.
>
> The migration team has scoped all three for rehosting.

What should the solutions architect recommend?

A. Repurchase: move to SaaS equivalents and remove all three from migration scope

B. Rehost all three as scoped, then evaluate SaaS later

C. Refactor the three applications to run on containers

D. Replatform the databases behind each to RDS

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Repurchase to SaaS

Commodity functions with no differentiating requirement, mature products available. Forty servers, three specialists and the licensing all leave scope.

B Rehost then evaluate

Migrates forty servers to postpone a decision the scenario has already made for you. Later evaluations rarely happen.

C Refactor to containers

Engineering investment in commodity software you did not write. It is the most expensive answer to the least strategic problem.

D Replatform the databases

Improves the database layer of applications you should not be operating at all.

THE SIGNAL

No differentiating requirement is the phrase. Building or hosting something that gives you no advantage is the definition of repurchase.

> A logistics application runs on Windows Server with a self-managed SQL Server database on EC2 after an earlier lift and shift. The DBA team spends significant time on patching, backups and failover testing. The application uses standard SQL Server features with no unsupported extensions.
>
> There is no deadline pressure and a modest engineering budget.

What should the solutions architect recommend?

A. Replatform the database to Amazon RDS for SQL Server, leaving the application unchanged

B. Refactor the application to use Aurora and microservices

C. Leave it as it is; it already runs on AWS

D. Repurchase a SaaS logistics platform

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Replatform to RDS for SQL Server

The canonical replatform. Patching, backups and failover become AWS's job, the application does not change, and no unsupported features block it.

B Refactor to Aurora

A far larger project. Aurora is not SQL Server, so this is an engine migration plus a rewrite, on a modest budget.

C Leave it

Running on EC2 is not the same as being done. The DBA effort the scenario describes continues indefinitely.

D Repurchase

Logistics is likely a differentiating capability for a logistics company, unlike the commodity systems in scenario three.

THE SIGNAL

No unsupported extensions is the planted detail. It is the thing that normally blocks a move to a managed engine.

> A company runs 400 VMware virtual machines and must exit its data centre in five months. Its infrastructure team is deeply skilled in vSphere and has no AWS experience. Retraining before the deadline is not realistic, and the applications include several the vendor forbids modifying.
>
> Longer term, the company intends to adopt native AWS services.

What should the solutions architect recommend?

A. Relocate to VMware Cloud on AWS to meet the deadline, then modernise into native services over time

B. Refactor all 400 workloads to native AWS services now

C. Retain the estate and negotiate a longer lease

D. Rehost all 400 with Application Migration Service and retrain the team simultaneously

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Relocate, then modernise

The operating model is unchanged, so the vSphere team stays productive. Five months is achievable, and modernisation follows without a deadline.

B Refactor now

Four hundred workloads refactored in five months, by a team with no AWS experience, including applications nobody may modify.

C Negotiate the lease

Changes the constraint rather than working within it, and the scenario presents the exit as fixed.

D Rehost and retrain

The strong alternative. Rehosting works, and it hands a team with no AWS experience 400 EC2 instances to operate from day one.

THE SIGNAL

The vSphere skills detail is what separates relocate from rehost here. Relocate keeps the team's existing expertise useful.

> A retailer's order management monolith cannot scale for peak trading, takes six weeks to release a change, and is the single largest constraint on the business's growth. It is the company's core differentiator. There is executive sponsorship, a dedicated team, and an eighteen-month horizon.
>
> A consultant has proposed rehosting it to AWS first.

What should the solutions architect recommend?

A Rehost it first, as proposed, then decide

B Refactor it, since the architecture itself is the constraint and the conditions for refactoring are met

C Replatform the database and leave the monolith intact

D Repurchase a commercial order management product

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Rehost first

Usually right, and here it moves the constraint to AWS unchanged. Six-week releases and no scalability follow it across.

B Refactor

All four conditions are present: the architecture is the constraint, it is a differentiator, and there is sponsorship, a team and time.

C Replatform the database

Addresses a component that is not the problem. The release cycle and scalability are architectural, not database, issues.

D Repurchase

Replaces the company's core differentiator with the same product its competitors use. Sound for commodity, wrong here.

THE SIGNAL

This is the rare case where refactor is right, and the scenario supplies every precondition deliberately. Most scenarios do not.

What to carry out of this module

- **The constraint chooses the R**, not a preference. Find it before you compare strategies.
- **Retire, retain and repurchase first.** Every application out of scope is one you never plan, test or cut over.
- **Refactor needs four things:** the architecture is the constraint, it is strategic, and there is sponsorship and time.
- **Relocate keeps the operating model**, which is its value and its cost. It defers modernisation rather than answering it.

> ON THE EXAM
>
> Two questions settle most of these: is there a deadline, and is anyone allowed to change the code?

MODULE 35 · DOMAIN 4 - ACCELERATE WORKLOAD MIGRATION AND MODERNIZATION

Server and Application Migration

Replication is the easy part. Cutover is where migrations fail.

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The service converts the **machine**. Whether the **application** works is a separate question, and you answer it by testing.

The lifecycle

01

Replicate

Agent copies blocks continuously into a staging area.

02

Test

Launch a test instance. Source keeps running, untouched.

03

Cut over

Brief pause, final sync, launch the cutover instance.

04

Finalise

Confirm, then stop replication and archive the source.

ON THE EXAM

Testing is non-disruptive and repeatable. Replication continues throughout, so you can test as many times as you like.

How replication works

- **Continuous block-level replication** into a staging subnet, using small replication servers and staging EBS volumes.
- **Outbound TCP 443 and TCP 1500** from source to AWS. No inbound access to your data centre is required.
- **Sources can be anywhere:** on-premises physical, VMware, Azure, Google Cloud, Oracle Cloud or Alibaba.
- Traffic is compressed with LZ4, typically cutting transit size by 60 to 70 per cent.

> ON THE EXAM
>
> The staging area is deliberately cheap. It is small instances and EBS volumes, not a running copy of your production fleet.

Agent-based or agentless?

AGENT-BASED

The default path

- Replication agent on each source server
- Works from any supported environment
- Physical, virtual, or another cloud
- Needs install permission on each server

AGENTLESS

VMware only

- MGN vCenter Client in the data centre
- Snapshot-based, vCenter 6.7, 7.0 and 8.0
- Same OS support as agent-based
- MGN only, not Elastic Disaster Recovery

THE TIEBREAKER

Agentless exists for estates where agents cannot be approved. Outside VMware vCenter, it is not an option at all.

Testing, and why it is skipped

- **A test instance launches from replicated data while the source keeps running.** Users notice nothing.
- Replication continues during and after the test, so you can test repeatedly as configuration changes.
- **What testing finds:** hard-coded IP addresses, hardware-bound licences, on-premises dependencies, missing firewall rules.
- Test into an isolated subnet, so a test instance cannot talk to production systems it should not.

> ON THE EXAM
>
> Every objection to testing is about time. It costs a launch and an hour, against discovering the same faults during a cutover window.

Cutover and after

- **Cutover means a brief write pause, a final sync, then launching the real instance.** Minutes, not a maintenance weekend.
- **Post-launch actions** run automatically on migrated instances: install agents, join a domain, run validation.
- Predefined actions exist, including **App2Container for replatforming** and installing observability agents.
- **Finalising archives the source and stops replication.** Do it once you are confident, not on cutover day.

> ON THE EXAM
>
> Post-launch actions are how you avoid 800 servers each needing a human to install the same three agents afterwards.

Cost, and a naming note

- **Free for 2,160 hours per source server**, which is 90 days of continuous replication, starting when the agent is installed.
- **Staging infrastructure is charged throughout**, including during the free period. The service is free; the EC2 and EBS are not.
- After 90 days, roughly $0.042 per server-hour. A migration that drifts across many servers adds up.
- **Naming:** the service has been rebranded **AWS Transform MGN**. The exam's list still says Application Migration Service.

> ON THE EXAM
>
> The free period starts at agent install, not at cutover. Installing agents early on servers you will migrate last wastes it.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Migrate servers from another cloud	MGN. Sources can be anywhere
No agents may be installed, VMware estate	Agentless replication, vCenter Client
Cutover failed and nobody knew why	Test instances were never launched
Install agents on every migrated server	Post-launch actions
Convert Oracle to PostgreSQL	DMS and SCT, not MGN
Replication has run for six months	The 90-day free period has expired

ON THE EXAM

MGN moves a machine as it is. Anything that changes the engine, the schema or the architecture is a different tool.

> Meridian Freight must move 300 servers from another public cloud provider to AWS within four months, following a commercial dispute. The servers run a mix of Windows and Linux, and the applications must not be modified.
>
> The team assumes AWS migration tooling only works from on-premises data centres.

What should the solutions architect recommend?

A. Application Migration Service, installing the replication agent on the source servers in the other cloud

B. Export each server as an image, transfer it to S3, and import with VM Import/Export

C. Rebuild all 300 servers natively in AWS

D. Use AWS DataSync to copy the servers

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A MGN from the other cloud

MGN works anywhere the agent can reach AWS on TCP 443 and 1500, including Azure, Google Cloud and others. The team's assumption is simply wrong.

B Export and VM Import

A point-in-time copy, so the source keeps changing while you transfer. It also means a long outage per server at cutover.

C Rebuild natively

The applications must not be modified, and 300 rebuilds in four months is not achievable regardless.

D DataSync

Moves file and object data. It does not migrate servers, operating systems or applications.

THE SIGNAL

The team's stated assumption is the question. MGN is a cloud-to-cloud migration tool as much as an on-premises one.

> A financial services firm must migrate 450 VMware virtual machines running on vCenter 8. Its change control board will not approve installing any agent on production virtual machines, and the exception process takes six months.
>
> The migration must begin within four weeks.

What should the solutions architect recommend?

A. Deploy the MGN vCenter Client and use agentless replication

B. Begin the exception process for agent installation

C. Use Elastic Disaster Recovery in agentless mode instead

D. Export the VMs as OVA files and import them individually

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Agentless via the vCenter Client

Exactly the case agentless exists for. One client in the data centre, nothing on the VMs, and vCenter 8 is supported.

B Exception process

Six months against a four-week start. It also spends political capital to obtain something agentless replication already provides.

C DRS agentless

Agentless replication is available in MGN only. Elastic Disaster Recovery does not offer it, and DRS is for disaster recovery anyway.

D Export OVA files

450 point-in-time exports, manually imported, with the source drifting throughout. It does not scale and does not sync.

THE SIGNAL

The vCenter version matters. Agentless supports 6.7, 7.0 and 8.0, so vCenter 8 is deliberately in scope here.

> A cutover of 40 servers over a weekend failed. Twelve applications would not start, citing licence errors and unreachable database hosts. The team rolled back after nine hours. Replication had reported healthy for weeks, and no test instances were ever launched.
>
> The next cutover window is in three weeks.

What should the solutions architect recommend before the next attempt?

A. Launch test instances into an isolated subnet, resolve the licence and connectivity failures, and repeat until clean

B. Extend the cutover window to a full weekend so there is time to fix problems

C. Re-install the replication agents and restart replication from scratch

D. Migrate the 12 failing applications last, and cut over the other 28 servers

◆ PAUSE AND DECIDE

What should the solutions architect recommend before the next attempt?

A Test instances, iterate until clean

Non-disruptive and repeatable, with the source still serving users. Licence binding and unreachable hosts are exactly what a test finds.

B Longer window

Turns a nine-hour failure into a longer one. Debugging licence activation under cutover pressure is the worst possible time.

C Restart replication

Replication was healthy and is not the fault. Restarting discards weeks of transferred data for no benefit.

D Defer the twelve

Cuts over 28 servers that were also never tested. The same class of fault is waiting in them.

THE SIGNAL

Replication healthy for weeks is the planted detail. It tells you the copying worked and the application never did.

> A migration includes an Oracle database that must become Amazon Aurora PostgreSQL, and 60 application servers that must move unchanged. A junior architect proposes using Application Migration Service for all of it, reasoning that MGN replicates whole disks including the database files.
>
> The application code will be updated separately.

What should the solutions architect advise?

A MGN for the 60 application servers, and DMS with the Schema Conversion Tool for the Oracle to Aurora PostgreSQL move

B MGN for everything, as proposed

C DMS for everything, including the application servers

D Rebuild the application servers and use MGN only for the database

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A MGN for servers, DMS and SCT for the engine change

MGN moves machines as they are. Oracle to Aurora PostgreSQL is an engine conversion, which needs schema conversion and data migration.

B MGN for everything

MGN would faithfully replicate an Oracle database onto EC2 running Oracle. The requirement was Aurora PostgreSQL, which it cannot produce.

C DMS for everything

DMS migrates databases. It does not migrate application servers or operating systems.

D Rebuild the servers

The servers must move unchanged, and rebuilding 60 of them is work the requirement explicitly avoids.

THE SIGNAL

The junior architect's reasoning is correct and irrelevant. MGN does replicate database files, faithfully, onto the same engine.

> After migrating 200 servers, the operations team must install the CloudWatch agent, the Systems Manager agent and a security agent on every instance, and join each to Active Directory. Doing this manually took three days for the first wave of 40.
>
> Six more waves are planned.

What should the solutions architect recommend?

A. Configure post-launch actions in MGN so each migrated instance is configured automatically at launch

B. Write a Run Command document and execute it after each wave

C. Build a golden AMI containing the agents and migrate onto it

D. Accept the three days per wave as unavoidable

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Post-launch actions

Built for this. Each migrated instance is configured automatically as it launches, including test launches, so problems surface during testing.

B Run Command per wave

The strong near miss. It automates the work and still needs a human to trigger it, after launch, per wave.

C Golden AMI

MGN replicates the source disk. A migrated instance boots from replicated data, not from an AMI you prepared.

D Accept it

Three days times seven waves is three working weeks of manual configuration, with the errors that implies.

THE SIGNAL

Post-launch actions also run on test launches, so configuration failures appear during testing rather than at cutover.

> A team installed replication agents on all 600 servers in month one to "get replication out of the way". Migration waves run monthly, and the final wave completes in month seven. Finance queries an unexpected Application Migration Service charge appearing from month four.
>
> The team believed the service was free.

What should the solutions architect explain, and recommend?

A. The 90-day free period starts at agent installation; install agents per wave, shortly before each wave migrates

B. The charge is for staging EC2 and EBS, which cannot be avoided

C. The free period covers the whole account, not per server

D. Request a service credit, since the service is advertised as free

◆ PAUSE AND DECIDE

What should the solutions architect explain, and recommend?

A Free period starts at agent install

2,160 hours per server from install, so agents fitted in month one expire in month four. Install per wave and most servers stay free.

B It is staging infrastructure

Staging EC2 and EBS are charged, and separately and from day one. A charge appearing in month four is the free period expiring.

C Account-wide free period

The free period is per source server, not per account. That is what makes staggered installation worthwhile.

D Request a credit

The terms were met. The service was free for 90 days per server, and the team consumed that on servers it was not yet migrating.

THE SIGNAL

Month four is the arithmetic. Ninety days from a month-one install lands exactly there, which is the clue the scenario gives you.

What to carry out of this module

- **Replication converts the machine; testing proves the application.** They are separate questions.
- **Test instances are non-disruptive and repeatable.** The source keeps serving users throughout.
- **Agentless is VMware only, and MGN only.** Elastic Disaster Recovery does not offer it.
- **The free period starts at agent install**, not at cutover. Install per wave, not all at once.

> ON THE EXAM
>
> MGN moves a machine as it is. An engine conversion is DMS and SCT; MGN handles the application tier alongside it.

MODULE 36 · DOMAIN 4 - ACCELERATE WORKLOAD MIGRATION AND MODERNIZATION

Data Migration

Over the wire or in a box. The arithmetic decides, not the instinct.

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An honest warning, again

- **Snowmobile retired in 2024. Snowcone was discontinued in 2024**, with support ending in 2025.
- **Snowball Edge closed to new customers on 7 November 2025**, the same day as Migration Hub and Application Discovery Service.
- **AWS will discontinue Snowball support in all commercial Regions on 31 December 2026.** This one has an end date.
- **The Snow Family remains on the exam's in-scope list**, so it remains the examinable answer. Answer with it.

> ON THE EXAM
>
> AWS points new customers at DataSync, AWS Data Transfer Terminal, partner appliances, and Outposts for edge compute. None is in scope.

Divide the data by the bandwidth. If the answer is longer than the deadline, no amount of tooling helps.

The arithmetic, done

LINK	PER DAY, AT 100%	500 TB TAKES
100 Mbps	About 1 TB	Well over a year
1 Gbps	About 10.8 TB	About 46 days, perfect
10 Gbps	About 108 TB	About 5 days, perfect
10 Gbps at 60%	About 65 TB	About 8 days, realistic

ON THE EXAM

Always discount for real utilisation. A link that is also carrying production traffic is not available to you at 100%.

AWS DataSync

- **Online transfer between file systems and object stores**, on-premises to AWS, AWS to AWS, or from other clouds.
- **Handles the hard parts:** integrity validation, retries, scheduling, incremental sync and bandwidth throttling.
- Sources include NFS, SMB, HDFS and object storage. Targets include S3, EFS, FSx and other Regions.
- **Incremental after the first run**, which is what makes it a synchronisation tool rather than a one-off copy.

> ON THE EXAM
>
> This is what people rebuild badly with rsync and cron. Validation, retry and throttling are the parts a script always lacks.

The Snow Family, as examinable

- **Snowball Edge Storage Optimized: 210 TB of NVMe**, transferring up to 1.5 GB/s locally. The data migration device.
- **Snowball Edge Compute Optimized: 104 vCPUs and 28 TB**, for processing at the edge rather than bulk transfer.
- Order, receive, copy locally, ship back, AWS imports to S3. The delay is shipping, and it is predictable.
- Cluster several devices for larger datasets. Snowmobile, which handled 100 PB, no longer exists.

> ON THE EXAM
>
> The exam still tests this. A remote site with poor connectivity and a hard deadline is the classic Snowball scenario.

Transfer Family and Transfer Acceleration

- **AWS Transfer Family** gives managed SFTP, FTPS, FTP and AS2 endpoints backed by S3 or EFS.
- Its purpose is **ongoing exchange with third parties** who will not change their process, not bulk migration.
- **S3 Transfer Acceleration** routes uploads over the CloudFront edge network onto the AWS backbone.
- Acceleration helps distant clients over long distances. It does nothing for a client already close to the Region.

> ON THE EXAM
>
> Transfer Family replaces an SFTP server you were operating. It is a steady-state integration, not a migration tool.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Petabytes, poor connectivity, hard deadline	Snow Family. Do the arithmetic first
Ongoing sync from on-premises NFS	DataSync, incremental after first run
Partners upload by SFTP and will not change	AWS Transfer Family
Global users uploading to one bucket, slowly	S3 Transfer Acceleration
Copy between AWS Regions or services	DataSync also does AWS to AWS
A script with rsync and cron	DataSync: validation, retry, throttling

ON THE EXAM

Do the division before choosing. A scenario giving both a data size and a link speed is asking you to calculate.

> Meridian Freight must move 480 TB of archived shipment imagery from a regional depot to Amazon S3. The depot has a 200 Mbps internet connection shared with operations. The data centre closes in ten weeks.
>
> The team has proposed copying it over the internet using the AWS CLI.

What should the solutions architect advise?

A. Do the arithmetic: at 200 Mbps the transfer exceeds the deadline, so use Snow Family devices

B. Proceed with the CLI, using multipart upload for parallelism

C. Use S3 Transfer Acceleration to speed up the upload

D. Use DataSync over the existing connection

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Arithmetic, then Snow

200 Mbps is about 2 TB a day at perfection, so 480 TB is roughly 240 days. Ten weeks is 70. The link cannot do it.

B CLI with multipart

Parallelism uses the link more efficiently. It does not make the link faster, and the link is the constraint.

C Transfer Acceleration

Improves throughput over long distances by using the AWS backbone sooner. It cannot exceed the depot's 200 Mbps.

D DataSync

The best online tool available, and still bounded by the same 200 Mbps. Better software, same physics.

THE SIGNAL

Three options optimise the transfer and one questions whether it can finish. The deadline against the link speed is the question.

> A media company must move 90 TB from an on-premises NFS share to Amazon FSx for Lustre, and then keep the two synchronised for three months while applications cut over in waves. The site has a 10 Gbps Direct Connect link. Integrity must be verified for every file.
>
> An engineer has proposed a nightly rsync job.

What should the solutions architect recommend?

A. AWS DataSync, with scheduled incremental tasks and built-in integrity validation

B. The nightly rsync job as proposed

C. Snow Family devices for the initial 90 TB, then rsync

D. S3 Transfer Acceleration into S3, then copy to FSx

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A DataSync, scheduled incremental

90 TB over 10 Gbps is a day or two. DataSync then handles the three months of incremental sync, with validation and retries built in.

B Nightly rsync

Rebuilds DataSync badly. No integrity validation, no managed retries, no throttling, and somebody owns the script for three months.

C Snow then rsync

Ships a box for data that transfers in under two days over the existing link, then still needs the ongoing sync solved.

D Transfer Acceleration to S3

Adds a hop through S3 that the requirement does not need, and Transfer Acceleration does not target FSx.

THE SIGNAL

The arithmetic runs both ways. Here it rules Snow out, because 90 TB over 10 Gbps finishes long before a device could be shipped.

> After migrating to AWS, 340 external partners must continue sending daily files. They currently connect to an on-premises SFTP server with credentials issued years ago. Renegotiating integration with each partner is not feasible.
>
> The files must land in S3 for processing.

What should the solutions architect recommend?

A. AWS Transfer Family, providing a managed SFTP endpoint backed by S3

B. Run the SFTP server on EC2 in AWS

C. Ask partners to upload via an API Gateway endpoint

D. Use DataSync to collect files from each partner

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Transfer Family

Partners keep using SFTP exactly as they do now, files land directly in S3, and AWS operates the endpoint. No partner changes anything.

B SFTP on EC2

Preserves the protocol and keeps you patching, scaling and securing an SFTP server, which is what Transfer Family removes.

C API Gateway

Requires 340 partners to build new integrations. The scenario says renegotiating with each is not feasible.

D DataSync

Moves data between systems you control. It does not receive inbound uploads from third parties.

THE SIGNAL

This is not a migration question at all. It is a steady-state integration that happens to appear during a migration.

> A company collects large video files from field teams in Australia, Brazil and South Africa, uploaded to an S3 bucket in eu-west-1. Uploads are slow and frequently fail partway. Teams in Ireland and the Netherlands report no problems.
>
> Changing the bucket's Region is not possible for data residency reasons.

What should the solutions architect recommend?

A. Enable S3 Transfer Acceleration on the bucket, so distant uploads enter the AWS backbone at a nearby edge location

B. Create buckets in each Region and replicate to eu-west-1

C. Ship Snow Family devices to each field team

D. Use DataSync agents at each field location

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Transfer Acceleration

Distant clients are slow and nearby ones are fine, which is the exact profile. Uploads reach the AWS backbone at a nearby edge instead of crossing the internet.

B Regional buckets and replicate

Stores the data in Regions the residency requirement excludes, even briefly. It also multiplies buckets and transfer cost.

C Snow devices

For bulk one-off movement, not continuous uploads from field teams. Shipping a device per upload is not a process.

D DataSync agents

Suits a site with a file system to synchronise. Field teams uploading individual files are not that shape.

THE SIGNAL

The geography is the diagnosis. Nearby teams fine, distant teams failing, means the problem is the internet path, not the bucket.

> A research institute must move 2 PB from a remote observatory with a 50 Mbps satellite link. The site closes in eight months. A junior architect has proposed AWS Snowmobile, having read about it in a study guide.
>
> The organisation has used Snowball Edge devices before.

What should the solutions architect advise?

A. Snowmobile no longer exists; cluster multiple Snowball Edge Storage Optimized devices, each holding 210 TB

B. Order a Snowmobile as proposed

C. Transfer over the satellite link with DataSync bandwidth throttling

D. Compress the data and transfer over the satellite link

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Cluster Snowball Edge devices

Snowmobile was retired in 2024. Ten 210 TB devices cover 2 PB, and the organisation is an existing Snow customer so can still order them.

B Order a Snowmobile

It no longer exists. Study guides written before 2024 still describe it as the answer for petabyte-scale transfers.

C DataSync over satellite

50 Mbps is about 0.5 TB a day. 2 PB would take roughly eleven years. The tool is irrelevant at that ratio.

D Compress and transfer

Even at an implausible 50% ratio, five years instead of eleven. Compression does not bridge three orders of magnitude.

THE SIGNAL

Snowmobile is the most commonly cited dead answer in migration study material. Petabyte-scale is now clustered Snowball devices.

> A company must move 700 TB from a data centre to S3. It has a 10 Gbps Direct Connect link currently running at about 20% utilisation, and a nine-week window. Security prefers data never leaves the private link.
>
> The team is debating Snowball devices against an online transfer.

What should the solutions architect recommend?

A. DataSync over Direct Connect, throttled to leave headroom for production traffic

B. Snowball Edge devices, since 700 TB is large

C. S3 Transfer Acceleration over the internet

D. Split the data and use both Snowball and DataSync

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A DataSync over Direct Connect

80% of 10 Gbps is roughly 86 TB a day, so 700 TB takes about nine days against a nine-week window. It also stays on the private link.

B Snowball devices

Chooses by data size rather than by arithmetic. Four devices, shipped and returned, to do what the existing link does in nine days.

C Transfer Acceleration

Routes over the internet, which the security preference is specifically against. The private link is already there and faster.

D Both

Two workstreams, two sets of tooling and reconciliation, to solve a problem one of them handles comfortably alone.

THE SIGNAL

700 TB sounds like a Snow scenario and is not. The link speed, not the data size, decides, and the private path is a bonus.

What to carry out of this module

- **Divide data by bandwidth before choosing anything.** Roughly, TB per day equals Gbps times ten, then discount for reality.
- **Snowmobile is retired and Snowcone is gone.** Snowball support ends in commercial Regions on 31 December 2026.
- **DataSync is the ongoing-sync answer**, with validation, retries, scheduling and throttling a script will not have.
- **Transfer Family is steady-state integration**, not migration. Transfer Acceleration helps distant clients only.

> ON THE EXAM
>
> The Snow Family remains examinable while being closed to new customers and dated for retirement. Answer with it; know the reality.

MODULE 37 · DOMAIN 4 - ACCELERATE WORKLOAD MIGRATION AND MODERNIZATION

Database Migration

Schema and data are two migrations. The effort hides in the first.

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Moving the rows is the solved problem. **Converting the logic is where migrations overrun.**

Homogeneous or heterogeneous?

HOMOGENEOUS

Same engine, both ends

- Oracle to Oracle, MySQL to MySQL
- No schema conversion needed at all
- DMS runs it serverless, using native tools
- Carries functions, triggers and partitions

HETEROGENEOUS

The engine changes

- Oracle to Aurora PostgreSQL, and similar
- Schema must be converted first
- Code objects need review and rewriting
- The data move is the easy half

THE TIEBREAKER

Homogeneous migrations need no conversion step at all, which is why they are dramatically faster to plan and to execute.

The three task modes

- **Full load** copies existing data once. Suitable when the source can be frozen for the duration.
- **Full load plus CDC** copies everything, then keeps applying changes. This is the minimal-downtime pattern.
- **CDC only** applies ongoing changes to a target you populated some other way, such as a native backup restore.
- You cut over when replication lag is near zero, not when the full load finishes. Those are different moments.

> ON THE EXAM
>
> Change data capture is what turns a maintenance weekend into a few minutes. Without it, downtime equals the whole transfer.

Converting the schema

- **AWS SCT** is the standalone client you install locally, and it has **more features**.
- **DMS Schema Conversion** is built into the DMS console, and has **fewer features** than the client.
- AWS's own wording: it converts **"a majority of the database code objects"**. A majority is not all of them.
- The assessment report lists what could not be converted, with an effort estimate. Read it before committing to a date.

> ON THE EXAM
>
> Read as "everything", the remaining work looks like review. Read correctly, it is rewriting, and that changes the plan.

Proving it arrived

- **Premigration assessment** checks the source and target for problems before you start, such as unsupported data types.
- **Data validation** compares source and target rows during and after migration, reporting mismatches per table.
- A completed task is not a validated task. Completion means DMS finished; validation means the data agrees.
- Keep the source running and replicating after cutover, so reversing is possible if something is wrong.

> ON THE EXAM
>
> Validation is the answer whenever a scenario says row counts differ or data looks wrong. Do not reach for a manual reconciliation script.

Two honest notes

- **DMS Fleet Advisor support ended on 20 May 2026.** It stopped taking new customers a year earlier.
- **AWS names Migration Evaluator as its successor** for database portfolio assessment, as in module 33.
- **DMS itself is unaffected and fully supported.** Fleet Advisor was one capability, not the service.
- **DMS Serverless** removes replication instance sizing for heterogeneous work, scaling capacity automatically.

> ON THE EXAM
>
> Fleet Advisor answered which databases to migrate and how hard each would be. Migration Evaluator now answers that.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Oracle to Aurora PostgreSQL	Heterogeneous: SCT converts, DMS moves
Oracle to RDS for Oracle	Homogeneous: no conversion needed
Minimal downtime at cutover	Full load plus CDC
Target already populated from a backup	CDC only
Row counts do not match after migration	DMS data validation
Which databases should we migrate, and how hard?	Migration Evaluator, not Fleet Advisor

ON THE EXAM

Ask whether the engine changes first. That single fact decides whether there is a conversion project at all.

> Meridian Freight must move a 4 TB Oracle database to Aurora PostgreSQL. The application is being rewritten separately. Downtime must not exceed 30 minutes, and the database serves continuous freight bookings.
>
> The team has scheduled a full load over a weekend.

What should the solutions architect recommend?

A. Convert the schema with SCT, then run a DMS full load plus CDC task, cutting over when replication lag is near zero

B. Run the full load over the weekend as planned

C. Use DMS CDC-only, since the target is empty

D. Export with Oracle Data Pump and import into Aurora PostgreSQL

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A SCT, then full load plus CDC

The engine changes, so conversion comes first. Full load plus CDC keeps the target current, so cutover is minutes rather than the transfer time.

B Weekend full load

Downtime equals the whole transfer, and bookings are continuous. A weekend is far more than 30 minutes.

C CDC only

CDC-only applies ongoing changes to a target populated another way. An empty target has nothing for changes to apply to.

D Data Pump export and import

Oracle's native tools produce Oracle format. They cannot load into PostgreSQL, and there is no ongoing replication.

THE SIGNAL

Two requirements: the engine changes, and downtime is capped. The first needs SCT and the second needs CDC.

> A company is moving a self-managed PostgreSQL database on EC2 to Amazon RDS for PostgreSQL. It contains 200 stored procedures, 40 triggers and several partitioned tables. A junior architect has scoped two months for schema conversion using SCT.
>
> The database is 800 GB.

What should the solutions architect advise?

A. This is homogeneous, so no schema conversion is required; use DMS homogeneous migration, which carries secondary objects

B. Proceed with the two-month SCT conversion as scoped

C. Rewrite the stored procedures for RDS compatibility

D. Use full load plus CDC with SCT running in parallel

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Homogeneous; no conversion needed

PostgreSQL to PostgreSQL means the dialect is identical. DMS homogeneous migrations are serverless and carry procedures, triggers and partitions.

B Two-month conversion

Two months of effort converting PostgreSQL into PostgreSQL. The schema already runs on the target engine unchanged.

C Rewrite the procedures

They are already PostgreSQL procedures. RDS runs the same engine, so there is nothing to make compatible.

D CDC with SCT in parallel

Correct on the data movement, and still budgets conversion work that this migration does not require.

THE SIGNAL

Same engine at both ends is the whole answer. Self-managed to managed is a hosting change, not an engine change.

> An SCT assessment report for an Oracle to Aurora PostgreSQL migration shows that 78% of code objects converted automatically. The project plan allocates two weeks for "reviewing the converted schema" and assumes the remaining objects need minor adjustment.
>
> There are 1,400 code objects in total.

What should the solutions architect advise?

A. Roughly 300 objects need manual rewriting, not review; re-estimate the plan from the assessment report's effort ratings

B. Accept the plan; 78% is a good conversion rate

C. Re-run SCT with different settings to raise the conversion rate above 95%

D. Migrate the unconverted objects with DMS instead

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A About 300 need rewriting; re-estimate

22% of 1,400 is roughly 300 objects requiring human work. The assessment report rates each by effort, which is what a plan should be built on.

B 78% is good

It is a good rate, and the plan misreads the other 22% as adjustment rather than rewriting. Good rate, wrong estimate.

C Re-run with new settings

Settings do not make an Oracle package with no PostgreSQL equivalent convertible. The gap is dialect, not configuration.

D Move them with DMS

DMS migrates data, not code objects. It has no mechanism for converting a stored procedure.

THE SIGNAL

AWS says "a majority" of code objects convert. This scenario is what that phrase means once you multiply it out.

> A DMS task reports as complete. Spot checks by an analyst find that three tables have fewer rows in the target than the source, and one has NULL values where the source had data. The team is preparing a manual reconciliation script.
>
> The migration is Oracle to Aurora PostgreSQL.

What should the solutions architect recommend?

A Enable DMS data validation, which compares source and target rows and reports mismatches per table

B Write the reconciliation script as planned

C Re-run the full load from the beginning

D Accept the differences, since DMS reported the task complete

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A DMS data validation

Built for exactly this: it compares rows between source and target and reports mismatches per table, continuously, without anyone writing code.

B Reconciliation script

Rebuilds a feature DMS already has, on a deadline, and a hand-written comparison across a large schema is itself a source of errors.

C Re-run the full load

Repeats the migration without knowing what went wrong, so the same unsupported data types or missing keys produce the same result.

D Accept it

Complete means DMS finished. It does not mean the data agrees, which is precisely the distinction being tested.

THE SIGNAL

A completed task and a validated task are different claims. Complete is about the tool; validated is about the data.

> A 20 TB SQL Server database must move to RDS for SQL Server. Restoring a native backup to the target would take about four hours, which the business accepts. However, the source will continue receiving writes for two weeks afterwards while an application is tested.
>
> The two databases must stay in step during that fortnight.

What should the solutions architect recommend?

A. Restore the native backup, then run a DMS CDC-only task to apply ongoing changes for the fortnight

B. A DMS full load plus CDC task instead of the backup restore

C. A DMS full load task, repeated nightly for two weeks

D. Restore the backup and re-restore it at the end of the fortnight

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Native restore, then CDC-only

CDC-only exists for exactly this: a target populated by other means, kept in step by DMS thereafter. It is the fastest path for 20 TB.

B Full load plus CDC

Would work, and DMS full load of 20 TB is slower than a native restore the business has already accepted.

C Nightly full loads

Reloads 20 TB every night for a fortnight, and the target is stale for most of each day between runs.

D Restore twice

Leaves the target two weeks out of date throughout the testing, which is the requirement it fails.

THE SIGNAL

This is the scenario CDC-only exists for, and it appears rarely. Native tooling loads faster; DMS keeps it current.

> A company must assess 240 databases across its estate to decide which to migrate, what each would cost on AWS, and how complex each conversion would be. An architect has proposed AWS DMS Fleet Advisor, having used it on a previous project.
>
> The project starts next month.

What should the solutions architect advise?

A. Fleet Advisor support ended in May 2026; use Migration Evaluator for the portfolio assessment

B. Proceed with Fleet Advisor as proposed

C. Use DMS Schema Conversion assessment reports for all 240 databases

D. Assess manually with a questionnaire to each database owner

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Fleet Advisor has ended; use Migration Evaluator

Support ended 20 May 2026, and AWS names Migration Evaluator as the successor. It also produces the cost comparison the scenario asks for.

B Use Fleet Advisor

The console and resources are no longer accessible. It stopped accepting new customers a year before support ended.

C Schema Conversion reports

Gives conversion complexity per database, and answers neither which to migrate nor what each would cost.

D Questionnaires

Asks owners to estimate conversion complexity and AWS cost, which is what the tooling exists to calculate.

THE SIGNAL

DMS is fine. Fleet Advisor was one capability within it, and only that capability has gone.

What to carry out of this module

- **Ask whether the engine changes.** Homogeneous needs no conversion at all, which changes the timeline completely.
- **Schema conversion handles "a majority" of code objects.** Multiply the remainder out before agreeing a date.
- **Full load plus CDC is the minimal-downtime pattern.** CDC-only is for a target populated some other way.
- **Complete is not validated.** Data validation compares the rows; task completion only means DMS finished.

> ON THE EXAM
>
> Fleet Advisor's support ended in May 2026, and Migration Evaluator replaces it. DMS itself is unaffected.

MODULE 38 · DOMAIN 4 - ACCELERATE WORKLOAD MIGRATION AND MODERNIZATION

Modernisation During Migration

Patterns outlive products. This module is deliberately pattern-first.

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An honest note on the tooling

- **On 7 November 2025, Refactor Spaces, App2Container, Porting Assistant, .NET Refactoring and Microservice Extractor all closed to new customers.**
- **Mainframe Modernization's managed runtime closed too.** The self-managed runtime remains, for Rocket Software and Blu Age.
- All point at **AWS Transform**, the agentic modernisation service. It is not on the exam's in-scope list.
- **So learn the patterns.** Strangler fig, containerise in place, replatform or refactor. Those are what questions test.

> ON THE EXAM
>
> When an entire tooling layer moves at once, the durable knowledge is the pattern each product implemented.

Migrate then modernise, or modernise then migrate. **Doing both at once doubles the risk and the unknowns.**

The strangler fig pattern

- **Put a routing layer in front of the monolith**, then extract one capability at a time behind it.
- **Each extraction is independently reversible.** Route back to the monolith and nothing else changes.
- The monolith shrinks gradually until what remains is small enough to retire or leave alone.
- It needs the routing layer to be reliable, and it takes longer than a rewrite would if a rewrite ever finished.

> ON THE EXAM
>
> The value is that every step ships. A big-bang rewrite delivers nothing until it delivers everything, which is why so many never do.

Containerising in flight

- **Containerising is a replatform, not a refactor.** The application is unchanged; only its packaging differs.
- It genuinely helps with density, consistent deployment, and a path towards further change later.
- **A containerised monolith is still a monolith.** It does not become scalable or independently deployable by being packaged.
- Tooling existed to automate this and has largely closed. The pattern is unaffected, and can be done by hand.

> ON THE EXAM
>
> Containerising buys deployment consistency and density. It does not buy the things people expect from microservices.

Mainframe: replatform or refactor?

REPLATFORM

Keep the code, change the runtime

- COBOL keeps running, on AWS
- Rocket Software runtime
- Faster, lower risk, exits the hardware
- You still employ COBOL developers

REFACTOR

Convert the code itself

- COBOL and PL/I converted to Java
- AWS Blu Age tooling
- Slower, and removes the skills problem
- Output needs review and testing

THE TIEBREAKER

The deciding question is usually people, not technology. If nobody can maintain COBOL in five years, replatforming defers that.

When not to modernise

- **The application is being retired.** Modernising something with eighteen months left is spending on a sunk asset.
- **A hard deadline dominates.** Modernisation is unbounded work; a lease expiry is not.
- **The application is commodity.** Repurchase it rather than modernise something that gives you no advantage.
- **Nobody understands it.** Modernising an application no one can explain means rediscovering its behaviour by breaking it.

> ON THE EXAM
>
> Modernisation earns its place when the current architecture is genuinely the constraint. Otherwise it is expensive tidying.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Extract services incrementally, reversibly	Strangler fig with a routing layer
Hard deadline, and leadership wants microservices	Migrate first, modernise after
COBOL, and nobody left who writes it	Refactor to Java, not replatform
Exit the mainframe hardware quickly	Replatform, keeping the COBOL
Application retires in eighteen months	Do not modernise it
Containerise the monolith to make it scale	Correct the expectation first

ON THE EXAM

Ask what the modernisation is for. If the answer is because we are moving anyway, that is not a reason.

> Meridian Freight's order monolith must eventually become services, but it also must leave the data centre in seven months. Leadership wants both, arguing that migrating twice is wasteful. The application team has never built a microservice.
>
> The monolith works and is well understood.

What should the solutions architect recommend?

A. Rehost to meet the deadline, then apply the strangler fig pattern to extract services incrementally in AWS

B. Decompose into microservices as part of the migration, to avoid migrating twice

C. Containerise the monolith during migration, which delivers the microservice benefits

D. Delay the migration until the decomposition is complete

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Rehost, then strangler fig

The deadline is fixed and decomposition is not. Rehosting meets it, and extraction afterwards happens without a lease expiring behind it.

B Decompose during migration

Two simultaneous changes, an inexperienced team and a fixed deadline. When something breaks, nobody can tell which change caused it.

C Containerise for the benefits

A containerised monolith is still a monolith. It changes packaging, not architecture, and the expectation is wrong.

D Delay the migration

The data centre exit is not negotiable. Delaying it for an unbounded decomposition risks having nowhere to run.

THE SIGNAL

Migrating twice is not wasteful; it is sequencing. Leadership's argument sounds efficient and ignores that the deadline is fixed.

> A retailer wants to extract the pricing engine from a large monolith already running on AWS. The monolith must keep serving all other traffic throughout, and any extraction must be reversible within minutes if pricing behaviour differs.
>
> There is no deadline pressure.

What should the solutions architect recommend?

A. A strangler fig approach: a routing layer in front, pricing requests routed to a new service, reversible by changing the route

B. Rewrite the monolith as microservices in a single release

C. Duplicate the monolith and remove everything except pricing

D. Leave the monolith intact and scale it vertically

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Strangler fig with routing

One capability moves, everything else is untouched, and reversal is a routing change measured in minutes. Every requirement is met directly.

B Single-release rewrite

The opposite of reversible. A big-bang rewrite cannot be undone in minutes, and it changes everything at once.

C Duplicate and strip

Produces a second monolith carrying all the original coupling. It is a copy, not an extraction.

D Scale vertically

Declines to extract anything. It may be reasonable elsewhere, and it answers none of the stated requirements.

THE SIGNAL

Reversible within minutes is the phrase. Only a routing change is that fast, which is precisely what the pattern provides.

> A bank runs core account processing on a mainframe in COBOL. Its four remaining COBOL developers retire within three years and the skill is not replaceable in its market. The business accepts a multi-year programme and has funded it.
>
> The mainframe hardware contract has four years to run.

What should the solutions architect recommend?

A. Refactor: convert the COBOL to Java with automated tooling, then review, test and operate it with available skills

B. Replatform: keep the COBOL and run it on a managed runtime on AWS

C. Retain the mainframe and recruit COBOL developers

D. Repurchase a commercial core banking platform

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Refactor to Java

The constraint is skills, not hardware. Only refactoring removes the COBOL dependency, and the funding and timescale to do it exist.

B Replatform

Exits the hardware and keeps the COBOL, so in three years the code runs on AWS and nobody can maintain it.

C Recruit COBOL developers

The scenario says the skill is not replaceable in its market. Recruiting into a shrinking pool defers the problem at rising cost.

D Repurchase

Core account processing is a bank's most differentiated and regulated system. It is the least suitable candidate for a package.

THE SIGNAL

Read which constraint has the deadline. The hardware has four years; the people have three, and only one option addresses people.

> During migration planning, a team proposes containerising a Java monolith with App2Container, expecting it to become horizontally scalable and independently deployable. The monolith holds session state in memory and writes to a local filesystem.
>
> The migration deadline is five months.

What should the solutions architect advise?

A. Correct the expectation: containerising changes packaging, not architecture, and the state assumptions must be fixed first

B. Proceed; containerising will make it horizontally scalable

C. Containerise and run a single replica to avoid the state problem

D. Refactor the monolith into microservices within the five months

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Correct the expectation

In-memory session state and local filesystem writes prevent horizontal scaling regardless of packaging. Externalise state first, or accept it does not scale.

B It will scale

A second replica loses sessions and diverges on local files. Packaging does not remove state assumptions.

C Single replica

Honest about the constraint and delivers none of the benefits claimed. It is a container running a monolith on one node.

D Refactor in five months

Overcorrects. A full decomposition inside a migration deadline is exactly what module 34 warns against.

THE SIGNAL

The state details are planted deliberately. In-memory sessions and local file writes are what actually block scaling.

> A migration portfolio includes a claims processing application scheduled for replacement by a purchased platform in eighteen months. The migration team has scoped nine months of work to containerise it and move it to ECS during the data centre exit.
>
> A simple rehost would take three weeks.

What should the solutions architect recommend?

A. Rehost it in three weeks; do not modernise an application being retired in eighteen months

B. Containerise as scoped, since it will run better on ECS

C. Retain it on-premises until the replacement arrives

D. Accelerate the replacement so the application never migrates

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Rehost in three weeks

Nine months of modernisation on something with eighteen months to live, then thrown away. Rehosting spends three weeks and meets the exit.

B Containerise as scoped

Nine months of engineering with roughly nine months of use afterwards. The investment never returns.

C Retain on-premises

The data centre is closing. Retain needs somewhere to retain it, which is the same objection as module 34.

D Accelerate the replacement

Changing another programme's timeline to avoid three weeks of rehosting work. It is not an architectural recommendation.

THE SIGNAL

Eighteen months against nine months of work is the arithmetic. Modernisation must have enough remaining life to repay it.

> A company must exit its mainframe within twenty months because the hardware lease ends and will not be renewed. It employs a stable team of COBOL developers who are not retiring. The application is well documented and works.
>
> A consultant has proposed a full refactor to Java.

What should the solutions architect advise?

A Replatform: run the existing COBOL on a managed runtime on AWS, meeting the deadline and keeping the team productive

B Refactor to Java as proposed

C Renegotiate the hardware lease

D Repurchase a packaged equivalent

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Replatform, keep the COBOL

The constraint is hardware, not skills. The team is stable, the code works, and replatforming exits the mainframe well inside twenty months.

B Refactor to Java

Solves a skills problem this company does not have, on a fixed deadline, and produces Java the COBOL team must then learn to maintain.

C Renegotiate the lease

The scenario says it will not be renewed. Changing a stated constraint is not an architectural answer.

D Repurchase

Replacing a working, well-documented core system inside twenty months, with all the process change that implies.

THE SIGNAL

Pair this with scenario three. Same two options, opposite answers, because the binding constraint changed from people to hardware.

What to carry out of this module

- **Migrate then modernise** unless the change is small and bounded. Doing both at once doubles the unknowns.
- **Strangler fig ships every step** and reverses with a routing change. A big-bang rewrite delivers nothing until it delivers everything.
- **A containerised monolith is still a monolith.** In-memory state and local files block scaling regardless of packaging.
- **Mainframe: replatform keeps the code, refactor keeps the skills.** Decide by which constraint binds first.

> ON THE EXAM
>
> The modernisation tooling layer largely closed in November 2025. Patterns are what the exam tests, and what survives a rename.

MODULE 39 · DOMAIN 4 - ACCELERATE WORKLOAD MIGRATION AND MODERNIZATION

Cutover and Validation

Written for people who will read it tired, at three in the morning.

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Every decision you have not made in advance, you will make at 3am, tired, under pressure. **Decide first.**

What a cutover runbook contains

- **Every activity, in sequence, with planned start and end times and a named owner.** A person, not a team.
- **The outage window**, and which tasks can run in parallel to compress it.
- **A rollback plan for exceeding the window**, not only for failing. Running late is the more common problem.
- AWS advises **at least two walkthrough meetings** with every task owner, including integration testers who sign off.

> ON THE EXAM
>
> "The networking team updates DNS" is not an owner. At 3am you need a name and a phone number, and that person needs to expect the call.

All at once, or phased?

ALL AT ONCE

One switch, everybody moves

- Flip DNS or the load balancer
- Necessary when both cannot run
- Hostname conflicts, licences, domain auth
- Rollback is another full switch

PHASED

A percentage at a time

- On-premises capacity stays available
- Rollback is a routing change
- Load increases gradually, so you can watch
- Both environments run, and cost, together

THE TIEBREAKER

AWS names three things that force all-at-once: hostname conflicts, licensing, and domain authentication. Look for those specifically.

The DNS problem

- **A record with a 24-hour TTL takes up to 24 hours to move traffic**, and the same to move it back.
- **Lower TTLs to 60 or 300 seconds days before cutover**, not on the night. The change itself needs the old TTL to expire.
- Raise them again once the migration is stable, since short TTLs mean far more queries.
- Some resolvers ignore TTLs. Plan for stragglers, and keep the old environment answering for a while.

> ON THE EXAM
>
> This is the single most common cutover mistake, and it is entirely preventable a week earlier. Module eight's caching problem, at the worst moment.

Rollback, decided in advance

- **Define checkpoints with criteria**: at this time, if this is not true, we roll back. Written down beforehand.
- **Name who decides.** A rollback argued about by six people at 4am is a rollback that happens too late.
- **A dual-write strategy** keeps both systems current, so reverting does not lose the transactions taken since cutover.
- **The point of no return** is a real moment. Know when it passes, and make sure everyone present knows too.

> ON THE EXAM
>
> The hardest part of a rollback is admitting it is needed. Predefined criteria remove that judgement from the tired people in the room.

Validation, and hypercare

- **Script the checks.** DNS answer, service ports, one real transaction. Nobody validates well from memory at 3am.
- **Validate functionally, then on performance, then on data.** Working is not the same as fast, and neither is the same as correct.
- **Hypercare** is a defined period of heightened support afterwards, with the migration team still engaged rather than dispersed.
- The application owner signs off, not the migration team. They know what correct looks like.

> ON THE EXAM
>
> A scripted check that runs in two minutes gets run. A checklist of forty manual steps gets skipped at the end of a long night.

Decommissioning, later than you think

- **Keep the source through at least one full business cycle.** Month-end and quarter-end jobs surface weeks after cutover.
- **Power off, observe, back up, then delete.** In that order, with a defined gap between each step.
- Exit hypercare before decommissioning begins. They are separate decisions, taken at separate times.
- Destroying the rollback in week one is the classic error. It converts a recoverable problem into an incident.

> ON THE EXAM
>
> A migration that went perfectly for three weeks can still fail at month end, on a job nobody remembered existed.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
Traffic did not move for hours after cutover	DNS TTL was not lowered in advance
Hostname conflict or licence bound to a host	All-at-once cutover is forced
Gradual load increase and easy rollback	Phased cutover
Rollback was debated during the incident	Predefined checkpoints and a named decider
Month-end job failed after decommissioning	Source kept for less than a business cycle
The runbook says "the DBA team"	Name individuals, per AWS guidance

ON THE EXAM

Ask what was decided in advance. Almost every cutover failure traces back to a judgement left until the night itself.

> Meridian Freight cut over its booking site at 02:00 by updating a Route 53 record. At 09:00 the on-premises servers were still receiving roughly 60% of traffic, and the two environments had diverged. The record's TTL was 86,400 seconds.
>
> The next cutover is in three weeks.

What should the solutions architect recommend?

A. Lower TTLs to 60 seconds at least a day before the next cutover, and keep the old environment answering during transition

B. Lower the TTL to 60 seconds immediately before the next cutover

C. Use a shorter maintenance window so divergence has less time to occur

D. Shut down the on-premises servers at cutover so traffic has nowhere to go

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Lower TTLs a day ahead

The lowering itself only takes effect once the old 86,400-second value expires. Doing it a day early means the new value is live at cutover.

B Lower it immediately before

The sharpest distractor. Resolvers caching the old record still hold 86,400 seconds, so the change is invisible for up to a day.

C Shorter window

The maintenance window was not the problem. Traffic kept arriving for seven hours because of caching, not because the window was long.

D Shut down on-premises

Turns a divergence problem into an outage for 60% of users, and destroys the rollback at the same moment.

THE SIGNAL

Option B is the trap: lowering a TTL is itself subject to the old TTL. That recursion is what makes advance planning mandatory.

> During a cutover, response times were three times normal. The team spent 90 minutes debating whether to roll back, with the migration lead arguing to continue and the application owner arguing to revert. By the time they decided, the point of no return had passed.
>
> No rollback criteria had been written down.

What should the solutions architect recommend for future cutovers?

A. Define rollback checkpoints with measurable criteria and a single named decision-maker, agreed before cutover night

B. Give the migration lead sole authority to continue

C. Extend the maintenance window so there is more time to debate

D. Roll back automatically whenever response times exceed normal

◆ PAUSE AND DECIDE

What should the solutions architect recommend for future cutovers?

A Checkpoints, criteria, named decider

Moves the judgement to a calm moment. The debate happens in a planning meeting, and cutover night executes a decision already taken.

B Migration lead decides

Fixes the deadlock and picks the person with the strongest incentive to continue. Authority without criteria is still a judgement at 4am.

C Longer window

More time to have the same argument. The debate was not short of time; it was short of an agreed threshold.

D Automatic rollback on slowness

Overcorrects. Some slowness is expected as caches warm, so an automatic trigger would revert healthy cutovers.

THE SIGNAL

The failure is not that they chose wrongly. It is that a judgement requiring calm analysis was scheduled for 4am.

> An application's licence is bound to a specific hostname, and the vendor will not issue a second licence for a parallel environment. The application also uses a fixed NetBIOS name that cannot exist twice in the domain.
>
> The team's standard practice is a phased cutover with gradual traffic shifting.

What should the solutions architect recommend?

A. An all-at-once cutover, since the environments cannot run in parallel

B. A phased cutover as usual, using weighted DNS records

C. Rename the migrated application to avoid the conflict

D. Negotiate a second licence with the vendor

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A All-at-once

AWS names hostname conflicts, licensing and domain authentication as exactly the constraints that force this approach. Two of the three are present.

B Phased as usual

Phased requires both environments running. The licence and the NetBIOS name each independently prevent that.

C Rename it

The licence is bound to the hostname, so renaming invalidates the licence. It solves one constraint by breaking the other.

D Second licence

The scenario says the vendor will not issue one. Worth asking, and not something to plan the cutover around.

THE SIGNAL

Phased cutover is better in almost every way and requires both environments to coexist. When they cannot, the choice is made for you.

> A cutover completed successfully. Three weeks later, the month-end reconciliation job failed, because it depended on a file share on a decommissioned server. The source environment had been deleted in week one to save cost.
>
> Recovery took four days and required restoring from an off-site backup.

What should the solutions architect recommend for future cutovers?

A. Retain the source through at least one full business cycle, then power off, observe, back up, and only then delete

B. Improve the dependency mapping so file shares are not missed

C. Take more frequent backups of the source before decommissioning

D. Delete the source only after the application owner signs off at cutover

◆ PAUSE AND DECIDE

What should the solutions architect recommend for future cutovers?

A One full business cycle, staged deletion

Month-end and quarter-end jobs are exactly what a three-week retention misses. The staged sequence keeps recovery cheap throughout.

B Better dependency mapping

Worth doing and it would not have caught this. A job that runs monthly makes no network connections during the discovery window.

C More backups

They had a backup; recovery still took four days. The problem is time to recover, not whether a copy existed.

D Sign-off at cutover

The owner did sign off, and correctly, because everything worked. The fault appeared three weeks later.

THE SIGNAL

Option B is the tempting one, and a monthly job is invisible to discovery that ran for a fortnight. Retention covers what mapping cannot.

> A cutover runbook lists 60 tasks. Owners are recorded as "Network team", "DBA team" and "App team". On the night, three tasks stalled because nobody on shift knew they were responsible, and one owner was on annual leave.
>
> The runbook had been circulated by email but never walked through.

What should the solutions architect recommend?

A. Name individual people per task, and hold at least two walkthrough meetings with every task owner present

B. Add more detail to each task description

C. Assign a single cutover manager to perform all 60 tasks

D. Automate all 60 tasks so no owners are needed

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Name people, hold walkthroughs

Exactly AWS's guidance. A named person can be checked for availability, and a walkthrough is where the leave clash surfaces beforehand.

B More task detail

The tasks were understood. Nobody knew they owned them, which is an assignment problem rather than a documentation one.

C One person does everything

Sixty tasks across networking, databases and applications, performed by one person, in one window. It removes parallelism and creates a bottleneck.

D Automate everything

Desirable over time and not achievable before the next cutover. Some tasks are approvals and validations by design.

THE SIGNAL

Both failures came from naming teams rather than people. A team cannot be on annual leave, which is exactly why the gap went unnoticed.

> A retailer is cutting over an order platform and wants to increase load on AWS gradually while watching performance, reverting quickly if anything degrades. Both environments can run simultaneously, and the database is being kept in sync by DMS with change data capture.
>
> The business is nervous after a previous failed cutover.

What should the solutions architect recommend?

A. A phased cutover shifting a growing percentage of traffic, with rollback by reverting the routing weights

B. An all-at-once cutover during a maintenance window

C. Run both environments indefinitely with no cutover

D. Cut over during the quietest hour to reduce exposure

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Phased, with weighted routing

Both environments can run, so the constraint that forced all-at-once in scenario three is absent. Rollback becomes a routing change.

B All-at-once

Available, and it discards the gradual load increase and the fast rollback the business specifically asked for.

C Never cut over

Pays for both environments forever and leaves the data centre exit unresolved. Phased is a transition, not a destination.

D Quietest hour

Sensible scheduling, and it is still all-at-once. A quiet hour reduces how many people notice, not the blast radius.

THE SIGNAL

Pair this with scenario three. Same choice, opposite answer, because here nothing prevents the two environments coexisting.

What to carry out of this module

- **Decide in advance what you will not want to decide at 3am.** Success criteria, rollback triggers, and who calls it.
- **Lower DNS TTLs days beforehand.** Lowering a TTL is itself subject to the old TTL, which is why it cannot wait.
- **Name people, not teams**, and walk the runbook through at least twice with every owner present.
- **Keep the source through a full business cycle.** Month-end jobs are invisible to dependency mapping.

> ON THE EXAM
>
> Phased cutover is better wherever it is possible. Hostname conflicts, licensing and domain authentication are what make it impossible.

MODULE 40 · DOMAIN 4 - ACCELERATE WORKLOAD MIGRATION AND MODERNIZATION

Hybrid and Coexistence

Closing Domain 4. The transition period is an architecture, not an inconvenience.

DOMAIN 4 of 4 EXAM WEIGHT 20% MODULE 40 of 44

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The hybrid period is not a gap between two architectures. **It is one, and it will outlive your estimate.**

Three things must span both estates

- **Network.** Direct Connect for predictable bandwidth, with VPN as backup, and Transit Gateway as the hub. Modules six and seven.
- **Identity.** One directory both sides trust, so a person has one account. AD trusts or Managed AD, from module four.
- **DNS.** Resolution must work in both directions, which is Route 53 Resolver inbound and outbound endpoints. Module eight.
- Get any of these wrong and every wave afterwards inherits the problem, because every wave depends on all three.

> ON THE EXAM
>
> These are foundation work, done once, before wave one. Retrofitting identity across a half-migrated estate is considerably worse.

Data gravity, and the chatty dependency

- **Split a chatty pair across estates and every call now crosses the link**, adding latency and cost to each one.
- **Data has gravity.** Applications want to be near their data, so moving the database usually decides the wave.
- Dependency mapping from module 33 is what tells you which pairs are chatty. Guessing produces the problem.
- A latency-sensitive dependency may add milliseconds per call. At thousands of calls per transaction, that is seconds.

> ON THE EXAM
>
> The application that worked fine on-premises and is slow in AWS is usually still calling something on-premises, thousands of times.

What a long hybrid costs

- **Two of everything.** Two monitoring stacks, two identity paths, two runbooks, two on-call rotas.
- **Data transfer across the link**, charged per gigabyte, for every conversation you split. Module twenty two.
- **The data centre does not get cheaper as it empties.** Rent, power and support contracts are largely fixed.
- Attention is the hidden cost. Teams running two estates are not improving either of them.

> ON THE EXAM
>
> A half-empty data centre costs nearly what a full one does. That is why finishing matters more than starting well.

When something genuinely cannot move

- **AWS Outposts** puts AWS infrastructure and APIs in your own facility, for workloads with local dependencies or residency requirements.
- AWS names **migration of applications with local system interdependencies** as an Outposts use case directly.
- Second-generation racks run C7i, M7i and R7i, roughly 40% faster than first-generation. Servers cover branch and retail sites.
- It needs real facilities: floor loading, 10 kVA power, cooling, and a service link of at least 500 Mbps to the parent Region.

> ON THE EXAM
>
> Outposts is for the workload that cannot leave, not for the workload nobody wants to move. Those are very different problems.

One operational model

- **Systems Manager hybrid activations** make on-premises servers managed nodes, so patching, inventory and access are identical. Module 26.
- **One observability stack**, with on-premises telemetry flowing to CloudWatch, so an incident has one timeline.
- One set of runbooks, one on-call rota, one place to look. The alternative is two of each for years.
- This is the strongest practical argument for hybrid tooling, and it is usually made too late to help.

> ON THE EXAM
>
> An incident spanning both estates with two toolsets means two timelines and no correlation. That is module 23's problem, doubled.

Reading the scenario

THE SCENARIO SAYS	IT IS POINTING AT
An app is slow after migration, calls on-premises	A chatty dependency split across estates
Users need accounts in both environments	One directory, with a trust
On-premises cannot resolve AWS private names	Route 53 Resolver outbound endpoint
Must sit beside factory equipment	AWS Outposts
Two monitoring stacks, two on-call rotas	One operational model, hybrid activations
The hybrid has run for three years	Finish it. Half-migrated is the worst state

ON THE EXAM

Most coexistence problems are one of three: something split that should not be, something that cannot span, or something never finished.

> Meridian Freight migrated its quoting service to AWS in wave two. It now takes 4 seconds to produce a quote, against 300 milliseconds before. The service makes roughly 1,200 calls to a pricing database that remains on-premises, per quote.
>
> The Direct Connect link adds about 3 milliseconds each way.

What should the solutions architect recommend?

A. Migrate the pricing database into the same wave as the quoting service, keeping the chatty pair together

B. Upgrade the Direct Connect link to a higher bandwidth

C. Add a second Direct Connect connection for redundancy

D. Increase the quoting service's instance size

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Move the database with it

1,200 calls at 6 ms round trip is about 7 seconds of pure latency. Only removing the crossing fixes it, which means the pair moves together.

B More bandwidth

Bandwidth is throughput, not latency. A wider link does not shorten the distance the packets travel.

C Second connection

Resilience, not speed. Two links of the same latency give the same round trip.

D Bigger instance

The service is waiting on the network, not short of CPU. A larger instance waits at the same speed.

THE SIGNAL

The arithmetic is given to you: 1,200 calls times 6 ms. Bandwidth and latency are different properties, and only one is the problem.

> During a three-year migration, staff will need to access applications in both the data centre and AWS. The proposal is to create separate AWS-native accounts for every user, with a manual process to keep passwords aligned.
>
> There are 6,000 staff and an existing on-premises Active Directory.

What should the solutions architect recommend?

A. Extend the existing directory: AWS Managed Microsoft AD with a trust to on-premises, so each person keeps one identity

B. Create the separate accounts as proposed, with password synchronisation

C. Wait until the migration completes, then build identity in AWS

D. Give every user an IAM user with an access key

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Managed AD with a trust

One identity per person across both estates, with existing groups and policies intact. Joiners and leavers stay a single process.

B Separate accounts, synced

12,000 identities and a manual sync for three years. A leaver disabled in one directory keeps access through the other.

C Wait until the end

Staff need access to AWS applications from wave one. Deferring identity means every wave invents its own arrangement.

D IAM users with keys

Long-lived credentials for 6,000 people, which module 28 spent a module removing. It also does not authenticate to on-premises applications.

THE SIGNAL

Two identities per person means a leaver can be disabled in one and still have access through the other. That is the security failure.

> After wave one, on-premises applications cannot resolve the private DNS names of migrated services, and migrated services cannot resolve on-premises hostnames. Teams have been adding entries to hosts files as a workaround, and several are now stale.
>
> There is a Direct Connect link and a Route 53 private hosted zone.

What should the solutions architect implement?

A Route 53 Resolver inbound and outbound endpoints with forwarding rules in both directions

B A script that distributes an updated hosts file to all servers nightly

C Move all DNS to the on-premises servers

D Use public DNS names for the migrated services

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Inbound and outbound endpoints

Inbound lets on-premises query AWS; outbound lets AWS query on-premises. The problem is bidirectional, so both are needed.

B Distributed hosts files

Automates the workaround. Hosts files bypass DNS entirely, so every IP change becomes a fleet-wide update that is stale by definition.

C All DNS on-premises

On-premises servers cannot resolve Route 53 private hosted zones natively. It also makes the data centre a dependency you are trying to remove.

D Public DNS names

Exposes internal service names publicly, and private IP addresses in public DNS do not resolve usefully anyway.

THE SIGNAL

Module eight's distinction: inbound means on-premises asks AWS, outbound means AWS asks on-premises. Both directions fail here, so both endpoints.

> A manufacturer must close its data centre, but one application controls production line equipment over a protocol requiring sub-millisecond response, physically wired within the plant. Everything else can migrate.
>
> The plant has a suitable room, power and cooling, and a 1 Gbps connection.

What should the solutions architect recommend?

A. AWS Outposts in the plant for the control application, migrating everything else to the Region

B. Migrate the control application and accept the latency over Direct Connect

C. Retain a small data centre for the control application indefinitely

D. Use a Local Zone for the control application

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Outposts in the plant

AWS infrastructure physically beside the equipment, with the same APIs and operational model as the Region. The facility requirements are met.

B Accept the latency

Sub-millisecond over any wide-area link is physically impossible. The requirement is not a preference to be negotiated.

C Keep a small data centre

Meets the latency need and keeps a facility, a support contract and a second operational model for one application.

D Local Zone

Local Zones sit in metropolitan areas, not in your plant. Closer than a Region, and still not on the factory floor.

THE SIGNAL

Outposts is for what cannot leave. Sub-millisecond to wired equipment is a physical constraint, not an architectural preference.

> An incident spanning both estates took six hours to diagnose. On-premises servers are monitored by a legacy tool, AWS resources by CloudWatch, and correlating the two timelines was done manually in a spreadsheet. Patching and access follow two different processes.
>
> The hybrid period has two more years to run.

What should the solutions architect recommend?

A. Register on-premises servers as managed nodes with SSM hybrid activations, and send their telemetry to CloudWatch for one timeline

B. Deploy the legacy monitoring tool into AWS as well

C. Accept two toolsets, since the hybrid is temporary

D. Accelerate the migration so the hybrid ends sooner

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Hybrid activations plus one timeline

One operational model across both estates: the same patching, inventory and session access, and a single correlated incident timeline.

B Legacy tool into AWS

Defensible, and it means licensing and running that tool across a growing estate while losing AWS-native integration.

C Accept two toolsets

Two more years is not temporary. It is longer than most teams keep the same tooling voluntarily.

D Accelerate the migration

Changes the programme timeline to avoid an operational problem that has a direct fix.

THE SIGNAL

Two more years is the planted detail. It converts "temporary inconvenience" into a design decision that needs making.

> A migration begun three years ago is 70% complete and has stalled. The remaining 30% are applications nobody wants to own. The data centre still costs $2.1m a year, barely less than when it was full, and two operational models are still being run.
>
> Leadership has asked whether to simply stop where things are.

What should the solutions architect advise?

A. Finish it: assign the remaining 30% to the seven Rs, retiring and repurchasing aggressively, because half-migrated is the most expensive state

B. Stop where things are and run permanently hybrid

C. Migrate the remaining 30% as-is with no assessment, to finish quickly

D. Move the migrated 70% back on-premises to consolidate

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Finish, with aggressive retire and repurchase

The data centre cost is fixed until it closes, so only finishing removes it. Applications nobody wants to own are prime retire and repurchase candidates.

B Stop and stay hybrid

Locks in $2.1m a year and two operational models permanently, to avoid finishing 30% of the work.

C Migrate as-is, no assessment

Fast, and it migrates applications that should be retired or repurchased. Speed without assessment moves the waste rather than removing it.

D Move back on-premises

Discards three years of work and re-provisions capacity in a facility the business wants to exit.

THE SIGNAL

The data centre bill barely fell at 70% migrated, which is the whole argument. The saving arrives only when the doors close.

What to carry out of this module

- **Design the transition deliberately.** Network, identity and DNS must span both estates before wave one.
- **Never split a chatty pair.** Bandwidth is not latency, and a wider link does not shorten the distance.
- **Run one operational model**, using hybrid activations and one observability stack. Two of everything for years is the real cost.
- **A half-empty data centre costs nearly what a full one does.** The saving arrives when it closes, not before.

> ON THE EXAM
>
> Outposts is for what cannot leave. Applications nobody wants to own are usually retire or repurchase candidates, not migration ones.

MODULE 41 · EXAM STRATEGY - ALL DOMAINS

Confusion Pairs

Thirty pairs, each divided by one property. Learn the divider, not the definitions.

DOMAIN Cross-cutting EXAM WEIGHT All four MODULE 41 of 44

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For each pair, one property decides. **Learn the divider**, and the definitions stop mattering.

Governance and identity

PAIR	THE ONE PROPERTY THAT DIVIDES THEM
SCP vs RCP	SCP follows the identity; RCP follows the resource
Boundary vs session policy	Boundary is permanent on the identity; session policy is per-session
Preventive vs detective	Preventive refuses the call; detective reports afterwards
AD Connector vs Managed AD	Connector stores nothing and cannot be shared; Managed AD can hold trusts
User pool vs identity pool	User pool is who you are; identity pool grants AWS credentials
RAM vs VPC sharing	RAM shares a resource; VPC sharing shares subnets, owner keeps the plumbing

ON THE EXAM

The SCP and RCP divider is the one to over-learn. Cross-account access needs both sides to allow it, plus the KMS key.

Networking

PAIR	THE ONE PROPERTY THAT DIVIDES THEM
Peering vs Transit Gateway	Peering is non-transitive; TGW is a router with route tables
Association vs propagation	Association is which table I read; propagation is which tables learn about me
Inbound vs outbound Resolver	Inbound: on-premises asks AWS. Outbound: AWS asks on-premises
Gateway vs interface endpoint	Gateway is a route entry, VPC-only, free. Interface is an ENI address
Geolocation vs latency routing	Geolocation is a legal guarantee; latency is a performance preference
CloudFront vs Global Accelerator	CloudFront caches; Global Accelerator does not, and handles any protocol

ON THE EXAM

Association versus propagation is the most missed pair on the exam. Say it aloud until it is automatic.

Building

PAIR	THE ONE PROPERTY THAT DIVIDES THEM
Nested stacks vs cross-stack refs	Nested share one lifecycle; exports let stacks deploy independently
DeletionPolicy vs UpdateReplacePolicy	One covers stack deletion, the other covers replacement on update
Custom resource vs hook	A custom resource does work; a hook judges whether to proceed
Reserved vs provisioned concurrency	Reserved is capacity contention; provisioned is cold-start latency
Queue vs stream	A queue message is consumed and gone; a stream record stays
SNS vs EventBridge	SNS filters attributes; EventBridge routes on the event body

ON THE EXAM

Reserved concurrency is a ceiling and a floor at once. That dual nature is why one setting solves a noisy-neighbour problem.

Operating

PAIR	THE ONE PROPERTY THAT DIVIDES THEM
Governance vs compliance mode	Governance can be overridden by a privileged user; compliance cannot
GuardDuty vs Detective	GuardDuty detects; Detective explains a finding something else produced
Run Command vs State Manager	Run Command is now, once; State Manager is continuously, forever
Canaries vs RUM	Canaries run with no users; RUM needs real users
Control plane vs data plane	Control plane creates things; data plane uses what exists
Coverage vs utilisation	Low utilisation: bought too much. Low coverage: bought too little

ON THE EXAM

Coverage and utilisation point in opposite directions. Reading only one is how finance teams reach confident wrong conclusions.

Migrating

PAIR	THE ONE PROPERTY THAT DIVIDES THEM
MGN vs DRS	MGN migrates permanently; DRS protects production that stays put
Agent vs agentless discovery	Only the agent sees processes, which is what dependency mapping needs
Rehost vs relocate	Rehost changes where it runs; relocate changes only the hypervisor
Replatform vs refactor	Replatform changes a component; refactor changes the architecture
Homogeneous vs heterogeneous	Same engine means no schema conversion at all
Full load + CDC vs CDC-only	CDC-only needs a target already populated by something else

ON THE EXAM

MGN and DRS share most of their technology and answer opposite questions. Ask whether production is moving or staying.

How to use these under exam pressure

- **Name the pair the question is testing** before reading the options. It is usually one of these thirty.
- **Find the word that resolves it.** Never, guarantee, automatically, ongoing, unattended, or a number.
- **Two plausible options usually means one pair.** Two implausible ones are there to make it four.
- If both members of a pair still seem right, you have found the wrong pair. Read the requirement again.

> ON THE EXAM
>
> Most questions reduce to two credible options. The other two exist to fill the page, and eliminating them is fast.

> A regulator requires that audit logs cannot be deleted by anyone, including the cloud team, and that the control cannot be removed by an administrator in the workload account. The account is a member of an organisation with the management account excluded from SCPs.
>
> The logs are written to S3 in a dedicated Log Archive account.

Which combination is correct?

A. Object Lock in compliance mode, plus an RCP on the bucket so the control follows the resource

B. Object Lock in governance mode, plus an SCP on the workload account

C. Object Lock in compliance mode, plus an SCP on the management account

D. Versioning, plus a permissions boundary on administrators

◆ PAUSE AND DECIDE

Which combination is correct?

A Compliance mode plus an RCP

Two pairs resolved: compliance cannot be overridden by anyone, and an RCP follows the resource so it applies regardless of which identity calls.

B Governance plus SCP

Both halves wrong. Governance mode exists to be overridden, and an SCP follows the identity rather than the bucket.

C Compliance plus SCP on management

Right on Object Lock, and SCPs do not apply to the management account, which the scenario states explicitly.

D Versioning and a boundary

Versioning keeps old copies that a delete-capable administrator can also remove. A boundary caps an identity, not the resource.

THE SIGNAL

Two pairs in one question: governance versus compliance, and SCP versus RCP. Each half eliminates two options.

> One Lambda function occasionally consumes most of the account's concurrency, throttling a payment handler. Separately, the payment handler's p99 latency spikes to 3 seconds on its first invocation after idle periods, and the business needs consistent latency.
>
> Both problems must be solved.

What should the solutions architect implement?

A. Reserved concurrency on both functions, and provisioned concurrency on the payment handler

B. Provisioned concurrency on both functions

C. Reserved concurrency on the payment handler only

D. Request a concurrency quota increase, and enable SnapStart

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Reserved on both, provisioned on the handler

Reserved caps the noisy function and guarantees the handler a share; provisioned keeps environments warm to remove the cold start. Two problems, two settings.

B Provisioned on both

Solves latency twice and capacity contention never. The noisy function can still consume the account's concurrency.

C Reserved on the handler only

Solves the contention and leaves the 3-second cold start. It answers one of two stated problems.

D Quota increase and SnapStart

A bigger shared pool fails the same way, and SnapStart is Java, Python and .NET only, with no runtime stated.

THE SIGNAL

Two problems that sound like one. Capacity contention and cold-start latency need different settings with confusingly similar names.

> On-premises servers cannot resolve the private DNS names of migrated services. Separately, a chatty application split across the two estates is now slow. The team proposes upgrading the Direct Connect link from 1 Gbps to 10 Gbps to fix both.
>
> There is a Route 53 private hosted zone.

What should the solutions architect advise?

A. A Route 53 Resolver inbound endpoint for the DNS problem, and moving the chatty pair into one estate for the latency

B. Upgrade the link to 10 Gbps, which resolves both

C. A Route 53 Resolver outbound endpoint, and upgrade the link

D. Distribute hosts files, and increase the application's instance size

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Inbound endpoint, and stop the crossing

On-premises querying AWS is inbound. And latency is distance, so only removing the crossing fixes the chatty pair.

B Upgrade the link

Bandwidth is not latency, and no link speed makes AWS private zones resolvable from on-premises. It fixes neither problem.

C Outbound endpoint plus upgrade

Outbound is AWS querying on-premises, which is the opposite direction. And the upgrade still does not address latency.

D Hosts files and bigger instances

Bypasses DNS entirely, and the application is waiting on the network rather than short of CPU.

THE SIGNAL

Two pairs again: inbound versus outbound Resolver, and bandwidth versus latency. Both halves of option B are wrong.

> A security team receives many related GuardDuty findings over a day but cannot see them as one incident. Once an incident is confirmed, they also need to determine which other resources the compromised role touched over the previous month.
>
> GuardDuty, Security Hub and Detective are all enabled.

What addresses each need?

A Extended Threat Detection correlates the findings; Detective answers what the role touched

B Detective correlates the findings; Security Hub answers what the role touched

C Security Hub correlates the findings; GuardDuty answers what the role touched

D A composite CloudWatch alarm correlates; CloudTrail Lake answers what the role touched

◆ PAUSE AND DECIDE

What addresses each need?

A Extended Threat Detection, then Detective

Attack sequences correlate signals into one critical finding, and Detective builds the behaviour graph for what a principal touched.

B Detective correlates

Detective investigates a finding that already exists. It does not produce the correlated finding in the first place.

C Security Hub correlates

Security Hub aggregates findings into one place, which is not the same as correlating them into one incident.

D Composite alarm and Lake

Composite alarms combine alarms, not findings. CloudTrail Lake is also in maintenance and not the investigation tool here.

THE SIGNAL

Three services, three verbs. GuardDuty detects, Security Hub aggregates, Detective explains. Aggregating is not correlating.

A company must move a self-managed MySQL database on EC2 to Amazon RDS for MySQL with under ten minutes of downtime. Separately, an Oracle database must become Aurora PostgreSQL, and the team has budgeted two weeks of schema work for each.

Both are 500 GB.

What should the solutions architect advise?

A MySQL is homogeneous and needs no schema conversion; Oracle to Aurora PostgreSQL needs SCT and far more than two weeks for code objects

B Both need schema conversion, and two weeks each is reasonable

C Neither needs schema conversion, since DMS handles both

D MySQL needs conversion because the host changes; Oracle does not, since DMS converts automatically

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Homogeneous needs none; heterogeneous needs much more

Same engine means the dialect is identical, so nothing converts. An engine change means code objects need rewriting, which is rarely two weeks.

B Both need it

MySQL to MySQL is a hosting change, not an engine change. Budgeting conversion work for it spends effort on nothing.

C Neither needs it

DMS moves data. It does not convert an Oracle schema into PostgreSQL, which is SCT's job.

D Inverted

Exactly backwards on both halves. Changing host is not changing engine, and DMS does not convert schemas.

THE SIGNAL

Self-managed to managed feels like a big change and is only a hosting change. The engine is what decides.

> A disaster recovery design keeps a database replicating continuously to a second Region, with compute defined in launch templates but not running. Auto Scaling will launch instances during a Region failure. Separately, an application's configuration keeps drifting back after engineers edit it during incidents.
>
> The RTO is two hours.

What describes each situation correctly?

A That is a pilot light, and it depends on the control plane; the drift needs a State Manager association

B That is warm standby, and it is statically stable; the drift needs Run Command

C That is a pilot light, and it is statically stable; the drift needs Run Command

D That is warm standby, and it depends on the control plane; the drift needs a State Manager association

◆ PAUSE AND DECIDE

What describes each situation correctly?

A Pilot light, control plane dependent, State Manager

Nothing is serving, so it is pilot light. Launching instances at failure is a control plane action. And "keeps drifting" means continuous, which is State Manager.

B Warm standby, statically stable, Run Command

Wrong on all three. Nothing is running, launching is control plane work, and Run Command applies once.

C Pilot light, statically stable, Run Command

Right on pilot light. Statically stable would mean nothing needs creating, and this design creates instances at failure.

D Warm standby, control plane, State Manager

Right on the last two. Warm standby requires a scaled-down copy actually serving, and nothing is running here.

THE SIGNAL

Three pairs in one question, and every option gets some right. Resolve each independently rather than picking the familiar shape.

What to carry out of this module

- **Name the pair before reading the options.** Most questions are one of about thirty distinctions.
- **Learn the divider, not two definitions.** One property decides each pair, and everything shared is noise.
- **Composite questions resolve independently.** Each pair eliminates two options, so two pairs leave one answer.
- If both members still seem right, you have the wrong pair. Re-read the requirement rather than the options.

> ON THE EXAM
>
> The five most-missed: association versus propagation, reserved versus provisioned, governance versus compliance, coverage versus utilisation, and Detective not detecting.

MODULE 42 · EXAM STRATEGY - ALL DOMAINS

What Changed

Where your study material is confidently wrong, and how to tell.

DOMAIN Cross-cutting EXAM WEIGHT All four MODULE 42 of 44

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First, a date you need

- **The last day to take SAP-C02 is 16 November 2026.** Registration for SAP-C03 opens on 27 October.
- **SAP-C03 moves from four domains to five**, adding generative and agentic AI, post-quantum cryptography and resilience engineering.
- If you are prepared, sit C02 before the deadline. Your certification stays valid for its full term either way.
- This course targets C02. Most of it transfers, and the new C03 material is genuinely new.

> ON THE EXAM
>
> Recorded on 2 September 2026. Check the AWS certification page before booking, because dates and details do move.

The exam lags the console, and study material lags the exam. **This module maps that gap.**

Renamed, and still examinable

NOW CALLED	YOU WILL SEE IT AS
AWS Security Hub CSPM	AWS Security Hub. The exam list uses the old name
Amazon Quick / Quick Suite	Amazon QuickSight
Aurora Serverless	Aurora Serverless v2, renamed April 2026
Amazon Data Firehose	Kinesis Data Firehose
AWS Transform MGN	AWS Application Migration Service
CloudWatch Database Insights	RDS Performance Insights

ON THE EXAM

Answer with the behaviour, not the branding. A rename never changes what a service does, and the exam tests what it does.

Closed to new customers, still examinable

SERVICE	STATUS
AWS Migration Hub	Closed 7 Nov 2025. Still on the in-scope list
Application Discovery Service	Closed 7 Nov 2025. Still in scope
Snowball Edge	Closed 7 Nov 2025. Support ends 31 Dec 2026
AWS App Runner	Closed 30 Apr 2026. Successor is ECS Express Mode
Simple AD	Closed 30 Jul 2026
Systems Manager Change Manager	Closed to new customers

ON THE EXAM

All still correct exam answers. Recognise the shape a scenario describes and answer with the service, even though you could not buy it today.

Answers that changed

THE OLD ANSWER	THE CURRENT ONE
Above 16,000 IOPS needs io2	gp3 reaches 80,000 IOPS and 64 TiB
Retain, delete, re-import to split a stack	CloudFormation stack refactoring, in one operation
Provisioned capacity is cheaper for steady DynamoDB	On-demand is now AWS's recommended default
Savings Plans cover compute only	Database Savings Plans arrived Dec 2025
Only unattached Elastic IPs are charged	All public IPv4 charged hourly since Feb 2024
Snowmobile for petabyte transfers	Retired. Cluster Snowball Edge devices

ON THE EXAM

Each of these was correct for years. That is what makes them dangerous: they are remembered confidently and rarely re-checked.

Newer capabilities worth recognising

- **GuardDuty Extended Threat Detection** correlates many findings into one critical attack sequence, automatically and at no extra cost.
- **Centralized root access management** deletes root credentials across member accounts. New accounts have none by default.
- **Just-in-time node access** removes standing permission to connect, with Cedar approval policies.
- **CloudFront VPC origins** reach a private ALB directly, replacing the secret custom header pattern.

> ON THE EXAM
>
> These four are the most likely to appear as the correct answer while being absent from older material entirely.

How to tell when your knowledge is stale

- **Specific numeric limits** age fastest. IOPS ceilings, retention periods, throughput caps and instance sizes all move.
- **Pricing-derived rules** age next. "Provisioned is cheaper" was a conclusion, not a fact, and the price changed.
- **Behavioural facts age slowest.** A queue message is still consumed and gone. An SCP still follows the identity.
- So anchor on behaviour, and treat any remembered number as something to sanity-check.

> ON THE EXAM
>
> If your answer depends on a specific number or on which option is cheaper, that is exactly where to be least confident.

> A design review proposes io2 Block Express for a database needing 30,000 IOPS on a 6 TiB volume, with single-digit millisecond latency and no clustering. The reviewer's stated justification is that gp3 cannot exceed 16,000 IOPS.
>
> Cost is a stated concern.

What should the solutions architect say?

A gp3 now reaches 80,000 IOPS and 64 TiB, so it meets every stated requirement at lower cost

B Approve io2 Block Express as proposed

C Stripe three gp3 volumes in RAID 0

D Use io1, which is cheaper than io2

◆ PAUSE AND DECIDE

What should the solutions architect say?

A gp3 meets it, cheaper

30,000 IOPS and 6 TiB are both inside gp3's current limits, and the scenario rules out every remaining reason for io2.

B Approve io2

Would work, at higher cost, on a justification that stopped being true in September 2025.

C RAID 0 striping

The workaround from when one volume could not deliver enough. Unnecessary complexity and a wider failure surface.

D io1

An older generation of the same idea. It also does not question the premise, which is where the error is.

THE SIGNAL

Notice what the scenario rules out: no sub-millisecond need, no Multi-Attach, no five-nines requirement. Every io2 reason is closed off.

> A team must split a monolithic CloudFormation stack into three, with no downtime and no resource replacement. A senior engineer describes the procedure: set DeletionPolicy to Retain, remove resources from the template, delete the stack, then import into new stacks.
>
> They have done this successfully before.

What should the solutions architect advise?

A Use CloudFormation stack refactoring, which does this in one operation without replacement

B Follow the retain-and-reimport procedure as described

C Recreate the resources in three new stacks and migrate data

D Convert to nested stacks under a parent template

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Stack refactoring

Purpose-built since February 2025 to move resources between stacks and rename logical IDs, in one operation, with no replacement.

B Retain and re-import

The engineer is right that it works, and it is many manual steps with a window where nothing manages the resources.

C Recreate and migrate

Replaces every resource and requires data migration, contradicting both stated constraints.

D Nested stacks

Keeps one lifecycle, so the coupling that motivated the split remains.

THE SIGNAL

The engineer's experience is real and dated. "I have done this before" is evidence about the past, not about what is available now.

> A DynamoDB table has run at a steady 5,000 read units and 2,000 write units for two years, on provisioned capacity with auto scaling. A cost review recommends keeping provisioned capacity, citing the standard guidance that provisioned is cheaper for predictable workloads.
>
> The table was configured in 2023.

What should the solutions architect advise?

A Model it against on-demand, which AWS has recommended as the default since the November 2024 price cut

B Keep provisioned capacity as recommended

C Buy DynamoDB reserved capacity to deepen the discount

D Move the table to Aurora, which is cheaper at steady load

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Model against on-demand

AWS halved on-demand pricing in Nov 2024 and now calls it the default for most workloads. Note the wording: model it, rather than switch blindly.

B Keep provisioned

Correct guidance for years, and it is a conclusion drawn from a price that has since changed by half.

C Reserved capacity

Deepens a commitment to a mode that may no longer be cheaper. Measure before committing for a year.

D Move to Aurora

Replaces the database to change a billing mode, and Aurora would not serve this access pattern better.

THE SIGNAL

This is a pricing-derived rule, the second-fastest category to age. The rule was a conclusion; the price underneath it moved.

> A scenario describes a company needing a fully managed way to run a container image as a scaling HTTPS service, with no load balancer, target groups or scaling policies to configure. Four options are offered, one of which is AWS App Runner.
>
> The candidate knows App Runner closed to new customers in April 2026.

What should the candidate do?

A Choose App Runner: it matches the described shape and remains on the exam's in-scope service list

B Eliminate App Runner, since it cannot be adopted today

C Choose ECS Express Mode instead, as the successor

D Choose EKS, since it is the most current option

◆ PAUSE AND DECIDE

What should the candidate do?

A Choose App Runner

The exam tests the in-scope list, and App Runner is on it. Knowing more than the exam does is only useful if you answer the exam's question.

B Eliminate it

Over-applying real-world knowledge. Commercial availability is not what the exam is assessing.

C ECS Express Mode

The right real-world answer and probably not among the options. If it were, App Runner would still be the exam's answer.

D EKS

Contradicts the requirement. The scenario asks for nothing to configure, and EKS is the opposite.

THE SIGNAL

Currency knowledge is for the workplace and for confidence. On the exam, answer the in-scope list, and let the shape decide.

> A security review needs to confirm that the organisation's GuardDuty deployment, enabled in 2022, is still detecting everything it can. The console shows GuardDuty as enabled and healthy in all 90 accounts, with no findings suppressed.
>
> The reviewer concludes coverage is complete.

What should the solutions architect check?

A Which protection plans are enabled: plans launched after GuardDuty was first enabled are not turned on automatically

B Nothing; enabled and healthy in every account is complete coverage

C Whether the delegated administrator is correctly registered

D Whether findings are being exported to S3

◆ PAUSE AND DECIDE

What should the solutions architect check?

A Which protection plans are on

Documented behaviour: a plan released after you enabled GuardDuty stays off, and the console still reports healthy. A 2022 deployment lags by years.

B Coverage is complete

Enabled and healthy means it is running, not that it is watching everything it could. That is exactly the reviewer's error.

C Delegated administrator

Worth confirming, and it affects who sees findings rather than which findings are generated.

D Export to S3

Affects retention and downstream processing. It does not change detection coverage.

THE SIGNAL

This is a currency question wearing an operations costume. The service did not change; what it can see did, and nothing surfaced that.

> A candidate is revising from a well-regarded study guide published in 2023. It states that Savings Plans cover only EC2, Lambda and Fargate, that Snowmobile handles petabyte-scale transfers, and that unattached Elastic IPs are the only ones charged.
>
> All three statements were accurate when written.

How should the candidate treat this material?

A Trust the behavioural explanations, and re-check every specific limit, price and service status against current AWS documentation

B Discard the guide entirely, since it contains errors

C Trust it fully, since it was accurate when published

D Trust the numbers and re-check only the service names

◆ PAUSE AND DECIDE

How should the candidate treat this material?

A Trust behaviour, re-check numbers

Behavioural explanations age slowest and are the guide's real value. Limits, prices and service status age fastest and are cheap to verify.

B Discard it

Overcorrects. A good 2023 guide still explains why a queue differs from a stream, and that has not changed.

C Trust it fully

All three quoted statements are now wrong. Accurate when written is not the same as accurate.

D Numbers yes, names no

Exactly inverted. Numbers are the fastest-ageing category, and renames rarely change the correct answer.

THE SIGNAL

Option D is the interesting wrong answer because it sounds like a sensible policy. It just gets the ageing order backwards.

What to carry out of this module

- **Sit SAP-C02 by 16 November 2026** if you are prepared. C03 registration opens on 27 October.
- **Answer the in-scope list**, even for services closed to new customers. Knowing more than the exam can cost marks.
- **Numbers age fastest, then pricing rules, then behaviour.** Be least confident about a remembered limit.
- **Renames never change behaviour.** Answer with what the service does, whatever it is currently called.

> ON THE EXAM
>
> "I have done this before" and "the guide says" are both evidence about the past. On a platform this fast, that is a different claim.

MODULE 43 · EXAM STRATEGY - ALL DOMAINS

The Stale Catalogue

Services in every old study guide that no longer exist. Reclaim the revision time.

DOMAIN Cross-cutting EXAM WEIGHT All four MODULE 43 of 44

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Dead and unexaminable is revision time to reclaim. **Dead and still on the list is an answer you must still give.**

AWS publishes a lifecycle now

- **Maintenance:** closed to new customers, no new features. Existing customers unaffected, and AWS keeps operating it.
- **Sunset:** a published end date, typically about twelve months out, when AWS ends operations and support.
- **Full shutdown:** removed from the portfolio entirely, and no longer available or supported in any capacity.
- The **AWS Product Lifecycle page**, introduced May 2025, plus recurring Service Availability Updates, is where to check.

> ON THE EXAM
>
> This structure only exists because the 2024 closures were communicated so poorly. There is now a page you can check in thirty seconds.

Fully shut down: gone entirely

SERVICE	GONE SINCE
AWS OpsWorks Stacks	26 May 2024 - disabled for existing customers too
AWS CodeStar	25 July 2024
Amazon Elastic Transcoder	13 November 2025
AWS IoT Analytics	15 December 2025
Kinesis Data Analytics for SQL	27 January 2026
CloudWatch Evidently	17 October 2025

ON THE EXAM

OpsWorks is the one to notice: it was disabled for existing customers as well, which is rarer than closing to new ones.

Closed in 2024, still taught everywhere

SERVICE	STATUS AND REPLACEMENT
AWS CodeCommit	Closed 25 Jul 2024. Use GitHub, GitLab or similar
AWS Cloud9	Closed 25 Jul 2024. Use IDE toolkits or CloudShell
AWS Data Pipeline	Closed 25 Jul 2024. Use Step Functions or Glue
Amazon CloudSearch	Closed 25 Jul 2024. Use OpenSearch Service
Amazon QLDB	Discontinued. Use Aurora PostgreSQL
AWS App Mesh, FSx File Gateway	Closed October 2024

ON THE EXAM

The July 2024 wave also included SimpleDB, Forecast and S3 Select. None is on the C02 in-scope list, so none needs revising.

Dead but still examinable

SERVICE	WHY YOU STILL LEARN IT
AWS Migration Hub	Closed Nov 2025, still on the in-scope list
Application Discovery Service	Closed Nov 2025, still on the list
AWS Snow Family	Closed Nov 2025, support ends Dec 2026, still listed
AWS App Runner	Maintenance Mar 2026, still listed
Simple AD	Closed Jul 2026, still listed
Systems Manager Change Manager	Closed to new customers, SSM is listed

ON THE EXAM

This is the group module 42 covered. The test is not whether you can buy it. It is whether the exam guide names it.

Triaging your own material

- **Take the in-scope service list as your index.** Anything a guide covers that is not on it is optional at best.
- **A chapter on CodeCommit or Cloud9 dates the guide precisely.** It was written before mid-2024.
- **Check anything unfamiliar against the Product Lifecycle page.** It takes seconds and settles the question.
- Do not discard a dated guide. Its explanations of behaviour are usually still sound, as module 42 argued.

> ON THE EXAM
>
> The in-scope list is the syllabus. A guide's table of contents is one author's guess at it, made on a date you can often infer.

> A practice exam offers AWS OpsWorks as the correct answer to a configuration-management question about Chef recipes across a fleet. The explanation is detailed and technically accurate about how OpsWorks Stacks worked.
>
> The candidate has never heard of it.

What should the candidate conclude?

A The question predates May 2024; OpsWorks Stacks is fully shut down and is not on the in-scope list. Skip it

B Learn OpsWorks, since it appeared in a practice exam

C Learn Chef instead, as the underlying technology

D Assume the practice exam is authoritative and the exam guide is out of date

◆ PAUSE AND DECIDE

What should the candidate conclude?

A Dated question; skip it

OpsWorks Stacks reached end of life in May 2024 and was disabled for existing customers too. It is not on the in-scope list, so it cannot be a correct answer.

B Learn it

Hours spent on a service that no longer exists and cannot appear. A detailed explanation is not the same as a current one.

C Learn Chef

Configuration management concepts are useful, and Chef specifically is not what the exam tests. Systems Manager is.

D Trust the practice exam

The exam guide is the syllabus. A third-party practice bank is one author's interpretation, on a date you can often infer.

THE SIGNAL

The explanation being accurate is what makes it convincing. It was accurate. The service is gone, which the explanation cannot tell you.

> A candidate's study guide devotes a full chapter to AWS CodeCommit, covering repositories, triggers, approval rules and integration with CodePipeline. Another chapter covers Cloud9 as the recommended development environment.
>
> The candidate has three weeks before the exam.

How should the candidate allocate that time?

A Skip both chapters: neither service is on the in-scope list, and both closed to new customers in July 2024

B Study both, since CodePipeline integration is in scope

C Study CodeCommit but skip Cloud9

D Study both quickly, in case they appear

◆ PAUSE AND DECIDE

How should the candidate allocate that time?

A Skip both

Neither is on the in-scope list. CodePipeline is, and it is examined through its own behaviour rather than through a source-control service.

B Study both

CodePipeline being in scope does not put CodeCommit in scope. Pipelines source from many places, and the exam tests the pipeline.

C CodeCommit only

Splits the difference for no reason. Both are equally closed and equally absent from the list.

D Study both quickly

Three weeks is finite. Every hour on an unexaminable service is an hour not spent on the 26% of the exam that is Domain 1.

THE SIGNAL

Two chapters on services closed in 2024 tells you when the guide was written, and therefore what else in it to verify.

> A candidate revising migration content spends time learning AWS Migration Hub in detail, then notices it closed to new customers in November 2025. They consider dropping it to focus elsewhere.
>
> They have already skipped CodeCommit for the same reason.

What should the candidate do?

A Keep studying Migration Hub: it is on the in-scope list, unlike CodeCommit, and remains a correct answer

B Drop it, consistent with the CodeCommit decision

C Study AWS Transform instead, as the successor

D Study both Migration Hub and AWS Transform equally

◆ PAUSE AND DECIDE

What should the candidate do?

A Keep it; it is on the list

Closure is not the test. Migration Hub is on the in-scope list and CodeCommit is not, which is the only difference that matters.

B Drop it for consistency

Consistent application of the wrong rule. The rule is not "closed means skip"; it is "not on the list means skip".

C Study AWS Transform

The right service for the workplace and not on the in-scope list. It cannot be a correct answer on C02.

D Study both

Half the effort spent on a service that cannot appear. Migration Hub alone is the examinable half.

THE SIGNAL

Pair this with scenario two. Same reasoning, opposite conclusion, because the in-scope list is the test rather than the closure date.

> A practice question asks how to run SQL queries continuously over a streaming data source. The offered options include Amazon Kinesis Data Analytics for SQL, Amazon Managed Service for Apache Flink, AWS Glue streaming, and Amazon Athena.
>
> The candidate recognises Kinesis Data Analytics from an older course.

What should the candidate conclude?

A. Kinesis Data Analytics for SQL was fully shut down in January 2026; Managed Service for Apache Flink is its replacement

B. Choose Kinesis Data Analytics for SQL, as the purpose-built option

C. Choose Athena, since the question mentions SQL

D. Choose AWS Glue streaming, since Flink is too complex

◆ PAUSE AND DECIDE

What should the candidate conclude?

A It is shut down; Flink replaced it

Full shutdown on 27 January 2026, with Managed Service for Apache Flink named by AWS as the migration target. Recognition is not currency.

B Kinesis Data Analytics for SQL

It was the right answer for years, which is why it is recognised. It no longer exists in any capacity.

C Athena

Queries data at rest in S3. It does not run continuous queries over a stream.

D Glue streaming

A real option for streaming ETL, and the reasoning given is a preference rather than a requirement from the question.

THE SIGNAL

Recognising a service is the danger. Familiarity feels like knowledge, and here it is a memory of a service that has been removed.

> A candidate encounters a service name in a practice question that they cannot place at all, and cannot tell whether it is very new, very old, or invented by the question author. They have twenty minutes of revision time left in the session.
>
> The exam is in a fortnight.

What is the fastest way to resolve it?

A Check the exam guide's in-scope list first, then the AWS Product Lifecycle page if it appears there

B Search for tutorials and learn the service from scratch

C Ignore it, since unfamiliar means unimportant

D Ask in a forum and wait for a reply

◆ PAUSE AND DECIDE

What is the fastest way to resolve it?

A In-scope list, then lifecycle page

Two lookups, under a minute. Not on the list means stop. On the list means learn it, and the lifecycle page tells you its status.

B Learn it from scratch

Twenty minutes on a service that may not be examinable at all. Establish relevance before depth.

C Ignore it

Unfamiliar may mean genuinely new and genuinely examinable, which is precisely the gap this course exists to close.

D Ask a forum

Slower than looking it up, and forum answers are frequently out of date in exactly the way this module describes.

THE SIGNAL

Establish relevance before depth. The in-scope list answers the only question that matters, and it answers it in seconds.

> A team's runbook references AWS Data Pipeline for a nightly ETL job, and the job still runs. A new engineer proposes rebuilding it on Data Pipeline for consistency with the existing pattern.
>
> The account has used Data Pipeline since 2021.

What should the solutions architect advise?

A. Data Pipeline closed to new customers in 2024; build new work on Step Functions or AWS Glue, and plan to migrate the existing job

B. Rebuild on Data Pipeline, since the account is an existing customer

C. Leave the existing job and build nothing new

D. Migrate everything to Data Pipeline for consistency

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Build new work elsewhere, plan the migration

Being an existing customer permits continued use, and building new dependencies on a closed service deepens a problem with a known ending.

B Rebuild on Data Pipeline

Technically possible as an existing customer, and it adds new work to a service receiving no features and heading for sunset.

C Build nothing

Declines to solve the problem. The team needs the job; the question is where to build it.

D Consolidate onto Data Pipeline

The worst option: it increases exposure to a closed service across more workloads.

THE SIGNAL

Existing customers can keep using a closed service. That is permission to continue, not a reason to expand onto it.

What to carry out of this module

- **The in-scope list is the test, not the closure date.** Migration Hub is dead and examinable; CodeCommit is dead and not.
- **OpsWorks, CodeStar, Elastic Transcoder, IoT Analytics and Kinesis Data Analytics for SQL are fully shut down.** Not closed. Gone.
- **A chapter on CodeCommit or Cloud9 dates a guide to before mid-2024**, which tells you what else to verify.
- **Establish relevance before depth.** The in-scope list answers the only question that matters, in seconds.

> ON THE EXAM
>
> AWS's Product Lifecycle page, from May 2025, is where to check. Maintenance, sunset and full shutdown mean different things.

MODULE 44 · EXAM STRATEGY - ALL DOMAINS

Exam Mechanics and Strategy

Three hours, seventy-five long questions, and a clock that does not care.

DOMAIN Cross-cutting EXAM WEIGHT All four MODULE 44 of 44

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The format, precisely

- **75 questions in 180 minutes.** 65 are scored; 10 are unscored trials and are not identified.
- **Multiple choice** has one correct answer and three distractors. **Multiple response** has two or more correct among five or more.
- **Unanswered is scored as incorrect, and there is no penalty for guessing.** Never leave a question blank.
- A list of AWS service short names is available through the **Help button** during the exam.

> ON THE EXAM
>
> Ten unscored questions means about one in seven does not count, and you cannot tell which. Treat every question as real.

How it is scored

- **A scaled score from 100 to 1,000, with 750 to pass.** Scaling equates forms of slightly different difficulty.
- **Compensatory model:** AWS states you do not need to pass each section, only the exam overall.
- So a weak domain can be carried by strong ones. Domain 4 is 20%, and it cannot sink you alone.
- **Assume multiple-response needs every selection correct.** Sources disagree on partial credit; the conservative reading costs nothing.

> ON THE EXAM
>
> 750 out of 1,000 is not 75%. Scaled scoring means you cannot convert it to a number of questions, so aim comfortably clear.

The time budget

- **2.4 minutes per question** on average. Some take forty seconds; a few take five minutes.
- **Checkpoints beat averages.** Question 25 by minute 60, question 50 by minute 120, leaving 60 for the rest and review.
- **The most common failure is spending six minutes on question twelve**, then rushing forty easy questions at the end.
- Flag and move at three minutes. A flagged question you return to is cheaper than three you never read properly.

> ON THE EXAM
>
> You cannot feel elapsed time reliably under pressure. Two checkpoints, at questions 25 and 50, are enough to catch drift early.

Reading a long stem

- **Read the final question first**, then the scenario. You will read the scenario knowing what to look for.
- **Find the constraint sentence.** It is usually about cost, downtime, a deadline, or who may change what.
- **Note the superlative:** MOST cost-effective, LEAST operational overhead, MOST secure. It decides between workable answers.
- Planted details are there to eliminate options. "The team has no Kubernetes experience" is not background.

> ON THE EXAM
>
> Most stems contain one sentence that eliminates two options. Finding it is faster than evaluating four answers in full.

Eliminating rather than choosing

- **Two options are usually implausible** and can go in seconds. That leaves a pair, which is module 41's territory.
- **An option that fails any stated requirement is out**, however elegant the rest of it is.
- **Watch for absolutes:** never, guarantee, ensure. Those demand preventive controls, not detective ones.
- If two options remain and both seem right, re-read the requirement rather than the options.

> ON THE EXAM
>
> Choosing between four is slow. Eliminating two, then resolving a pair on one property, is how the course has worked throughout.

The day itself

- **Breaks are unscheduled and the clock keeps running.** Decide in advance whether you will take one.
- Online proctoring needs a quiet room, a clear desk and a room scan. A test centre needs ID and early arrival.
- **A preliminary result appears immediately.** The official score report follows within a few business days.
- Fatigue is the real adversary at three hours. Sleep matters more than a final evening of revision.

> ON THE EXAM
>
> The section-level table on your score report is diagnostic, not a pass condition. Compensatory scoring means the total is what counts.

One date, restated

- **The last day to take SAP-C02 is 16 November 2026.** SAP-C03 registration opens on 27 October.
- If you are prepared, book now. Your certification is valid for three years regardless of which version you sit.
- You can retake after a 14-day wait, so leave room for a second attempt before the deadline if you book late.
- Check the AWS certification page before booking. This was recorded on 2 September 2026.

> ON THE EXAM
>
> If you are within a fortnight of ready, the retake window is the thing to plan around, not the exam date itself.

> At the 100-minute mark you have completed 38 questions. Six are flagged for review. You are finding the scenarios long but not conceptually difficult.
>
> There are 80 minutes and 37 questions remaining.

What should you do?

A Speed up: aim for roughly 2 minutes per remaining question and defer the flagged six until the end

B Go back and resolve the six flagged questions now, while they are fresh

C Continue at the current pace and accept leaving some unanswered

D Skip ahead to answer the easiest remaining questions first

◆ PAUSE AND DECIDE

What should you do?

A Speed up, defer the flagged six

38 in 100 minutes is 2.6 each, slightly behind. 37 in 80 is 2.16 each, which is achievable, and unseen questions are worth more than flagged ones.

B Resolve the flagged six now

Spends time on questions you have already attempted, while 37 unseen ones wait. An unseen question has a much higher expected return.

C Continue as you are

At 2.6 minutes each, 37 questions needs 96 minutes and you have 80. That guarantees unanswered questions, which score as incorrect.

D Cherry-pick the easy ones

You cannot tell which are easy without reading them, and scanning to choose costs the time you were trying to save.

THE SIGNAL

Do the arithmetic at the checkpoint rather than trusting the feeling. Slightly behind is recoverable; discovering it at question 60 is not.

> A question says "Select THREE." You are confident about two options and genuinely cannot decide between the remaining three, all of which are plausible. You have used two minutes.
>
> You do not know whether partial credit applies.

What should you do?

A Pick the most likely third, select three, flag it, and move on

B Select only the two you are sure of, to avoid an incorrect selection

C Spend another three minutes reasoning it through properly

D Leave it unanswered and return at the end

◆ PAUSE AND DECIDE

What should you do?

A Guess the third, flag, move on

There is no penalty for guessing, and selecting two when three are asked cannot be fully correct under either scoring rule. A one-in-three guess is free.

B Select only two

Guarantees the answer is incomplete. If partial credit exists you gain nothing over guessing; if it does not, you have thrown the question away.

C Three more minutes

Five minutes on one question, when you have already narrowed it to a one-in-three guess. The marginal return is very low.

D Leave it blank

Unanswered is scored as incorrect. Blank is strictly worse than a guess, always.

THE SIGNAL

No penalty for guessing makes this arithmetic, not judgement. An incomplete selection can never beat a complete one with a guess in it.

> Two options both appear to satisfy the scenario. One uses a Config rule with automatic remediation; the other uses a service control policy. You have re-read both options twice and cannot separate them.
>
> You have not re-read the scenario.

What should you do?

A Re-read the requirement, looking for an absolute like "never" or "guarantee" that forces the preventive control

B Choose the Config rule, since detective controls are more flexible

C Choose the SCP, since preventive controls are always stronger

D Guess between them and move on immediately

◆ PAUSE AND DECIDE

What should you do?

A Re-read the requirement

The discriminator lives in the stem, not the options. Re-reading options you have already read twice cannot produce new information.

B Default to Config

Applies a general preference instead of reading the specific requirement. Sometimes right, for no reason.

C Default to the SCP

Preventive is not always correct. Module 25's scenario one showed a case where prevention broke legitimate incident response.

D Guess immediately

Reasonable if time is critical, and here one targeted re-read is likely to resolve it. Guess only after that fails.

THE SIGNAL

You have read the options twice and the scenario once. The information you are missing is almost certainly in the part you read least.

> With three minutes left you have four unanswered questions, all flagged, all long. Reading one properly would take most of the remaining time.
>
> There is no penalty for guessing.

What should you do?

A. Answer all four immediately with best guesses, then use any remaining seconds to reconsider one

B. Read and answer one carefully, leaving three blank

C. Read all four quickly and answer the two you can

D. Leave them; a wrong answer looks worse than no answer

◆ PAUSE AND DECIDE

What should you do?

A Guess all four, then reconsider

Four guesses have positive expected value; blanks have none. Securing all four first means running out of time costs nothing.

B One carefully, three blank

Trades three free guesses for one likely-correct answer. The arithmetic favours four attempts over one certainty.

C Read all four quickly

Risks the clock expiring mid-question with several still blank. Answer first, then read with whatever is left.

D Leave them blank

Factually wrong. Unanswered is scored as incorrect, and nobody sees which questions you guessed.

THE SIGNAL

Secure the guesses first, then improve them. Reversing that order means the clock can take questions off you entirely.

> A practice exam returns 68%. The breakdown shows Domain 1 at 55%, Domain 2 at 74%, Domain 3 at 71% and Domain 4 at 70%. The exam is in two weeks.
>
> Domain 1 is 26% of the real exam.

How should the remaining two weeks be spent?

A Concentrate on Domain 1, since it is both the weakest and the second-heaviest domain

B Spread revision evenly across all four domains

C Concentrate on Domain 2, since it carries the most weight at 29%

D Take more practice exams until the score improves

◆ PAUSE AND DECIDE

How should the remaining two weeks be spent?

A Domain 1

Weakest by 15 points and worth 26%. Weight times gap is the highest of the four, so improvement there moves the total score most.

B Even spread

Treats a 55% domain and a 74% domain as equally urgent. Even effort produces uneven returns when the gaps differ this much.

C Domain 2

Reads weight alone and ignores the gap. Improving 74% costs more effort per point than improving 55%.

D More practice exams

Measures the problem repeatedly without addressing it. Practice exams diagnose; they do not teach.

THE SIGNAL

Prioritise by weight multiplied by gap, not by either alone. Domain 1 at 26% and 15 points behind is the clear target.

> A candidate is consistently scoring 82% on practice exams and feels ready. It is 20 October 2026. They are considering waiting for SAP-C03 to certify on the newer content instead.
>
> SAP-C03 registration opens 27 October; the last SAP-C02 date is 16 November.

What is the sound reasoning?

A. Sit C02 now: they are prepared, the credential is valid three years either way, and C03 content is not yet studied

B. Wait for C03, since a newer certification is more valuable

C. Wait for C03 and sit it in the first week, before question banks exist

D. Book C02 for 15 November, to maximise revision time

◆ PAUSE AND DECIDE

What is the sound reasoning?

A Sit C02 now

Prepared for the exam that exists, credential valid three years regardless, and C03 adds GenAI, agentic AI and post-quantum content they have not studied.

B Wait for C03

The credential is the same certification either way. It means studying substantial new content while already prepared for the current one.

C Sit C03 in week one

A brand-new exam with no preparation material and unstudied content, chosen for a reason that does not help the candidate pass.

D Book 15 November

One day of margin before the deadline. A 14-day retake wait means a failure leaves no second attempt.

THE SIGNAL

Option D is the subtle error: it looks prudent and removes the retake window entirely. Book early enough to fail once.

What to carry out of this module

- **Never leave a question blank.** Unanswered scores as incorrect and guessing carries no penalty.
- **Checkpoints, not averages.** Question 25 by minute 60, question 50 by minute 120. Flag and move at three minutes.
- **Read the question first, then the stem**, and hunt for the constraint sentence and the superlative.
- **Prioritise revision by weight times gap**, and remember scoring is compensatory: you pass overall, not per domain.

> ON THE EXAM
>
> Book early enough to fail once. With the C02 deadline on 16 November, the 14-day retake wait is what to plan around.

MODULE 45 · EXAM STRATEGY - ALL DOMAINS

Practice Exam: Domain 1

Fifteen questions on organisational complexity. Pause, decide, then watch.

DOMAIN 1 of 4 EXAM WEIGHT 26% SET 1 of 5

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How to use this set

- **Give yourself 2.4 minutes per question.** That is the real exam's budget, and fifteen questions is 36 minutes.
- **Write your answer down before unpausing.** Reading the walkthrough first teaches you nothing about what you knew.
- **Note which ones you guessed**, even when you guessed correctly. Those are the gaps.
- The final slide maps every question to a module, so a miss tells you exactly what to rewatch.

> ON THE EXAM
>
> A question you guessed right is a question you got wrong. Track guesses separately or the score flatters you.

> A company must guarantee that no principal, in any member account, can share an S3 bucket with an AWS account outside the organisation. Several teams legitimately share buckets with partner accounts that have been added to the organisation. The management account must also be covered.
>
> Control Tower is deployed.

Which approach meets the requirement?

A. An RCP denying s3 actions where aws:PrincipalOrgID does not match the organisation

B. An SCP denying s3:PutBucketPolicy across all OUs

C. A Config rule detecting externally shared buckets, with automatic remediation

D. S3 Block Public Access enabled account-wide in every account

◆ PAUSE AND DECIDE

Which approach meets the requirement?

A RCP on PrincipalOrgID

RCPs apply to the resource, so access from outside the org is refused regardless of caller. RCPs also apply to the management account, unlike SCPs.

B SCP denying PutBucketPolicy

Blocks legitimate internal sharing too, and SCPs do not apply to the management account.

C Config plus remediation

"Guarantee" is preventive. Detection leaves a window, however brief.

D Block Public Access

Stops public access. Sharing with a named external AWS account is not public access.

THE SIGNAL

Two clues: "guarantee" forces preventive, and "the management account must also be covered" rules out SCPs. Module 02.

> An organisation has 400 accounts. Developers currently receive IAM users in each account they need. Onboarding takes four days, and an audit found 61 users belonging to people who left. The company uses Okta as its corporate identity provider.
>
> Access must be revocable centrally within minutes.

What should the solutions architect recommend?

A. A central identity account with cross-account roles assumed from IAM users

B. A SAML identity provider configured in each of the 400 accounts

C. IAM Identity Center with Okta as the identity source, and permission sets assigned to groups

D. AWS Managed Microsoft AD with a trust to Okta

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Central account with IAM users

Better than 400 user sets, and still long-lived credentials that must be deprovisioned manually.

B SAML provider per account

400 configurations to maintain and 400 places to update when a role changes. Identity Center exists to remove this.

C Identity Center with Okta

One identity source, permission sets deployed across all 400 accounts, and disabling in Okta revokes everywhere. No long-lived credentials.

D Managed AD with a trust

Adds a directory the company does not need. Okta federates to Identity Center directly.

THE SIGNAL

Leavers retaining access is the signal for federation, not for better process. Module 04.

> Three VPCs are attached to a Transit Gateway. VPC A and VPC B must reach each other. VPC C is a shared services VPC that both must reach, but A and B must not reach each other through it. An engineer has associated all three with the default route table and enabled propagation from all three.
>
> Traffic currently flows between all three VPCs.

What change achieves the requirement?

A. Use separate route tables: associate A and B with one that has C propagated, and associate C with one that has A and B propagated

B. Add security group rules denying traffic between A and B

C. Remove the propagation from VPC C's attachment

D. Replace the Transit Gateway with VPC peering between A-C and B-C

◆ PAUSE AND DECIDE

What change achieves the requirement?

A Separate route tables

The near-miss. Read the stem again: A and B must reach each other. This isolates them, which the requirement does not ask for.

B Security group rules

Security groups have no deny rules. This is not something they can express.

C Remove C's propagation

Stops A and B learning routes to C, which breaks the shared services access they need.

D VPC peering A-C and B-C

Peering is non-transitive, so A and B cannot reach each other through C, while both reach C directly. That is exactly the stated requirement.

THE SIGNAL

Read the requirement twice. A and B MUST reach each other, and must not do so THROUGH C. Non-transitivity gives you both. Module 06.

A company connects to AWS with a single 10 Gbps Direct Connect connection from one location, terminating on one device. It requires resilience against device failure, location failure and connection failure, and has budget for additional connections.

The workload cannot tolerate falling back to the internet.

What is the MINIMUM configuration that meets all three requirements?

A Two connections at the same location, on separate devices

B Two connections at two different locations, each on a separate device

C One connection plus a Site-to-Site VPN as backup

D Four connections across two locations, two at each

◆ PAUSE AND DECIDE

What is the MINIMUM configuration that meets all three requirements?

A Two at one location

Covers device and connection failure. A location failure takes both out.

B Two locations, separate devices

Covers all three failure modes with the fewest connections, which is what MINIMUM asks for.

C One connection plus VPN

The VPN traverses the internet, which the scenario rules out explicitly.

D Four connections

Meets the requirements and exceeds them. MINIMUM makes this wrong despite being more resilient.

THE SIGNAL

MINIMUM is doing work here. Option D is not incorrect architecture; it is the wrong answer to the question asked. Module 07.

> On-premises applications must resolve records in a Route 53 private hosted zone in AWS. AWS-hosted applications must resolve internal names held by on-premises DNS servers. A Direct Connect link exists between the environments.
>
> Both directions currently fail.

What is required?

A. A Route 53 Resolver inbound endpoint only

B. A Route 53 Resolver outbound endpoint only, with forwarding rules

C. Both inbound and outbound Resolver endpoints, with forwarding rules for the on-premises domains

D. Conditional forwarding configured on the on-premises DNS servers alone

◆ PAUSE AND DECIDE

What is required?

A Inbound only

Solves on-premises querying AWS. AWS querying on-premises still fails.

B Outbound only

Solves AWS querying on-premises. The other direction still fails.

C Both, with forwarding rules

Both directions are stated as failing, so both endpoints are needed. Inbound receives queries; outbound forwards them.

D Conditional forwarding alone

On-premises servers need an inbound endpoint IP to forward to. Configuring forwarding with no target does nothing.

THE SIGNAL

"Both directions currently fail" is the whole question. Answering with one endpoint is the common half-correct response. Module 08.

> A security team must retain CloudTrail logs from 200 accounts for seven years, in a form that no administrator in any account, including the Log Archive account, can alter or delete. Auditors must be able to query the logs.
>
> The company uses Control Tower.

Which combination meets the requirement?

A. An organisation trail with Object Lock in governance mode, queried with Athena

B. An organisation trail to a central bucket with versioning and MFA delete

C. Per-account trails to per-account buckets, with a nightly copy to a central bucket

D. An organisation trail to a central S3 bucket with Object Lock in compliance mode, queried with Athena

◆ PAUSE AND DECIDE

Which combination meets the requirement?

A Governance mode

The near-miss. Governance mode exists to be overridden by a privileged user, which the requirement forbids.

B Versioning and MFA delete

Versioning keeps old copies that a sufficiently privileged principal can still remove. It is not immutability.

C Per-account trails plus copies

A member-account administrator can disable their own trail before the copy runs. The gap is in the source.

D Org trail, compliance mode, Athena

An organisation trail covers all 200 accounts including new ones. Compliance mode cannot be overridden by anyone, including the root user.

THE SIGNAL

"Including the Log Archive account" is the discriminator between governance and compliance mode. Modules 10 and 19.

> A platform team publishes an internal API. Consumers are in 40 accounts across two Regions. The API must be reachable privately, must not traverse the internet, and consumers must not need routes to the provider's VPC or non-overlapping CIDRs.
>
> The provider wants to control which accounts can connect.

What should the solutions architect recommend?

A AWS PrivateLink: an endpoint service behind an NLB, with allowed principals

B VPC peering from each consumer VPC to the provider VPC

C Transit Gateway attachments for all 41 VPCs

D An internet-facing ALB restricted by security group

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A PrivateLink endpoint service

One-way access to a single service, no routing between VPCs, overlapping CIDRs are fine, and allowed principals control who may connect.

B VPC peering

Requires non-overlapping CIDRs and routes on both sides, which the requirement explicitly excludes. Also 40 peerings to manage.

C Transit Gateway

Provides network-level routing between all attached VPCs, which is far more access than publishing one API requires.

D Internet-facing ALB

Traverses the internet, which the requirement forbids.

THE SIGNAL

"No routes and overlapping CIDRs are acceptable" is the PrivateLink signature. It connects services, not networks. Module 09.

> Finance cannot determine which of 30 business units is responsible for a $180,000 monthly increase. Resources carry a BusinessUnit tag applied by every team since the organisation began. Cost Explorer does not offer the tag as a grouping dimension.
>
> The increase spans EC2 and data transfer.

What is the MOST likely cause, and what should be done?

A Cost Explorer cannot group by tags; use the Cost and Usage Report instead

B The tag has not been activated as a cost allocation tag; activate it, noting that activation is not retroactive

C Tag policies must be enabled before tags appear in billing

D The tag must be applied in the management account to be visible

◆ PAUSE AND DECIDE

What is the MOST likely cause, and what should be done?

A Cost Explorer cannot group by tags

It can, once the tag is activated. The CUR offers more granularity but the same activation prerequisite applies.

B Not activated; activate it

A tag exists on resources but is invisible to billing until activated in the management account, and activation applies only from that point forward.

C Tag policies required

Tag policies standardise tag keys and values. They are unrelated to billing visibility.

D Must be applied in the management account

Tags are applied to resources wherever they live. Activation happens centrally; application does not.

THE SIGNAL

The painful half is that activation is not retroactive, so this month can be attributed going forward but not backwards. Module 11.

A company is deploying Control Tower into an existing organisation with 120 accounts. Several accounts already have their own CloudTrail trails, Config recorders and custom SCPs. Leadership wants a standard landing zone without disrupting running workloads.

Some accounts are in Regions Control Tower will not govern initially.

What is the correct expectation to set?

A Existing accounts can be enrolled into Control Tower, but enrolment requires meeting prerequisites and will apply the landing zone's controls

B Control Tower can only be deployed into a new organisation; the accounts must be migrated

C Control Tower will automatically remove conflicting SCPs and trails without action

D Control Tower governs all Regions automatically, so ungoverned Regions are not a concern

◆ PAUSE AND DECIDE

What is the correct expectation to set?

A Enrolment with prerequisites

Existing accounts enrol, and enrolment has prerequisites and applies the landing zone's controls. Setting that expectation is the architect's job.

B New organisations only

Control Tower can be set up on an existing organisation and can enrol existing accounts.

C Conflicts removed automatically

Conflicting configuration is something you must reconcile. Assuming otherwise is how enrolment fails midway.

D All Regions governed

Regions are governed selectively. Ungoverned Regions are a real gap and a common audit finding.

THE SIGNAL

Ungoverned Regions are the detail most people miss. Resources there are outside the landing zone's controls entirely. Module 03.

A central networking account owns subnets that application teams in 25 accounts must deploy into. The networking team must retain sole control of route tables, NAT gateways and network ACLs. Application teams must see and manage only their own instances.

Each team must be billed for its own resources.

What should the solutions architect recommend?

A Cross-account IAM roles granting teams full VPC access in the networking account

B A VPC per team, peered to the networking account's VPC

C VPC sharing through AWS RAM: the owner keeps the VPC and network configuration; participants deploy resources into shared subnets

D Transit Gateway with a shared services VPC per team

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Cross-account roles

Teams work inside the networking account, so resources are billed there and teams can see each other's work.

B VPC per team

25 VPCs, 25 sets of route tables and NAT gateways. The networking team loses the central control the requirement demands.

C VPC sharing via RAM

Participants deploy into shared subnets and see only their own resources; the owner alone manages route tables, NAT and ACLs. Costs land in each participant's account.

D TGW plus per-team VPCs

Same objection as B, with a Transit Gateway added. It multiplies rather than centralises network configuration.

THE SIGNAL

"Owner keeps the plumbing, participants deploy into it" is the VPC-sharing signature, and billing lands in the participant account. Module 05.

> A regulator requires that customer data is never processed or stored outside two European Regions, and asks for continuous evidence. Engineers currently have access to all Regions. A previous proposal to deploy a Config rule that deletes non-compliant resources was rejected.
>
> The company uses AWS Organizations.

What should the solutions architect implement?

A. An SCP denying actions where aws:RequestedRegion is outside the two Regions, plus a Config rule for evidence

B. IAM policies on each role restricting Regions, plus Security Hub

C. Disable all other Regions in every account

D. A tag policy requiring a Region tag on all resources

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A SCP plus Config rule

The SCP refuses the API call so nothing is created elsewhere; the Config rule supplies the continuous record. "Never" plus "evidence" is always this pair.

B IAM policies per role

Applies only to roles you remember to update, and any new role starts unrestricted. The ceiling is in the wrong place.

C Disable other Regions

A genuine control, applied per account rather than organisation-wide, and it produces no compliance evidence.

D Tag policy

Standardises tag keys and values. It does not prevent a resource being created in another Region.

THE SIGNAL

The rejected proposal is a hint, not decoration: detect-and-delete was refused, so the answer must be preventive. Modules 02 and 25.

An organisation reports Savings Plans utilisation of 99% and coverage of 38%. Compute usage has been flat for eighteen months and is forecast to remain so. Finance concludes the strategy is optimal and proposes no change.

The uncovered usage runs continuously in production.

What should the solutions architect advise?

A Agree; 99% utilisation indicates the commitment is correctly sized

B Increase the commitment: 62% of eligible usage is paying On-Demand rates unnecessarily

C Reduce the commitment to improve flexibility

D Move the uncovered usage to Spot Instances

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Agree, 99% is optimal

Utilisation of 99% says the commitment you bought is fully used. It says nothing about the 62% running at full price beside it.

B Increase the commitment

Flat, continuous, forecast to continue: exactly the profile a commitment exists for, and 62% of it is undiscounted.

C Reduce it

Backwards. High utilisation with low coverage means the commitment is too small.

D Spot

Continuous production workload is the least suitable candidate for interruption.

THE SIGNAL

Two metrics, opposite directions. Reading utilisation alone is how a good number produces a wrong conclusion. Modules 11 and 32.

> Developers in a sandbox OU must be able to create IAM roles for their applications, but must never be able to create a role that grants administrative access, and must not be able to remove the restriction from roles they create.
>
> An SCP already denies iam:CreateUser across the OU.

What should the solutions architect implement?

A. An SCP denying iam:CreateRole in the sandbox OU

B. A permissions boundary that developers must attach, enforced by an IAM condition requiring the boundary on role creation

C. A session policy applied when developers assume their role

D. An RCP on the IAM service in the sandbox OU

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Deny CreateRole

Removes the ability the requirement says developers must have.

B Enforced permissions boundary

The classic delegation pattern: developers create roles, the boundary caps what those roles can do, and the condition stops them creating unbounded roles.

C Session policy

Limits the developer's own session. It does not constrain roles they create for applications.

D RCP on IAM

RCPs govern access to resources from outside. This is about capping what a created principal may do.

THE SIGNAL

"Must be able to create roles, but the roles must be capped" is the permissions boundary signature. Module 02.

A security team wants GuardDuty, Security Hub and Config enabled in all 200 accounts and in every account created in future, with findings visible in one place. They must not have to log into member accounts to enable anything, and the management account must not be used for daily operations.

The company uses AWS Organizations.

What should the solutions architect recommend?

A Enable each service manually and document the runbook for new accounts

B Enable each service from the management account and operate from there

C Use CloudFormation StackSets to enable each service per account

D Register a security account as delegated administrator for each service, and enable auto-enable for new accounts

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Manual plus a runbook

200 accounts by hand, and a runbook depends on someone following it for every new account.

B Operate from the management account

The requirement explicitly excludes daily operations there, and it is the account with the widest blast radius.

C StackSets

Would deploy configuration, and these services have native organisation integration that handles aggregation and auto-enable properly.

D Delegated administrator plus auto-enable

Central enablement without touching member accounts, findings aggregate to the security account, and new accounts are covered automatically.

THE SIGNAL

"Not the management account for daily operations" is the delegated administrator signature, and it appears constantly. Module 10.

> Two teams need access to the same DynamoDB table in a shared data account. Team A is in one account and needs read access; Team B is in another and needs read and write. The table is encrypted with a customer managed KMS key.
>
> Team A can assume the role but every request fails with an access denied error.

What is the MOST likely cause?

A Team A's account needs to be added to the organisation

B The DynamoDB table needs a resource-based policy

C The KMS key policy does not grant Team A's role permission to use the key

D The role's trust policy is missing Team A's account

◆ PAUSE AND DECIDE

What is the MOST likely cause?

A **Add to the organisation**

Cross-account access does not require shared organisation membership.

B **Resource-based policy needed**

The role already exists in the data account and can be assumed. Access is being denied after assumption.

C **KMS key policy**

Encrypted data needs both data-access and key-use permission. The key policy is the third grant people forget in cross-account access.

D **Trust policy**

The scenario states assumption succeeds. A trust policy failure would prevent assuming the role at all.

THE SIGNAL

"Can assume the role but requests fail" localises the fault after assumption, and with a CMK the key policy is the usual culprit. Module 05.

Where to revisit

IF YOU MISSED	REWATCH
Q1, Q11, Q13	Module 02 - Guardrails: SCPs, RCPs, boundaries
Q9	Module 03 - Control Tower and landing zones
Q2	Module 04 - Identity at scale
Q10, Q15	Module 05 - Cross-account access and sharing
Q3, Q4, Q5, Q7	Modules 06-09 - Connectivity, DNS, private access
Q6, Q8, Q12, Q14	Modules 10-11 - Org-wide logging and cost

ON THE EXAM

12 or more out of 15 is on track for this domain. Below 9, rewatch before attempting the next set.

MODULE 46 · EXAM STRATEGY - ALL DOMAINS

Practice Exam: Domain 2

Fifteen questions on designing new solutions. The heaviest domain, at 29%.

DOMAIN 2 of 4 EXAM WEIGHT 29% SET 2 of 5

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> A shared networking stack creates a VPC and subnets. Thirty application stacks, owned by different teams, need the VPC and subnet IDs. Teams deploy on their own schedules and must not be blocked by, or able to affect, the networking stack's deployments.
>
> The networking team updates its stack roughly monthly.

How should the application stacks obtain the IDs?

A Make the application stacks nested stacks of the networking stack

B Export the values from the networking stack and import them with Fn::ImportValue

C Hard-code the IDs in each application template

D Pass the IDs as parameters supplied manually at each deployment

◆ PAUSE AND DECIDE

How should the application stacks obtain the IDs?

A Nested stacks

Nested stacks share one lifecycle, so every application deploys when the parent deploys. That is precisely the coupling the requirement forbids.

B Export and ImportValue

Independent lifecycles with a real dependency. Note the trade: an exported value cannot be changed while it is imported.

C Hard-code

Thirty templates to edit whenever anything changes, and no relationship CloudFormation can protect.

D Manual parameters

Works and depends on a human supplying the right value thirty times. Nothing prevents a wrong one.

THE SIGNAL

"Deploy on their own schedules" is the divider. Nested means one lifecycle; exports mean many. Module 12.

A new ECS service on Fargate must deploy with zero downtime, shift traffic gradually, and roll back automatically within two minutes if the 5xx rate rises. The team wants to avoid managing additional deployment tooling.

The service sits behind an Application Load Balancer.

What should the solutions architect recommend?

A Rolling update with a longer health check grace period

B Blue/green through CodeDeploy, with a manual rollback runbook

C ECS built-in blue/green deployments with a CloudWatch alarm configured as a rollback trigger

D Deploy to a second cluster and switch with a weighted Route 53 record

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Rolling update

Replaces tasks in place, so a bad version is partially live and rollback means another rolling deployment.

B CodeDeploy with a manual runbook

Manual rollback cannot meet two minutes reliably, and it adds the extra tooling the team wanted to avoid.

C ECS native blue/green with alarm rollback

Native since 2025, with canary and linear shifting, and an alarm trigger reverses automatically. AWS now recommends it for new ECS deployments.

D Second cluster plus DNS

DNS caching means you cannot guarantee when a shift or a rollback takes effect.

THE SIGNAL

Automatic rollback within a stated time always means an alarm trigger, never a runbook. Module 13.

> A nightly genomics pipeline processes 40,000 independent tasks. Individual task failures can be retried without consequence, and the pipeline has a twelve-hour window to complete a four-hour workload. Cost is the primary concern.
>
> The workload runs on Linux and is architecture-agnostic.

What should the solutions architect recommend?

A. An EC2 Auto Scaling group of Spot Instances using the price-capacity-optimized allocation strategy, across several instance types

B. On-Demand Instances with a three-year Compute Savings Plan

C. Spot Fleet with the lowest-price allocation strategy

D. Reserved Instances sized to peak nightly demand

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Spot via ASG, price-capacity-optimized

Fault-tolerant, flexible timing, architecture-agnostic: the Spot profile exactly. Price-capacity-optimized is AWS's recommended strategy, and Spot Fleet is legacy.

B On-Demand with a Savings Plan

Commits for three years to a workload that runs four hours a night and tolerates interruption. Spot is far cheaper here.

C Spot Fleet, lowest-price

Two dated choices. AWS labels Spot Fleet legacy and advises against lowest-price, which chases cheap pools with poor capacity.

D Reserved Instances

Pays for capacity around the clock for a workload using it four hours a night.

THE SIGNAL

Option C is right about Spot and wrong on both details. The strategy and the service have both moved on. Module 14.

A company is designing a new container platform. It has four years of Helm charts, custom operators and Kubernetes-native tooling, and its platform team is deeply skilled in Kubernetes. A consultant recommends ECS on the grounds that it has lower operational overhead.

The workloads are standard long-running services.

What should the solutions architect advise?

A Adopt ECS, since lower operational overhead is the deciding factor

B Adopt ECS but keep Kubernetes on-premises for the operators

C Adopt App Runner, which removes container orchestration entirely

D Adopt EKS, because the existing Kubernetes investment and skills outweigh ECS's operational simplicity

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Adopt ECS

True in general and wrong here. Helm charts and custom operators do not run on ECS, so the whole investment is discarded.

B ECS plus on-premises Kubernetes

Two platforms, two operational models, and a hybrid estate created deliberately to avoid using EKS.

C App Runner

Also in maintenance since March 2026, and it cannot run custom operators at all.

D Adopt EKS

Existing artefacts and skills change the answer. Four years of Helm charts and operators transfer to EKS unchanged.

THE SIGNAL

ECS is simpler in the abstract. Existing Kubernetes artefacts and skills are the standard exception, and they appear often. Module 15.

A new application on Windows Server instances requires a shared file system accessible over SMB, with Active Directory integration for per-user access control and Windows ACL support. It must be available across two Availability Zones.

Throughput requirements are modest.

Which storage service should be used?

A Amazon EFS with a mount target in each AZ

B Amazon FSx for Windows File Server, Multi-AZ

C Amazon FSx for Lustre with an S3 data repository

D An EBS Multi-Attach volume shared between instances

◆ PAUSE AND DECIDE

Which storage service should be used?

A **EFS**

EFS speaks NFS, not SMB, and does not provide Windows ACLs or native AD integration.

B **FSx for Windows File Server**

Native SMB, Windows ACLs and AD integration, with a Multi-AZ option. Every stated requirement maps directly.

C **FSx for Lustre**

A high-performance parallel file system for HPC and analytics. Wrong protocol and wrong problem.

D **EBS Multi-Attach**

Attaches one volume to several instances in one AZ, and needs a cluster-aware file system. It is not a shared file service.

THE SIGNAL

SMB plus Active Directory plus Windows ACLs is the FSx for Windows signature, and all three appear together. Module 16.

> A new mobile backend needs single-digit millisecond reads on key lookups, unpredictable traffic that may go from zero to 50,000 requests per second within minutes, and no capacity planning or patching. The access pattern is a single lookup by user ID.
>
> The team has no database administrators.

What should the solutions architect recommend?

A Amazon RDS for PostgreSQL with read replicas

B Aurora Serverless with a read-only endpoint

C Amazon DynamoDB in on-demand capacity mode

D Amazon ElastiCache for Valkey in front of RDS

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A RDS with read replicas

Relational capability the access pattern does not need, plus instance sizing and replica management for a team with no DBAs.

B Aurora Serverless

The strongest distractor. It scales and it is still a relational engine sized in ACUs, for a pure key lookup.

C DynamoDB on-demand

Single-key access, single-digit millisecond reads, instant scaling with no capacity planning, nothing to patch. On-demand is now AWS's recommended default.

D ElastiCache plus RDS

Two systems to operate and a cache-coherence problem, to serve an access pattern DynamoDB handles natively.

THE SIGNAL

"A single lookup by user ID" tells you no relational capability is needed. That sentence is the whole question. Module 17.

> An order system must process events for each customer strictly in the order they were produced, and must never process the same event twice. Events for different customers may be processed concurrently, and throughput is around 2,000 messages per second.
>
> Consumers are Lambda functions.

What should the solutions architect recommend?

A. An SQS FIFO queue, using the customer ID as the message group ID

B. An SQS standard queue with a deduplication table in DynamoDB

C. An SNS standard topic fanning out to Lambda

D. An EventBridge bus with a rule per customer

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A FIFO with customer ID as group ID

Ordering and exactly-once are guaranteed within a message group, and separate groups process in parallel. Both requirements, one setting.

B Standard queue plus dedupe table

Handles duplicates and not ordering, and builds a mechanism FIFO provides natively.

C SNS standard topic

SNS offers neither ordering nor exactly-once delivery in its standard form.

D EventBridge rule per customer

EventBridge does not guarantee ordering, and a rule per customer does not scale to a real customer base.

THE SIGNAL

The message group ID is what makes per-customer ordering and concurrency compatible. Without it, FIFO serialises everything. Module 18.

A new payments platform requires an RTO of 15 minutes and an RPO of 5 minutes in a second Region. The business will fund standing infrastructure but wants to avoid running full production capacity twice. Failover must not depend on launching large numbers of instances during a Regional event.

The database is Aurora.

Which strategy fits?

A Backup and restore, with cross-Region snapshot copies

B Pilot light: database replicating, compute defined but not running

C Multi-site active/active across both Regions

D Warm standby: a scaled-down but running copy, with Aurora Global Database

◆ PAUSE AND DECIDE

Which strategy fits?

A Backup and restore

Hours of RTO. It is an order of magnitude away from fifteen minutes.

B Pilot light

The strong near miss. RPO is fine, and recovery depends on launching compute during the event, which the scenario rules out.

C Multi-site active/active

Meets the objectives and means running full production capacity twice, which the business explicitly declined to fund.

D Warm standby with Aurora Global

Something is always running, so failover means scaling up rather than creating. Aurora Global Database gives roughly one-second RPO.

THE SIGNAL

"Must not depend on launching instances during the event" is a static-stability requirement, and it eliminates pilot light. Modules 19 and 30.

A new application connects to Amazon RDS for PostgreSQL. Security requires that the database password rotates every 30 days, that no developer ever sees it, and that rotation does not require an application deployment or restart.

The application is written in Python.

What should the solutions architect recommend?

A Store the password in Parameter Store as a SecureString and rotate it with a scheduled Lambda

B Use AWS Secrets Manager with managed rotation for RDS, and retrieve the secret at runtime

C Store the password in an encrypted environment variable, redeployed monthly

D Use IAM database authentication with a token refreshed every 15 minutes

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A. Parameter Store plus custom Lambda

Parameter Store has no built-in rotation, so you write and maintain the rotation logic Secrets Manager provides.

B. Secrets Manager with managed RDS rotation

Managed rotation changes the password in the database and the secret together. Runtime retrieval means no redeploy and no developer ever sees it.

C. Environment variable

Requires a deployment to rotate, which the requirement forbids, and the value is visible to anyone who can read the configuration.

D. IAM database authentication

A genuinely strong option that eliminates the password entirely, and it has connection-rate limits and changes how the application authenticates.

THE SIGNAL

Rotation without redeployment is the divider between Secrets Manager and Parameter Store. One has it built in. Module 20.

> A new multiplayer game server uses UDP. Players worldwide report inconsistent latency and long recovery when a Regional endpoint degrades. The company needs static entry-point IP addresses for allowlisting on corporate networks, and failover in seconds.
>
> Content is dynamic and not cacheable.

What should the solutions architect recommend?

A. Amazon CloudFront with a custom origin

B. Route 53 latency-based routing with health checks

C. AWS Global Accelerator with endpoint groups in each Region

D. An Application Load Balancer with cross-zone load balancing

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A CloudFront

Caches HTTP content. It does not carry UDP, and the scenario says the content is not cacheable anyway.

B Route 53 latency routing

The near miss. It directs to the nearest healthy Region, and failover waits for DNS TTLs rather than taking seconds.

C Global Accelerator

Any protocol including UDP, two static anycast IPs for allowlisting, traffic on the AWS backbone, and failover in seconds without DNS.

D ALB with cross-zone

Regional and HTTP-only. It addresses neither the global latency nor the UDP requirement.

THE SIGNAL

Three signals point the same way: UDP, static IPs, and seconds. Any one of them rules CloudFront and DNS failover out. Module 21.

A new data pipeline in private subnets will read and write roughly 90 TB per month to Amazon S3 in the same Region. The architecture routes all outbound traffic through NAT gateways. A cost review flags NAT data processing charges as the largest line item in the design.

The pipeline also calls a small number of third-party APIs.

What is the MOST effective change?

A Add a gateway VPC endpoint for S3, so that S3 traffic bypasses the NAT gateways entirely

B Add an interface VPC endpoint for S3

C Move the pipeline into public subnets with public IP addresses

D Compress objects before writing them to S3

◆ PAUSE AND DECIDE

What is the MOST effective change?

A Gateway endpoint for S3

Gateway endpoints cost nothing and remove 90 TB a month from NAT data processing. The third-party calls still use the NAT.

B Interface endpoint for S3

Also bypasses the NAT, and charges hourly plus per GB. A gateway endpoint does the same job for free.

C Public subnets

Removes NAT charges by removing network isolation. It trades a cost problem for a security one.

D Compression

Worth doing and it reduces the volume rather than eliminating the charge. Gateway endpoints remove it entirely.

THE SIGNAL

Gateway endpoints are free; interface endpoints are not. For S3 and DynamoDB in-Region, that is the whole comparison. Modules 09 and 22.

A CloudFormation template creates a production RDS instance. The team must ensure the database survives two situations: the stack being deleted, and a template change that causes CloudFormation to replace the resource rather than update it in place.

A previous incident lost data during a stack update.

What should the template specify?

A DeletionPolicy set to Snapshot

B UpdateReplacePolicy set to Retain

C A stack policy denying updates to the database resource

D Both DeletionPolicy and UpdateReplacePolicy, since each covers a different event

◆ PAUSE AND DECIDE

What should the template specify?

A DeletionPolicy only

Covers stack deletion. A replacement during an update is a different event and is not covered.

B UpdateReplacePolicy only

Covers replacement during update. Deleting the stack would still remove the database.

C Stack policy

Prevents the update rather than protecting the data, and blocks legitimate changes. It also does nothing on stack deletion.

D Both policies

Two events, two attributes. The previous incident was a replacement during update, which is exactly the half people forget.

THE SIGNAL

The scenario names both events deliberately. Setting only DeletionPolicy is the commonest production mistake here. Module 12.

> A new financial application needs a relational database with strongly consistent reads and transactions in its primary Region, low-latency reads for users in three other Regions, and a cross-Region RPO of about one second with failover in under a minute.
>
> The schema is relational with foreign keys.

What should the solutions architect recommend?

A DynamoDB global tables with strongly consistent reads

B Aurora Global Database with secondary Regions for reads

C RDS Multi-AZ with cross-Region read replicas

D Aurora DSQL for active-active multi-Region writes

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A DynamoDB global tables

Not relational, no foreign keys or joins, and global tables replicate with eventual consistency between Regions.

B Aurora Global Database

Relational with transactions, sub-second cross-Region replication, secondary Regions serve local reads, and failover typically under a minute.

C RDS cross-Region replicas

Replication lag is higher and less predictable, and promotion is slower than Aurora Global's managed failover.

D Aurora DSQL

Genuinely impressive for active-active, and it does not support foreign keys, and it is not on the exam's in-scope list.

THE SIGNAL

"Relational with foreign keys" rules out both DynamoDB and DSQL in one phrase. Module 17.

> A new account will run a critical payment Lambda function alongside a batch reporting function that occasionally scales to thousands of concurrent executions. The payment function must always have capacity available, and the batch function must never be able to consume the whole account's concurrency.
>
> Cold starts are not a stated concern.

What should the solutions architect configure?

A Provisioned concurrency on the payment function

B A concurrency quota increase for the account

C Reserved concurrency on both functions

D SnapStart on the payment function

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Provisioned concurrency

Solves cold-start latency, which the scenario explicitly sets aside. It does not stop the batch function consuming the pool.

B Quota increase

A bigger shared pool that the batch function can still exhaust. It raises the ceiling without partitioning it.

C Reserved concurrency on both

Reserved is a ceiling and a floor at once: it caps the batch function and guarantees the payment function its share.

D SnapStart

Also a cold-start feature, and it is limited to Java, Python and .NET. No runtime is stated.

THE SIGNAL

"Cold starts are not a stated concern" is planted to eliminate the two latency answers. Reserved is the contention answer. Module 15.

> A company runs its production ERP on-premises and intends to keep it there. It needs a disaster recovery capability in AWS with an RPO measured in seconds and an RTO under 20 minutes, testable without disrupting production.
>
> A colleague has suggested AWS Application Migration Service.

What should the solutions architect advise?

A. Use AWS Elastic Disaster Recovery, since production stays on-premises and this is DR rather than migration

B. Use Application Migration Service as suggested, since the replication is identical

C. Use AWS Backup with cross-Region copies

D. Rehost the ERP in AWS, then use Multi-AZ for resilience

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Elastic Disaster Recovery

DRS protects production that stays put, giving RPO in seconds and RTO of 5-20 minutes, with non-disruptive drills. MGN migrates permanently.

B MGN as suggested

The replication technology is largely shared, and MGN is built for a one-way permanent move, not ongoing protection and repeated failback.

C AWS Backup

Hours of RPO and RTO. It is an order of magnitude away from seconds and twenty minutes.

D Rehost the ERP

The scenario states production stays on-premises. This answers a question nobody asked.

THE SIGNAL

MGN and DRS share most of their technology and answer opposite questions. Ask whether production is moving or staying. Modules 19 and 35.

Where to revisit

IF YOU MISSED	REWATCH
Q1, Q12	Module 12 - Infrastructure as code
Q2	Module 13 - Deployment strategies and rollback
Q3, Q4, Q14	Modules 14-15 - Compute, containers and serverless
Q5, Q6, Q13	Modules 16-17 - Storage and databases
Q7, Q8, Q15	Modules 18-19 - Integration and continuity
Q9, Q10, Q11	Modules 20-22 - Security, performance and cost

ON THE EXAM

Domain 2 is 29% of the exam, the largest single block. 12 or more out of 15 is on track; below 9, rewatch before continuing.

MODULE 47 · EXAM STRATEGY - ALL DOMAINS

Practice Exam: Domain 3

Fifteen questions on improving what already exists.

DOMAIN 3 of 4 EXAM WEIGHT 25% SET 3 of 5

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A connector on-premises publishes a custom heartbeat metric to CloudWatch every minute while it is healthy. An alarm is configured to fire if the metric drops below one. The connector's host lost power for nine hours overnight and the alarm never fired.

The threshold and evaluation period were both reviewed and are correct.

What should the solutions architect change?

A Increase the number of evaluation periods

B Publish the metric at high resolution

C Set the alarm's missing data treatment to breaching

D Add a second alarm on the same metric with a lower threshold

◆ PAUSE AND DECIDE

What should the solutions architect change?

A More evaluation periods

Evaluating over longer still evaluates nothing. The absence of data points is the problem, not their number.

B High resolution

Publishes more often while healthy. When the host is off, there is nothing to publish at any resolution.

C Missing data treated as breaching

With no host there is no data point, so the alarm sits in INSUFFICIENT_DATA and never pages. Breaching makes absence the alarm condition.

D Second alarm, lower threshold

A second alarm on the same absent metric behaves identically. Two silences instead of one.

THE SIGNAL

A metric that exists only while something is healthy disappears exactly when you need it. Default treatment is the trap. Module 23.

> An Auto Scaling group must scale on memory utilisation, because the workload is memory-bound and CPU stays below 25% throughout. The team enabled EC2 detailed monitoring and still cannot find a memory metric to build a scaling policy on.
>
> The instances run Amazon Linux 2023.

What is required?

A Install and configure the CloudWatch agent to publish memory as a custom metric

B Enable Compute Optimizer to expose memory metrics

C Switch the scaling policy to a target tracking policy on CPU

D Enable Container Insights on the Auto Scaling group

◆ PAUSE AND DECIDE

What is required?

A CloudWatch agent

Memory lives inside the guest OS, which the hypervisor cannot observe. Only an agent on the instance can publish it.

B Compute Optimizer

Makes rightsizing recommendations and can consume memory data. It does not generate the metric, and it is not a scaling source.

C Scale on CPU

The scenario says CPU stays below 25% while memory is the constraint. Scaling on CPU would never trigger.

D Container Insights

For ECS and EKS container metrics. Nothing suggests containers are involved.

THE SIGNAL

Detailed monitoring buys frequency, not coverage. It changes five minutes to one and adds no new metric. Module 23.

A checkout API has a p99 latency of six seconds against a one-second target. The request passes through eight microservices and two third-party calls. Every owning team reports that its own service's average latency is well within target, and three incident calls have ended without a cause.

The services are instrumented with nothing beyond CloudWatch metrics.

What should the solutions architect implement?

A. A CloudWatch dashboard combining all eight services' latency metrics

B. Distributed tracing with AWS X-Ray, instrumented through ADOT, and review the service map

C. Detailed monitoring on every service's instances

D. Increased log verbosity across all eight services

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Combined dashboard

The strong near miss. Eight averages on one screen still cannot connect the slow requests to one another.

B X-Ray tracing via ADOT

A trace follows one slow request through every hop, including the third parties, and attributes the milliseconds. Only tracing answers where.

C Detailed monitoring

Higher resolution on the same per-service averages, which every team already reports as healthy.

D More logging

Eight log streams with no shared request identifier. Correlating one journey stays manual.

THE SIGNAL

Healthy averages plus a bad p99 is not a contradiction. Each service is slow for a small, different subset of requests. Module 24.

> A retailer's checkout failed for two hours at 03:00. Every internal health check passed: instances healthy, target groups healthy, database responsive. The cause was an expired certificate on a third-party payment script loaded by the browser. Customers reported it at 08:00.
>
> The company already runs CloudWatch RUM.

What would have detected this?

A More aggressive ALB health check thresholds

B CloudWatch RUM, which is already deployed

C X-Ray tracing on the checkout service

D A CloudWatch Synthetics canary completing a checkout every five minutes

◆ PAUSE AND DECIDE

What would have detected this?

A Tighter health checks

Health checks probe your targets, which were genuinely healthy. No threshold detects a broken third-party asset.

B RUM, already deployed

The planted trap. RUM would catch this class of fault, and it needs real users. At 03:00 there were none, which is why it did not.

C X-Ray

Traces server-side request paths. A certificate failing in the browser never reaches your services.

D A checkout-flow canary

Runs on a schedule regardless of traffic, exercises the full journey from outside, and would have failed within five minutes.

THE SIGNAL

"RUM is already deployed" is planted to tempt you. 03:00 is the word that rules it out: real user monitoring needs real users. Module 24.

> An audit finds 512 resources that an AWS Config remediation rule should have corrected over the past four months. The remediation runbook has been failing since an IAM permission change, and no alert was raised. The Config console showed the rule as configured with automatic remediation throughout.
>
> Weekly compliance reports were being reviewed.

What should the solutions architect implement?

A. An EventBridge rule matching SSM Automation executions with status Failed or TimedOut, routed to an SNS topic

B. An increased remediation retry count and retry interval

C. Daily rather than weekly compliance report reviews

D. AdministratorAccess on the remediation role to prevent permission gaps

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Alarm on failed automations

Closes the fourth step of the remediation loop. A failing runbook becomes visible immediately rather than at the next audit.

B More retries

Retrying a call the role has no permission to make fails identically, more often. The failure is deterministic.

C Daily reviews

The console showed the rule as configured throughout, which is why nobody noticed. Reading a configuration view more often changes nothing.

D AdministratorAccess

Closes this gap by removing least privilege from something that modifies production resources.

THE SIGNAL

Detect, route, act, verify. The fourth step is the one teams skip, and it fails silently for months. Module 25.

A bank uses Session Manager successfully. A new policy requires zero standing privileges to production nodes: every connection must be explicitly approved and time-bound at the moment of access. Non-production access must remain frictionless, and operators must not wait for approval there.

The estate includes on-premises managed nodes.

What should the solutions architect implement?

A. A permissions boundary limiting operators to non-production instances

B. Remove ssm:StartSession from all policies and grant it manually on request

C. Just-in-time node access, with approval policies requiring human approval for production and auto-approval elsewhere

D. A shorter Session Manager idle timeout on production

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Permissions boundary

Removes production access entirely rather than gating it. Operators legitimately need it sometimes.

B Manual IAM grants

Reaches the right outcome by the worst route: a human editing IAM at 3am, with no timeout, policy or audit structure.

C JITNA with per-environment policies

Approval policies can deny, auto-approve or require a human, so production is gated and non-production stays frictionless. Covers on-premises nodes too.

D Shorter idle timeout

Ends idle sessions sooner. The standing permission to start a new one at any moment is untouched.

THE SIGNAL

Two requirements pulling against each other, resolved by one mechanism with different policies per environment. Module 26.

> Newly launched instances in a private subnet do not appear as managed nodes. The AMI includes the SSM Agent and the process is running. The instance profile has AmazonSSMManagedInstanceCore attached. There is no NAT gateway in the VPC. Identically configured instances in a public subnet register normally.
>
> Security will not permit a NAT gateway.

What is required?

A Attach AdministratorAccess to the instance profile

B Create VPC interface endpoints for ssm, ssmmessages and ec2messages

C Update the SSM Agent to the latest version

D Enable Default Host Management Configuration for the account

◆ PAUSE AND DECIDE

What is required?

A AdministratorAccess

The managed instance core policy is exactly what AWS provides for this. More permission does not create a network path.

B Three interface endpoints

The agent connects outbound to Systems Manager. With no NAT and no endpoints, a private subnet has no route at all. Note there are three, not one.

C Update the agent

The identical AMI registers in a public subnet, so the agent version is not the variable.

D Default Host Management

An alternative way to grant permissions. The instances already have an instance profile.

THE SIGNAL

Same AMI, same role, different subnet: one variable. Missing ssmmessages is the classic partial fix that registers but cannot start a session. Module 26.

> A platform team publishes a hardened golden AMI monthly using EC2 Image Builder. An audit finds that only 44% of running instances were launched from an approved image; the rest use images up to three years old. The company runs AWS Organizations with SCPs already in place.
>
> Teams must still be able to launch instances.

What should the solutions architect recommend?

A. Email the current AMI ID to all teams monthly

B. Delete all non-approved AMIs from every account

C. Rely on Patch Manager to bring older instances up to date

D. An SCP denying ec2:RunInstances unless the AMI carries the approved tag, plus a Config rule detecting unapproved images already running

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Email the AMI ID

Depends on every team reading and acting on an email monthly. The audit shows 56% already do not.

B Delete other AMIs

Breaks running Auto Scaling groups whose launch templates reference them, turning a compliance gap into an outage.

C Patch Manager

Patches the OS and does nothing about the hardening, agents and configuration the golden image carries.

D SCP to prevent, Config rule to detect

Producing a standard and enforcing it are different projects. The SCP stops new non-compliant launches; the Config rule reports what is already running.

THE SIGNAL

The mention of existing SCPs is a hint that the preventive half is available to you. Detect plus prevent, again. Module 27.

An organisation has 180 member accounts. Each has root credentials with a password in a shared vault and an MFA device in a safe. An audit could not locate the MFA devices for 15 accounts, and nobody knew the password for 6. The organisation has a dedicated security account.

The company wants to stop managing root credentials entirely.

What should the solutions architect recommend?

A. Enable centralized root access management, delete member account root credentials, and register the security account as delegated administrator

B. Reset all 180 root passwords and order replacement MFA devices

C. Apply an SCP denying all actions by the root user across the organisation

D. Record all root credentials in a central register reviewed quarterly

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Centralize and delete

Removes 180 sets of credentials rather than managing them. Accounts created afterwards have none by default, and privileged root actions run centrally.

B Reset and re-order

Restores the problem to working order: still 180 passwords, still 180 devices, and the same audit finding next year.

C SCP denying root

SCPs do not apply to the management account, and the credentials still exist to be stolen or misused.

D Central register

Improves record-keeping of credentials that should not exist. Documentation is not a control.

THE SIGNAL

"Stop managing root credentials entirely" is the phrase. The best security improvement is usually deletion, not better custody. Module 28.

> A review finds 74 IAM roles carrying AdministratorAccess, granted between 2019 and 2023. The owning teams refuse removal because they cannot prove nothing would break, and the security team cannot prove the permissions are unused. The company needs to reduce this without causing outages.
>
> CloudTrail has been enabled organisation-wide throughout.

What should the solutions architect recommend?

A. Remove AdministratorAccess from all 74 roles and restore it where things break

B. Apply a permissions boundary to all 74 roles and consider the finding closed

C. Use Access Advisor and IAM Access Analyzer unused access findings to establish what is actually used, then generate least-privilege policies from CloudTrail

D. Leave them in place and monitor with GuardDuty

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Remove and see what breaks

Discovers dependencies through outages, and it is exactly why the teams are refusing, so it will not be agreed.

B Boundary, finding closed

A useful ceiling that does not reduce what the roles are granted. The audit finding stands.

C Measure use, then generate policies

Replaces an argument about intent with evidence about use. CloudTrail has been on throughout, so policy generation has the history it needs.

D Monitor with GuardDuty

Detects misuse after it happens. Seventy-four administrator roles remain, which was the finding.

THE SIGNAL

The teams' objection is reasonable, and evidence answers it. The CloudTrail detail is what makes policy generation possible. Module 28.

A security review of a GuardDuty deployment enabled organisation-wide in 2022 finds no malware findings for S3 uploads and no runtime findings for EC2 or containers, despite both capabilities appearing in current AWS documentation. GuardDuty reports as enabled and healthy in all 90 accounts.

No suppression rules exist.

What is the MOST likely explanation?

A GuardDuty must be disabled and re-enabled to pick up new capabilities

B Protection plans released after GuardDuty was first enabled are not turned on automatically for existing customers

C Those capabilities require Security Hub to be enabled first

D The findings exist but fall below the default severity filter

◆ PAUSE AND DECIDE

What is the MOST likely explanation?

A Disable and re-enable

Would lose finding history and is unnecessary. Plans can be enabled individually at any time.

B New plans are not auto-enabled

Documented behaviour. On first enabling, all plans switch on except Runtime Monitoring; anything launched later stays off, while the console still reports healthy.

C Requires Security Hub

Security Hub aggregates GuardDuty findings. It is not a prerequisite for generating them.

D Severity filtering

Would hide low-severity findings, not whole categories. Malware findings are not low severity.

THE SIGNAL

Enabled and healthy means it is running, not that it is watching everything it could. A 2022 deployment lags by years. Module 29.

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> GuardDuty reports an EC2 instance communicating with a known command-and-control server. The runbook says terminate immediately. Legal requires evidence preserved for possible prosecution, and the response team needs to know what the attacker accessed. The instance has an instance profile with access to three S3 buckets.
>
> The instance is in an Auto Scaling group.

What should the response be?

A Terminate the instance immediately, as the runbook specifies

B Stop the instance and hand it to the forensics team

C Remove the instance profile and leave the instance running

D Detach it from the Auto Scaling group, apply an isolation security group, snapshot the volumes, revoke the role's active sessions, and copy evidence to a forensics account

◆ PAUSE AND DECIDE

What should the response be?

A Terminate

Stops the attack and destroys volumes, memory and every trace. Legal's requirement is not met.

B Stop the instance

The near miss. Volumes survive, and memory is lost the moment it stops, which is where much of the useful evidence lives.

C Remove the profile

Leaves the instance talking to the attacker, and credentials already stolen keep working until they expire.

D Detach, isolate, snapshot, revoke, preserve

Detaching stops the ASG replacing it mid-investigation. Isolation stops the traffic, and revoking sessions handles credentials already taken off the box.

THE SIGNAL

Two details: revoking sessions is what people forget, and detaching from the ASG stops it terminating your evidence. Module 29.

> An Auto Scaling group spans three Availability Zones and is sized so that all three together exactly meet peak demand. The recovery plan for an AZ failure is that Auto Scaling launches replacements in the surviving zones. During a recent AZ event those launches were throttled and the service was degraded for 40 minutes.
>
> The business will fund additional capacity.

What should the solutions architect recommend?

A. Increase the Auto Scaling group's maximum size

B. Add a fourth Availability Zone to the group

C. Pre-provision capacity so that any two zones can serve full demand, making the design statically stable

D. Reduce the health check grace period so replacements come into service sooner

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Raise the maximum**

The group was not at its ceiling. It was throttled, which is a different failure, and a higher maximum does not obtain capacity.

B **A fourth AZ**

Reduces the share lost to one zone, and still requires launching instances during the event. The dependency remains.

C **Pre-provision for N-1**

Nothing needs creating when a zone fails, so the control plane is off the recovery path entirely. That is static stability.

D **Shorter grace period**

Speeds up instances that did launch. The problem was launches being throttled, not being slow to become healthy.

THE SIGNAL

"Throttled" is the giveaway. Every customer's recovery plan fired at once and yours was queuing behind them. Module 30.

> An Aurora PostgreSQL cluster is reported as slow. Writer CPU utilisation sits at 18%. Database load is high, and the breakdown shows most sessions waiting on lock events rather than CPU or I/O. One UPDATE statement dominates the load.
>
> The team has proposed a larger instance class and additional read replicas.

What should the solutions architect recommend?

A Scale to a larger instance class as proposed

B Investigate and fix the contended UPDATE, since sessions are waiting on locks rather than resources

C Add read replicas to spread the load

D Increase the cluster's provisioned IOPS

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Larger instance

More CPU for a database using 18% of what it has. A session waiting on a lock waits exactly as long on a bigger machine.

B Fix the contended UPDATE

Lock waits are contention, not capacity. The wait-event breakdown has already named the statement, so the fix is in the query or the transaction.

C Read replicas

Move read traffic. The contention is on an UPDATE, which only ever runs on the writer.

D More IOPS

The wait events are locks, not I/O. Faster storage does not shorten a wait for another transaction to commit.

THE SIGNAL

Lock contention is the one bottleneck no hardware fixes. That is why wait events matter more than CPU. Module 31.

> A company bought a large three-year Compute Savings Plan in February, sized to its On-Demand spend at the time. In April it completed a rightsizing programme that cut compute consumption by 28%. Savings Plans utilisation has sat at 73% since, and finance is asking why savings are below forecast.
>
> The two projects were run by different teams.

What is the situation, and what should change?

A Over-committed: the plan was sized to pre-rightsizing usage. Size future commitments to the baseline that will remain after planned changes

B Under-committed: buy an additional Savings Plan to raise coverage

C Reverse the rightsizing so utilisation recovers

D Convert the Compute Savings Plan to an EC2 Instance Savings Plan

◆ PAUSE AND DECIDE

What is the situation, and what should change?

A Over-committed; size to the surviving baseline

Utilisation of 73% means 27% of a three-year commitment buys nothing. Two teams optimised independently and the second undercut the first.

B Buy more

Buying more commitment when a quarter of the existing one is unused makes it worse. Low utilisation means too much, not too little.

C Reverse the rightsizing

Runs oversized instances for three years to justify a commitment. It optimises the metric rather than the bill.

D Convert the plan type

Savings Plans cannot be converted between types. Convertible Reserved Instances can be exchanged; Savings Plans cannot.

THE SIGNAL

Two cost programmes run independently subtracted from each other. That coordination failure is the real lesson. Module 32.

Where to revisit

IF YOU MISSED	REWATCH
Q1, Q2	Module 23 - Observability foundations
Q3, Q4	Module 24 - Tracing, canaries and RUM
Q5	Module 25 - Automated remediation
Q6, Q7	Module 26 - Systems Manager operations
Q8, Q9, Q10	Modules 27-28 - Deployment and security posture
Q11, Q12, Q13, Q14, Q15	Modules 29-32 - Threats, reliability, performance, cost

ON THE EXAM

Questions 6, 9 and 11 test capabilities newer than most study material. Missing those is a currency gap, not a knowledge gap.

MODULE 48 · EXAM STRATEGY - ALL DOMAINS

Practice Exam: Domain 4

Fifteen questions on migration. The domain the question banks under-serve.

DOMAIN 4 of 4 EXAM WEIGHT 20% SET 4 of 5

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> Before planning waves for 1,400 servers, a team must establish which servers communicate with which, including dependencies that appear in no documentation. The existing CMDB is five years old. Security has approved installing software on production servers.
>
> The estate is a mix of physical servers and VMware.

What should the solutions architect recommend?

A. Deploy the Application Discovery Service Agentless Collector into vCenter

B. Deploy the Application Discovery Agent to the servers, collecting network connections and running processes

C. Export the CMDB and validate it in workshops with application owners

D. Run Migration Evaluator to produce a dependency map

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Agentless Collector**

Excellent for inventory and utilisation, and it is vCenter-only, so it misses the physical servers entirely and cannot see inside the guest OS.

B **Discovery Agent**

Only the agent sees network connections per process, which is what dependency mapping requires. Agents are permitted, and it covers physical servers too.

C **Validate the CMDB**

Asks people about undocumented dependencies. Undocumented usually means nobody knew, not that nobody wrote it down.

D **Migration Evaluator**

Builds a business case and a TCO model. It answers what it will cost, not what connects to what.

THE SIGNAL

Two details decide it: undocumented dependencies, and a mix including physical servers, which agentless cannot reach. Module 33.

> A board will not approve a migration programme without a comparison of current on-premises total cost against projected AWS cost, including Windows Server and SQL Server licensing options and more than one purchasing scenario. The company already holds three years of inventory and utilisation data from a third-party monitoring product.
>
> The board meets in six weeks.

What should the solutions architect recommend?

A. The AWS Pricing Calculator, sized from the existing inventory

B. Application Discovery Service, then a cost model built in a spreadsheet

C. Migrate a pilot application and extrapolate its cost across the estate

D. Migration Evaluator, importing the existing third-party data, producing TCO with what-if purchasing scenarios

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Pricing Calculator

Prices an AWS design. It does not know the on-premises cost, so it cannot produce the comparison the board asked for.

B Discovery plus a spreadsheet

Rebuilds Migration Evaluator by hand, and a hand-built model carries little external credibility with a board.

C Pilot and extrapolate

One application rarely generalises across a portfolio, and it will not complete within six weeks.

D Migration Evaluator with imported data

Built for this artefact: TCO against current cost, multiple purchasing scenarios, and Windows and SQL licensing analysis. It accepts third-party data.

THE SIGNAL

"Comparison" is the word. Anything that prices only the AWS side answers half the question a board is asking. Module 33.

A team configured Migration Hub and selected us-east-1 as the home Region during setup. Workloads will migrate into eu-west-1 and eu-central-1, and company policy requires all migration metadata to be stored within the EU. Discovery data has already been collected into the existing home Region.

The programme starts in three weeks.

What should the solutions architect advise?

A The home Region cannot be changed; use a different account and re-collect discovery data

B Change the home Region to eu-west-1 in the Migration Hub console

C Migration Hub data replicates automatically to the target Regions, so the policy is met

D Configure a second home Region for the EU

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A It cannot be changed

The home Region is set once per account and is permanent. Meeting the policy means a different account, and re-collecting discovery data.

B Change it in the console

There is no such operation. This assumption is exactly what the question tests, and it is why the choice matters on day one.

C It replicates

Migration Hub data stays in the home Region regardless of where migrations run. That is precisely the problem here.

D A second home Region

One home Region per account, by definition. There is no second.

THE SIGNAL

An irreversible day-one choice made in a minute during setup, costing weeks to undo. Module 33.

> A data centre lease expires in nine months and will not be renewed. There are 920 servers running 240 applications. Every development team is committed to a regulatory programme for the next fourteen months and cannot take on migration work. Leadership has asked for a modernisation-first approach.
>
> The applications currently work and are well understood.

What should the solutions architect recommend?

A. Refactor the applications to cloud-native services during the migration

B. Extend the data centre lease to create time for modernisation

C. Rehost the majority to meet the deadline, then modernise selected applications afterwards in AWS

D. Retain everything on-premises until development teams are available

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Refactor during migration**

Refactoring needs developers, and there are none for fourteen months. Leadership's preference does not create engineering capacity.

B **Extend the lease**

The scenario says it will not be renewed. Changing a stated constraint is not an architectural recommendation.

C **Rehost now, modernise later**

Rehosting needs no application change, so it is unblocked by the developer shortage, and it is the only strategy that fits nine months at this scale.

D **Retain everything**

There will be no data centre in nine months. Retain requires somewhere to retain it.

THE SIGNAL

A preference meets a constraint, and the constraint wins. Saying so is the architect's job. Module 34.

> Discovery across 1,600 servers finds 240 with no inbound or outbound network traffic over 90 days, no interactive logins, and no owner recorded anywhere. Several carry hostnames referencing projects that closed years ago. The migration budget assumes all 1,600 will move.
>
> Some business processes run quarterly.

What should the solutions architect recommend?

A Investigate the 240 for retirement: power off, retain recoverable for a defined window, then delete and remove from scope

B Rehost all 1,600, since migrating is safer than deleting

C Power off the 240 immediately and remove them from scope

D Retain the 240 on-premises indefinitely

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Investigate, then staged retirement

Retire is the cheapest strategy. 15% leaves scope, and the recoverable window covers the quarterly processes 90 days of data would miss.

B Migrate everything

Pays to plan, test, cut over and then run 240 servers nobody uses, indefinitely. Migrating waste preserves it.

C Power off immediately

The right instinct without the discipline. The scenario says some processes run quarterly, which 90 days of observation can miss entirely.

D Retain indefinitely

Keeps them powered on-premises forever. Retain is for workloads you will revisit, not ones nobody can identify.

THE SIGNAL

"Some business processes run quarterly" is planted to separate A from C. Ninety days is not a full business cycle. Module 34.

> A company must vacate its data centre in five months. It runs 380 VMware virtual machines. Its infrastructure team is deeply skilled in vSphere and has no AWS experience, and retraining before the deadline is not realistic. Several applications are vendor-supplied and may not be modified.
>
> The company intends to adopt native AWS services over the following three years.

What should the solutions architect recommend?

A. Refactor all 380 workloads to native AWS services now

B. Relocate to VMware Cloud on AWS to meet the deadline, then modernise into native services over time

C. Rehost all 380 with Application Migration Service and retrain the team in parallel

D. Retain the estate and negotiate a longer lease

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Refactor now**

380 workloads refactored in five months, by a team with no AWS experience, including applications nobody may modify.

B **Relocate, then modernise**

The operating model is unchanged, so the vSphere team stays productive. Five months is achievable, and modernisation follows without a deadline.

C **Rehost and retrain**

The strong alternative. Rehosting meets the deadline, and it hands a vSphere-skilled team 380 EC2 instances to operate from day one.

D **Negotiate the lease**

Changes the constraint rather than working within it, and the exit is presented as fixed.

THE SIGNAL

The team's vSphere skills separate relocate from rehost. Both meet the deadline; only one keeps the team productive. Module 34.

> Following a commercial dispute, a company must move 420 servers from another public cloud provider into AWS within five months. The servers run a mix of Windows and Linux, and the applications may not be modified. The team believes AWS migration tooling only works from on-premises data centres.
>
> Outbound internet access from the source environment is permitted.

What should the solutions architect recommend?

A Export each server as an image, transfer it to S3, and import with VM Import/Export

B Rebuild all 420 servers natively in AWS

C Application Migration Service, installing the replication agent on the source servers in the other cloud

D AWS DataSync to copy the servers into AWS

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Export and VM Import

A point-in-time copy, so the source keeps changing while you transfer, and cutover means a long outage per server.

B Rebuild natively

The applications may not be modified, and 420 rebuilds in five months is not achievable regardless.

C MGN from the other cloud

MGN works anywhere the agent reaches AWS on TCP 443 and 1500, including Azure, Google Cloud, Oracle Cloud and Alibaba. The team's belief is simply wrong.

D DataSync

Moves file and object data. It does not migrate servers, operating systems or applications.

THE SIGNAL

The team's stated belief is the question. MGN is a cloud-to-cloud migration tool as much as an on-premises one. Module 35.

> A weekend cutover of 55 servers failed. Eighteen applications would not start, reporting licence activation errors and unreachable database hosts. The team rolled back after eleven hours. MGN replication had reported healthy for six weeks, and no test instances had ever been launched.
>
> The next cutover window is in four weeks.

What should the solutions architect recommend before the next attempt?

A. Extend the cutover window to a full weekend so there is time to resolve problems

B. Re-install the replication agents and restart replication from the beginning

C. Cut over the 37 applications that worked, and defer the 18 that failed

D. Launch test instances into an isolated subnet, resolve the licence and connectivity failures, and repeat until clean

◆ PAUSE AND DECIDE

What should the solutions architect recommend before the next attempt?

A Longer window

Turns an eleven-hour failure into a longer one. Debugging licence activation under cutover pressure is the worst possible moment.

B Restart replication

Replication was healthy and is not the fault. Restarting discards six weeks of transferred data for no benefit.

C Defer the eighteen

Cuts over 37 servers that were also never tested. The same class of fault is waiting in them.

D Test instances, iterate until clean

Non-disruptive and repeatable with the source still serving users. Hardware-bound licences and unreachable hosts are exactly what a test finds.

THE SIGNAL

"Replication healthy for six weeks" tells you the copying worked and the application never did. Those are separate questions. Module 35.

A team installed MGN replication agents on all 800 in-scope servers during the first month, to get replication established early. Migration waves run monthly, with the final wave completing in month eight. Finance queries an unexpected Application Migration Service charge that first appears in month four.

The team believed the service was free for the duration.

What is the explanation, and what should change?

A The free period is 2,160 hours per server starting at agent installation; install agents per wave, shortly before each wave migrates

B The charge is for staging EC2 and EBS, which cannot be avoided

C The free period applies per account rather than per server

D Request a service credit, since the service is advertised as free

◆ PAUSE AND DECIDE

What is the explanation, and what should change?

A Free period starts at agent install

2,160 hours is 90 days from installation, so agents fitted in month one expire in month four. Install per wave and most servers stay free.

B Staging infrastructure

Staging EC2 and EBS are charged, separately and from day one. A charge first appearing in month four is the free period expiring.

C Per account

The free period is per source server, which is exactly what makes staggered installation worthwhile.

D Request a credit

The terms were met. The service was free for 90 days per server, and the team consumed that on servers it was not yet migrating.

THE SIGNAL

Month four is the arithmetic: ninety days from a month-one install lands exactly there. That is the clue given. Module 35.

A regional facility must move 620 TB of archived imagery to Amazon S3. Its internet connection is 300 Mbps and is shared with operations. The facility closes in twelve weeks. The team has proposed copying the data over the internet using the AWS CLI with multipart upload.

The company is an existing Snow Family customer.

What should the solutions architect advise?

A Proceed with the CLI, using multipart upload for parallelism

B Do the arithmetic: at 300 Mbps the transfer far exceeds the deadline, so use Snow Family devices

C Enable S3 Transfer Acceleration to speed up the upload

D Use DataSync over the existing internet connection

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A CLI with multipart

Parallelism uses the link more efficiently. It does not make the link faster, and the link is the constraint.

B Arithmetic, then Snow

300 Mbps is about 3 TB a day at perfection, so 620 TB is roughly 200 days against an 84-day deadline. Three Snowball Edge devices cover it.

C Transfer Acceleration

Improves throughput over long distances by reaching the AWS backbone sooner. It cannot exceed the facility's 300 Mbps.

D DataSync

The best online tool available, and still bounded by the same 300 Mbps. Better software, same physics.

THE SIGNAL

Three options optimise the transfer; one asks whether it can finish. Note the scenario confirms existing Snow customer status. Module 36.

A media company must move 110 TB from an on-premises NFS share to Amazon FSx for Lustre, then keep the two synchronised for four months while applications cut over in waves. The site has a 10 Gbps Direct Connect link. Integrity must be verified for every file transferred.

An engineer has proposed a nightly rsync job.

What should the solutions architect recommend?

A The nightly rsync job as proposed

B Snow Family devices for the initial 110 TB, then rsync for the ongoing sync

C AWS DataSync, with scheduled incremental tasks and built-in integrity validation

D S3 Transfer Acceleration into S3, then copy across to FSx

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Nightly rsync

Rebuilds DataSync badly: no integrity validation, no managed retries, no throttling, and somebody owns the script for four months.

B Snow then rsync

Ships a box for data that transfers in under two days over the existing link, and still leaves the ongoing sync unsolved.

C DataSync, scheduled incremental

110 TB over 10 Gbps is a day or two. DataSync then handles four months of incremental sync, with validation and retries built in.

D Transfer Acceleration to S3

Adds an S3 hop the requirement does not need, and Transfer Acceleration does not target FSx.

THE SIGNAL

The same arithmetic that chose Snow in Q10 rules it out here. Compare the two: link speed decides, not data size. Module 36.

A company is moving a self-managed PostgreSQL 15 database from EC2 to Amazon RDS for PostgreSQL 15. The database contains 260 stored procedures, 55 triggers and several partitioned tables. A junior architect has scoped ten weeks for schema conversion using AWS SCT.

The database is 1.2 TB and downtime must be under 30 minutes.

What should the solutions architect advise?

A Proceed with the ten-week SCT conversion as scoped

B Rewrite the stored procedures for RDS compatibility

C Use SCT in parallel with a full load plus CDC task

D This is homogeneous, so no schema conversion is required; use a DMS homogeneous migration, which carries secondary objects

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Ten-week conversion

Ten weeks converting PostgreSQL into PostgreSQL. The schema already runs on the target engine unchanged.

B Rewrite the procedures

They are already PostgreSQL procedures. RDS runs the same engine, so there is nothing to make compatible.

C SCT plus CDC

Correct on the data movement, and it still budgets conversion work this migration does not require.

D Homogeneous; no conversion

Same engine means the dialect is identical. DMS homogeneous migrations are serverless and carry procedures, triggers and partitions.

THE SIGNAL

Self-managed to managed is a hosting change, not an engine change. That single distinction saves ten weeks. Module 37.

A 26 TB SQL Server database must move to Amazon RDS for SQL Server. Restoring a native backup to the target takes about five hours, which the business accepts. However, the source will continue receiving writes for three weeks afterwards while an application is validated, and the two databases must stay in step throughout.

The engine and version are unchanged.

What should the solutions architect recommend?

A. A DMS full load plus CDC task instead of the native restore

B. Restore the native backup, then run a DMS CDC-only task for the three weeks

C. A DMS full load task, repeated nightly for three weeks

D. Restore the backup now, and restore it again at the end of the three weeks

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A. Full load plus CDC

Would work, and a DMS full load of 26 TB is slower than the native restore the business has already accepted.

B. Native restore, then CDC-only

CDC-only exists for exactly this: a target populated by other means, kept in step by DMS thereafter. Fastest path for 26 TB.

C. Nightly full loads

Reloads 26 TB every night for three weeks, and the target is stale for most of each day between runs.

D. Restore twice

Leaves the target three weeks out of date throughout validation, which is the requirement it fails.

THE SIGNAL

CDC-only needs a target already populated by something else. Combining native tooling with DMS replication is a real production pattern. Module 37.

A cutover performed at 01:00 updated a Route 53 record to point at AWS. By 10:00 the on-premises servers were still receiving about 55% of traffic, and the two environments had accepted conflicting writes. The record's TTL was 86,400 seconds.

The next cutover is in a month.

What should the solutions architect recommend?

A. Lower the TTL to 60 seconds immediately before the next cutover

B. Shut down the on-premises servers at cutover so traffic has nowhere else to go

C. Lower TTLs to 60 seconds at least a full TTL period before the next cutover, and keep the old environment answering during transition

D. Use a shorter maintenance window so divergence has less time to occur

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Lower it immediately before

The sharpest distractor. Resolvers caching the old record still hold 86,400 seconds, so the change is invisible for up to a day.

B Shut down on-premises

Turns a divergence problem into an outage for 55% of users, and destroys the rollback at the same moment.

C Lower TTLs a full period ahead

Lowering a TTL is itself subject to the old TTL, so it must happen at least one full period in advance to be live at cutover.

D Shorter window

The window was not the problem. Traffic kept arriving for nine hours because of caching, not window length.

THE SIGNAL

Option A is the trap: lowering a TTL is itself a DNS change subject to the old TTL. That recursion makes it a planning task. Module 39.

A pricing service migrated to AWS in wave three now takes five seconds to return a quote, against 400 milliseconds before. It makes roughly 1,500 calls per quote to a rates database that remains on-premises. The Direct Connect link adds about 3 milliseconds each way.

The team proposes upgrading the link from 1 Gbps to 10 Gbps.

What should the solutions architect recommend?

A Migrate the rates database into the same wave as the pricing service, keeping the chatty pair together

B Upgrade the Direct Connect link to 10 Gbps as proposed

C Add a second Direct Connect connection for redundancy

D Increase the pricing service's instance size

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Move the database with it

1,500 calls at 6 ms round trip is about 9 seconds of pure latency. Only removing the crossing fixes it, which means the pair moves together.

B More bandwidth

Bandwidth is throughput, not latency. A wider link does not shorten the distance the packets travel.

C Second connection

Resilience, not speed. Two links of the same latency give the same round trip.

D Bigger instance

The service is waiting on the network, not short of CPU. A larger instance waits at the same speed.

THE SIGNAL

The arithmetic is handed to you: 1,500 calls times 6 ms. Bandwidth and latency are different properties. Module 40.

Where to revisit

IF YOU MISSED	REWATCH
Q1, Q2, Q3	Module 33 - Portfolio assessment and wave planning
Q4, Q5, Q6	Module 34 - The seven Rs
Q7, Q8, Q9	Module 35 - Server and application migration
Q10, Q11	Module 36 - Data migration
Q12, Q13	Module 37 - Database migration
Q14, Q15	Modules 39-40 - Cutover, and hybrid coexistence

ON THE EXAM

Compare Q10 and Q11: identical arithmetic, opposite answers. If you got one and missed the other, that is the lesson.